

THE EMERGENCE OF ENEL OR THE IMMERGENCE OF LANDMARKS

INTRODUCTION TO THE STUDY OF FORMS AND COHERENCE FIELDS

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ARK'ALL

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Leon Sprink

SHAPES AND ELECTRIC FIELDS

by J. RAVATIN

(France)

Léon Sprink was born in Russia at the end of the 19th century, he died in France in 1967. He is buried in the Russian cemetery of Sainte-Geneviève-des-Bois. His family was of Baltic origin. Very young, he was very imaginative and curious about what interested him; it is necessary to announce the observation which it made of a solar eclipse whereas it was in a field belonging to a family property, rather particular eclipse, the solar plasma had been put in the form of cross of Saint-Andrew. His father, before the war of 1914-1918, took him to visit beer factories in Moscow. The beer obtained in these factories was of rather poor quality and the fermentation of the hops required too much time, according to the technicians present. They had no remedy to decrease the fermentation time and increase the quality. The young Sprink, having returned home, asked his father for permission to work on the fermentation of beer. His father gave him the elements and he imagined (he must have been about 16 to 17 years old) forms of fermentation tanks as well as the alloy which was to compose them. His process was applied and the quality of the beer was singularly increased as well as the speed of fermentation of the hops. But during the revolution, in 1917, the crowd entered the factories and smashed these vats with hammer blows.

Anyway, even if the tricks imagined by the young Sprink were lost (we could find equivalent ones now), the beers made in our time are quite good and the fermentation speeds sufficient not to pose the problem again (we should point out that some beers now contain dyes and other chemical products and that it would still be 'better to treat them by a Sprink-type process, this would further enhance their taste and make a very healthy drink). The Russian Revolution did not spare the Sprinks. He was locked up in the "Peter and Paul" prison in Saint Petersburg. There he was tortured. (He has, moreover, in this prison,



Leon Sprink and his wife



Léon Sprink behind one of his cameras

witnessed the agony of a very famous Pope from Russia). It should be noted that he was one of the rare survivors of this prison; indeed there were only 3 or 4 people who were not murdered there. He arrived in France and was declared disabled at 99Z.

He met shortly after a young widow, Marie-Thérèse Roussel with whom he married. She died in 1951 of cancer. She was remarkably intelligent and they began their research work. '

In 1923 L. Sprink was in Strasbourg, in a chemical industry factory: a stearin factory. He was the owner of a process for manufacturing soaps with high detergent power that the factory in question had adopted. There was a research laboratory, modest indeed, but the factory was at his disposal for industrial checks.

The results of the application of these soaps, especially in the raw wool washing factories (in Alsace and in the North of France) were more than satisfactory.

But now, after two years, there were declines and irregularities in performance. First of all, the cause was sought in the difficulties specific to the industrial scale-up of a process established in the laboratory. Time was wasted verifying the process itself as well as its theoretical bases. (*)

(*) What relates to the life of L. Sprink and his wife as well as their attempts has been reported in the notes of the couple as well as their "notebook" which they kept very well throughout of their research career. We will not mention here in detail the scientific personalities whom they touched to make known and share their work, their approaches would be the subject of a volume - hopes and disappointments have followed one another as for all these researchers who worked in non-Orthodox areas. Let us quote the meeting of L. Sprink and A. Einstein which took place in Germany before the departure of the latter from this country. This meeting was memorable: it lasted an afternoon. Einstein proposed to the Sprinks to go to the USA where he was already thinking of going. The Sprinks during the 39-45 war remained in France and worked near Grenoble). Einstein gave to Sprink his thought on his work: it is summarized thus: "I work by decomposing and you (L. Sprink), you work by assembling". Einstein had really seen Sprink's approach and that is what is essential because in this area the approach is different from the usual approach.

This factory still had a boss, born in this factory and master at home. One day he reproached L. Sprink for "looking too much from noon to two o'clock" and he added "You know, manufacturing recipes only last for a short time, without there being any need to modify them".

There we recognize, in this reflection, the point of view always too hasty and too poor in solid bases of understanding of the mechanisms, of the industrialist. It is the contribution of research laboratories (in basic research) that then becomes valuable.

It therefore emerged from this interview that the manufacturing process as such was perhaps not the cause of these yield disturbances. We had to look elsewhere.

At the time, the manufacture of soap in this factory was too recent to establish a scale of comparison of the presumed variations. On the other hand, it was not the chemical analyzes of its composition that could provide information. In the end, there were only the results obtained among users. There, we knew that the soap was more or less good; therefore variable in quality.

This soap was made with fatty acids produced at the factory. L. and MT Sprink therefore began by studying the results of the laboratory control of these products. Immediately, they were presented with the averages per month and per year, as well as the figures for the output of the factory. The averages per month, taken step by step, were not very clear; on the other hand, the averages per year turned out to be much stronger for certain periods with a decline before and after. This was especially striking for the years 1911 and 1912.

Autoclave study :

The first processing operation in a stearin factory is saponification: decomposition of triglycerols into glycerine and fatty acids. In the factory in question, this operation was carried out with autoclaves, under the

2 constant pressure of 10 kg/cm with heating, direct to 180° by steam. It was carried out in two stages: five hours under pressure - a rest - change of water (if not, the concentration of these waters in glycerin would have slowed down the reaction), a rest of twelve hours and a re-pressurization for three hours. This gait was found empirically and gave very good results at the time. After the first five hours of walking and at the

end of the operation, the results were checked in the laboratory. As there were four autoclaves in operation every day, there were 16 index determinations per day. This made it possible to group the saponification indices and the acid indices by four, which initially avoided excessive differences, due to anomalies which could occur due to fortuitous causes. It is for this reason, and also because saponification was closely dependent on the yields of subsequent manufacturing operations, that the Sprinks subjected these data to close study.

Immediately, they had to realize that with the system of averages per month, there was nothing to do.

Faced with this semi-failure, they fell back on statistical methods, while maintaining a certain mistrust of them.

The construction of frequency curves of a discrete numerical series, which gives the number of cases for a given result, their a. obviously provided dispersion curves of the results (Gaussian bell curve).

But the Sprinks have therefore attached themselves to a statistical law to deduce the results to be exploited. It should be noted that their apparatus does not function entirely in a rational manner and that the view advocated by M. Beigbeder [1]: a statistic depends on the mode of reasoning chosen and the statistics used up to the usual field of coherence, was not known to them., the book of M. Beigbeder coming out in 1979. the use of statistics on the operation of his device. On the other hand, he had to get industrialists to accept his results and his manuscript at the Academy of Sciences. On the other hand, a statistical law can be applied to the phenomena studied by the Sprinks, in particular the saponification in a stearinerie This digression was necessary to show the caution with which these phenomena must be approached and to know the field of coherence in which one must place oneself.

The first surprise of L. Sprink and his wife was to note that, in spite of a statistical mean (the most frequent value) and which was by no means what they considered important (which was the considerably lower arithmetic mean) there There was on each curve not only one, but two or three characterized maxima. But what surprised them even more, this was the observation, absolutely unexpected, that from month to month, there

(*) L. Sprink deposited a manuscript at the Academy of Sciences composed of 2 parts. The first was only descriptive and the second an attempt at a theoretical approach involving conformal transformations and propagation results of an electromagnetic field

had as a coming and going of the location of these maxima. In other words, from a statistical point of view, the reaction oscillated more or less far depending on the period of the year. It was then that the idea came to them to lay out the daily results in the form of a curve, where the ordinates were brought in proportion to their numerical value and they finally saw a certain periodicity appear.

Obviously, we knew "by experience" that, for example, the yields of autoclaves varied with the seasons and that yields were better in winter than in summer. We even found for all these irregularities a perfectly plausible explanation by the low winter temperature - autoclaves obviously cool much more than in summer, so there is more steam entering an autoclave in winter than in summer, hence a greater agitation of the materials in reaction and a greater dilution of the glycerinous waters. However, this explanation does not correspond to the real state of the conditions. However, as often happens in similar cases, this had prevented further research.

To begin with, a quick harmonic analysis revealed a period that corresponded to the 28 days of the lunar period. Under these conditions, grouping the results by month obviously could not give anything very clear.

The simple fact that the two consecutive maxima were in reverse order in spring and autumn demonstrated to the Sprinks that their autoclave-related curves followed the rhythm of the tides. But to neutralize the tides, there is no way!

There was still an anomaly that put them on the right track. We have just said that autoclaves generally work better at the winter solstice than at the summer solstice. However, the tidal indices are the same at these two times of the year. It was therefore necessary to find one more variable which was periodic and this period had to be the same as that of the operation of the autoclaves. Among the curves drawn up by them, there was one which related the results obtained from the autoclaves and the hours of their pressurization. The results followed the curve of variations in the earth's electric field. It is known that the field is relatively more intense in winter, in dry and cold weather, and is therefore much more stable than in summer.

This last observation left them very hesitant for a long time.

An autoclave being a completely closed container with walls approximately 50 mm thick in pure copper, it is therefore a Faraday cage. The Sprinks still wondered if it was not another phenomenon that was alone, the real responsible.

Fortunately, they were able to consult the original publications about the Faraday cage. This is very interesting because going back to the sources of the works one finds detailed experiments and also the reflections of their authors. Publications and works were still, fifty years ago, much richer in detail. This "academic" style presentation that we now find makes researchers believe they are working on solid concepts, and even some do not know the reason for certain hypotheses. A hypothesis, an axiom, are often an admission of ignorance and their role is to bring together various intuitions. Knowing these intuitions, bringing them out and plunging them into the midst of one's personal thoughts allows them to be given new life; and also makes it possible to pinpoint the weakness of certain theories, or to feel other approaches. The machines built in the 19th century have for some, come very close to the relocation of their functionality; this is how Tesla even in the twentieth century was able to obtain rather curious effects. He had observed electrical actions which appeared to him outside the usual laws. If we consider electric motors, there were more than a thousand models of electric motors made by multiple inventors and only a few have survived (such as the so-called 'fen squirrel cage') that very few people had imagined strange windings, and then in this case, at the level of the axis of rotation can appear decalaires or channel and emissions with high Taofel fields.

The search for localized principles has resulted in the unorthodox cases being dismissed or forgotten. As it was believed that only the principles attached to localization (the concept of localization and escape from landmarks being ignored) were the only ones making it possible to describe the universe, technical objects were made according to them and the others. objects that were not of the class of technical objects, have disappeared or have remained memories of no interest.

By consulting Faraday's original publications, the Sprinks realized that, as the author himself pointed out: the Faraday cage is only valid for electrical charges in equilibrium.

In an autoclave, heated by steam, this is not the case because there is a continual supply of electrical charges. They have had conclusive proof of this) P. Smirne has developed an alternator, of small volume and weight, whose performances are without common measure with those already existing, all this by carrying out bizarre windings. We will not develop this realization here

ante by removing these fillers inside the autoclave which greatly reduced the degree of saponification.

They therefore placed the autoclaves in an electrostatic field in such a way that its value was large enough to neutralize as far as possible the variations of the earth's field and so that the values which appeared to be values could be exceeded. "threshold" for high efficiency.

The success was overwhelming and spectacular (it happened in 1933-34). After a few months, with the new regime, it was possible for them to draw some provisional conclusions.

1) From the point of view of the quality of the goods there was a notable progress. The saponification and acid indices were higher than with the old step and joined the saponification indices of fats and oils when they were loaded into the autoclaves. This drop in the indices at the exit of the autoclaves was intriguing, which is why the Sprinks had made spectroscopic measurements before and after the passage of the material in the autoclaves and had demonstrated that it was due to chemical condensations . fatty acids and oxidations. The suppression of these reactions gave them a clearer product but above all a less acidic pH. Their products thus benefited from a new application: cold paving. On the other hand, the fatty acids now crystallized much better than before, which, in a short time, increased the yield at the presses (hot presses) and thus considerably increased the production capacity of the factory in which they were experiencing.

The amplitude of the variations qualified as lunar by their care was considerably reduced. On the other hand, for the annual curve, it was less clear.

2) The autoclaves: for the autoclaves, they obviously adopted a different operation from that before the tests. We will not develop their method; it would lead to too technical details.

They continued their work. Mrs. Sprink studied the formation of cements at low temperatures. She had arrived at a very fine process of manufacture with ammonia. Unfortunately, this process required special model kilns which the industry could not afford to build, especially at the time of the great industrial crisis of the 1930s, which had seriously affected the cement industry.

but note that a French electrical appliance factory initially refused

They therefore thought that the difficulty could be circumvented and obtain a massive saving of fuel by using the installations which L. Sprink had just applied to the autoclaves.

There were two reasons that encouraged them to try the experiment:

a) the decarbonation of limestone is from the same group of reactions as the saponification of fatty substances,

b) L. Sprink had observed a very strong increase in temperature in the autoclaves. Very troublesome to autoclaves, since it hindered circulation steam, this exotherm represented an enormous advantage for the ovens.

This increase in exotherm had at first puzzled them. It may seem that the principle of conservation of energy is no longer valid , but it should be remembered that the thermal efficiency of a furnace is only 20 to 25% of the energy supplied. It could therefore be admitted that quite simply the efficiency of the oven had been increased. This is the explanation they adopted. It doesn't really explain anything, but at least it was found satisfying for the mind. This is often the case in research where an explanation that is not correct but reassures the mind, allows the researcher to stay on "the beaten track". Fortunately Sprink had fully sensed the danger of such an attitude.

This "exothermy" remained variable, as did the yields and the results at the stearinage. It even seemed that the deviations from the new average had increased. It was therefore clear that an additional factor of which they were unaware was at play. But it was certainly not the earth's electric field or the lunar tides.

Previously we saw that these same variations had also manifested themselves in autoclaves but in a much less clear-cut way and moreover not very inconvenient for the application of the new process in the fatty substances industry, the desired goal having been achieved.

Such would not have been the situation in the cement industry. Indeed the qualities of the cements, in their kiln, and especially the resistance were very variable and followed the inverse curve of the fuel consumption.

Thanks to a few observational data, few in number, but which could not be considered as fortuitous coincidences, and discarded for this reason, the Sprinks arrived at the quasi-conviction that these disturbances

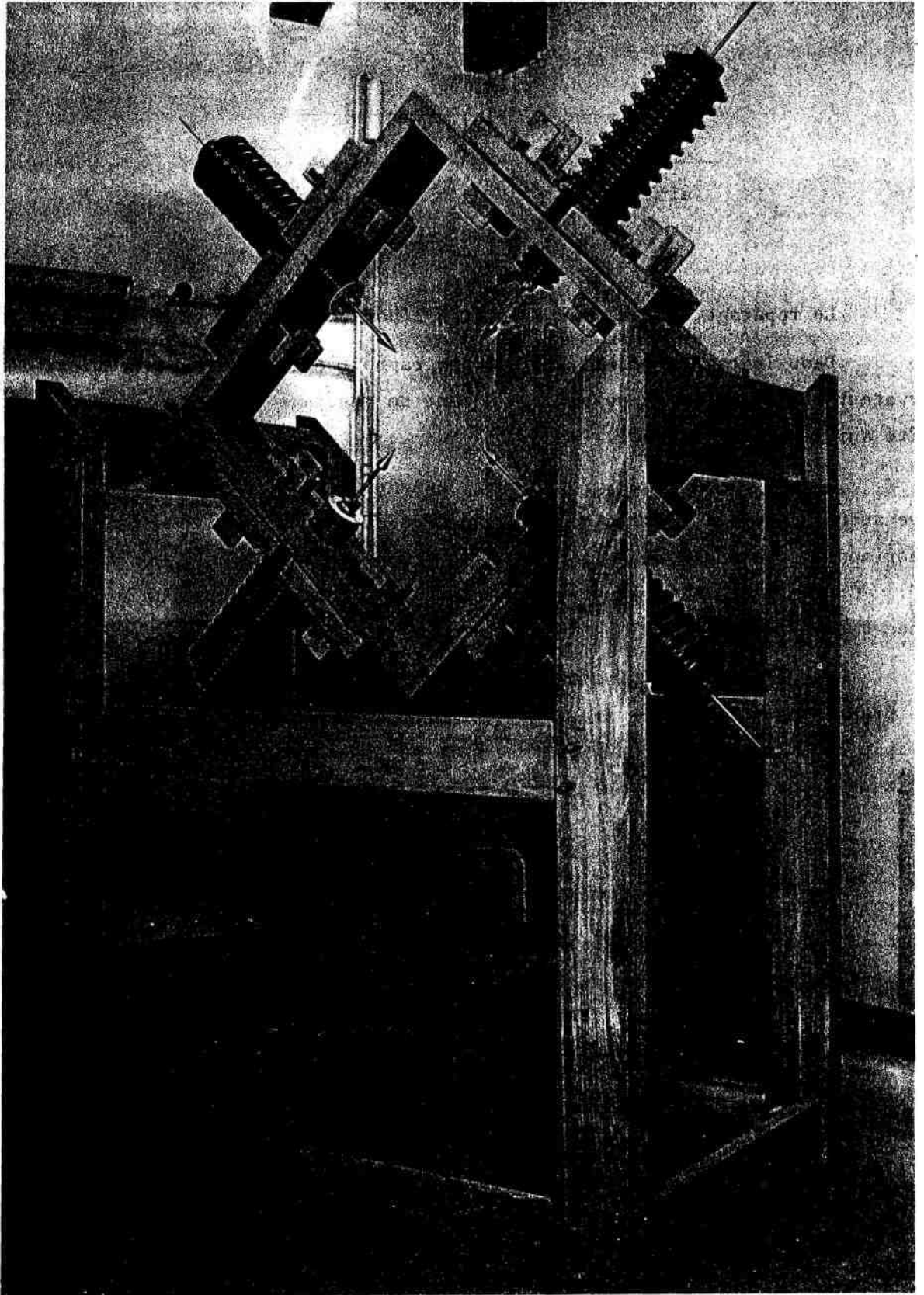
buildings depended on solar activity, made more apparent by the suppression of much shorter period lunar effects. This last result can obviously be obtained by harmonic analysis, thus demonstrating the existence of these larger periods. But at that time, harmonic analysis had a very great defect: it eliminated all accidents. But they had only those to guide themselves! Based on certain observations of the solar corona, they quickly realized that this was where the solution of the problem that fascinated them would come from.

Until the 1939-1945 war, the solar corona could only be observed during solar eclipses. Since then, observations have been made as regular as possible but of a still insufficient duration (covering two to three solar periods) for them to be able even in 1950 (these studies were carried out since that date but the Sprinks had thought about it for a long time) to bring to their studies decisive elements. (It should be noted that the Lyot coronagraph only dates from 1937).

They were able to see that the best returns in the industry occurred 3-4 years after peak internal activity. This excluded the influence of sunspots in either direction.

It has long been known that the appearance of the solar corona changes during a period of solar activity. Very intense at the time of maximum solar activity, it is then uniformly distributed around the sun and especially at its poles. At least it is very developed in the direction close to the equator. In the intermediate periods, its form is intermediate. At certain times, it has the shape of a cross of Saint-André. And we will focus particularly on such times.

Here we are faced with a form which presents itself in a certain way to the observer; it will obviously not be the same for all observers (example: an observer in space in a direction at 90° from the earth-sun line). So the eifs detected will not be the same. Objectivity and subjectivity here disappear. It is enough to make a simple experiment to be convinced some: if one considers a horizontal magnetic field and that one puts in this magnetic field produced by 2 cylindrical magnetic bars a cross of Saint André, one will observe certain eifs. If we rotate this cross of Saint-André by 45° , we will observe other eifs. But if the observer turns 45° , instead of turning the cross of Saint-André, all the apparatus remaining fixed, he does not



Self-assembled Sprink device

will not find the eifs of the cross of Saint-André.

plan of the cross
of St. Andrew

magnet j	magnet
north pole	South Pole

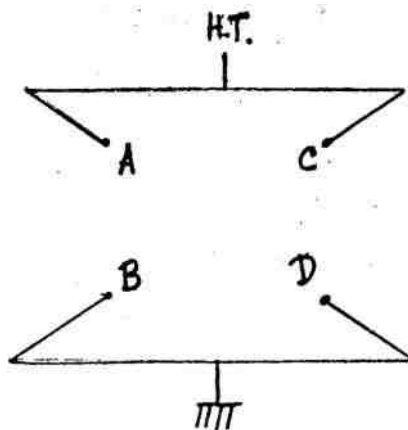
The registration is here relative to the horizontal for example.

In the case of the sun, it is relative to the line passing through the - observer and the center of the sun. We therefore have correct identification and no ambiguity on the observed cross.

When we speak of the cross of Saint Andrew, without attaching ourselves to a device with which it can be associated, it is because we tacitly assume the notion of horizontal in relation to which it is placed. And in the field of EF - C. de T. and associated states, this tacit knowledge must always be taken into account.

Let us recall that L. Sprink, child, in Russia, walking on the lands of the paternal properties had had the chance to contemplate, during an eclipse of the sun, the solar corona but presenting itself in the form of a cross of Saint-Andrew.

It was therefore, as we have already indicated, shortly before the war ie 1914 and this in conditions of extraordinary visibility. He drew from this memory the image of two crossed discharges between four points, and being made between the diagonals of the square joining the points. (Naturally the spikes are assumed to be placed so as to distribute the correct symmetry). In electrostatics, we learn that in this case, without going into details, the discharge takes place along 2 sides of the square AB and CD.



It is especially on the insistence of MT Sprink who relies as much on his intuitions as on his knowledge, that they themselves produced a device inspired by this remark. It consisted of four electrodes, arranged on a circumference and connected to a stable high voltage electrical source; however, the results have proved to be incredibly violent and often contrary to the desired result. They called it their "little sun". It was quite by chance that they were able to co-opt that the phenomenon they observed was very similar to those caused by the activity of the sun. It was MT Sprink who had the joy of noticing it; and it was on the unforeseen occasion, and for good reason, of the aurora borealis on the night of January 25, 1938.

This aurora borealis covered all of Europe. She was a grand beauty. It has been seen in France but also in almost all the countries of Europe, as far as Portugal, northern Italy, Greece, and with exceptional intensity. In France, the Jura, the north of the country, the western provinces, Normandy, Anjou have been particularly favored by this majestic phenomenon, so rare in our centers, and as far as Provence where the astronomer of Kerolya saw her in Forcalquier.

Gabrielle Camille Flammarion wrote an article in the illustration [2]. We reproduce his description of the phenomenon.

"However, more than two hours after sunset, the clouds are breaking up. Around 6:30 p.m. the northern horizon lights up, as if the day was about to be born again. It is a nocturnal dawn whose light comes from the North. A blue-greenish glow invades the north and travels towards the northwest. Gradually, the northern horizon is set ablaze. A kind of purple cloud appears in the northeast and moves towards the northwest. It folds up undulates, expands, dies out, revives, while huge jets, whose color changes from blood-red to orange-red and yellow, soar towards the zenith. The apotheosis occurred around 7:50 p.m. magnetic disturbance was recorded by Perrotey, at Juvisy: it began on the 26th at 3 a.m.; the violence was such that the luminous spot left the field of the recording paper for more than an hour". It is even reported that this aurora was seen, even further south in Europe, from a ship passing off the coast of the Rio de Oro, that is to say at a latitude of 28° [3] "When the sun reappeared, there was nothing abnormal, apparently on its surface^ \ However, a round spot, of a particular blackness (*)

(*) This documentation was provided by Jean-François Boédec.

intensely intense had just passed through the central meridian of the star".

By testing the phenomenon, we find that there was an anti-decalar appearance. There were issues at C. de T.: $1 \text{ ' } \text{-----}' 1.2 * i^2$, $3 i^{-3}$, $4 i^{-14}$.

On the other hand, a sunspot corresponds to a decalar. Some may be anti-decals but ours are not channels. There are no emissions at high C. de T. from these spots.

The anti-shift associated with the aurora borealis is not necessarily that of a sunspot. An aurora borealis is not associated with an anti-decalar in general, nor with emissions at high C. of T..

There was a very interesting effect reported by L. and MT Sprink as they worked with their device. Their device was placed near a heating boiler; during the aurora borealis, the boiler had its water brought to a boil although the automatic air shut-off functioned perfectly. The next day, L. Sprink learned that several central heating boilers had been damaged and that chimneys had burst (*) in the city where he was. In 1949, he spoke about it to Cotton. Cotton sends him a little-known text by Buffon on his observations relating to blast furnaces.

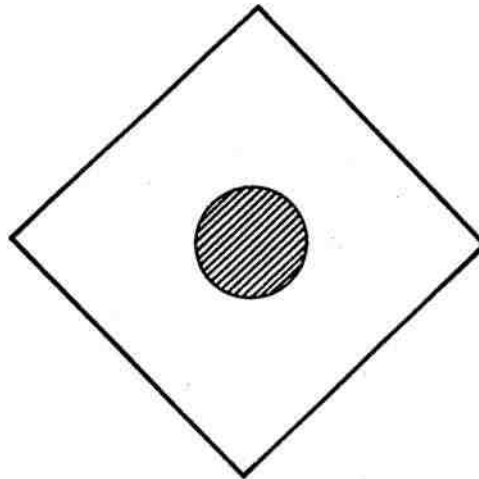
In the 18th century, the aurora borealis were frequent over France and the masterful observations reported by Buffon are in all respects equivalent to those observed by L. Sprink.

L. Sprink also cites Weyprocht's observations of the aurora borealis during Tegetheff's expedition to François-Joseph Land. It was observed during this expedition that during some very mobile aurora borealis, the magnetic needle moved towards the east, the horizontal component decreased and the vertical intensity increased! However, this is what L. and MT Sprink were able to verify on their device in January 1938 (except for the horizontal component of the earth's magnetic field because they did not have an appropriate device). The Sprinks had concluded that the field emitted by their device was identical to that emitted at the time of the aurora borealis phenomenon.

In reality in the case of that of January 1938, we can see that there

(*) Mr and Mrs Cotton have produced a precision balance that bears their name

had the C. of T. that came out of the Sprink assembly but that there were 3 others in addition. On the other hand, the Sprink device makes it possible to obtain a decalar and not an anti-declar. L. Sprink also points out that when observing Arosa, it has been noted that the "clear nights" return periodically and always after the maxima and before the minima of solar activity. We have not checked whether this luminescence of the night sky is interpretable in the same way as the aurora borealis, that is to say whether there is a decalar or an antideclar and emissions at C. de T. students. It should be remembered that all the aurora borealis do not have the characteristics of that of 1938 [4], Let us point out a curious observation which occurred in 1913 [5] . She was observed from a boat on January 21, ^{1913k} . This is a square halo around the moon. The color of the halo was yellowish. It looked like this: the length of the side of the square was



This was not known to the Sprinks, but it is interesting to quote this unusual figure and also to notice this position of the square halo which is found somewhat in this form in one of the assemblies used by the Sprinks (♦*)

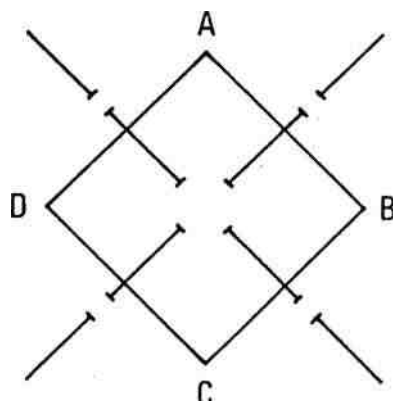
They actually made a square frame put in the position below. This frame was placed on a support. In the middle of the sides of the square and perpendicularly were fixed the electrodes which produced a cross of St.

(*) This observation was taken up in the work by William R. Courliss entitled - "Handbook of Unusual Natural Phenomena, Eds. The Source of Book Project, Glen Arm Md. 12057, USA (1977).

Additional details on times, latitudes and longitudes are given in the book.

(♦) The first experiments took place with a circular wooden frame and two diametrically opposed electrodes. The frame was sometimes laid horizontally.

André with respect to the horizontal plane. There is a resemblance between the 2 figures.



If we consider the shape composed of the moon and this halo, we see that it was a shape neither testable nor untestable. But we can make it testable. There are no Taofel fields or decalaires. We see that the broadcasts are not the same but it is good to point out this oddity which is roughly what the Sprinks had produced in one of their montages.

Human history is full of these observations which do not seem to fall within the standards usually accepted: we can also cite that of June 15, 1965 where we saw from the deck of a boat, and this was photographed, two suns in the almost perfectly clear sky. The passengers were on the deck of the ship. They saw the sun and the false sun; the false sun had a little lower luminosity, its diameter was slightly smaller, it was at the same height above the horizon, that is $1^{\circ}8'$ of arc, and its distance from the sun was $3^{\circ}36'$ to the left of the true sun. The phenomenon has disappeared.

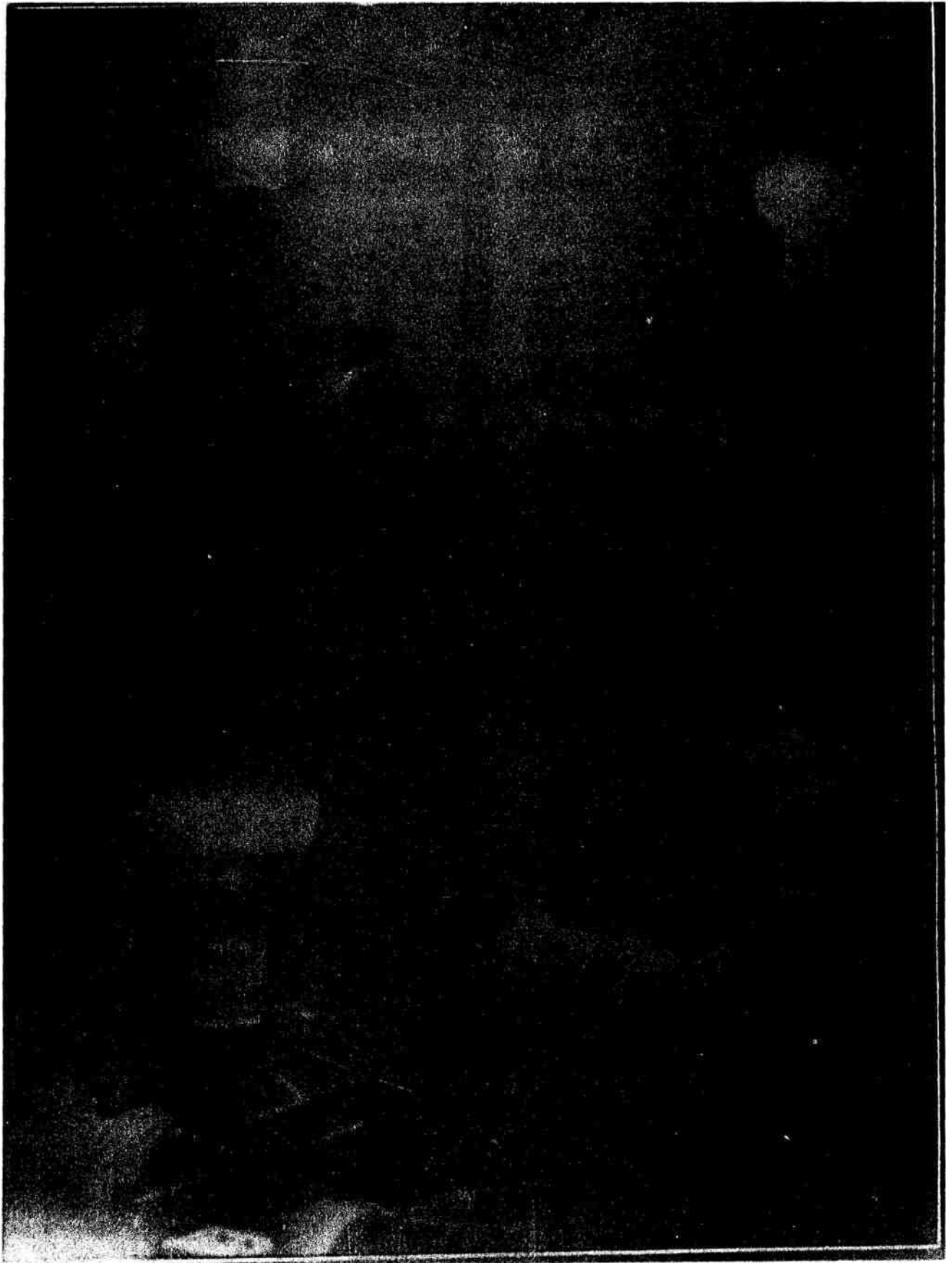
If we test, we find Taofel fields from $1 > \text{-----} 1$ to $7^1 \rightarrow 7$.

The false sun emerges from a decalaire. There is no magic state of any type. We must not believe that every bizarre phenomenon has a magical state. A magic state always corresponds to an inconsistency in relation to a human being, inconsistency with a will behind it or inconsistency due to the fact that a problem is poorly considered.

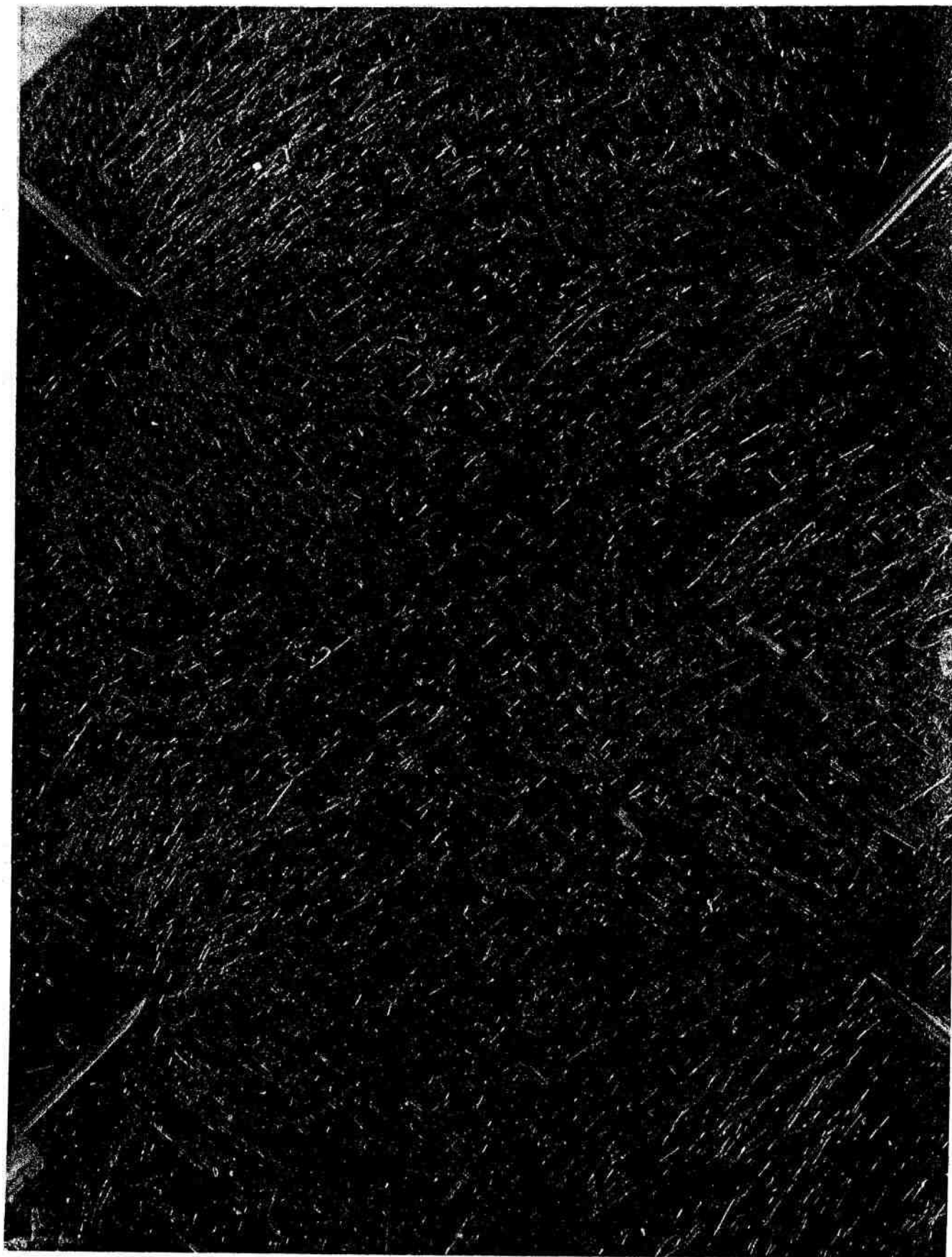
A phenomenon like a form can very well be only an emergence and then there are other impacts in the local. Emergence leads to a "trace" in the room which is confused with the phenomenon itself if it is completely localized. But if it is not completely in the local then there will be some kind of echoes of this phenomenon for localization. Sprink's work continues. It was not until 1944 that they found in a cement factory (where

Part of an electrical assembly

L.Sprink



Lines of force between device electrodes



they were doing their tests on the device developed by them in Strasbourg before the 2nd World War) really complete archives, including daily analyzes of continuous production. It goes without saying that this documentation was of exceptional value to them. For more than forty years neither the manufacturing methods nor the construction of the furnaces in the factory in question had changed. The coal always came from the same mine; the quarries were of perfect and invariable homogeneity for more than a century as to their composition and their physical state.

They were therefore able to plot the tensile strength of cements from 1898 to 1939. They were greatly surprised when they realized that the exceptional maxima corresponded to the years 1911 and 1930.

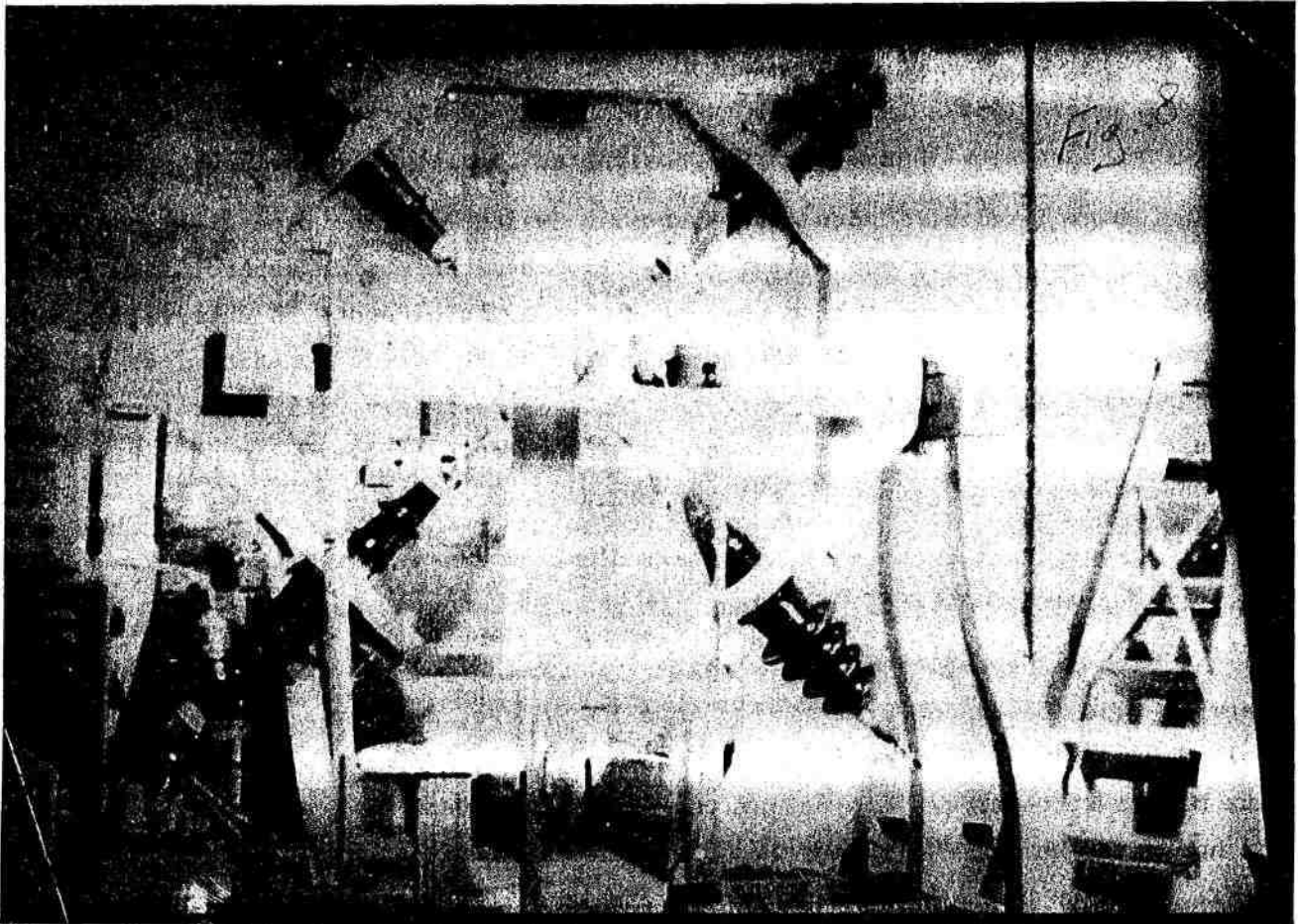
In the fatty substances industry they had already been able to pinpoint the year 1911. Now they had two points of reference at their disposal, without knowing, however, whether they involved any physical phenomenon whatsoever. The interval of 18 years corresponds to the saros of astronomers. The Chaldeans already knew this period of 18 years and 11 days which approximately regulates the returns of the eclipses. It includes 223 lunations or 19 times the interval of 346.6 days which separates two passages of the sun through the lunar node.

For their research, it was of extreme importance to find by grouping, this period in natural phenomena.

Yet the periodicity of oceanic transgressions offers a typical example of this. They have their maximum every 18 years, with a secondary maximum at the half-period. Curiously, the years 1911 and 1930 are located exactly at the secondary maxima. The Sprinks also found that on these two dates the Wolfers-Wolf number for the sun was three years from its minimum. It was said above that it is very probably above all the coronary phenomena which intervene in the physicochemical balances.

The Sprinks thought that the true concordance of the solar and saros cycles can only occur every 120 years. Oceanographic observations confirm this supposition on the whole, but the situation is complicated by the fact that the solar periods have no fixed data and can vary from 9 to 15 years. It is therefore not surprising that the meetings of the moon and the sun, the latter being in a determined state, only occur in an apparently random way, but each time they occur, the manifestations of these meetings take extraordinary magnitude. The chronicles of

A montage by L. and M.-T. Sprinkle



. A montage by L. and M.-T. Sprinkle



fisheries provide impressive examples of this, describing not only monster tides, but miraculous catches as well. As soon as the action of the sun was no longer in doubt, it was normal to return to the annual variations of the results obtained in the factories. And it is there that the curves of the monthly averages of the resistances of the cements for the periods from 1899 to 1917 were invaluable. The choice of this period was determined for two reasons:

1) During this time, the laboratory premises have not been changed or remodeled and the working methods have remained the same; it was therefore reasonable to think that these factors were not involved.

2) This period is sufficiently long so that, on the averages, the variations due to large periods are canceled out.

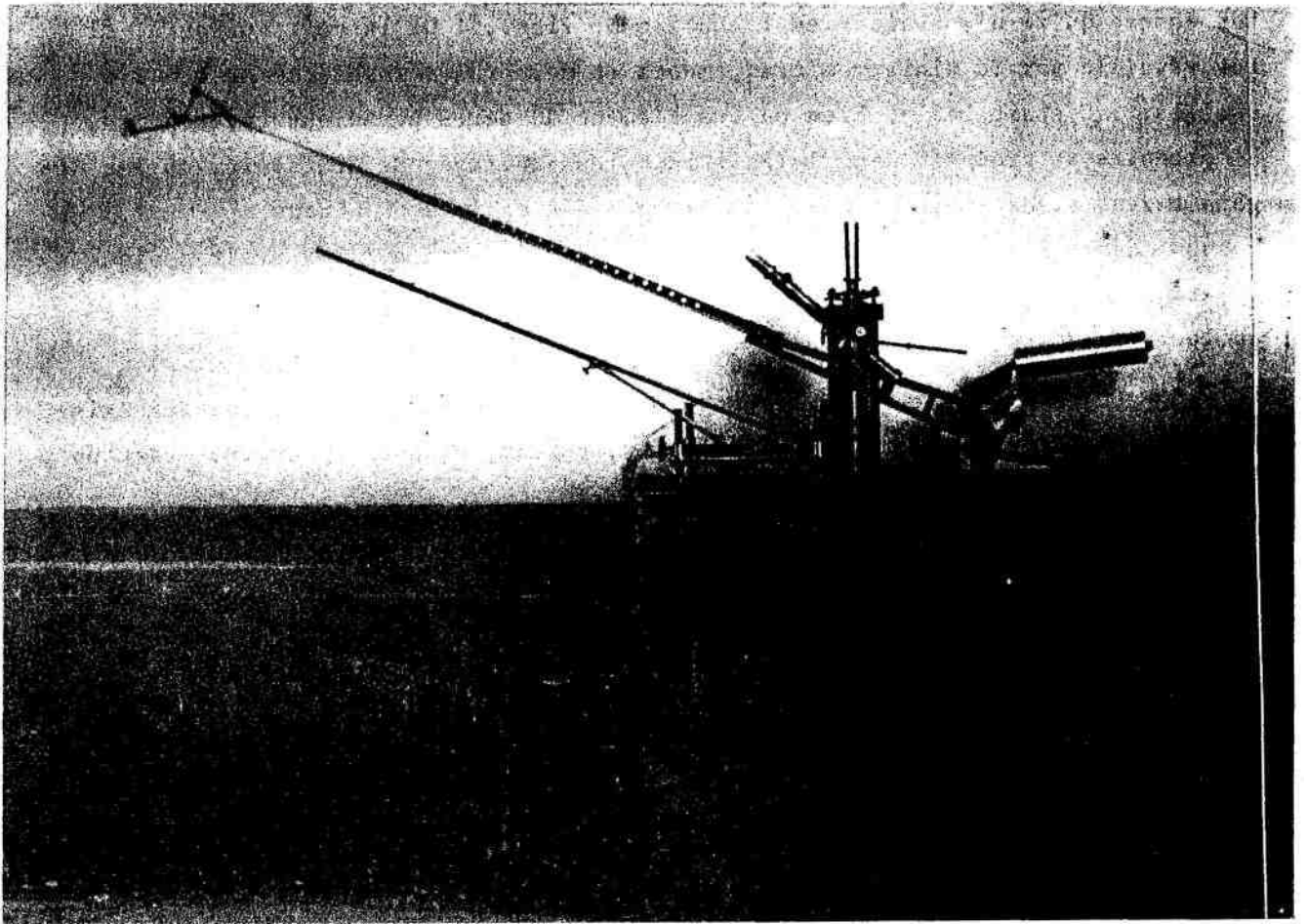
For the clarity of the presentation which will follow, it is necessary to open a parenthesis. For a long time the hardening of cement had been considered either as a colloidal process or as a crystallization. Both points of view had their partisans and the theories of Le Châtelier and Michaelis shared the favors of specialists. It must be admitted that in these quarrels nationalism did not lose its rights. But it's worth acknowledging that both were somewhat right. Indeed, when it sets, the cement behaves like a colloid; it is only then that crystallization, if you prefer, "devitrification" takes place. We will see later that by their behavior in the face of external conditions, cements fully justify this point of view.

To observe these behaviors, the Sprinks were well placed, having had the opportunity to study in stearinization and the variations of processes of a colloidal nature and those which depended only on crystallizations.

By superimposing a cement resistance curve on the solar actinometric curve, the Sprinks found that the two curves are inverse to each other.

After a certain number of remarks on which we will pass, the y Sprink, by reconsidering the results obtained at the stearinerries, had come to the conclusion that the yields of limestone followed the same curve as the hardening of cements. In 1932, they had noticed that the curve of yields at the presses and that of the variation of the vertical component of microseismic movements (roughly in the plain of Alsace) were identical. At the Strasbourg factory, they had even tried to reproduce these small earthquakes, without obtaining any effect. This is normal because causing

One of the gravimeters made by L. Sprink



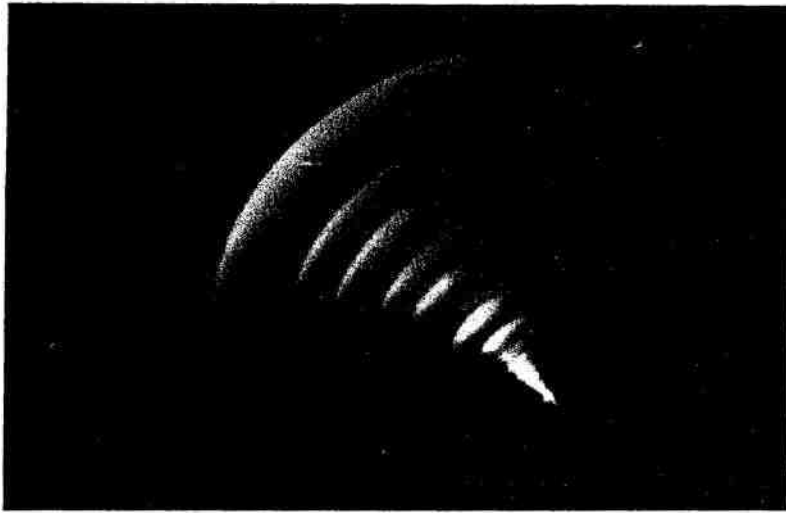
ground shaking by explosives in another way has nothing to do with a real earthquake which is much more of a localization character. In a natural earthquake, there are often chasms that appear.

Chance put the Sprinks on the trail of the real solution to the problem . One day, the Strasbourg factory had an extraordinary output at the presses. L. Sprink immediately telephoned the University's Institute of Geophysics to find out if there was increased microseismic activity. The answer was given to him that there was nothing! absolutely nothing...! But the following night, he was awakened by sinister creaks and strange noises and his bed rocked. It was indeed a clearly characterized earthquake. The epicenter was in the Palitinate, near Stuttgart, where the damage was quite extensive and the church bells rang from the shaking. It was therefore concluded that there was some immediate relation between the preceding phenomena, the earthquakes and their yields at the presses.

We know that the warning signs of earthquakes exist . The behavior of animals, horses, dogs, birds and others before an earthquake is well known.

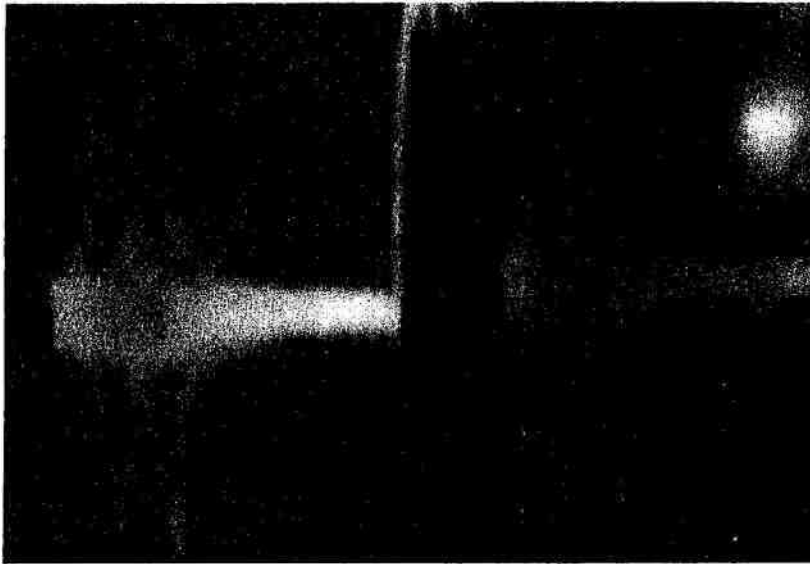
About some earthquakes, we can detect eifs 4 hours to 6 hours (and even 12 hours) before the shaking occurs. These are Taofel fields. These are certainly felt by the animals . Even human beings are able to sense the earthquake. Indeed in Persia, in Armenia, in the North of the Caucasus, certain natives of these regions feel a vague uneasiness that a long experience has taught them to relate to an imminent earthquake. Animals and human beings may feel the Taofel Fields but there is a feeling that is due to something else. Everything happens as if something foreign is coming into our universe and to which some people and animals are sensitive. This something is presented as an "unfolding" which allows the establishment of the usual field of coherence, which will make it possible to find the turbulence of the earth's crust etc...

Let's open a parenthesis on the prediction of earthquakes, (naturally this will be the usual point of view of earthquakes).



Ondes stationnaires





Acoustic resonance

100 kc



Electromagnetic resonance

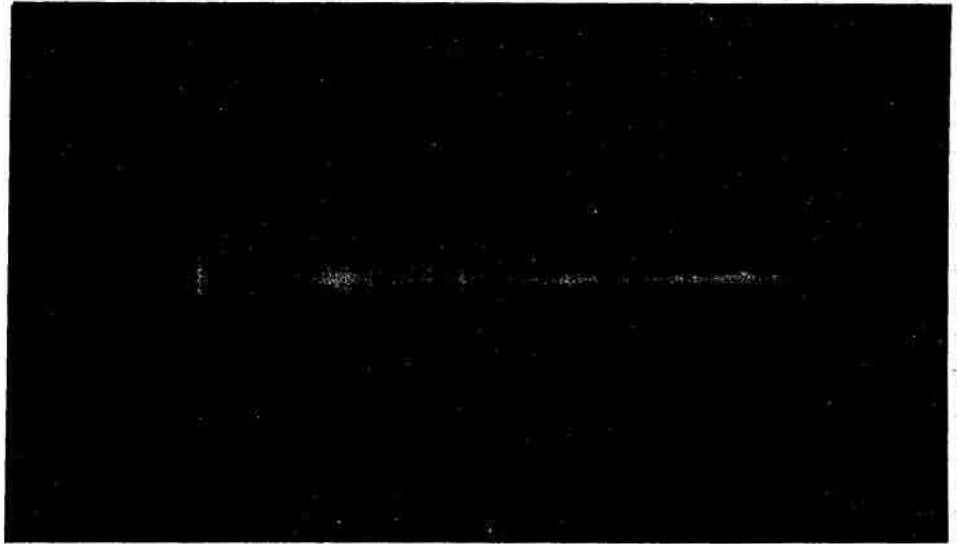
25 kc



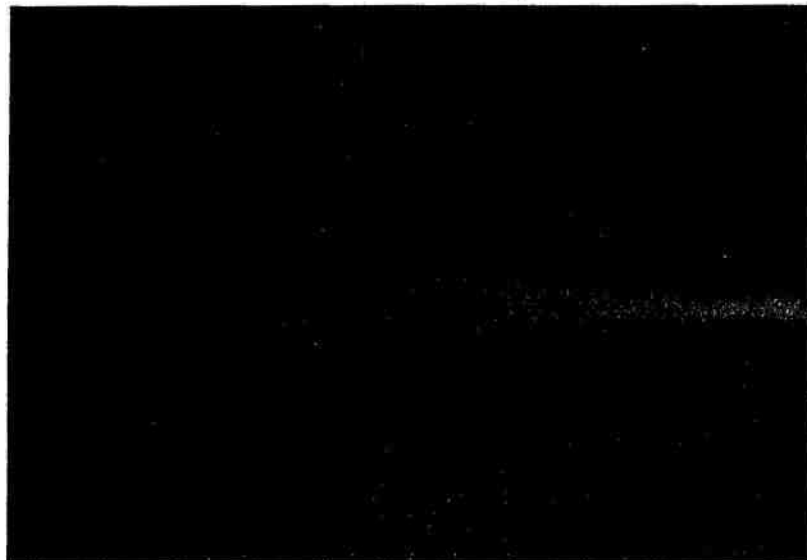
Electromagnetic resonance

25 kc

Standing waves, natural size



Damped oscillation like capacitor discharge



Resonance with the low frequency of the magnetic field

For this, consider the case of China, which has been hard hit by these disasters. It was in China that the deadliest earthquake of all time took place, that of Hushsien, in Chen-si, which caused approximately 820,000 deaths in 1956. There were many others and that which struck Hsingtai in Ho-pei in 1966 led Prime Minister Chan-En-Lai to give seismology priority.

Today, China has 10,000 professional seismologists and 100,000 amateurs working in an organization comprising 17 centers and nearly 300 regional stations. By enlisting volunteer observers (peasants, workers, teachers, telephone repairers, meteorologists, radio personnel) the Chinese wanted the ordinary citizen to be on the side of the scientist. They enjoyed trying to predict earthquakes.

At first suspicion fell on the province of Leao-hing. The specialists noted that the epicentres of the small tremors consecutive to another earthquake, that of Hsingtai, seemed to move towards the region of Leao-ning, a densely populated industrial region. In 1970 it was decided to start monitoring this province very carefully. All the techniques existing in the arsenal of seismology were implemented, starting with the study of past earthquakes and local geodesy. The geodetic study of a major fault, made from September 1973 to June 1974, showed that the surface of the ground there rose twenty times faster than in normal times and inclined towards the northwest. During the same period, a change in the local magnetic field was recorded and the tide stations on the coasts of Leao-tong Bay observed a rise in sea level. Finally, in 1974 there were five times more small earthquakes in Leao-ming than in previous years.

All this led the government to warn the population that a dangerous earthquake was brewing. However no one could yet say when it would take place. Other symptoms were therefore sought, for example changes in the behavior of the animals.

A few hours before the earthquake that took place in Po-hai on July 18, 1969, a guard at the T'ien-tsien Zoo noticed that the tigers were behaving strangely; he warned the seismologists of the place and reported that a

great earthquake may well be approaching. In the province of Seu-Tcb'ouan, in September 1972, a forecasting team was able to announce the outbreak of another earthquake after observing "the panic of chickens, pigs refusing to enter their piggeries, horses and sheep starting to run frantically in all directions. For the Chinese, people in the streets and fields can provide valuable information, whether on the behavior of animals or on the level of groundwater.

Precisely such symptoms were observed in Leao-ning in December 1974. In four communes, the water in the wells became muddy; bubbles burst to the surface. Rats and mice left their holes, just like the wintering snakes that were seen venturing on the ice. An earthquake then occurred seventeen kilometers from Hai-ch'eng, leading the authorities to , . > (*) thinking that a bigger tremor was brewing.

This case is interesting because, although there may have been emissions already appearing on that date, and disappearing shortly after to reappear a few hours before the earthquake, it seems that the localization is felt well before the earthquake. .

At the beginning of February, the animals continued to behave abnormally, and the wells began to show other symptoms of disorder: twenty-two became artesian. The opposite was observed in a hot springs station: three times the outflow of water was interrupted. According to the report, the most remarkable event took place in Chaozihe commune: Yu-nan province, where water and gas wells were seen to spring from the ice like geysers.

The pace of events quickened. On February 2, a seismological station recorded a sudden drop in ground electricity; another observed that the tilt of the ground was moving towards the southwest.

All of these remarks made it possible to avoid a catastrophe: on February 4 , 1975 at 7:36 p.m. the earthquake of Hai-ch'eng took place, but most of the inhabitants, warned, had been able to leave their houses and the animals had also been taken out.

But other disasters that are not earthquakes have also been perceived by animals. It is well known in France that sliding

(*) The UNESCO Courier, May 1976 ((XXIX year)).

seroent of land which occurred at night on the Fourvière hill in Lyon around the 1900s, was sensed by the horses which the night before did not want to enter their stables. There too, perhaps some eifs could have been felt by the animals, but something else, which is a contribution to our localized world, is felt. There is an instability of the physics of the localized and gas bubbles, mud in the waters, geysers on the ice etc... are manifestations of this.

Thus, all these phenomena with quite catastrophic effects seem to be only a localized view of a deeper effect because not localized; it's almost an attempt at localization.

With a form like the planet earth are associated many other forms which are not localized, which cannot be localized. They are in a channel. But they can have localization tendencies. There are also "echoes" to the forms.

These other forms are not in our field of coherence. In the case of animals either they project another field of coherence which gives them access to another reality, or they feel the disturbance of their field of coherence. One can wonder if one is in a localization or in the disturbance of a field of coherence by another field of coherence in the case of an earthquake.

The Cartesian interpretation of these phenomena is an unfolding in the room on which causes and effects overlap.

Let us point out a device known since the highest antiquity and called "wind harp". (*) (* **)

During numerous observations, we have been able to realize that the lowest sounds of these instruments precede the rain by a few days and that the highest sounds precede the rain by a few hours. As a result of these observations, special instruments called "meteorological harps" have even been constructed. Here is the description given by a chronicle of the time, of a harp built in 1789 by Ventan, provost of Burkly, near Basel in Switzerland.

"This harp consisted of 15 iron wires 320 feet long, placed in a garden, 2 or 3 inches apart, and stretched with the help of large cylinders . _The_ thinnest rope was about a line thick, the strongest

(*•) L. Sp.rink was aware of this experience and mentioned it in his report

(**) We will come back to this instrument later. We point this out here for the first time because we must be careful not to model the effects of delocalization relating to earthquakes on meteorological harps.

6 lines. They were arranged in the direction from north to south, and inclined so as to form an angle of 20 to 30° with the horizon. When the weather changed, the harp gave out sounds so powerful that they drowned out all the noises in the house and almost inspired a feeling of terror; sometimes its quiverings imitated the rapid murmurs of boiling water; sometimes the distant playing of a harmonica, a chime or an organ. They tried to stretch these iron cords in the direction from east to west; it no longer produces any effect".[6]

More recent results say that the wires must be stretched in the direction perpendicular to that of approach of the disturbance not yet sensitive and that on the other hand, it would have been found that on two lines stretched perpendicular to each other there is always only one that reacts.

The remark about the wires stretched in the direction perpendicular to the direction of displacement of the disturbance is vague because the harp is not symmetrical because of the different diameter of each string; then it seems that the harp oriented in a well determined direction, that of the north of form (if it is isolated enough in nature) must then give the most powerful effect;

This meteorological harp also expresses an effect in the process of - localization. At the time of the songs it emits, there are fields of Taofel which are present. That is to say that we are faced with a phenomenon where the local is deformed but there is no substitution and therefore no appearance of OF as a support. This harp is a localized form but it still has an Ext.

In the case of earthquakes, we have shifts in the terrestrial globe, not measurable in position since the landmarks are on the run, but through these shifts other benchmarks are formed and these are modeled on the pressure forces, they , localized and other parameters. It turns out that the earthquake is caused by a form that cannot be located. This form happens to exist by a localized structure or something else. The localized structure may be a disappearance of rocks with pressures or something else. In the case of the "meteorological harp", it is not a form which is not localizable which intervenes. It is a completely localized form which has an Ext. A fleeing form uses the local to realize itself but as its name As we say, it is made up of fleeing pieces, which cannot be indexed by a sequence of integers. There appears a shift in the numeration. But a fleeing form relocates. In the case of the meteorological harp, there is has in

more than one channel.

L. and MT Sprink immersed themselves in the work of V. Bjerknes [7] and noticed that this physicist noted before the First World War the formal equivalence between an electrodynamic field and a hydrodynamic field . V. Bjerknes used this equivalence in his calculations of "physical hydrodynamics" without drawing any other conclusions. However, it was only after the Second World War, and faced with the fact that the speed of the advancement of solar prominences increased during eruptions, and that this acceleration seemed to be due to a mysterious and unknown force, that we began to glimpse the extraordinary significance of this analogy. At the present stage of research ^o, we are still at the simple observation of the existence of the hydrodynamic fields of Bjerknes. V. Bjerknes wrote [7] (in volume 1, p.25) "we must indeed regard as a secret of nature, of which we are still very far from knowing the scope, the deep analogies of the hydrodynamic field and the hydrodynamic field electricity and, likewise, with the magnetic field. There must be between the two series of phenomena a relationship little known today for a system of fluids as general (permanent hydrodynamic field) as the one we considered, can only move according to the formulas of the electrostatic and the magnetic". And further "We will leave aside the possible significance of these analogies from the physical and philosophical points of view. We only look at the objective facts having in practice a considerable scope. The equations of hydrodynamics indeed make it possible to use the same results, the same methods and the same modes of intuitive representation in very different fields of physics".

This analogy was limited to the study of solar prominences.

The Sprinks have considered it reasonable to admit the existence of analogous fields also on the terrestrial globe and to seek there the cause of the mysterious duration of the phenomenon which is about to be described hereafter.

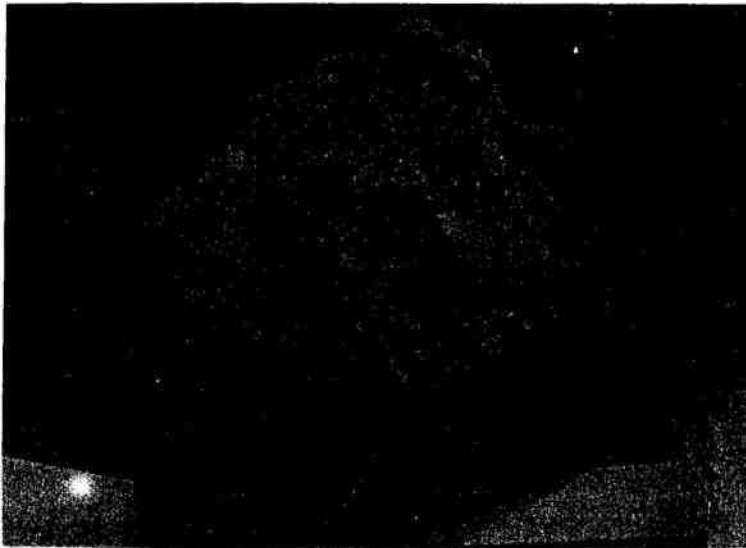
It is not known whether L. Sprink tried to explain earthquakes by this analogy between fields. However, the concepts we are developing were not known to the Sprinks and they tried to express them in the usual language despite the feeling that they were facing something else...

Let's see their study on the structuring of stones. The Sprinks, in the course of their experiments, removed from the lime kilns fired blocks having

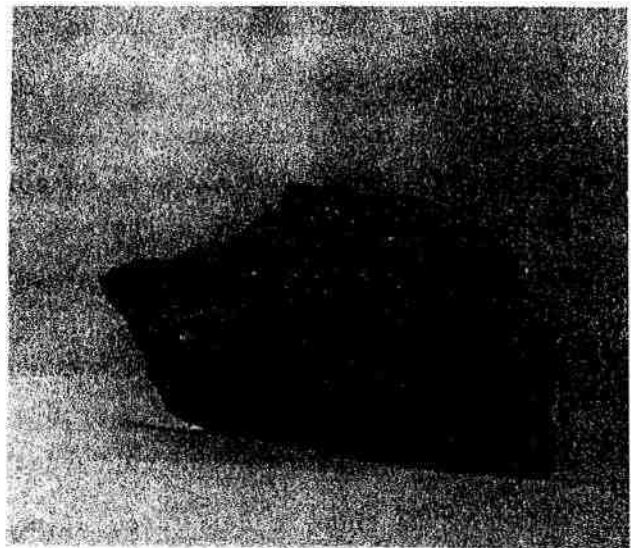


Honeycomb structure

Rod structure



Cut of a rudist



Radiated structure



a geometric spatial structuring that did not exist before firing. Naturally their device operated near the ovens and these observations were observed by them only during the operation of the device.

They were able to observe two types of structuring: one in rods (generally sections in pentagon or hexagon, kind of basaltic structuring "in organs"), the other in "honeycombs", (kind of structured land of the Tundra). The first is identical to that of band clouds, the second to that of "lacunar" clouds. The structures could be reproduced in laboratories with smoke. The necessary and sufficient aerodynamic conditions for their formation are well known.

J. Coulomb and J. Loisel have given a description of convection phenomena which make it possible to establish certain equivalences between clouds with a 'lunar structure' and the structured stones of L. and MT Sprink. Let us see what J. Coulomb and J. Loisel.

"As early as 1900, Bénard had studied experimentally, in a more or less definitive way, the convective vibrations in a layer of viscous liquid, uniformly heated from above. The layer breaks up into prismatic cells of more or less uniform dimensions, in the center of which the liquid rises to descend along the walls. Bénard gave this organization the name of "cellular eddies". In steady state, the cells are almost regular hexagons. If the liquid is made to flow on a hot plate (terada) the cells are replaced by rolls parallel to the lines of slopes which are called "band eddies".

"The theory of "thermoconvective" circulations in viscous liquids was initiated by Rayleigh, starting from Boussinesq's equations, and perfected among others by Jeffrays".

"Tests to verify Rayleigh's theory have generally given fairly good results. Verifications can relate to the minimum transverse dimensions of stable cells and also to the amount of heating required for start-up. In fact, for moderate heating, if the liquid's calorific viscosity and conductivity are high enough, a particle moving under the action of a difference in density would do so so slowly that it would have time to settle. temperature equilibrium with its surroundings before having traveled an appreciable path, and thus canceling out this difference in density. The particle therefore does not move

noticeably, and we then obtain a curious stable preventive state. In support of this, excessive heating would lead to the sudden replacement of cellular movement by disorganized turbulent movement, but the theory does not yet provide for this latter phenomenon".

"Convective circulations in air are much less well known than circulations in liquids".

The Sprinks, in addition to the structuring mentioned above, had often found in their samples taken from the ovens, stones with a radiated structuring. Contrary to what is observed on oolites with radiated structure, here the rays are not in general straight lines, but are curved parabolically and one does not find in the center of the geometric system the slightest trace of a crystal or of a fossil. This spatial structuring is due, for the Sprinks, to the existence of an auxiliary electric field, out of connection with the stone itself and to the polarization of the particles electrified by this field. This structure is reminiscent, as L. Sprink points out, of that of the debris of a racing car that fell in 1947 in Sikhoté-Alinsk, in Siberia. It was also the first time that such a structure was encountered for a meteor.

In the case of the "Pierres de tonnerre" of the chalk of champagne and the oolites with radiate structure of Cettigien of the Ardèche, it seems that there is a field which structures the stone but the starting point of this field is localized by the debris of a crystal or a fossil. This fossil can be whole as is often the case in Champagne.

L. Sprink made a very interesting point about stones structured in "honeycombs". He had been struck by the resemblance of their appearance to that of the surface of the moon. He said: "For stones, the origin of these craters is beyond doubt: it is due to stationary vortices , generated by mass transverse vibrations".

"It would be quite plausible to admit that the craters of the moon are not due to meteoric bombardments, because the structures of the lunar surface offer disturbing analogies, if not a complete identity with the structural shears which determine the flakings surfaces of our stones. It seems likely to us that the formation of sunspots with their powerful magnetic fields goes back to analogous causes, that is to say to stationary waves, generating more or less stable vortices of the electrified matter. As to the origin of these waves (in the two

latest demonstrations) we have for the moment the slightest indication.

Naturally L. Sprink did not know that his device was emitting eifs at the BCM level and the Taofel Field 1 i -----i 1 . When such a field of Taofel appears from the decalaire which is at the point of intersection of 2 cross discharges between the spark gaps (which only happened very rarely in the case of the assembly of the Sprinkles since they did not know the optimal conditions for the success of the phenomenon) privileged zones appear in the space surrounding the device, which are like weakened echoes of the decalaire. They constitute a localization neighborhood. The shifter and all of these zones behave like a_form_in_leak.

In the case of the structured pebbles of L. and MT Sprink, there is a disturbing fact at first sight, it seems that the ores behave as if they were plastic and yet in the ovens at the temperature of 1100 to 1200°, they are certainly not and keep their primitive forms. The same phenomenon is found in nature for oolites and modules which interpenetrate without any mechanical deformation. Here is what Mr. Cayenne, who has extensively studied the plasticity of oolites, says:

"Deformation by impression has long been reported and often invoked to demonstrate the plasticity of oolites. Juxtaposed oolites penetrate more or less deeply into each other, like "impressed pebbles". separate, their surface is hollowed out with small cupules, giving the measure of the impression".

by another oolite. plasticity of matter. In my opinion, this matter of plasticity does not necessarily evoke that of soft matter, and this is so true that we have only ever observed oolites in the process of formation on the edge of the reefs and on the shores of the Salt Lake were soft. Moreover, I have noted many examples of ferruginous oolites deeply set in the debris of molluscs that we do not imagine to be plastic at all" [8].

The plasticity of rocks and crystals is well known. It is well known that, without the action of a continuous and immutably directed force any (*)

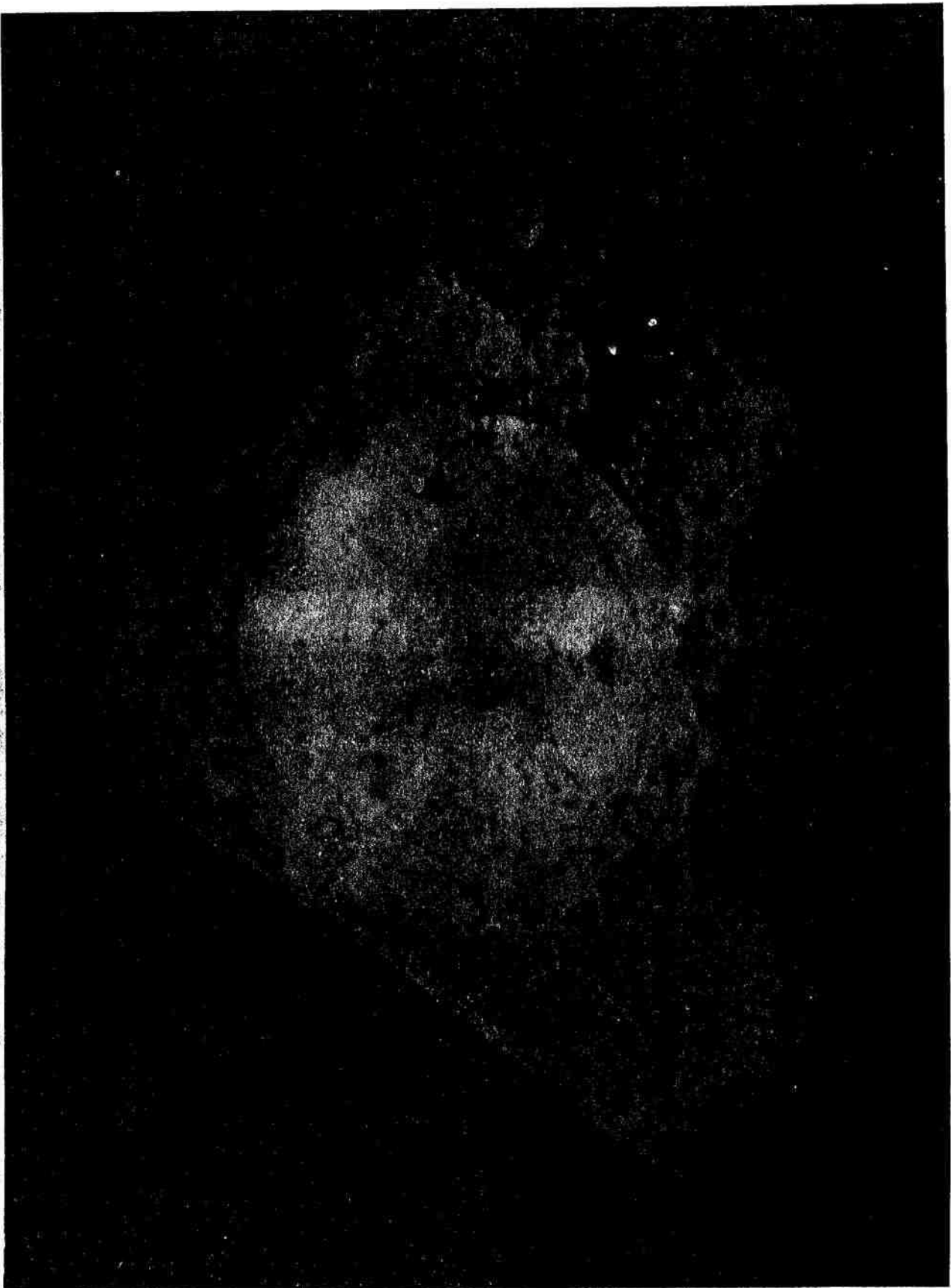
(*) Location neighborhood or voiloc.



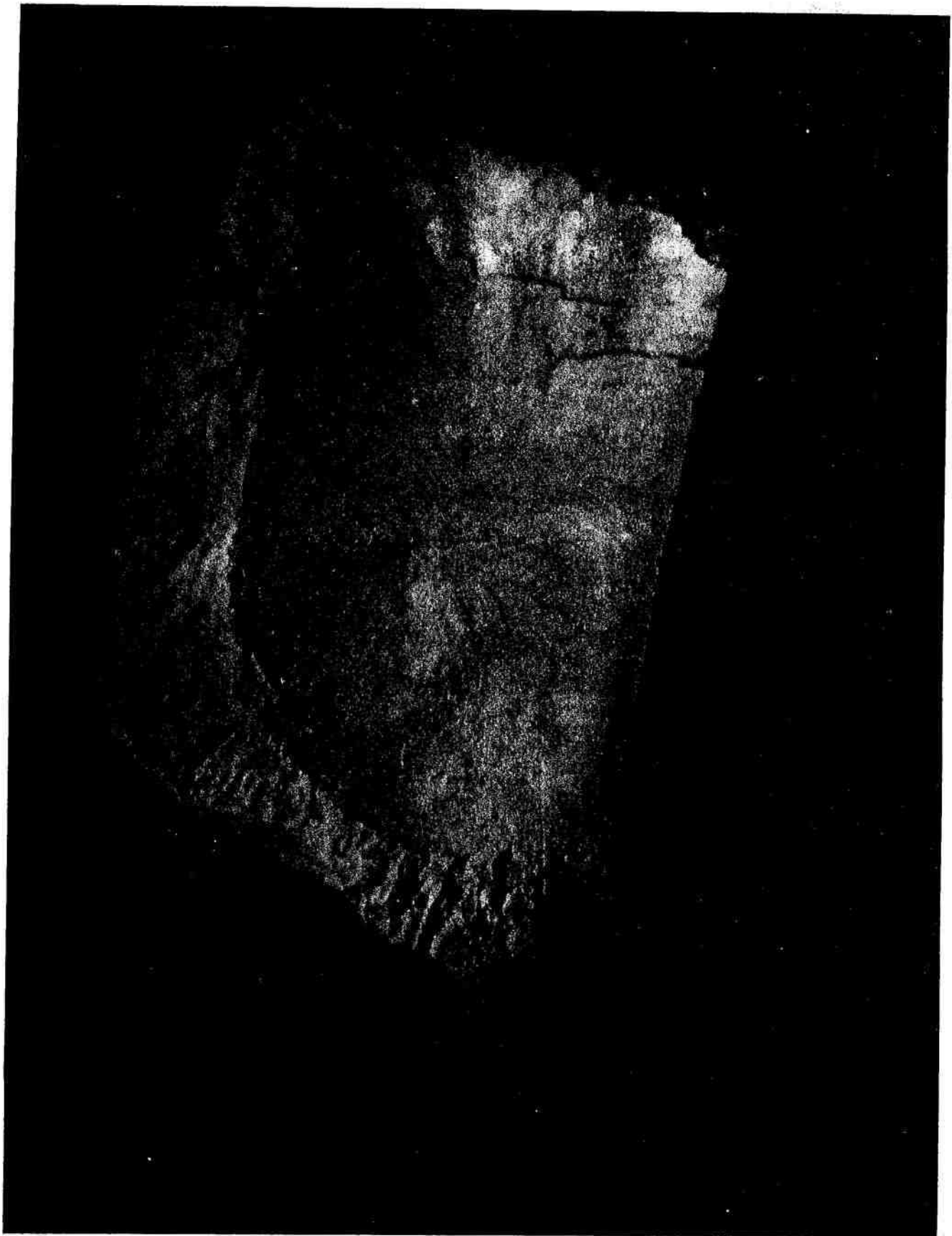
Structuration nid d'abeilles



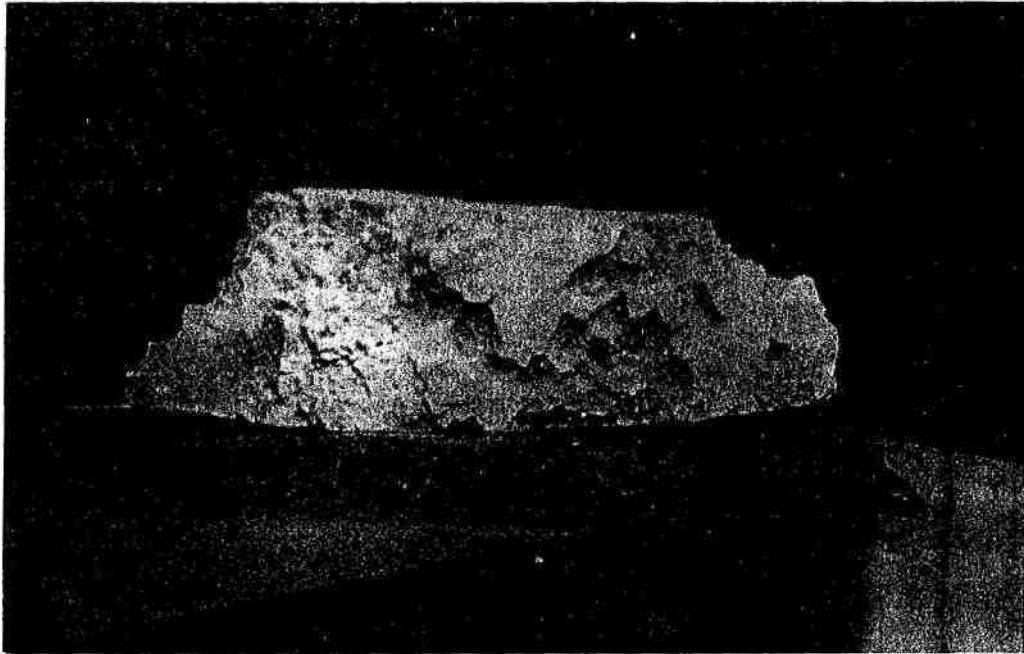
Radiated structure



Radiated structure



honeycomb structure



Structuring struck off



solid deforms slowly. What surprised the Sprinks in their structured stones was the speed with which these structures appeared, and the fact that the sheet structure of the limestone remains without influence on the structure. These two structures interpenetrate without destroying each other. They could give no explanation for this phenomenon, but Mr. Cayeux's observations confirmed the existence of analogous facts.

L. Sprink thought of hydrodynamics to explain such phenomena and he sought what could be the nature of the wave capable of giving similar vortices in very viscous media; it is a wave across the. He had first thought with his wife of an ultrasonic wave. Then he fell back on an electromagnetic wave.

From where if we follow his approach, the Champ de Taofel 1 ' *1
(since it is about him), behaves like an electromagnetic wave. And we fall back into the points of view of Bélizal-Chaumery before the last war, who also thought of electromagnetic waves for Emissions due to Forms. This point of view was followed by J. de la Foye and others a few years ago. Can we axiomatize the existence of transverse waves which are not of a known nature? and posit that these transverse waves are what we call the eifs at the BCM level and the Taofel Fields? We cannot talk about these localized notions because we use a language and representations where we try as much as possible to escape localization. But about the stones emerging from the ovens, the Sprinks were able to discern three different structures:

- the "honeycomb" structure: the imprints are generally in the planes parallel to the planes of the rock deposits and are arranged in almost parallel rows, which thus gave a very clear image of the elastic vibrations of a solid under the influence of a transverse wave. And if we express ourselves in the language of vibrations, we would have to deal with lines of hyperbolic currents.

- the structuring of stones that peel like an onion; especially in the kidneys of the limestone: what is remarkable is that whatever the origin of the kidney, we always obtain the same thickness of the peels. This was the starting point used by L. Sprink for the calculation of the acting wavelength. In the language of vibrations we would have to deal with elliptical current lines.

- the structuring in rods, the length of which can reach several tens of centimeters. Mathematically, it should be possible to achieve this structuring for the two preceding cases, on condition that the limits of the two opposite cases of a vibrating cell are rejected ad infinitum. Obviously this is only possible in the direction parallel to that of the layers of the deposits, otherwise there are structural discontinuities and reflective surfaces. Bjerknes pointed it out that way. Except for the honeycomb structure, the two others can be obtained naturally, although in a completely exceptional way. Now, with their devices, in operation, the Sprinks had them regularly; but of course, let's remember, the correct operation of their device was obtained only after long start-up times, and when they cut off the high voltage, there was a very long recovery time - after this high voltage had been restored. In conclusion, we detect with the phenomenon mentioned above on stones a decalaire and the C. of T. 1 i -----1 1 .

In an attempt to determine the nature of these famous emissions, the Sprinks carried out laborious tests from the years 1951 to 1953 and they had added to their device (comprising an octagonal frame, spark gaps, stabilized high voltage , frame support), a field variable magnetic field and this in 1952. The frequency of the magnetic field was not very large: two megacycles at the most. Some sort of resonance seemed to occur with certain frequencies of the magnetic field. The resonant frequencies formed a discrete 1-2-3 series... They concluded that their device was both an amplifier and an acoustic resonator. Here we have analogies with usual devices; this is also what is found when using a fairly cylindrical tree trunk of radius r ; one can find a propagation of the eifs which resembles a kind of rectilinear optics, with however different laws (propagation between the circle of radius r and the coplanar and concentric circle of radius $2 r$) [9] .

Naturally this analogy was reinforced by remarks made by L. Sprink, such as the following: acoustic resonators (a Helmholtz sphere or the pots of Romanesque churches) respond only to the vibrations of the notes for which they have been tuned, whereas a electromagnetic resonator only responds a little better to the frequency which is its own.

Mars L. Sprink says "Our device is not a self-contained transmitter, but a resonator whose exciter is elsewhere. It also makes a dis

tion following that he used the low frequencies of the magnetic field (up to 25,000 kg cycles) where he says that then it is the acoustic resonance whereas between the 2 there would be a mixture of the 2 types of resonances. Above 100,000 kg cycles he noticed that he had hydrodynamic waves .

Let us not forget, L. Sprink saw the world only in its localized presentation despite that "his approach was to assemble whereas the usual scientific approach is to divide" Einstein had told him during a visit that L. Sprink returned to him before Einstein's departure from Germany.

But the Sprinks had clearly realized, despite the fact that they were thinking of resonances, that the effects, in particular, in vertical furnaces could not be attributed to electrostatic fields because an electrostatic field due to a spatially localized electric charge, could not have a range at least equal to that which they had observed on rows of ovens separated by several tens of meters and this, without the effect being weakened. Because we know that the electrostatic field decreases as the square of the distance. In addition, the effects on the aligned ovens were not, over time, felt in order of distance from the source but the first oven got carried away then the third, the fourth, then the second etc... and this reoccurred in this order after shutting down the device and turning the high voltage back on. Thus, thanks to the high frequencies of the magnetic field, they had the assurance of a new phenomenon, the effects of which appeared as waves of a hydrodynamic nature.

They measured the thickness of the "peels" of the structured stones in the case of elliptical currents and this gave them a wavelength (if they assumed that they were dealing with "massive" waves) of about 7mn.

The "weather harp", too, can be considered from the point of view of L. Sprink as a resonator, the frequency of which is audible. The measured wavelength still justifies the name wave form given to these emissions, as had been supposed by Chaumery, Belizal and others. It must be admitted that the temptation is great to confuse them with waves. But if we consider that the universe has in it a delocalized existence, we notice that certain notions of the delocalized partially resemble notions of the localized world.

Let us take this opportunity to say a few more words about ancient harps . They have been little reported by the muciciens. A book by Georges Kastner [11] deals in part with the following subjects: The Aeolus harp - origin of music

that - Origin of the myth of the Harmony of the Spheres - Sound phenomena, examples taken from the Bible, from Greco-Roman antiquity - stories of travellers-Echoes - Harmonies of caves, rocks, waterfalls - Vegetable Harmonies. We cannot cover all these topics here. Some, like many echoes, are explained in physics by the reflection of sound on walls, but others, like the music of the spheres, are much more difficult to study because they are heard by a few people at certain times or can be heard during duplications.

. (*)
Samuel Franeric had such an experience. L. Spnnk seems to have heard this music one night near his camera. Naturally, it does not appear **that it** was the wind that blew on the spark gaps to produce these sounds, the device being enclosed. We will nevertheless, here, look at these Aeolus harps and so-called meteorological harps.

The origin of the Aeolus harp is not known. ... "Legends and even true accounts, says Kastner, reveal the existence of natural Aeolus harps, produced by the rustle of foliage, by the friction of reeds and tall grass waving on the breath of the breeze.. .". Nature was able to inspire researchers, who used the wind to excite the strings of instruments they made.

Kichner has a slight tendency to reject any attempt at a non-rational approach - which makes him very fond of instruments that work with the wind.

The Jesuit Father Kircher and shortly before him Jean-Baptiste Porta (Neapolitan born around 1540 approximately) studied treatises by researchers who preceded them on Aeolian harps and trumpets and sound pipes. Likewise Father Bartoli, also a Jesuit, after Kircher and Porta, studied all these works; an English physicist also, William Jones and later Matthew Young (from Dublin) mathematician treated the harp of Aeolus scientifically. It is obvious that this list is not exhaustive; for more precise details it is good to relate to the work of Kichner and certainly that anonymous researchers also obtained results; often on the death of the researcher, the writings and apparatus are put back in attics and after several moves end up in the fire or in the trash and we have even seen in some cases that the families, as soon as the brilliant discoverer dies, hurry to everything_destroy_by_reason for fear of witchcraft which could have interfered.

(*) Researcher of the ARK'ALL Foundation.

in these works. Under the pretext of prudence and prudery, it is an **attitude** of profound stupidity.

We have already seen that L. Sprink had knowledge, perhaps through the work of Kastner, of a meteorological harp, that of Ventan, prior at Bulki near Beles.

The meteorological name was given by Father Gattoni who had made a gigantic harp (he called it: meteorological harmonica).

It consisted of 15 metal ropes and guts, attached to the top of a high tower about 85 meters and 150 paces from his house to which he fixed the ropes so as to form an inclined plane. The instrument initially gave results of a pleasant harmony, but soon the agreement was broken by the atmospheric influences and it is then that the inventor of this instrument imagined to make an instrument of meteorological observation of it in the means of intonation of the strings resulting from the various states of atmosphere.

Let us see the remarks of Canon Gattoni:

- 1) It does not matter whether the metal wires are insulated.
- 2) You don't hear the sound quivering when the wind blows violently and shakes the wires, but only when the air is relatively calm.
- 3) No sound is obtained when the wires are less than 10 m long, annealed wires are more favourable.
- 4) The wires produce this same quivering, but less sonorous, when stretched with glass supports; it is still less noticeable when they are stretched with silk cords.

They become electrified at the approach of storm clouds, like any other insulated metal. One cannot however allot these sounds to electricity, ccnune some wanted to support it without any foundation adds the canon (it is a question here of static electricity). Nevertheless adds Bellani (in a footnote reported by Kastin) if it is about dynamic electricity, one can obtain senses but let us notice in this case there was dynamic electricity.

- 5) The variation in the state of the atmosphere for Gattoni is the cause of the effects produced by the harmony of the threads: for if the sky is serene or rainy for a long time, no quivering is observed in the threads.

- 6) It is proven that sounds are not produced by accidental causes, such as the impact of birds, insects...

7) It has also been proven that changes in temperature only influence metal wires by successive expansions or condensations, producing quiverings which depend on the relaxation or stretching of the wire.

In summary, it emerges from these remarks of the chamonine Gattoni, that the meteorological harp is not excited solely by the wind as the aeolian harp must be, although certain aeolian harps certainly have the characteristics of the meteorological harps, but that the weather changes that are to occur are signaled by this instrument. It therefore seems that a change of weather is necessary and that the phenomenon which interests us occurs before the change of weather. It is for the observer attached to the harp, an annunciator.

It appears from the high-field eifs of Taofel. We are going to find ourselves with a partially delocalized process which is localized. Note that these meteorological harps have yn ext (a usual harp does not have one; moreover, a harp does not emit Taofel fields except for certain very specific tunes). Could a variation in atmospheric pressure initiate the phenomenon of the singing of the strings? It is not impossible. But it is interesting to think back to the noises reminiscent of those of jets of pressurized steam which were heard near the Keely bottles (the noises are different in sound from the point of view of those of the weather harps but can be linked to them by the phenomenon of delocali station). If we consider the occultist point of view, we can relate these phenomena to the "ether of sounds". This ether is in reality the Aether of which L. Boutard speaks. The alchemists also speak of "heat ether"; they distinguish several "ethers". This corresponds to their unexpressed feelings, and very vague for most of them (vague in the sense that the concept of ether for them was isolated, not attached neither to a coherence, nor to a field of coherence).

The aether can contain the small snakes of which L. Boutard speaks but also more elaborate notions. The small snakes are only the result of a projection of a fundamental structure as for the usual field of coherence, the atoms, and fundamental particles are the result of a search for the materials of assembly of the matter. The sounds that came out of the bottles associated with Keely's engine, such as those of the meteorological harps, **David's Harp**, are on the same level as the production of glycerine, colors, mass etc. which come out of the generator in the shape of Boutard's rhombus. It is the same, moreover, with the music of the spheres, which is a localization. At times, **which** L. Sprink could not control and very rarely, at night, his camera played **the** role of L. Boutard's rhombus. The music of the spheres can be heard equal—

lies during a duplication; it is the one that doubles which is the equivalent of the device with a rhombus; it is the state obtained by doubling which gives this possibility but certain individuals can hear it while looking at the starry sky, on a summer evening, in an atmosphere of solitude. (*) [11]

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(*) S. Franeric actually heard the music of the spheres under such conditions.

On the work of Léon and MT SPRINK

Where

Historical report relating to a set of various devices intended to improve certain chemical, thermal, metallurgical reactions and whose goal is an energy and financial saving in the future world.

by P. SMIRNE

CRAVE

(France)

Intro .

We present in this report the validity of physical observations and industrial results covering a period of nearly 130 years. All of this work relating to changes in the state of matter began with analyzes and scientific remarks made around 1850 and published in various reports by the Academy of Sciences. Practical applications on various industries were carried out in France from 1920 to 1955 on an experimental basis. Currently, a resumption and continuation of previous developments is considerably strengthened thanks to the technological fallout from the measurement of certain physical fields in space following NASA experiments and astrophysical information collected by Soviet satellites.

In addition, newly experimented effects at the level of emissions , absorption, resonances due to forms oriented in space, now make it possible to clarify certain variables or phenomena considered as difficult to reproduce for lack of being able to control the parameters even in fine physics measurements. classic.

Chapter 1 . Principles of Sprink-Ravatin devices made experimentally in operational models and practical advantages .

- 1) Summary of patent n° 1,005,532 issued on December 26, 1951 to Mr. Léon Sprink and Mrs. MT Roussel.

... "Improvement of certain remarkable chemical reactions such as combustion reactions, reduction, hydrogenation, etc., by the use of metal points mounted on an insulating support in the form of a frame and connected to the two poles of a current generator high voltage continuous...

Experience proves that such a device put into operation near autoclaves, cement and ore kilns allows reductions in operating time from 24 hours to just 3 hours with a calorie saving of 50% maximum.

2) Summary of patent no. 966,735 issued on April 8, 1950 to Mr. Léon Sprink and Mrs. née MT Roussel.

... "Recognition that a large number of changes in the state of matter or reactions depend, between the classic factors of temperature, pressure, etc., on electrical or magnetic phenomena produced nearby from a certain degree of activation of effluvium of crossed electrons and laid out according to a determined orientation."...

The experiment shows that for reactions such as the clinkerization of cements, the decarbonation of lime, saponifications, the reduction of ores, and when using the device near the products to be transformed, we always observe a gain on operating time and on the fuel used.

3) Patent application no. 78 - 21083 of July 13, 1978 by Mr. Jacques Ravatin, Doctor of Science, ESCIL Engineer.

Summary : The subject of the invention is an apparatus making it possible to amplify, according to a so-called "localization" principle, the emissions due to simple geometric shapes by associating them with a device generating electric, magnetic and electromagnetic fields, as well as to a closed polygonal frame with the character of right and left Moebius strips. Are associated with the volume generated by the devices of natural or artificial minerals such as

- Pyrargyrite or Arggyrthrosis
- Proustite or Red Silver
- Obsidian or volcanic vitreous paste with low water content
- Carborandum or Silicon Carbide: CS i

of which it was noted the amplifying power of the effects obtained as well as of

confederate a character called "Fractal* ¹ (well known in mathematics) to the parameters of the close physical fields.

The practical interest of the variation in the value of close physical fields is directly linked to changes in the state of matter with a considerable acceleration of the chemical and thermal reactions sought for obtaining conventional industrial products or new steel products with a reduction of the energy potential usually required by at least 30%.

The additional advantage of J. Ravatin's device is positively materialized by the instantaneous obtaining of the desired phenomenon or phenomena, whereas with L. Sprink's device, the start-up time could sometimes require a few days to several weeks.

Chapter II . Different types of observations and experiences that have brought patent applicants .

L. and MT Sprink have made experimental devices capable of saving fuel time and obtaining better or new manufactured products.

1) General remark . The observation of certain natural phenomena in physics , chemistry, biology, sometimes makes it possible, by applying a practical analogy, to realize on a small scale what nature does on a large scale, or at least to draw a conclusion which will help man in art, industry or science. Thus the knowledge of the interstellar radiation in the vacuum allowed the realization of the radio lamps, the comprehension of the radiations of the radio activity brought the realizations of the "nuclear" and the sending of electric current in the ground allowed the brothers Schlumberger to build an empire based on the search for oil since 1912.

2) How L. Sprink was brought to be interested in the problems of the Forms , their action and the fields of force compared to bacterial or chemical reactions at various temperatures.

Léon Sprink comes from an old white-Russian Baltic family allied to the Tolstoi. Very early on he showed qualities of observation, analysis and realization whose originality challenged him to pragmatism. For example: in his 15th year his father took him to visit a brewery in his native country where it was difficult to obtain by the then known processes a beer of mediocre quality.

tee that gave a lot of tablature to professional brewers.

Barely returned from this visit, the young L. Sprink imagined a model of tanks whose alloy and especially the forms allowed the accelerated fermentation of the hops, giving a beer of much better quality and more quickly than by the usual artifacts.

In view of the intelligent execution of this model and in view of the positive results obtained, his family directed him to the Polytechnic Institute of Saint Petersburg from which he graduated before the First World War.

Around 1913 L. Sprink had the chance to observe an unusual phenomenon of the solar corona during an eclipse. The emissions he observed gave the image of a cross of Saint Andrew outside the halo, that is. an extremely rare and unexplained phenomenon at that time.

The main work of observations and attempts to correlate phenomena related to solar plasma, and the aurora borealis were made by Gascendi in 1621. Have done work on the subject in order to "understand" the interliaison of phenomena, Cassini 1686, Halley 1716, De Mairan 1726, Arago 1855, Angot in 1897 etc... a special place goes to Auguste Bravais, author of the theory of networks for the intimate structure of crystals who established correlations between these structures and the elisation of phenomena relating to halos, parhelia, aphelia and unusual diffractions observable in the Earth's atmosphere.

This French physicist, member of the Academy of Sciences in 1854 tried to connect the close and distant observable effects by similarities such as Sprink will use some later.

It should be specified here that from Gassendi until 1937, the total time of the multiple observations of the solar corona during the eclipses did not exceed a few hours due to the very short time of the possibility of observing the total eclipse (1).

L. Sprink therefore had not only the possibility of making an exceptional observation but above all the intuition of bringing closer to the vision of these - plasma efflu ves, the then brand new notions in electricity and propagation of electromagnetic waves (developed by Rhumkonf and Hertz) a possibility of acting at the level of the earth on the phenomena of combustion and heat by combining all of the thermal reactions of the currents of static electricity

emitted by a stabilized generator.

At that time, fertile in multiple and primordial discoveries, it turned out that many intuitions could be crowned with success at the level of technological reality, while mathematical theory could only account for the reality of phenomena for a few years. later. This was the case for Sprink's devices in its first trials.

The events of the First World War as well as those of the Revolution led to imprisonment in the Saint Peter and Paul fortress for L. Sprink, who came out of it, but remained physically very ill.

He emigrated to France where he married Marie Thérèse Roussel, granddaughter of a famous French geologist.

The couple will then work together on all the theoretical and practical achievements aimed at obtaining new chemical reactions at high temperature, products with new structures, by synthesizing the effects of electric fields supposed to exist in the upper atmosphere with gradients more or less high thermics obtained by various combustions.

During this period, the Sprinks were in frequent contact with Maurice and Louis de Broglie, the Cottons, the Curies, who understood Sprink's work as the dawning era of all new chemistry, all new physics and the opening towards a fruitful understanding of the true field relationships existing in the solar system.

The Sprinks went to Germany in 1930 and met Einstein at length to explain their work to him.

The conclusion of the latter is summed up in this sentence: "I try, says Einstein, to analyze the universe and you, you recreate it before our eyes".

3) Sprink's fundamental experiments and first observations of facts not yet observed by the predecessors: Coulomb, Faraday, Gauss, Whimshurt... we saw above how Sprink, according to his observation of a spatial phenomenon, had connected the new possibilities of carry out chemical and thermal reactions by associating a stabilized high electrical voltage with it, which at the time we could assume that this voltage must have existed in interstellar space and in particular around the sun. It should be noted that the notion of solar electro-jet only came after the discovery of thermo-nuclear reactions.

res and measurements made by rockets and satellites.

L. Sprink therefore carried out a first rudimentary assembly consisting of a very dry circular wooden frame through which were fixed in a cross of Saint Andrew along the diagonals of the metallic conductive electrodes **themselves** connected to a stabilized high voltage generator .

When setting up the assembly on an insulating support, a first remark was made by Sprink: the electric discharge from the electrodes, instead of passing along the shortest path, the case of a classic arc, was heading towards the opposite electrode. Which means that the electrical discharges crossed in the axial extension of the 4 electrodes, also forming a cross of Saint Andrew. At the point of intersection of two effluvia then a new fact was produced, not observed until then and of which the known formulas or laws of electrostatics do not account. Only the so-called "power of the spikes" effect on a flame is known.

We insert here a remark made by Buffon and recorded by him in a little-known text that Mr. Cotton, the physicist who invented the famous balance , had shown to Sprink:

"In the 18th century the aurora borealis were frequent in France. The dates of observation of the auroras according to Buffon corresponded to frequent fires ravaging entire villages in the north of France and Europe, wooden constructions (the wood in the constructions being of use more consequent than now).

L. Sprink made a similar observation in 1938 during the aurora of January 25 visible in France accompanied by the destruction of boilers and chimneys.

The Sprinks experimentally carried out a certain number of chemical and thermal reactions in closed environments: autoclaves, furnaces, coke ovens, etc. by operating their device in a cross pattern nearby, noting very regularly and punctually the actual calories used in the classic reaction and the times spent to obtain the products manufactured industrially according to the demand of the national and international markets in order to be able to compare their own production times and the real economy achieved.

I? I

■
|> see normally spent fuel.

The first results were promising enough to have to be addressed to interested industries. We will briefly review some of them.

! elements.

4) Origin of Industrialization and Improvements to Sprink's Device from 1923 to

1950 . i I
I a) Fundamental discovery of autocorrelation of irregularities in yield

■ of a soap factory in relation to the activity cycle of the solar plasma and to the cycle

■ of the earth's electric field.

* In 1923 Sprink owned a manufacturing process in Strasbourg.

tion of soaps with high detergent power that a stearin factory had adopted.

J According to the factory boss himself born in this industry it was known that r
| the manufacturing "recipes" only "lasted" for a certain time with irregularities.
yield larities and then we had to find something else...

L. Sprink and his wife then carried out studies in autoclaves in which, with regard to production statistics, various cyclic laws could be established by a rapid harmonic analysis of the progress curves of reactions in autoclaves with regard to various daily, nocturnal , seasonal (equinoxes and solstices) annual, decennial etc... linked on the one hand to the positions of the moon and the sun and on the other hand to the terrestrial vertical electric gradient. A correlation could be established between the chemical thermal reactions in a vacuum with the aqueous and terrestrial tides as well as with an aerial "electric tide" (terrestrial electric vertical gradient) and the terrestrial horizontal electrotelluric activity which we currently know (since 1952) correlation formulas with terrestrial magnetism.

i Let us point out in parentheses that the world industrial developments of applied geophysics for the discovery of hydrocarbons have only been possible through the detailed analysis of several million measurement points of these various parameters, the laws of which we are now beginning in 1980 to master.

Commissioning of the first device .

The conclusions of the Sprinks were that to bring production back from 11 p.m. to 3 a.m. (!) it was necessary to remove the autoclaves and their reactions - from the action of the aforementioned fields. A "grounding" device for

press frames were made which had the effect that in 1936 the factory was running at 150% efficiency against 1932 taking into account the working hours reduced to 40 hours instead of 48 hours and without changing the "recipes".

However, although the application of the new process preserved the plant from the most troublesome fluctuations in its production, there were still peaks in the qualities and yields above the critical threshold.

b) Continuation of the industrial work of the Sprinks at their laboratory in Meaux (Seine et Marne) 1938-1940 and at the Cimenterie de Port de France from 1943 to 1944 with the aim of obtaining massive savings in fuel .

Around the 1930s, Mrs. Sprink was also studying the problem of the formation of cements at low temperature and had arrived at a very beautiful manufacturing process using the horizontal Mecker furnace. However, as in stearinage, the results and the yields presented differences linked to another unknown factor not linked to the terrestrial electric fields nor to the lunar and solar tides as regards the Newtonian action.

However, based on certain observations of the solar corona on the one hand and the appearance of the spots on the other hand, by correlating the anarchic deviations from the variations in the quality of the cements, it was established that the deviations were directly linked to the electro-magnetic activity of the corona (currently the electro-jet) and a terrestrial consequence translated into our upper atmosphere around 100 km altitude which are the aurora borealis.

In brief the law of interdependence of the phenomena of coronal radiation in relation to the two passages of the sun on the lunar nodes on the one hand and on the other hand the consultation of the variations in quality of the cements of Port de France (Isère) recorded over forty years of daily analyzes allowed cross-checking.

It appeared that the tensile strength of cements analyzed from 1898 to 1939 followed the cycle known by astronomers of 18 years 11 days under the name of Sarros. This cycle corresponds to 223 lunations or 19 times the interval of 346.6 days separating two passages of the sun through the lunar node.

It turns out that the periodicity of oceanic transgressions also partly follows the same law. As above, there appeared a correlation between these various phenomena and the physico-chemical balances of the reactions analyzed •

It was then that the Sprinks used their second device consisting of 4 cross-shaped aerial electrodes placed on a circle and powered by a stabilized DC high voltage source. This device was the subject of the patents mentioned at the beginning of our report, Chapter I, 1.2.

The coupling of the two devices was tried with profit at the glass works of Mers les Bains in 1949 where significant rises in temperature were observed for a reduction in the tonnage in combustion of pulverulent coal and fuel oil. Tests were also carried out in Brazil in 1951 in a lime potash plant. But the results remained inconsistent because the situation is complicated by the fact that the solar periods do not have a strictly fixed duration and can present random periods of encounter with the moon according to variations ranging from 9 to 15 years.

5) Attempt to develop a theory in the face of the complexity of phenomena to fully control chemical and thermal reactions, work from 1951 to 1955 .

In view of the achievements already obtained on the lime kilns and in the manufacture of cement, the important economic element was reflected in a gain in fuel (coming from the same coal mine) which could reach a rate of 50% without clinker. However, this economy did not follow a known law , which is acceptable considering the complex remote phenomena on which depend the thermal reactions in chemistry of high temperatures, the states of matter.

In short, the Sprinks carried out a series of tests which led to the following conclusions: at low frequencies of the magnetic field:

a) - up to about 25,000 KC/S the acceleration effects are related to electromagnetic resonance; at higher frequencies an acoustic resonance is established from 100,000 KC/S.

b) - the Sprink device is not an autonomous transmitter, but a local amplifying resonator whose excitation is found elsewhere.

1. - At low frequency, it is an electromagnetic resonator, moreover thwarted by terrestrial radio broadcasts.

2. - For high frequencies, the effective waves are of a mass nature with a hydrodynamic character.

After the tests on rows of vertical cement kilns, the only hypothesis of the effects observed simply due to electrostatic fields was insufficient, a spatially localized electric charge not being able to have an action over several tens of meters without any weakening, the field electrostatic decreasing with the square of the distance. So there was something else, because the device in its applications was only effective when oriented perpendicular to the earth's magnetic field .

There remained a fundamental point to be elucidated: what was the primary radiation source ? the device being a "relay-resonator-amplifier".

During the tests on rows of lime kilns, another curious fact shows experimentally that the rise in temperature, the intense drafts and the unusual thermal reactions first appear at well-determined distances from the apparatus arranged in circular areas analogous to known images of Saturn's rings with similarities of gaps reminiscent of asteroid belts around the sun.

A phenomenology causing such effects was unknown until now and the Sprinks seem to be the first to have observed all these facts, to have analyzed them with the means at their disposal and above all to immediately make industrial use of them .

For the nature of this radiation, the Sprinks were reduced to hypotheses - approaching the theses of Louis de Broglie on wave mechanics, in particular in the two different states: the vector and the pseudo-scalar, also valid for "mesons" or heavy electrons than for "photons". According to Maurice de Broglie, with whom the Sprinks were linked, interactions between particles via pseudo-scalar photons would be spin interactions.

It would therefore seem that the extreme ease of establishing homopolar links in the presence of our device in operation confirms this hypothesis.

In the conclusions and the mathematical developments carried out by Sprink it appeared according to him that the very electric and magnetic fields of the earth , other their derivatives and known slow variations, were subjected to other rapid variations at very high frequency.

Sprink admitted that the rhythm of the device's discharges was regulated by the magnetic field which is perpendicular to them, that is to say the field ter

remains whose various rhythms are those of the phenomena emanating from the solar plasma.

Sprink considered that the field of his oriented device could be described as a variable magnetic field at very high frequency, without distorting reality, especially since the intensity of the currents used remains very low.

Sprink considered to be in the domain of mass resonances which caused this radiation to be confused with simply acoustic waves, whereas at very high frequencies it is plasma vibrations which appear.

Moreover, in relation to the frequencies effective for the purpose sought, Sprink has made standing waves visible by extending the electrodes with strands of fine red copper soldered to the ends. On these small strands appear luminous points located at the nodes of the waves and correspond to the two lines of absorption of the spectrum of rotation of oxygen of 2.5 mm and 5 mm. This gas could be the resonator par excellence of the primitive radiation. There is an analogy at the level of the solar prominences with the strands of thin wires and their luminous phenomena; indeed local reinforcements of the brilliance appear on the photos of the paths that the pulsed plasma travels in the direction of the crown according to the laws of hydrodynamics.

In the hypothesis of a mechanism of the propagation of the mass field Sprink excludes the propagation by wave: electromagnetic because of the presence of the stationary waves EM visible on the photos of the wire strands quoted above since the OS do not radiate. It also excludes material vibrations because the transverse waves observed physically near the devices do not propagate in a gaseous medium. However, the radiation is linked to very significant ponderable effects, probably amplified by quantum resonance in the reacting materials. Sprink thought he was in the presence of quantized pulse energy flow. This way of seeing is in agreement with the development given by Einstein for the global weight-motive force.

In the analysis of the phenomena present there still remains that of the formation of a vortex magnetic field which can only be treated by the methods of wave physics. Sprink believed that this field could be described as a spinal field which could only be propagated by pseudo-scalar photons following L. de Broglie's theory of wave mechanics.

6) Stopping of the theoretical and practical work of the Sprinks and difficulties encountered with the collaborators.

As can be glimpsed in the summary of the Sprinks' scientific and industrial adventure, the unusual, the complex, the permanent contact with unknown variables were the daily companions of the Sprinks.

From what we can know of their relations with the great international names in science, they have always been excellent, whether it be Maurice de Broglie, the Curies, the Cottons, Einstein or others. contemporaries who, while having admiration for Sprink's work, were themselves too absorbed to devote themselves to a common collaboration.

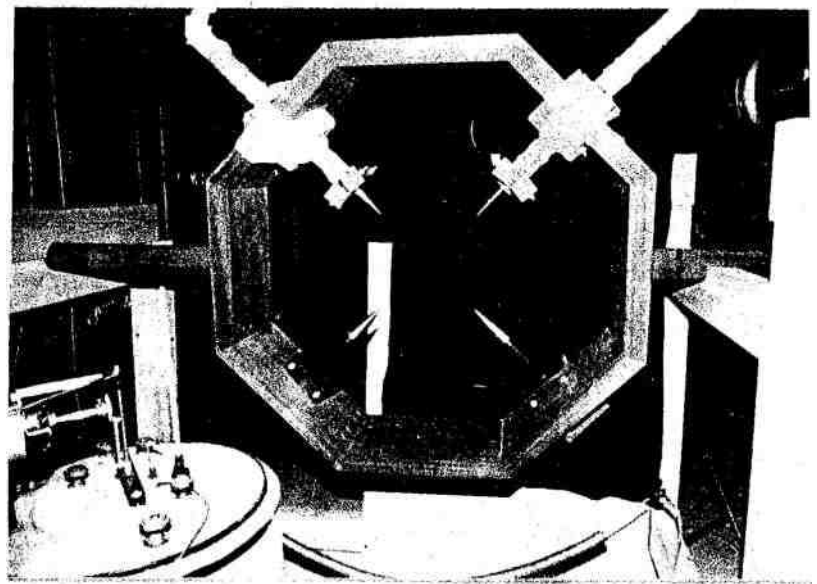
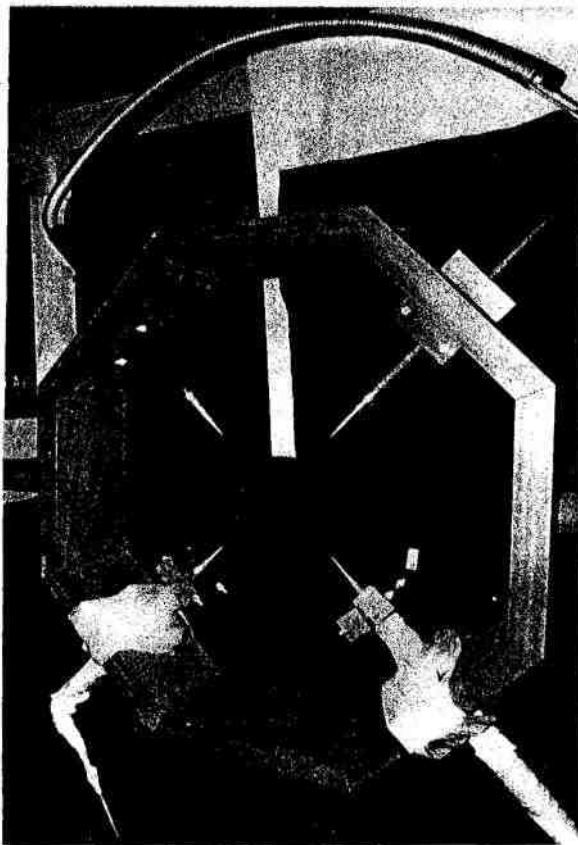
In their contacts with factory bosses, the latter not being at the head of a study or research company, very often these heads of industry saw themselves obliged to consider only the economic aspect of things and ask It was up to the engineers or technicians of their company to make their willingness to collaborate available to inventors.

At this level, the Sprinks have not always met with the understanding necessary to be perfectly followed. It should also be considered that their working hypotheses, especially before 1945 (Hiroshima), could seem incomprehensible to classically trained engineers working in factories at the time. On the other hand, with regard to the foremen and the workers, very often enthusiasm reigned in the face of the results obtained.

After the death of Marie Thérèse Sprink in 1951, Léon Sprink devoted himself more to theoretical laboratory tests and to writings intended to leave a satisfactory concretization of the ten industrial patents taken out between 1930 and 1954.

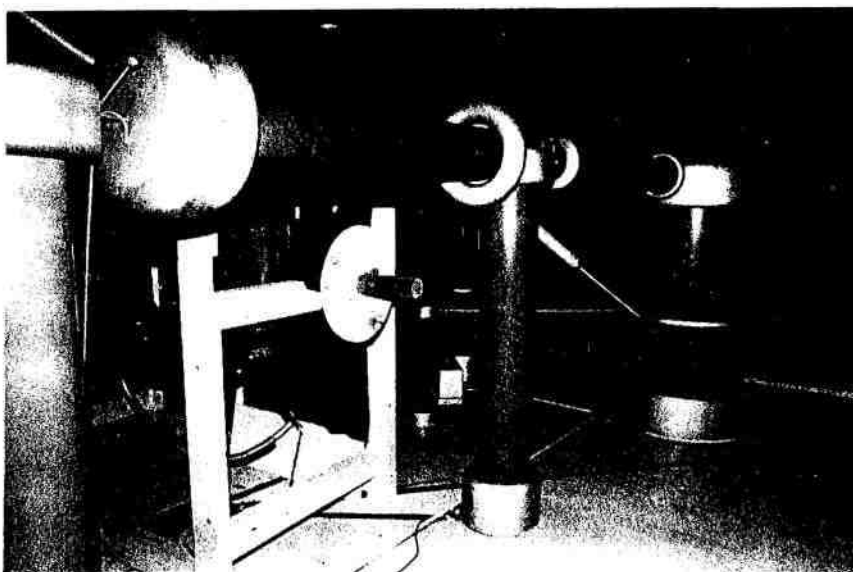
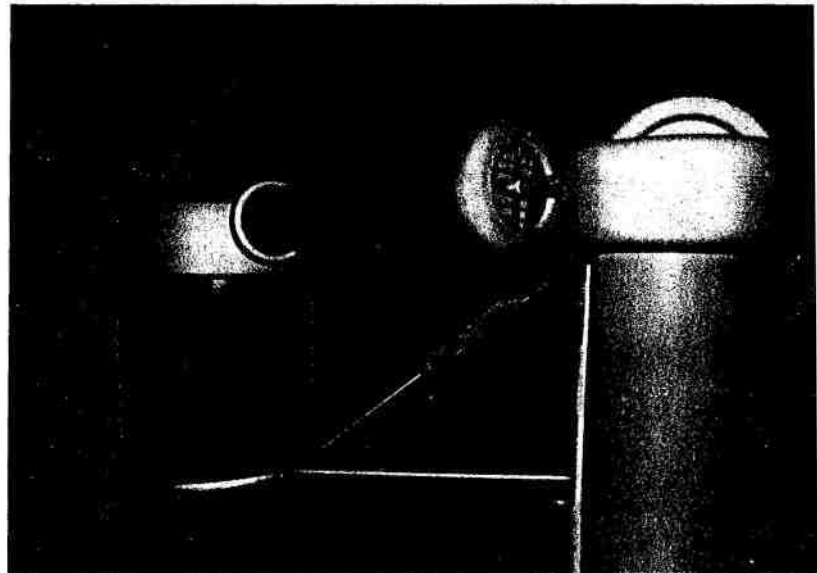
Sprink's last industrial practical work ended in 1965 for various reasons: either the factories had disappeared, changed production or had been reabsorbed by groups having opted for a different policy or range of products. Moreover, Sprink had always been a relatively independent researcher and financed all his research and experiments himself in his laboratory in Seine et Marne in Meaux over the last years of his life.

L. Sprink built gravimeters derived from those of lacoubovitch for the study of the horizontal component of the terrestrial "g". Sprink perfected

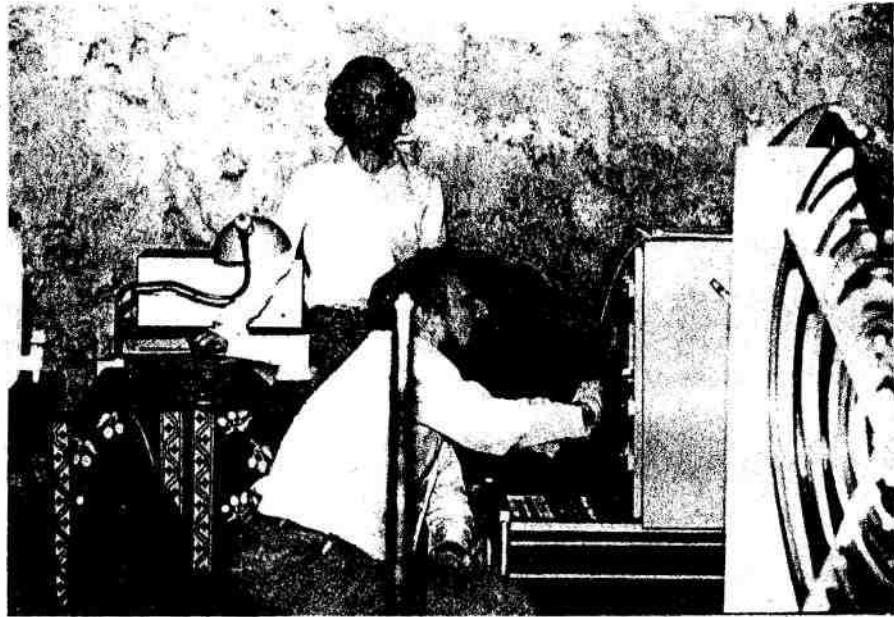


Sprinkle device. Editing by P. Smirns
M. Simonneau, J.-P. Béranger, C. Naville,
L. Baranger, J. Ravatin and other researchers
from the ARK'ALL Foundation.

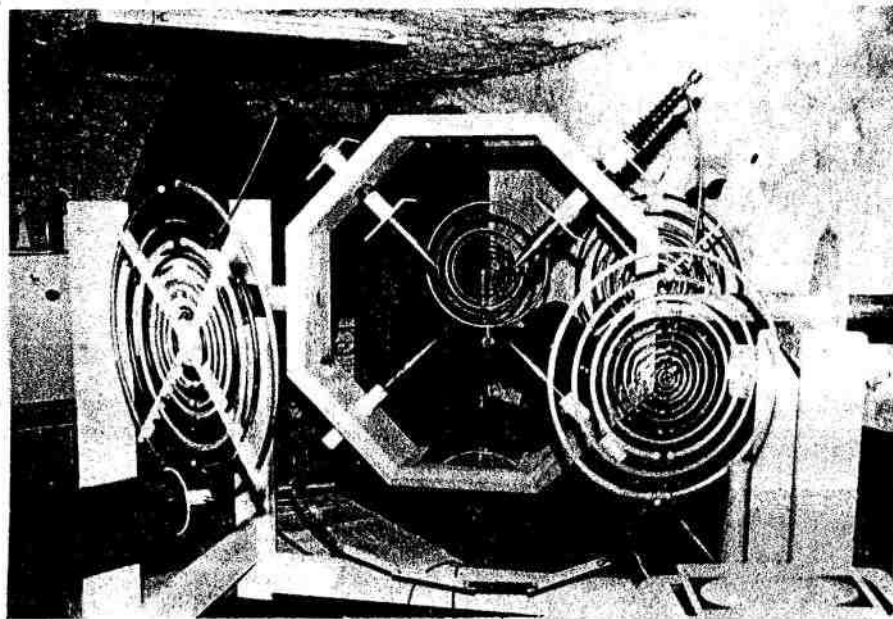
The basic shape used here is
octagonal. We could use other
forms, but not just any.



Electrostatic generator
must be taken
that its shape does
not interfere with that of the

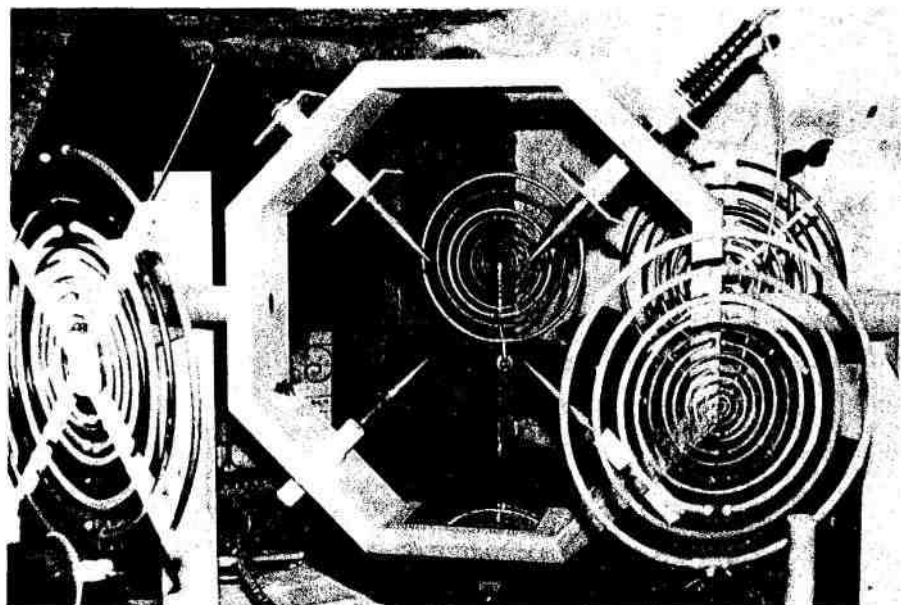
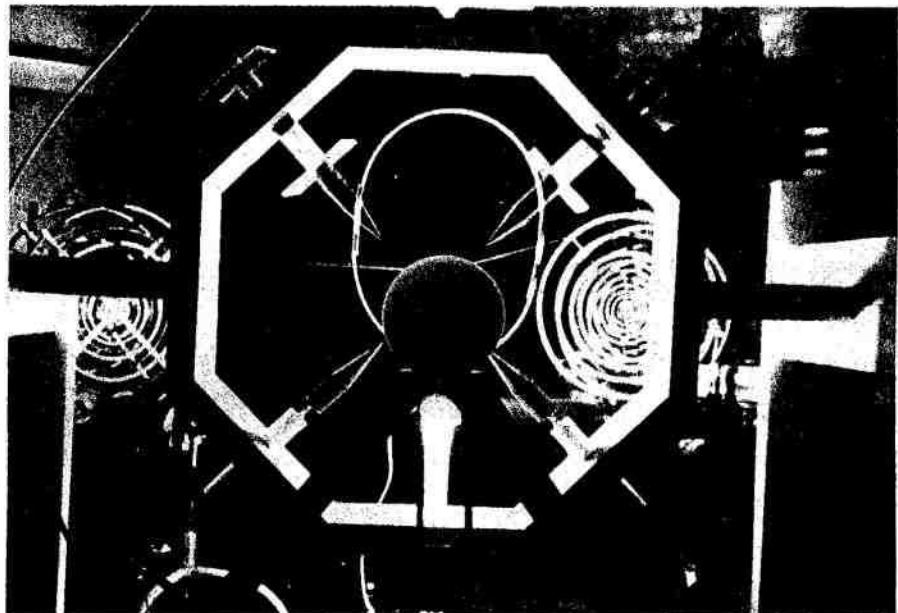
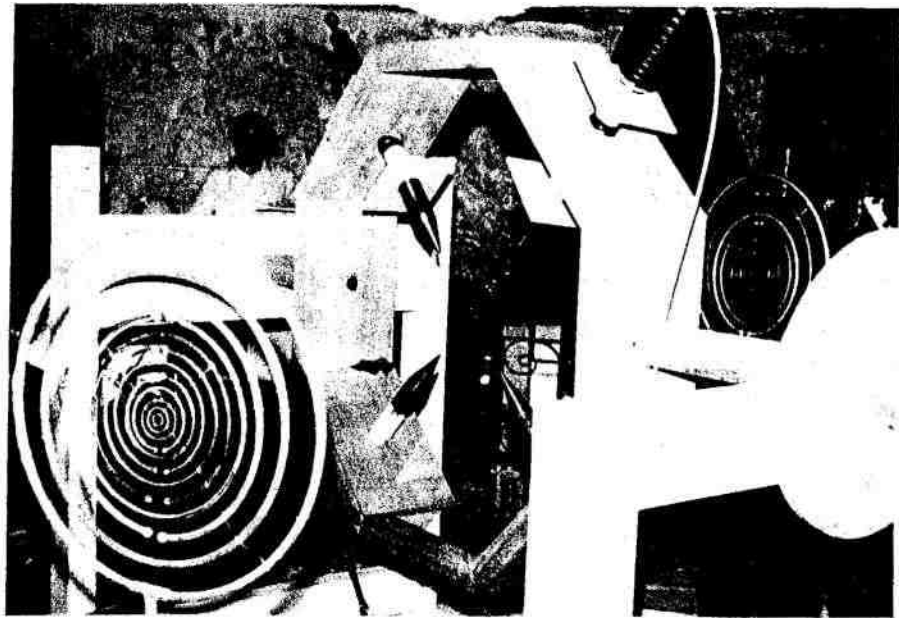


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Coupling Sprink devices and two Lakhovsky multi-wavelength oscillators

Produced with P. Smirne, P. Godart, A. Cheyssial, M. Simonneau, A.-G. Cheniere, K. Fidaali.



the equipment by installing an optical effect multiplier and adding a photographic recording. For these studies a collaboration was made with Mr. and Mrs. Labrouste of the Institute of Physics of the Globe and members of the International Gravimetric Bureau. The harmonic analysis revealed outside the periods like the luni-solar tides other periods and variations of horizontal "g" during the operation of the cross device.

In 1967 Sprink passed away as the last descendant of an old Baltic family of Protestant origin. His origin earned him a burial in the Cime tière de Sainte Geneviève des Bois, although he is not buried in the part reserved for the Orthodox.

Chapter III .

1) Continuation of the work of Sprink - New developments .

After his death, a succession of training not being assured, Sprink's work would have disappeared if a few particularly well-informed people had not put on the trail of these discoveries, the future of which could be considerable, a chemical engineer, doctor of physics-mathematics which took up the known theoretical part of Sprink's devices and continued it by completing it masterfully with the contribution of new discoveries, in particular on the notions of emissions and absorption due to oriented forms as well as by the notions of "Location", "Global", "Decalar", "Cumular" [1].

This is Jacques Ravatin, experienced experimenter and theoretician who took over Sprink's experiments and had Professor Mesnard of the University of Lyon, director of the electronics laboratory of this University, note the amplifying power of the effects obtained. as well as the Fractal character of these effects whose physical consequences Jean Perrin has also developed on the parameters of the physical fields linked to certain states of matter at the corpuscular level.

The practical application directly usable in industry with some of the enormous consequences that it represents is summarily exposed in the patent application of July 13, 1978 under the n° 78 - 21083 deposited at the office Casalonga, 8 avenue Percier, 75008 Paris in the name of Mr. Jacques Ravatin.

■ 2) Descriptive summary and experimental principles of Mr. Jacques' invention Rayatin .

We know from experience that certain simple geometric shapes presenting a minimum of symmetry manufactured in very varied materials are, when they present appropriate proportions and a certain orientation in space, the seat of effects in their close environment which , for not being explained clearly, are ascertainable;

- either on living organisms or organically dead cells.
- or on metals on which a molecular rearrangement seems to occur .

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The first observers used to describe the phenomena a ill-adapted expression: "waves of form" . It is more exactly a resonator absorbing or capturing locally a more fundamental notion than the energy emitting measurable effects according to a certain orientation.

These industrially exploitable effects or more exactly become so thanks to the amplifying apparatus of Mr. Ravatin which considerably amplifies the phenomena according to a principle known as "localization" in which geometrical forms presenting a minimum of symmetry are associated with devices producing a electric field, a modulated or non-modulated magnetic or electromagnetic field.

The various parameters allowing the amplification of the phenomena expected for industrial exploitation have been developed to the maximum. The start-up of the device to obtain the desired effects requires a time of about one hour for a noted action zone of about thirty meters.

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The generator producing the electrical potential must deliver a current rant of about 100 microamperes. We make him gradually increase a tension electricity starting from zero to reach at least 45 KV.

3) Effects noted on Mr. Ravatin's operational model .

1. - Action on the local terrestrial magnetic declination of approximately 6° towards the west after half an hour of operation with variations of effects depending on the orientation of the frame.

2. - Local action on gravitation At 45 KV of potential it is found that the local gravitational field goes to 0.9 times its value at the close measurement point

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of the apparatus taking into account the drift corrections of the gravimeters and the luni-solar variation of the tides.

3. - Local action on the refractive index of the air which increases all the more as the positive potential applied to the serial electrodes is higher.
4. - Observation that in combustion reactions usually producing clinker, there is no formation of clinker during the operating time of the apparatus and even 4 days after the voltage cut at the electrodes.
5. - Reduction of the quantity of fuel necessary in the classic combustion reactions to obtain the same result. The saving of 30% of coal is possible with a high voltage of 100 KV -100 microamperes.

In the case of an internal combustion engine using gasoline, there is a reduction of up to 50/» in fuel consumption for identical performance. However , the engine is stationary and not mounted on a vehicle taking changes of direction in space.

6. - Increase from 1 to 2.3 of the solubility coefficient of a salt in water.
7. - Increase in plant growth rate and size by a factor of 2 to 10.
8. - Possibility of obtaining catalysis without catalyst.

4) Ravatin 's approach to biology .

This statement has been deliberately restricted from certain possible developments applicable to the industry . We have discarded the important observations made at the biological level and on human behavior from the point of view of fatigue and which have been noted by entire teams of workers.

Compared to the experimental model used for the first tests, the development of the next prototype - subject of the 1978 patent application made by J. Ravatin - is based on the data of his doctoral thesis in science presented in 1965 on the so-called "non-Cartesian" systems under the authority of Professor Mesnard.

5) Verification of the validity of the results obtained by Mr. Ravatin .

Professor Mesnard, director of the electronics laboratory of the University of Lyon, supervised the experiments resulting from the theories of these non-Cartesian systems and followed up the experiments carried out.

It should be noted in particular that this new experimental physics theoretically and practically reveals a notion to ponder: the usual physical measurements, however fine they may be, only indirectly account for certain observable facts and in a fragmentary way.

The approach to the phenomena observed and highlighted under the control of Professor Mesnard by J. Ravatin involves the notion of "Fractant".

It appears that taking into account the foregoing, the findings and all the facts proposed for our analysis by the developments advanced and obtained from Professor Mesnard by J. Ravatin cannot be analyzed solely from conventional measurement methods. accounting, so to speak, only for the "spillover" of integrated facts; from the origin of our galactic system.

6) New ways, appropriate semantics .

Understanding and mastering the desired effects requires a terminological redefinition developed by J-Ravatin in which the expressions "Global, Local, Cumular, Shift, Cumulo-Shift and the concept of Fractant appear.

The aim pursued in the construction of the future prototype is to demonstrate by amplifying them the "bursting" effects of the classical space-time structure . The practical results will have industrial consequences on the level of chemistry and metallurgy - to quote only these two industries - comparable to those of the spin-offs of the conquest of space with an initially more immediate ease and a saving of obvious energy. The progress made in relation to the work of Sprink from 1925 to 1965 was possible thanks to the integration of notions unknown at that time both in relation to a better knowledge of extraterrestrial space and to the improvements brought by J. Ravatin .

7) Consider the future of a post-industrial society .

It can be foreseen that following this work developed over decades, we find ourselves at the dawn of an era in which the conceptions mentioned above will take on more and more importance, if only due to the global energy situation. From Léon Sprink's little sun to Jacques Ravatin's fractants, don't we find ourselves face to face with a re

construction of a universe as Einstein expressed it to the Sprinks... but could we add the industrial universe of the world of tomorrow in the face of res-! planetary responsibilities never encountered.

(*) NDLD Note the stability of the phenomena obtained by J. Ravatin. and their reproducibility.

There are also thresholds which are given by the voltage, the dimensions of the apparatus, the shapes involved. If the imposed conditions are neglected for reasons of economy, it is not possible to obtain certain effects [2].

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THE ETHERIC ENGINE OF JW KEELY OF PHILADELPHIA [1]

by J. RAVATIN

Paris

Keely was still a child when he began to occupy himself with the various inventions which he was to accomplish later» he also tried to discover the ether. For eighteen long years he worked day and night with hands and head, struggling against discouragement and almost alone. He suffered all ridiculous accusations of deceit, charlatanism, madness, everything that the brains of rogues could imagine and admit that of imbeciles, all the miserable insinuations, all the wicked lies that prejudice, ignorance, stupidity , interests established or to be established, injustice, greed, hypocrisy and dishonesty could invent; these are the encouragements given by the world to this discoverer, and to many others besides, of truth and other representations of things that only a few circles of insiders suspected. In front of multiple people able to appreciate Keely's work, it was proven that he had sensed new principles and developed machines that could use them. These were brought almost to perfection, at least they were perfection next to the technical objects of our Cartesian civilization.

The Keely affair became known at a time when the American mining world was seized with an inexplicable emotion; suddenly the value of the gold mines began to increase. Mines that had been abandoned for a long time, because of the expense of labor there, came out of their lethargy and resumed their place on the Stock Exchange quotation. It had just transpired that the possessors of the strongest heads, and the largest coffers, among the mining capitalists, were quietly and cheaply buying up all the abandoned gold mines of which they could have knowledge; immediately everyone had to conclude that there was something fishy. As a result, speculators began to buy abandoned mine shares at an ever-increasing price, and after the small stock marketers had stuffed their wallets well, they wondered what was going to happen. No one knew exactly what it was, everyone had their eyes fixed on the union of big capita

lists to regulate his conduct according to what they were about to see him undertake:

the syndicate contented itself with waiting; after the attack of fever comes depression

. The small fry of the Stock Exchange not having bought to keep soon felt uneasy, then gradually became frightened, frightened, and finally panicked. The tide of gullibility ebbed faster than it had risen, in no time, leaving thousands of little studs dry and thrashing in agony.

The laws which govern the movements of the stock exchange make it easy to realize such a panic; but they do not at all explain

A few weeks before the catastrophe in question occurred, a dozen millionaires (of the time of course) had an appointment at a certain laboratory in Philadelphia to see them experiment with the disintegration of quartz by a new method. These people were magnates of the mining empire and, as such, very much interested in discovering the means of extracting gold quickly and inexpensively from quartz.

The inventor was content to put in contact, under their eyes, a few quartz balls with a small machine that he held in his hand, and each ball thus touched instantly fell into impalpable dust in the middle of which the pieces of gold appeared. like pebbles on a beach. Then the twelve pei c:;iï\$ said with one accord,

"Mr. so many dollars." We left for the "Catskill mountains" and arrived, there Keely took out his small camera saying "Gentlemen, please take note of the time that the operation will last". In 18 minutes, a tunnel about 5 meters deep by 0.80 meters wide was drilled in this mountain quartz. This done, Keely quietly returned to Philadelphia with the checks in his pocket, and the twelve men got worse at buying up every abandoned mine they found from New York to San Francisco, which produced the disastrous stock market effect of which we we talked. As these people were sworn to secrecy, this was the first time that there was a slowdown in the public by the massive purchase of the mines.

How was the quartz disintegrated? This is one of Kelly's secrets!

i The disintegration of rocks is only one of the lesser effects of what
Keely found out. This particular application is due to a remark on what he called
etheric currents and researching on these he saw a ¹ t
a block of granite disintegrating before his eyes. A few days later he had realized
what he called an "etheric disintegrator".

ts Surely, all the echoes of the press, of the tribune resounded from
the announcement of this good news.

In fact, scientists have remained in a persistent ignorance of this domain
of Nature into which Keely has entered, that of delocalization. Yet a member of the
Royal Society of Edinburgh, Major Ricarde-Seaver traveled to Philadelphia to examine
the reality of Keely's discoveries. He returned by saying that it seemed to wield
forces whose nature was unknown to modern science, which had not even suspected their
existence. On the other hand Lord Raleigh, while he was in the USA at that time,
considered that these works were of no interest. But back to Keely. Keely thought he
was dissolving molecules, atoms etc. into ether. This ether which for the Ancients
seemed to be the protoplasm of all things. And in the light of Keely's experiments,
gravitation is only the manifestation of a much more general law.

One of Keely's little experiments was to wrap a wire around an iron cylinder
that weighed at least a ton and when the so-called etheric force? by the wire (such was
Keely's representation) any person could lift this cylinder with one finger and carry it as
easily as if it were a cork. He himself carried one of his vibratory devices providing 500
horsepower with one hand, from one end of his workshop to the other without even scratching
the floor. The engineers of the time, astonished, declared that they could not have carried
out this transport without a crane, which would have required the removal of the roof of the
workshop.

He also applied his principle (which he never explained simply except by
speaking of etheric force) to optics and by means of three wires laid across them.
| towards the lens of a microscope, he multiplied the magnification by a factor
huge.

; Note that many of those who witnessed his experiences on
acoustics, often too ignorant to understand its meaning, or too imbued with an
outrageous rationalism and stubborn after having congratulated it, barely

had they come out of his studio-laboratory they denounced him as a magician at a fair.

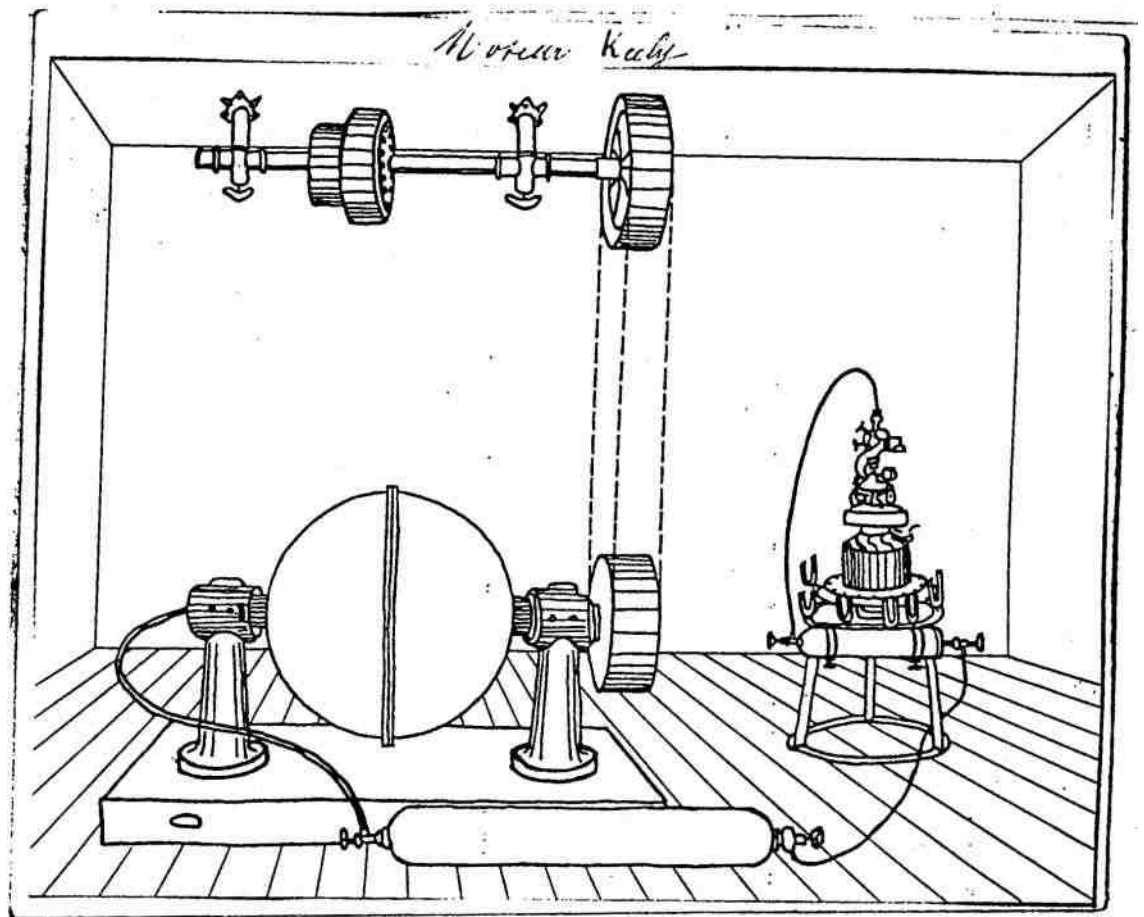
A company was created: the Keely Motor Company; but it was the stumbling block for the inventor. Greedy speculators, after bribing American newspapers, have played up and down on its shares. It no longer provided funds to Keely and it virtually disbanded. Mrs. Bloomfield Moore who helped him a lot denounces the attitude of the law gent towards Keely; so when he abandoned his old generator of etheric force, for lack of money, he was sued by certain members of the board of directors and certain shareholders of the Keely Engine Company who threatened him with imprisonment. He then destroyed a large number of his marvelous devices and took the firm resolution, if he were condemned to prison, to leave only his corpse there.

He did not find an individual generous enough to provide him with the necessary to obtain results which did not promise an immediate benefit. And this is found in all financiers and especially in French financiers.

Dr. Franz Hartmann visited Keely and reported on his visit. This happened sometime in March 1888; he found Keely busy with a trial that seemed to take all his time. Hartmann was granted only two short interviews and he understood nothing of what Keely explained to him; indeed Keely used terms invented by himself and he never gave definitions to them.

Hartmann first noticed that Keely had the characteristics of a strong medium, and Keely himself recognized that he could strike a table without making visible contact with it. He waited in vain for what Keely held out in liberators, multipliers, disintegrators, vibrometric resonators, sympathetic acoustic impulses, introducer minors, media chords etc. But he also hoped to learn what was the nature of Keely's "etheric force". . Unfortunately there too, this remained unanswered.

Hartmann made a pencil sketch of Keely's generator and engine. It was one of the first developed by Keely; afterwards the device was taken over by Keely and made much smaller. We reproduce here this drawing by Hartmann.



On the other hand, other reports have become known to the general public. Let us quote another meeting in agreement with Keely. At this were present Messrs. Buckanan, Hashell, Bissel, engineers of the American railroads. In all, there were about thirty people. We reproduce this report here using Keely's terminology. Keely presented one of his devices: the liberator "it is in this liberator that the "etheric vapour" or "inter-atomic force" is released, vitalized and preserved.

'The whole apparatus is approximately one meter high and weighs approximately 65 kilograms. It is placed on a movable wooden base, about 75 centimeters high, and is entirely isolated there, without any communication with the floor, the ceiling or the walls. No force can be brought in from without; we are all the more convinced of this as we have seen the apparatus move and carry it hither and thither across the room. Above the wooden base are symmetrically arranged the circular frames and the other parts of the liberator. There are many steel wire rods, from five to six hundred

meters long, fixed at one end and free at the other so that they can vibrate when struck, much like the tongues of a music box. Around the metal frames, many tubes are screwed, also in metal and arranged in rays. Each series thus resembles small cannons emerging from the embrasures of a circular fort. On this species of fortress is a similar construction, surrounded by about forty resonant tubes, 14 to 16 centimeters in length, arranged vertically and similar in miniature to a colonnade around an old Greek palace.

A small metal box, of very singular shape, crowns this strange device. This box constitutes the liberator itself, and contains a certain number of "resonators". These are hollow tubes. Beneath this apparatus is a large steel Chladni plate, approximately 40 centimeters in diameter; this plate fixed horizontally by means of a metal bar which crosses it from bottom to top, seems to play an important role in the generation or the release of the "force". (Let us not forget that this text was written according to the Keely's expressions). Around this plate and above are several fork tuning forks, firmly attached by their stems to the frame. Above this frame and beside the base hangs a horizontal cylinder of hollow metal (like a kind of bottle for diving), intended to receive the "etheric force", already "vitalized" and to store it. This force arrives at the "liberator" by means of a flexible copper pipe. Keely then replaced this pipe by a simple platinum wire which showed that the "etheric force" was not a gas. To cause the release of the "etheric force", Keely sounded three fork tuning forks, by means of a violin bow. One of these tuning forks was on a resonator, somewhere at a distance from the apparatus, with which it had no material communication.

As the tuning forks sounded like this in what Keely calls an "etheric" chord, he struck gently with a small hammer on the steel disk, in order to add its vibratory effects to those of the tuning forks. Almost immediately, Keely announced that the "etheric force" was "released", the tubes were "vitalized", and the receiver cylinder was charged with a .

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pressure of value at least equal to 10 kilograms per square centimeter, of which he was going to convince his entourage at the same time.

At some distance from the generating apparatus was a steel cylinder, the piston of which was fastened to a lever provided with a foot to the other.

end. The piston, whose surface was five square centimeters, was placed between the two extremities of the lever, so that the weight represented 750 kilograms. Keely repeatedly tried the accumulated strength, then added a weight of around 300 kilograms. To lift this weight, without taking into account that of the lever itself, a pressure of 9.5 kilograms per cm was needed ... on the piston (results obtained by calculations made with the engineers present; it was enough to compare the lengths lever arms). When all was ready, Keely forced the "etheric force" into the steel cylinder, and the plunger snapped out, carrying the heavy lever away like a wisp of awning straw. To ensure that the pressure reached 10,000 kg/cm , people in the audience hung on the end of the lever, without being able to get the piston into the cylinder.

Having repeated these experiments several times to the satisfaction of all present, Keely put the most important of his machines into operation: the Keely-motor. It is moreover by his engine that Keely made the most talk about him.

Keely's motor consists of a hollow polished metal sphere, 75 centimeters in diameter. At one of the poles is fitted a movable trunnion, rotating with it. At the end of this mobile trunnion, which exceeds the frame supporting the sphere, is adapted a pulley, from which starts a leather band which sets in motion saws, a lathe etc...

In the interior of the sphere extends the fixed trunnion, at the end of which are numerous sound tubes and other vibratory apparatus. But all these devices are attached to the ends of the trunnion and do not touch the rotating sphere.

Here is now the most spectacular side: the sphere began to turn on its axis with a terrible speed, as soon as Keely made pass the famous etheric vapor by the passage arranged through the fixed journal.

How does this machine move? This is Keely's explanation; Keely wanted it to be a vapor at all costs and this was perhaps due to the fact that when he opened the taps to supposedly fill his bottle, we heard in the room a hissing of gas which expands. So Keely explains that the "gas", once entered into the sphere, forms a whirlpool there which skims the interior with enormous speed and imparts a rotary motion to it. He then believed that this vapor, which he calls "inter-atomic" possesses such penetration that it finds hold on the molecular structure of the sphere and

can thus drag it along in its whirlwind. As all observers have remarked, the most extraordinary thing about this experiment is that there is no escape for the "steam, which seems to dissipate inside of the sphere, without the need for any exhaust duct.

The observers have verified that neither the apparatus nor its framework are in any communication with the floor, despite the fact that certain fair magicians retorted to the contrary; many of those present each time were able to make this statement. The machine works with an extraordinary speed by developing a very great mechanical power. The power of the machine is demonstrated by setting the lathe, the saw, etc. in motion by means of a strip of leather passed over the pulley fitted to one of the trunnions. Two spectators tried to stop the pulley by pressing with all their might on a plank that had been fitted to it; they failed.

For the Keely case, it was thrown two hundred thousand dollars at the time. There were various business people, especially bankers and scientific experts. The facts were observed over and over again and perfectly reproducible. This funding held for a period of fourteen years and provided him with the money for his costly experiments. So he was not betrayed from the start. It was only when it was judged in high places that its results were valid and even too valid that it was scuttled.

Here is an abridged extract from "The Scientific Arena" newspaper of September 1866; it relates experiments made with the Keely engine with a power of 250 horsepower.

"... The apparatus in question is exactly similar to that of which we have already given the description, but the capacity of the sphere is about ten times greater, its diameter being one meter twenty five centimeters; Keely first released the "etheric force" in the way we have described. He showed the presence of the phenomenon by opening the valve from which, according to him, the "etheric vapor" escaped with a noise similar to that of steam at high pressure, but without leaving traces in the atmosphere. Then the experiment of the lever and the piston was repeated, so as to demonstrate categorically that the

pressure exceeded $10,000 \text{ kg/cm}^2$, then Keely tried out his new machine.

The large copper sphere began to move; and the speed of rotation soon became so great that the whole building shook, and all the spectators ran and crowded into the farthest corner, for fear of an accident.

tooth. But Keely moderated and increased several times, at will, the speed of rotation of the sphere and proved by various experiments that the power developed even exceeded that of 250 horsepower which he had announced. He announced that he intended to fix this machine on a locomotive platform and thus run a train from Philadelphia to New York.

To match these accounts, the "British Mercantile Gazette" of February 15, 1887 published a short biography of Keely.

"Keely was born in 1827 in Philadelphia; he found himself alone and without resources at the age of sixteen. From his childhood he was impelled to study the relations which exist between sound and force; At the age of thirteen he made the first discovery which became the seed of his subsequent investigations".

^M Just as Newton was led to the theory of gravitation by the fall of an apple (at least that is the legend attached to it), so Franklin was led by his experiments with a kite to knowing the secrets of atmospheric electricity, as well as the vibration of window panes and crystals in response to certain musical chords, gave Keely the first idea of his theory of vibratory force. Pursuing his simple investigations, he found that certain chords always produced a certain vibration in distant objects; which put him on the track of the great discovery: that is to say what he calls the agreement of the "mass of all material bodies"; and it is by this means that he succeeds in producing disintegration. The first machine consists simply of a steel ring, with three hundred points of the same metal, rotating in a simple box.

For over sixteen years Keely studied his theory, experimenting in two media of different density, air and water, to produce the necessary change in equilibrium. But he discovered later that air alone gave better results and thus he achieved a great simplification of his apparatus. The extraordinary powers which Keely ascribes to this new force, which lies outside the universally accepted theories of gravitation and cohesion, have drawn upon the inventor the vigorous denunciations of men (* J of orthodox science. Many inventions , even in America, have always been greeted with the same disbelief. Recall Elias Home and the laughter that greeted the first sewing machine.

As regards the practical utility of the invention, Keely pierced a (*) Professor Pautrizel, who was one of the main protagonists of the "Prioré affair" (to which reference will be made at the end of the article). book) points out that science and the scientific community should not be confused. Indeed the scientific community is polluted by pretentious and ignorant people.

six-meter-deep, one-meter-diameter hole in quartz in Catskill Mountains; and the operation lasted only eighteen minutes!... (the values are slightly different from those previously quoted but here we faithfully reproduce the gazette).

The experiment was made in the presence of several mining engineers; and it was conclusive".

Finally, let's quote the report of two expert engineers about the Keely engine. These two honorable engineers were part of the Franklin Institute at the time.

"Having repeatedly observed Keely's experiments relating to the production of his 'etheric force'; having examined in detail the apparatus which he calls 'liberating', the various parts of which were assembled and adapted immediately after our examination; having seen the apparatus connected to the "receiver" and being satisfied that the whole machine contained nothing but atmospheric air at normal pressure, we declare to have seen a force of

more than 10,000 kg/cm² without there being any other agent for this purpose than the vibration of several fork tuning forks. There was no possible communication between the device and an external source of energy. There were no appreciable thermometric changes either at the time of the instantaneous development of the force or at the time of its instantaneous disappearance in the chamber.

"Being forced to eliminate known factors in the production of this force, the known agents, such as: heat, electricity, chemical combustion, etc....we are forced to conclude that the force develops in the way indicated by Keely; that is to say, it results from the disintegration of the air contained in an apparatus; and that this change produced by the vibration, determines the release of a vapor or an ether from a extreme tenuity."

"It is absurd to suppose that Keely employs compressed air or any other stored force. We have seen, at various times, attempts by Keely to use this force as motive power, and from our observations of it we believe that the obstacles which still prevent its practical success with locomotives, will be overcome by the inventor".

Such is the report of these two engineers. This was also reported in the British Mercantile Gazette by W. Bamett le Van and JA Linville.

The conclusion of the two engineers is interesting but naturally the appeal to the disintegration of air to explain the release of a vapor

or ether is not valid, because the researchers, engineers present only saw localized notions while Keely worked with other notions that he himself did not really foresee.

It emerges from this report that the force produced is not proportional to the mechanical force used to release it. We are faced with a fact so extraordinary, so contradictory with orthodox mechanical theories, that most researchers experience some difficulty in getting used to it, but this is not a sufficient reason to deny it. "Keely provided experimental proof that the greatest forces of nature come from a region beyond palpable matter and he overthrew the conventional form of conservation of energy at a stroke," concludes B. Keightley. in the September 1887 article of the magazine "Le Lotus Rouge" [2].

HP Blavatsky gave his opinion, in his book "La Doctrine Se crète" [3], on the work of Keely. She placed herself on the plane of occultism.

"According to the opinion of occultists, Keely is on the threshold of one of the greatest secrets of the Universe: the secret on which rests all the mystery of the physical forces as well as the esoteric meanings of the "egg of the world". The occult philosophy regarding the manifested Kosmos and the unmanifest Kosmos as forming a unity, it follows that the ideal conception of the first is expressed symbolically by this "golden egg" which has two poles. in the manifested world and the negative pole is lost in the absolute unknowable, in the "Sat". Blatvasky continues: "But that cannot be the doctrine of Keely who believes in a personal god. However, his ideas on the ethero-material constitution of the Universe bear a strange resemblance to ours; they are almost identical to them. Here is, for example, one of our explanations about the machine in question "In all the machines currently existing, we could not find the means of producing a "neutral center". Otherwise, the difficulties encountered by the researchers of the "perpentuum mobile" would have been overcome, and the said perpetual motion would be a commonplace fact..."

"Any construction requires a base in relation to the weight to be supported , but the foundations of the Universe rest on a "point" of "vacuum" much smaller than a molecule, or, to express this truth more exactly, on (*) an "inter-etheric point" To probe the depth of an etheric center is absolutely the same thing as to probe the immensity of the celestial ether to find one (*) Note that this point is not a point in the geometric sense of the term, but that it is a decalar.

the end, with this difference, that this one is the positive field, while the other is the negative field...",

Blatvasky adds "the force which Keely has just unconsciously discovered is none other than the terrible astral force known to the race of the Atlantians and which they called 'mash-mah'; it is the vril of Bullver Lytton's novel 'The Corning Race' (the race to come) and future races of our humanity. The word "vril" is perhaps an invention of the author; in any case, the force is not, because it is made of it. mention in all the secret books of India". "It is this same vibratory force which, directed on an army by means of a machine ("agni-roth") fixed to a "flying vessel" according to the instructions contained in the "Ashtar Vichya", reduced to ashes a hundred thousand men and elephants as easily as a straw. It is mentioned in the "Vishnu Purana" under the symbol of the "glance" of Kapila; the Sage whose glance reduced to a mountain of ashes the sixty thousand sons of King Sagara".

Thus Blavatsky had identified Keely's result with what I had apparently studied of human beings in India, perhaps 3,500 years ago. Naturally, she used her language which was inspired by this ancient knowledge, knowledge which she absorbed and which she tried, not without difficulty, to share and make known.

"Let us say a few words about Ernest Chladni, German physicist and musician (1756-1827) . point which was the center of the plate, the powder grouping on the nodal lines. He deduced mathematical laws and thus constants The action of a piezoelectric oscillator gives analogous results. It was Pierre Curie who discovered that by subjecting a crystal to pressure, an electric current could be drawn from it; we then had the effect called piezoelectric.

What is interesting here is to fixate on the frozen form. This was well observed by Keely. But the Ancients also observed the living form, therefore moving; the manifestation form of another representation; the partial or complete localization form. And from the localized form, the Ancients tried to trace the process and achieve delocalization. Often for them the form was interesting only as a starting point towards a delocalization or so to plunge it into another field of coherence.

Keely therefore used a Chladni disc in his engine. And what he

What is interesting to note is that his motor only worked with him, and the fact was ensured by the disc. This justifies the previous remark about Keely's mediumship gifts. We can make a Chladni disk that is in agreement with each individual, that is to say that if we change the manipulator, we have to change the disk. There is a transfer between the shape of the disc (its dimensions) and the person.

We find the same phenomenon during the conquest of the Americas by the Spaniards; they found two gold discs and were told by the Indians who had survived the massacre that the emperor and his wife each had their disc and that when these gold discs were struck in a certain way, they one rose in the air carrying the emperor and the other his wife who rested above. Naturally, for the conquering Spaniards these sayings were taken as stupid tales of ignorant savages. It should be noted that any piece of metal struck by a hammer in a certain way begins to emit Taofel fields which can be very high.

On the other hand, if this same metal is heated after it has been charged by these emissions, it ceases to emit from a certain temperature. The rhythm used changes with the metal, its shape, the place in which it is located and if you want to obtain effects of delocalization on certain parameters, you still have to take into account the moment when this operation is carried out. Emission at high Taofel fields was not enough to make the discs elevate. When this happened, other forms were certainly needed around, inert or living forms in order to create an operative "background" [4],

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APPENDIX

Let's see Leslie Shepard's thesis about Keely. [1]

Leslie Shepard points out that after Keely's death in 1898, several articles appeared in the press of the time pointing out that all of JW Keely's work was a fraud. Yet many renowned scientists had checked his assemblies piece by piece. They had moved his heavy machinery to override any power supply through the ceiling, floor, or walls of the rooms in which Keely worked. Among these people were JM Willcox, J. Leidy, DG Brinton; the latter was a professor at the University of Pennsylvania.

Newspaper articles claimed that with hermetic and mind-blowing jargon, Keely had fooled scholars, financiers, industrialists of the time who had believed in the "etheric force". The expressions "Musical sphere", "Vibrational accumulator", "globe machine", musical ball" were only used to increase the confusion by language in the minds of the observers.

This can be blamed on any researcher who departs from orthodox paths. Such a researcher does not manufacture words and expressions for pleasure, it is rather because he feels the need for them and thus expresses a deep feeling that he has in front of these phenomena. We find the same technique in Dufourg, in those who have manipulated EFs, in Moray etc... Often these researchers have mixed two languages: that of their creation and the scientific language which they have unfortunately chastised because the latter adapts harm to this research, which goes beyond the usual field of coherence.

It was also said that a few months before Keely's death, a friend had asked him "what would you like for an epitaph - on your grave" and Keely would have replied "Keely, the greatest boaster of the 19th Century".

From where L.Shepard adds: "it is clear that Keely was a charlatan". He admits that despite this, Keely was quite brilliant and that he emphasized something unknown which could be explained by nuclear physics or

ring fields and that Keely may have trapped himself with his jargon.

But if we look more closely at the description of certain indoor or outdoor experiments, indoors when observers could lift huge steel cylinders, or outdoors when it was pulverized in the Rocky Mountains, gold-bearing quartz, Keely didn't seem like a quack. On the other hand, the examination of the forms used shows that it produced high fields of Taofel, which does not produce a technical object even falsified, or if one, by its forms, produces some, it would be curious that all those made by Keely, if they had only been technical objects, would have caused them to appear.

But one wonders why, in his house, there were these clock mechanisms, these hollow wires and hidden under a trapdoor, the protuberance rubbers (which supposedly were discovered in Keely's cellar a few years ago). time after his death). Could all this have caused its spheres to rotate? Who was all this equipment for? We can assume many things. . "

You should know that Keely's montages often only worked with him. He had to orient certain parts of his fixtures in certain devices to achieve the desired effect and the orientation varied with the time of use. We must not forget that we are not in front of technical objects.

Perhaps the devices hidden in the ceilings, under the floors, etc,... were used at the beginning of Keely's experiments to launch his devices when these were not in focus, perhaps they were Already Phenist objects still technical enough for uninformed observers to confuse them with the usual machines of the time? Perhaps Keely had them in reserve in case a financier absolutely wanted to see his devices operate under conditions that Keely could not provide? The fact remains that now, thanks to the results obtained by other researchers and by our results, it seems that Keely has touched the music of the spheres and that Mrs. Blatvasky was not mistaken on her account. J

Bibliography :

[1] Clara Bloomfield Moore, "Keely and his Discoveries", University Books, Secaucus, NJ

ELECTROGRAPHY

by G. HADJO

CRAVE

(France)

Electrography is known particularly since the invention of the coils of Tesla and Rumkorf; thus one can manufacture generators which emit a field of HF under HT. By switching on the ignition, we observe a sort of "corona" light effect that it is very tempting to want to photograph.

Carlsten carried out a first experiment in Great Britain around 1842, then the Germans worked on it in 1884 and the Czechs in 1889. In 1893 Tesla made a demonstration of luminous effects in Chicago. In 1896 in Paris, Doctor Henri Baraduc and his collaborator Lodko Narkiewicz carried out electrophotography with one hand and numerous documents in this field. In 1939 Prat and Schlemmer publish a documentation on their work remained without echoes.

It would seem that all these researchers found themselves faced with a phenomenon , but there was no chaining of ideas to find the practical application. They did not appear to have carried out the transposition from one coherence field to another.

Around this time in the USSR Victor Adamenko rediscovered this technique and the Kirlian couple, unaware of the work of their predecessors, carried out research in their turn.

Semyon Kirlian was an electrician who was asked to do a repair in a hospital where there were many devices as we often see for therapy. It was an HF generator under HT, and he too had noticed this luminous effect by objects when switching on the ignition.

Returning home, with his wife Valentina, he studied the process and built his own device for his experiments. The great merit of the Kirlians was to have made methodical research and to have observed the variations

bioluminescences according to the state of living subjects, and this is how they gave their name to this effect.

They visualized this unknown force which the Soviets call bio-plasma and which they consider to be the 5th state of matter.

It should be added, for the story, that they received personalities to show their experiences, and that on the day of the visit, they did a control test just before the visitors arrived. At this very distressed moment, they noted that the image of the bio-luminescence was not at all similar to the usual. They thought their circuit had failed, but everything seemed to be in good working order. This would be the beginning of their observations on the electro-bioluminescent variations according to the psychic state of the living being. In 1960 the Soviet government grants credits to Kirlian.

In 1965 the Czech encyclopedia publishes an electrophotography; researchers are talked about like Smart and Milner for England and V. Inyouchin and V. Puchkin in the USSR. It was around 1968 that we began to know about the work of Eastern European countries in this field, although the USA already had a report on the Kirlians in September 1959.

It was in 1969 that the West took a serious interest in parapsychology . In the USSR there are no preconceived ideas for research, we systematically apply the experience and we see if it is conclusive or not.

According to a talk by Verovaki in 1981, on a trip he took with Kripner, it was Khrushchev who kick-started unorthodox research in the USSR, following his trip to India where he was received by Nehru and whom he asked to see two or three of the best yogis.

In Moscow the first tests begin in spite of the skepticism of in tourage, but the party decided it, in 1959 one founds an institute with teams, in 1960 the telepathic transmissions give results. There is an opening for the public and tests for recruitment, because subjects are needed. In 1964 the results seem too important, we form the institutes, the libraries having this specialty are prohibited to the public. In 1965 there is a survey of Soviets all over the world, and all the phenomena are studied and applied, in a more or less occult way. According to Verovaki, a large number of centers work in secret in this kind of research. In 1970, the book by Ostreder and Shroeder on parapsychological research appeared.

gics in the USSR, the style is journalistic, tending towards exaggeration, the facts are inflated, but all the same, this publication had an impact in interested circles who wondered where the Soviets were in this field, large chapters were devoted to the Kirlians, and on reading this work, the American psychologist Thelma Moss of the University of UCLA, was prodigiously interested, and wondered if one could finally visualize this unknown energy linked to living beings, this bio- energy known it is called in the west, that Reiche calls orgone, and that von Reichenbach calls odic force, each seeking to put a name to these energies.

A connection has been made between the Kirlian effect and the aura. If the phenomenon often behaves in the same way, I do not think that it is the complete aura which is formed of seven bodies, what we visualize would be the first two or three envelopes, in particular the etheric body.

Thelma Moss obtained an invitation to go to the USSR. As you can imagine, she was unable to meet the Kirlians (who died in 1971), but she was received by Victor Adamenko who was of great help; he defined the Kirlian effect to him as being "the conversion of the non-electric properties of the object, into electric properties..., with a direct transfer of charges from the object to the photographic plate". He spoke to him about acupuncture, still little known at that time, but the great idea of this collaborator of the Kirlians was to see the close relationship between the bio-plasmic body structure revealed by electrography and the energetic body traversed. chi noise acupuncture meridians . Thelma Moss was given a diagram of the Kirlian device, which was later found to be incomplete. Returning to the USA, she asked engineers to build her a device, but they claimed that it was impossible and dangerous; finally it was his collaborator Kendal Jhonson who made him his first device for his experiments. To obtain an electrophotograph, one starts from an HF generator under HT whose plate connected to the coil serves as an electrode; the photographic emulsion is placed on this plate, then the object on top. An electric wire starting from a mass serves as a second electrode that is placed on the object, all that remains is to switch on the contact. The human body is self-sufficient to form a mass if you want to electrophotograph your hands, for example. It is necessary to operate in lighting as in a photographic laboratory. At the University of Ucla Thelma Moss has a laboratory and a particularly suitable cabin for her experimental electrophotographic research, which is mainly centered on psychology.

Our team also read and closely studied the book by Ostreder and Schroeder. With Christian Malezieux and Jean Marc Conoir, we experimented with our first generator. Now the other team that we are forming with Guy Gruais has made it possible to produce a highly efficient device.

We have also taken different paths, in particular what we call electrophotonics and GIHV (Interior High Voltage graphics).

We have developed a transparent direct-reading screen which allows the effects of rehabilitation to be visualized directly; this material should be of effective help to acupuncturists.

All these new ideas and research, we do them by our own means, because in France we do not have subsidies. At the beginning of the works, the disturbances caused to the televisions of the building took on tragicomic proportions, as for the volunteers for the experiments, my family still shuns me. Beyond this anecdotal side, the devices are very developed and very reliable.

In 1978, André Laiigt presented a thesis on electro-biophotography, and at the Domus Medica in Paris was presented by two German doctors, Mendel and Lerner, a spectro-energetic diagnostic technique based on acupuncture points. These two practitioners have carried out some of the most interesting systematic research which confirms what Victor Adamoto had observed. Dr. Kovacks who chaired this conference asked me to present my electrophotographs (which appear to be of a different technique).

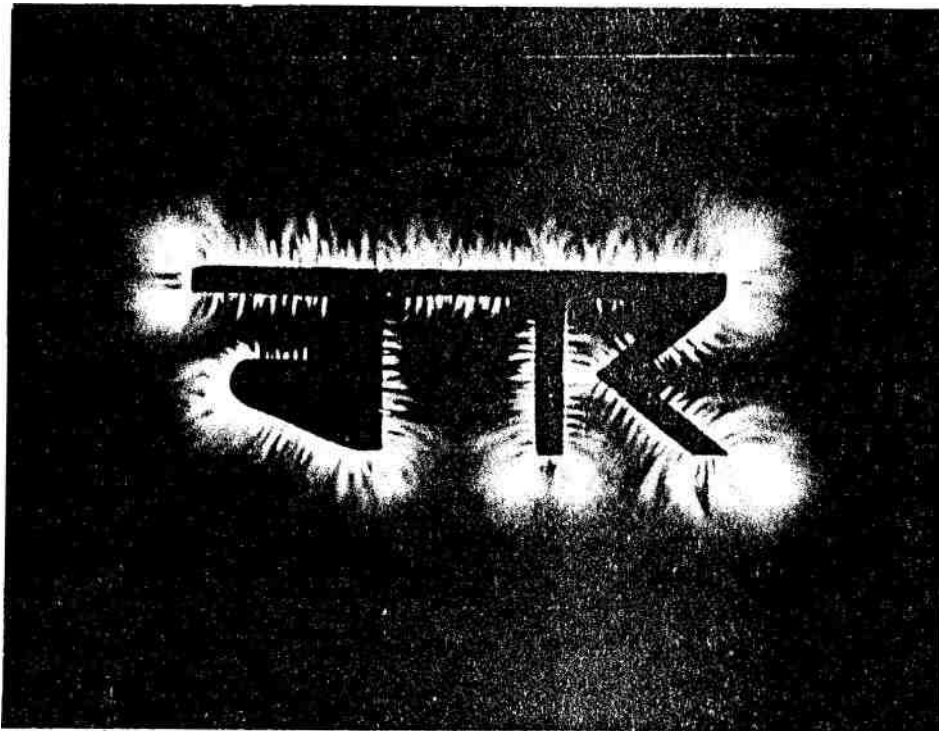
With Dr. Mendal we found ourselves some time later at a symposium on bio-energetics where there was also a presentation by Dr. Dimitrescu who is doing very specific research with his process which he calls electronography. Dimitrescu commented on some of my electrophotographs which he found completely unpublished. We thought that everyone uses different techniques and that nothing is standardized in this field.

In Bucharest, Dr. Dimitrescu had a whole department in a hospital dependent on the Ministry of Labour.

At the congress on bio-energetics in Brussels, I projected slides , then Dr. Mendel and Lerner presented their technique.

At this congress Professor Ravatin set us on new paths with the emissions due to the forms and currently we are experimenting in this

Pictures taken by G. Hadjo with his camera



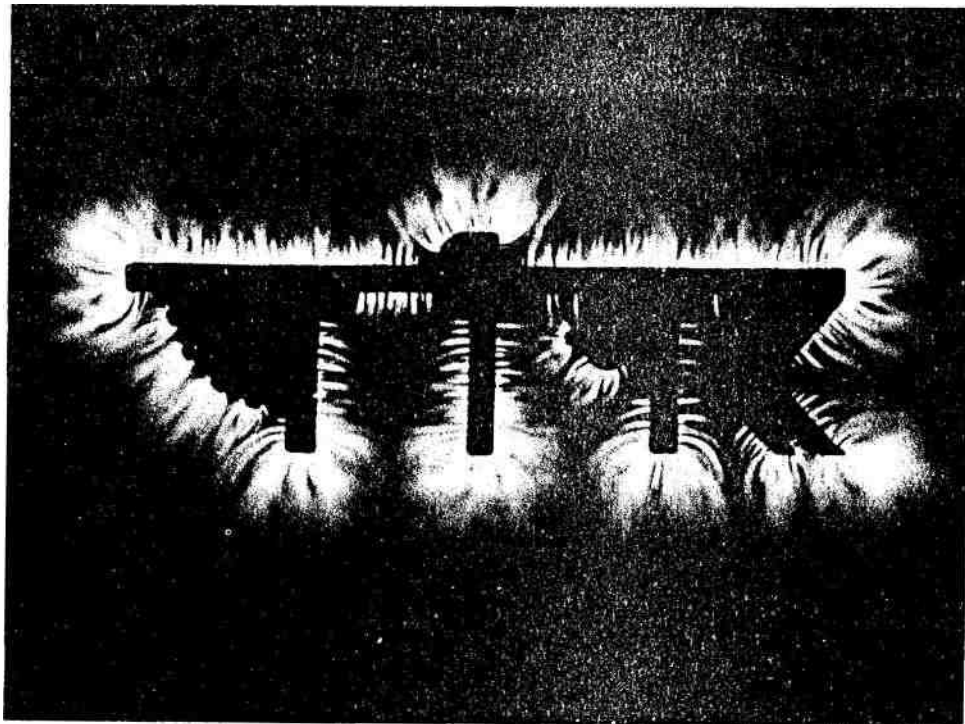
The world tells us we were created by the Word. But if we are to believe some Indian and Tibetan Temples, the Great Creation resided in three parts (this is of course only valid in the new C. de C.).

Here it is in the order in which they were transmitted.

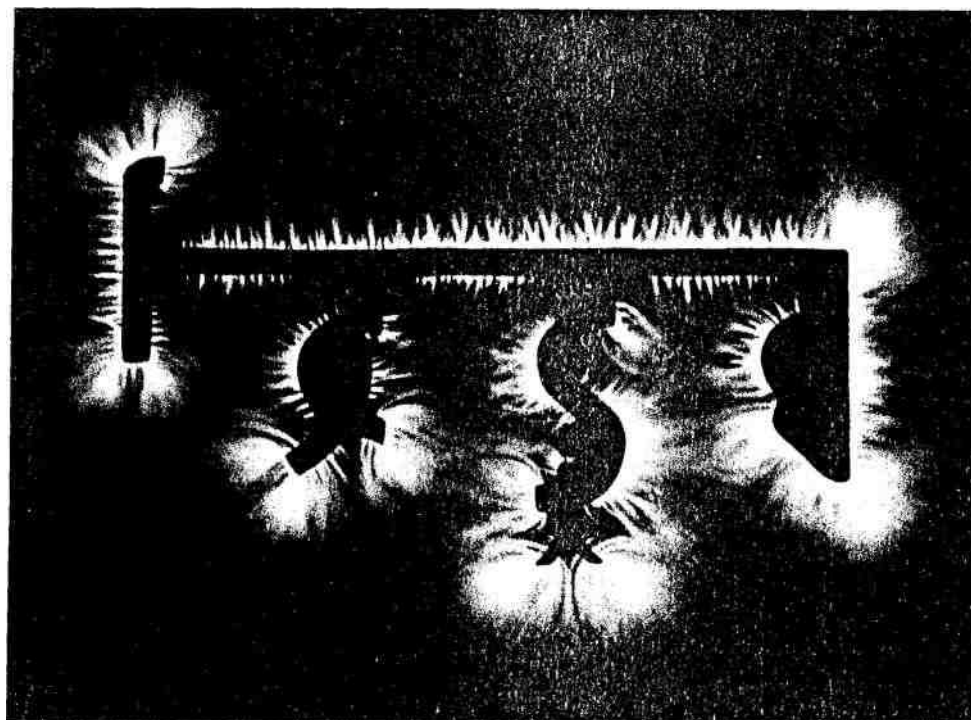
The signs have been magnified to be passed to the Kirlian effect [1]

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[1] Kamala-Inava, Edit.
Georges Charlet, Argentièrre
(Haute-Savoie) 1961



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branch which represents a completely separate field of investigation.

Note that the practical applications are made on organic and inorganic materials, metals, food, plants for ecology, insects, animals and finally human beings. The press often mentions these applications, but we don't often see the photographs of them, so we took pictures of the liquids, but apart from our shots, we didn't have the means to compare with others. researchers. As far as the medical field is concerned, the interpretations of the bioluminescences of the extremities of the hands and the feet can be related to a certain number of reference systems related to bio-energetic analyses.

If we take acupuncture, for example, we know that all symmetrical acupuncture meridians end or begin at the fingertips of the hands or feet.

According to recent hypotheses, they are traversed by an energy whose manifestations could be reflected in the different aspects of the images obtained and arrive at clinical and possibly prognostic conclusions based on the very nature of these meridians.

With regard to the more classic neurophysiological interpretation and of particular interest to physiotherapists, correlations can be established between the images and anomalies in the sectors of the main nerve bundles. This is how rehabilitation behaviors for

can be controlled practically and almost immediately.

On switching on the device, on triggering, the formation of a cloud of electrons creates a powerful accelerating field, these electrons will reach high speeds, and their energy will make it possible to further excite the gaseous mixture. The molecules of this mixture will reach an energy level higher than their ionization potential. There is then ejection of an electron when these molecules return to their fundamental level; there is emission of photons of different energies, the most powerful of which will ionize other molecules in the mixture. This is the source of light observed in the corona discharge.

We communicate only through radiation, traditionally thinking in visual terms. Optics is electromagnetic in nature. There are many forms of energy in nature. By isolating them and tuning by them, we could gather a lot of information

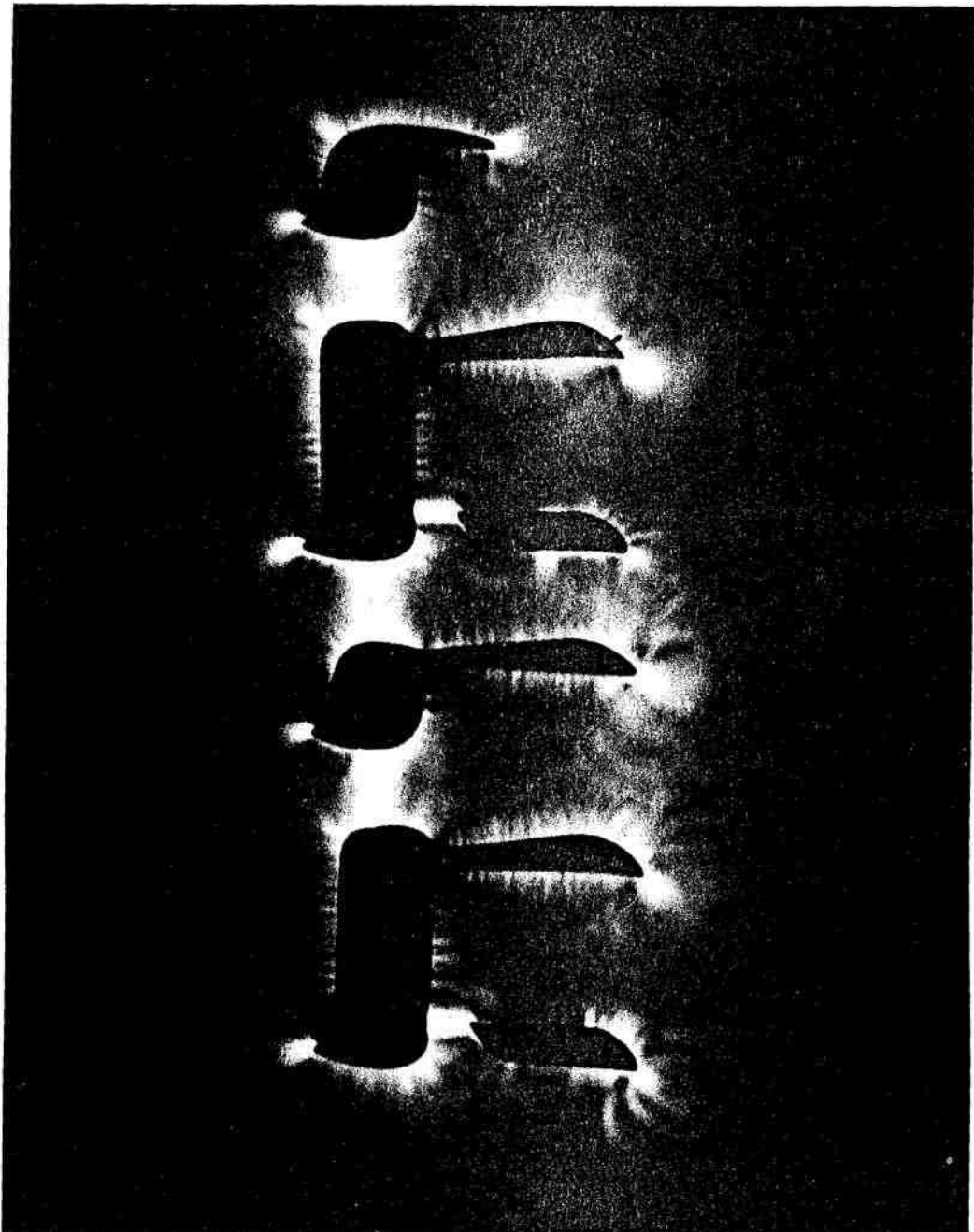


Photo taken by G. Hadjo with his
camera

on our environment. We could come back to the Aura and say that the seven planes or seven crowns which form it are substances with different modes of configuration. These radiations acting in spatio- temporal frameworks are distinct from each other. These seven substances can enter into interaction and by polarity generate energies. The more one searches in the microcosm, the more one gets lost in energy fields, to the point that currently the discovery of quarks requires a form of tunnel 7 km in circumference to be able to encircle it.

As William Teller had proposed, the interaction of these various fields of vibrational substances is quite loose. The spirit can intervene to operate modifications in the structural organization of these various levels to arrive from there on the physical plan; it would be a "training" action.

This is how with our devices we work on two different circuits, one pathological and the other psychological, by comparing the two, we better visualize the interaction of the different planes.

Our goal in electrophotography is to explore all possibilities of investigation and to develop the best application methods. The range of research is very wide and it should be multidisciplinary , but the financial problem is difficult to surmount. Fortunately, for the true seeker, there remains the exaltation of the creative state and a philosophy of the global that detaches itself from the local level. and [1].

Bibliography

[1] J. Ravatin, K. Fidaali and G. Hadjo, ARK'ALL Commun., Vol. 6, fasci.2, p.147 (198Q). (*)

(*) A rapid study was presented in [1],

by Pierre SMIRNE

PREFACE

These "reading notes" were presented in a special issue of the journal ARK'ALL Communications devoted to the research we carry out in "electro-culture"; they must be understood as a "working document". As much as possible, the results obtained over a hundred years by various researchers who have operated with different methods in different places have been brought together - without interpretation.

Our work is not exhaustive in this first presentation. Thus, for example, we have not mentioned the work of Mateo Tavera, founder of "Nature and Progress" and whose memory we honor here. ³

The working document we are proposing raises more questions than, perhaps, the experimenters have asked themselves. We are not asserting anything, we are not setting ourselves an example, we have experimented ourselves on several sites in France with very interesting results in silviculture and r". vegetable crops since 1963.

During experimental remarks made in the field, it was possible for us to establish a certain number of cross-checks with old work Little by little we were able to understand why the isolated efforts of researchers did not lead to a possible industrialization and why an abandonment of methods, however fruitful in quality , was the lot of these predecessors, the results being inconstant...

We were led to create a model that could account for all the observed phenomena. This MODEL now allows us to control and master seed and seed stimulation.

This model, acquired during tests, controls and measurements in spectrometry by Raman-Laser effect is not exposed in these notes; it will be the subject of another publication.

This model takes into account the definitions of the genetic code, the behavior of Deoxyribonucleic Acid and metals. chelating agents , but mastering the model also involves the FRACTANT properties of the energy spectra existing in radiation physics, organic chemistry and molecular biology, these fractant properties being the KEY to the problem. (*)

(*) Editor's note Other research is in progress in Belgium and is giving very good results, for example the obtaining of methaye from products of plant origin.

WARNING

Arrived at the end of these notes we propose a final development of a set of devices whose practical applications in electroculture affect the rebalancing of diseased plants, the stimulation of all seeds by appropriate settings and combined effects at the same time on living cells and on the trimer water of tissues.

The photos we produce represent a group of devices tested in France in different places.

The assemblage of devices photographed includes two Lakhovsky multi-wavelength oscillators, a low-frequency radio cellulo-oscillator, and a Leon Sprink oscillating device. The choice of shapes, orientation, coupling of the devices was guided by the application of the Model of which we spoke in the "preface-warning". Control of the stimulation of the regeneration of living tissues and their rebalancing for an optimization of the results could be obtained by taking into account the " fractonet " type applications of the physics of radiation and fields, by taking into account the phenomena absorption and emission of various materials, of appropriate forms and differently oriented in space. Although these devices designed in their time (1931-1952) were built with a classic theory of Ertzian fields, we were able at the level of plant molecular biology to bring out new facts and above all to control the action of the coupling of the devices. on the complete growth of vegetable seeds (from plant to harvest).

Thus, in relative proximity (25 meters) to the experimental laboratory and in the ground, we stimulated tomatoes, black radishes, celery, leeks, and various other legumes over two years from a distance and obtained without fertilizer or special competition a increase in production of 40% on average over an area of 200 square meters.

We propose in future texts to describe the implementation of the stimulation devices as well as the economic study of the possible industrialization of the processes resulting from the mathematical and physical MODEL developed. However, we specify before this report that in this period of energy saving, the use of the processes developed will result in a certain gain (*)

(*) 1980-1981)

tain of 19% minimum on the whole of the chain of the producer to the consumer.

We also insist on the fact that the main active agent in these processes is provided by forms of energy existing in the atmosphere in all places on the globe and that by channeling them it is possible to energize living substances such as if it was the triggering of a physical catalysis without chemical input. The system is moreover reversible and it is possible by the transfer of forms of oscillating radiations to prevent the normal growth and to arrive at destructive genetic mutations, not desirable as that could be carried out. Finally, we point out that our model also applies to cultivation in greenhouses, in layers, in bells for which appropriate farms have been studied.

WORKS EL REMARKS ON ELECTROCULTURE

(Reading Notes)^(HAS)

by P. SMIRNE

(Paris)

In 1910 the Librairie du XXe Siècle published a certain number of works by the author Ernest BOSCH de VEZE, including a work titled L'AIMANTATION UNIVERSELLE (25, rue Serpente) dealing with the action of electricity on plants (a these publications will be taken over by "Editions Le Courrier du Livre", 21 rue de Seine, 75006 Paris).

E. Bosch' de Vèze gives the following points of reference regarding work similar to his on the influence of electricity on vegetation:

1783 Bertholon demonstrates the action of electricity on plants (references or experiments are not given).

1878-79 Bertholon, Newmann, Thwaite, Loemstroem.

1885-1897 IODKO (from Minsk) also demonstrates the action, either of atmospheric electricity, or of electricity produced with the devices of this period and their invention on the vegetable kingdom.

Bose indicates: most of the experimenters have installed in fields (without specifying the nature of the soil or subsoil) "either call points or metal trellises (without indication of mesh) connected to devices of high voltage and high frequency. They recorded very diverse findings that they tried to explain or at least to interpret scientifically".

Electroculture trials have been made in California on fruit trees, and the results obtained by various methods have been remarkable for the beauty of the fruits and their flavor. (*)

(*) Editor's note The question marks throughout the text correspond to inaccuracies in the reports of the researchers whose work is presented summarized here. This article appeared in the journal ARK'ALL Communications, Vol.8, fasc.1 (1983).

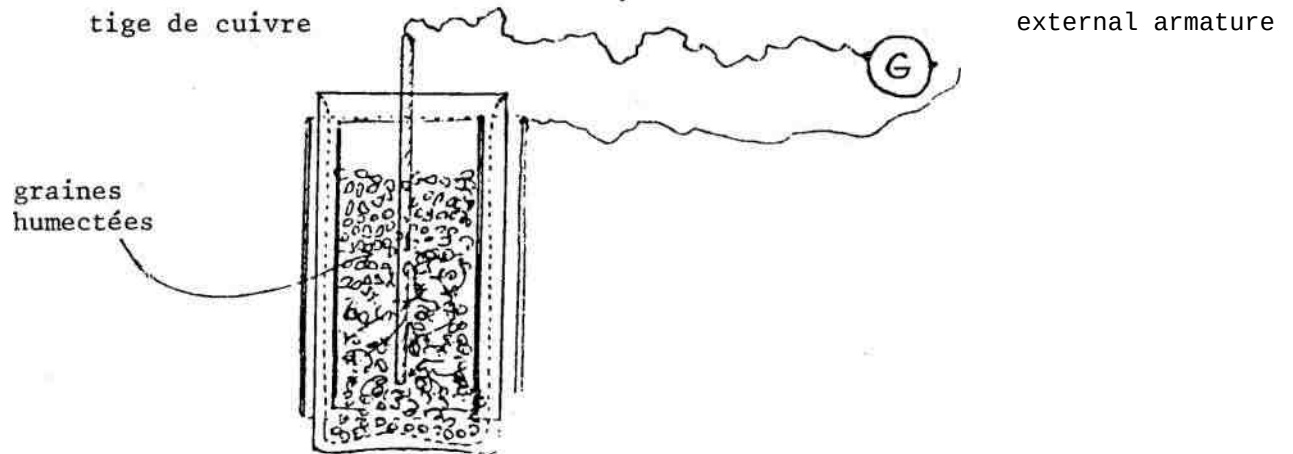
seeds

^>04 wool ?
glass <~" *:■■■,

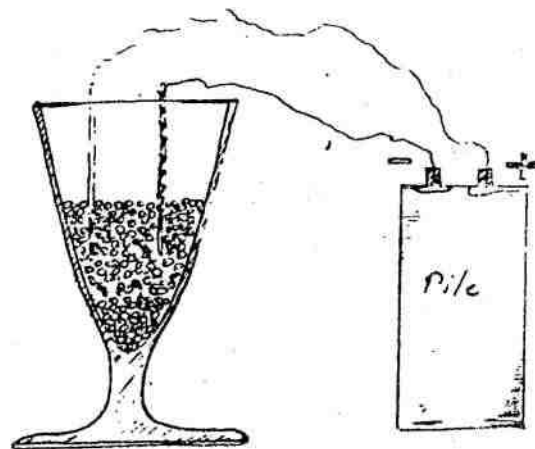
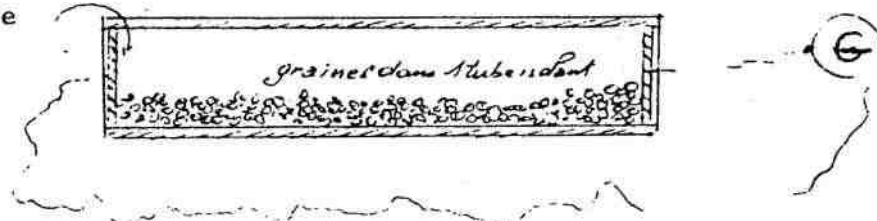
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electrostatic
generator



plaque de cuivre



Two electrodes from
the two poles of a battery
are immersed in the center of the
vase in the seeds moistened
with water.

"Electrification of Seeds" (E. Bosc)

- 1°) ... place the seeds on a glass plate connected to a conductor ...
(?) (who goes where?) (go to = where)(*)
- 2°) ... place the seeds in a jar covered inside and outside with tin foil. In the middle of the seeds, we make a stem (AA) ^t ^{tg} ^{nl} of copper placed in communication with an electric machine... (?) (is it an electrostatic machine?) (a dynamo?)
- 3°) ... place the seeds in a tube (wood? glass? other material?) whose ends (open) are closed by copper plates which are connected to a source of electricity; (the nature of the metals is not otherwise specified according to their varieties) finally, one can place the seeds in any vases (the idea of the role of the shape of the containers is absent from the texts as well as from their material. The notion of electric current of convection taken into account by Maxwell in 1865 is not mentioned either).
- 4°) ... the two electrodes of a battery are brought to the middle of the seeds... (the volume of the seeds is not indicated, neither is the temperature). (Also specify the notion of electrical equipotentials).
- 5°) ... we will add that: whatever the mode adopted, it is necessary to moisten them (the seeds) with ordinary water ... (serial tests with rainwater, storm water , spring waters are not specified...) nor are waters that have undergone treatment by visible or invisible light radiation (EM field) UV - IR - X etc...

SPECHNEFF :

Conical winding : the central airlock receives, like an hourglass, the seeds subjected to the EM field which becomes geometrically variable in strength and intensity. The bi-conical device of the winding ' * is made on two hollow pieces of turned wood the seeds flow ... in the field of the coil following 1 axis of the cells of the wood during the growth of the tree.

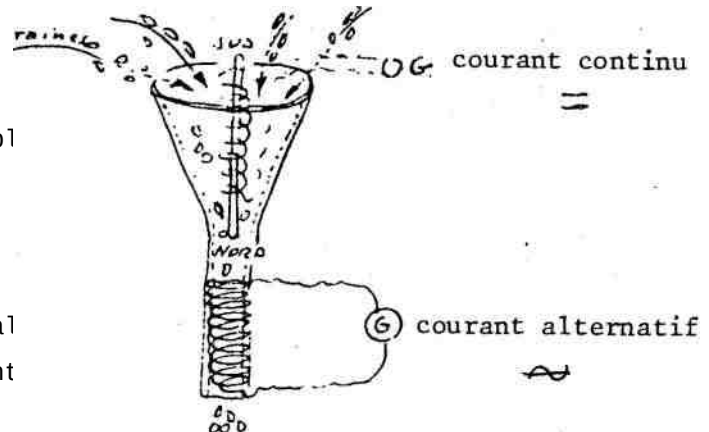
(*) See lodko (1897), transcendental photography, then Kirlian.

(**) See works Bignand, Violet, St. Gobain in Fresnes (1963)

Cylinder winding :

The stream of seeds is propelled into the cone of a braided funnel (wickerwork) in the center of which is placed a mild steel bar on which an insulated copper wire is wound.

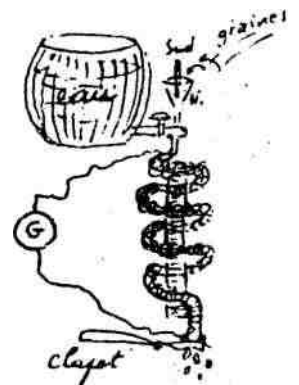
A DC voltage forms a bar magnet, the assembly is inclined vertical magnetic inclination of the working point



The hollow solenoid through which the seeds pass is powered by an alternator. The frequency to apply varies according to the seeds. The winding pitch of the lower solenoid (right or left as the case may be) is very important for the results, also in the bi-conical coil.

Spechneff sower

A wooden barrel filled with rainwater feeds through a cinnamon a serpentine passing around a cylinder. This rubber coil receives the mixed water and entrains the spilled seeds into the funnel as above with a permanent magnet. The coil serves as a support core for an insulated wire wound on it on which a "corkscrew" solenoid is applied an alternating voltage. A valve is open at the bottom of the coil for laying in the ground seeds moistened with "electrified" water (subject to 1 EM field)



Bosc indicates: One must electrify the seeds from hour to hour during several days... the hard seeds date palms, coconuts, can be left constantly in the electric current... one can make germinate very old seeds.

One of the first experimenters in electricity applied to germination was the Russian botanist SPECHNEFF (date?). he notes that the vegetation of the plant is faster, more vigorous by electrifying the seeds with an induced current (?), ^that the direct current (?/ ^or continuous action of a current?]^pi produces a more abundant harvest , that germination is more activated. _He_compare:

(*) See opening note.

Germination

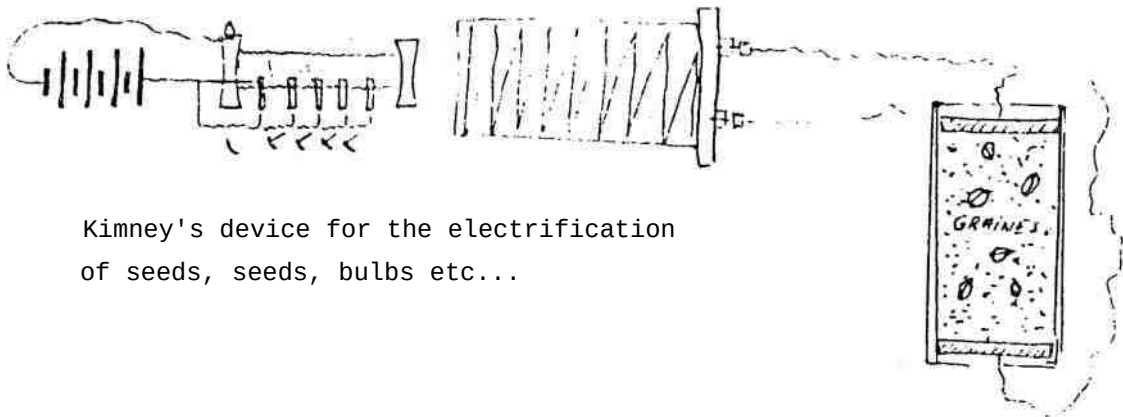
| | Not Electrified | Electrified |
|-----------|-----------------|-------------|
| Peas | 4 days | 2.5 days |
| Beans | 6 " | 3 " |
| Rye | 5 | |
| Sunflower | 15 " | 8.5 " |

[None of the conditions of the experiments are specified in Bose's text; the following questions can be asked:

- a) Origin and nature of the seeds ... selection?
- b) Experience season - indoor or outdoor?
- c) What water used? What light?
- d) Choice of time for non-electrified and electrified plant during humidification? Processing time.
- e) Electrification device and power supply.
- f) Nature of the terrain, pedology, geology, latitude?]

Bosc cites the name of another bean experimenter: Monsieur PAULIN.

KIMNEY



Kimney's device for the electrification of seeds, seeds, bulbs etc...

multi-output primary powered by 4 Leclanché and 2 Samson cell cells. The circuit is made by successive switches, (action of the non-negligible extra breaking currents).

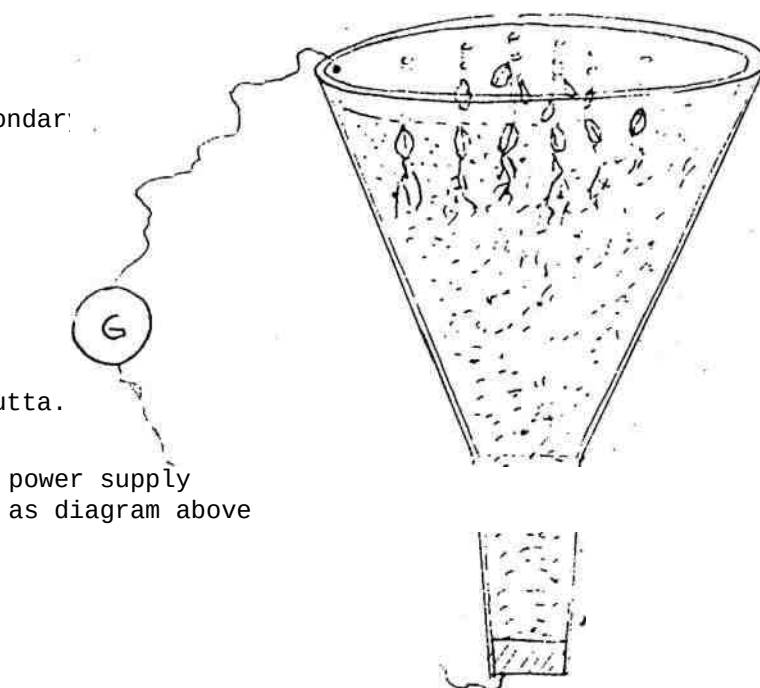
Secondary supplying two copper plates closing off a glass tube. The seeds which have just been moistened are packed in the cylinder between the 2 plates.

The two copper plates
are in contact with the
wet sand and connected to the secondary
as above.

These are electric shocks
similar to those that Sir
Jagadis Chunder Bose will use to
measure the reactions of
adult plants in 1900-1917 in Calcutta.

Kimney did not study the
electrical equipment
and their variations in the
funnel although equipotentials
on metal have been known
since the work of NOBILI

on batteries and electrolytes. It is true that this concept was not
mathematically analyzed during Nobili's lifetime.



power supply
as diagram above

Asa S. Kimney Experiments

Purpose: to determine the effect of electric current on various seeds and which is
the most favorable, (date of the experiments?)•

Kimney uses as a current source a kind of Rhumkorff coil modified in order
to vary the number of turns of the primary to obtain various voltages. The coil is
powered by four Leclanché elements or batteries. in series giving 4 to 5 volts and
two Samson elements giving 2.88 volts; he places the moistened seeds in a cylindrical
glass tube whose two open ends are sealed with copper washers (position of the tube?
underground?).

Kimney result:

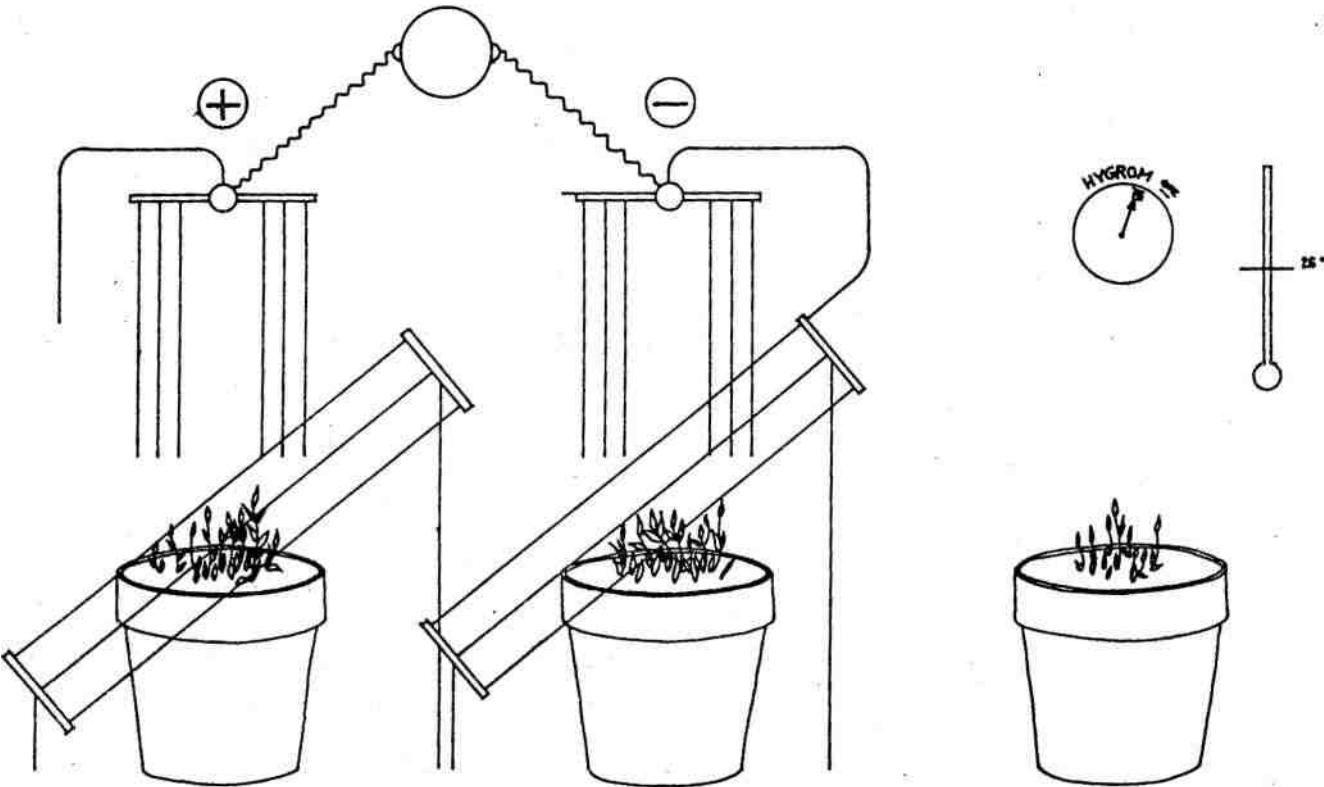
Germinated within 24 hours, 32.40 7, seeds (? species) controls? in 48 ", 21.05
7, " "
in 72 ", 7.35 7, " "

Kimney also used 0.17 cm glass funnels. Filled with moist (limestone?
siliceous? micaceous?) sand he seeded in each

funnel 12 seeds. The lower opening is obstructed by a small solid copper disc in contact with the sand; the upper opening has a large disc of the same metal in contact with the wet sand. The upper disc is pierced with 12 holes in the center of which are placed the seeds. The growth of the rootlets is observable in the cone of the funnel glass. Kimney results:

Germinated within 24 hours, 17.65% of seeds (species? strain?)

| | | |
|----|---|---------|
| 48 | " | 11.47 % |
| 72 | " | 2.38 % |
| 96 | " | 2.38 % |



Tests by Selim Lestroem of Helsinfors at constant temperature and humidity, principle of one of the devices on control pots. The whole experiment consisted of rows of pots.

Devices included parallel horizontal wires either superposed or in a plane. Other devices included wires stretched obliquely to the vertical or horizontal. Other devices have wires hung on insulating or metallic supports.

Conclusions drawn by Kimney from his experiences.

1°) The electrification of the seeds exerts a considerable influence on their - germination and on the development of the seedling.

(Lengths not specified by Bose)

2°) The application of an electric current for short periods accelerates germination by: 30% in 24 hours, 20% after 48 hours and only 6% in 73 hours.

3°) The maximum electro-motive force (?) which can be used is 1 volt for germination and 3 volts of induced current (?) for the growth of the stem and the root.

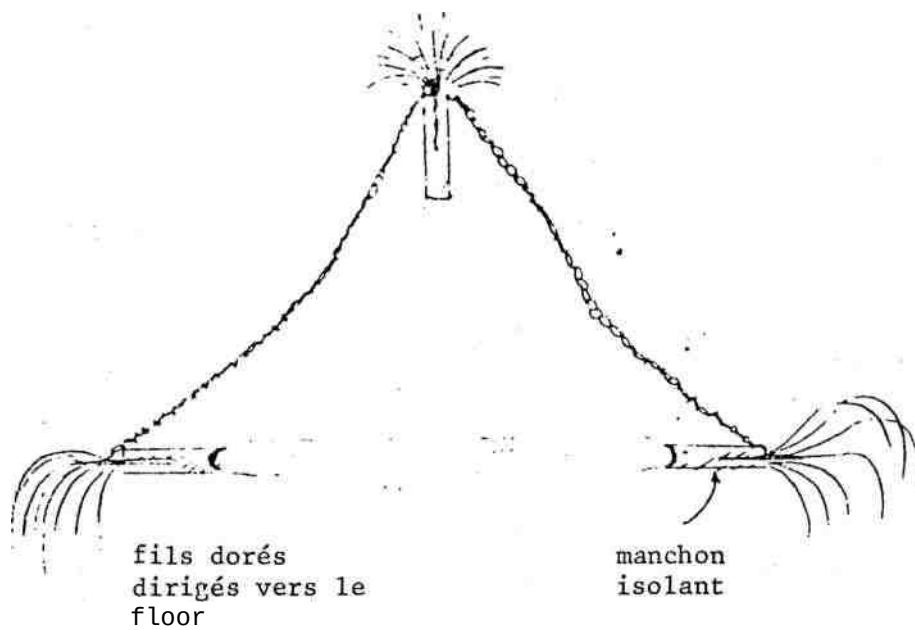
4°) The effect of the electrification of the seeds is especially noticeable near the first 24 hours (?).

5°) The effect on the development of stems and stems is 13% lower than that exerted on the growth and development of roots.

Bosc de Vèze quotes the tests of GRANDEAU and LECLERC of plants subjected to the effects of atmospheric electricity and subtracted from the same influence by Faraday cage, (which plants? which exposure devices?)

Note is made about the yields of rye and wheat in Lapland and northern Norway that one hectolitre yields 28 to 40 hecto without deep ploughing. Bose concludes with an influence of the aurora borealis on the vegetation of northern countries, which he calls Direct Influence.

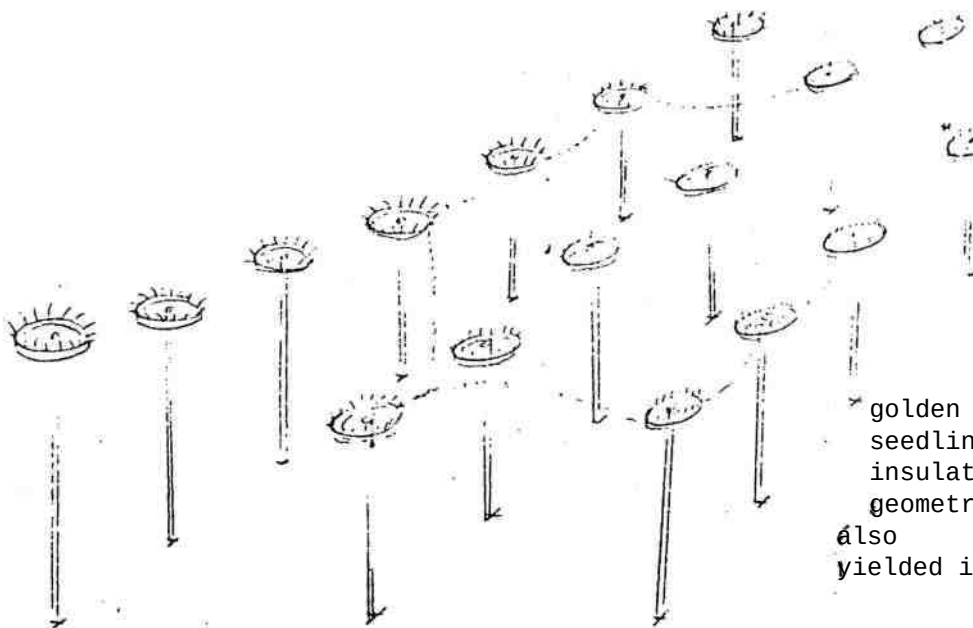
Mention of the experiments of a physicist from the University of Helsingfors: Selim LESTROEM: "The Professor separated into 3 batches terracotta vases filled with the same topsoil sown with the same seed (G?) under identical conditions of lighting, heat and humidity. Above the first batch, he placed a network of wires electrified positive (+) in the second batch, the wires were electrified negative (-), finally the 3rd batch, witness received no The experiment gave for the electrified batches a surplus of 10% (the orientation of the wires in the terrestrial field is not specified nor in which building or field the experiment was made.)



In 1783 Abbé Bertholan will try to plant devices with metal hedgehogs . Only one device of this type was installed for one field. (Clearly insufficient in static device).



floor



Spechneff's essays golden crowns in seedlings on insulating poles. Various geometric arrangements have also yielded interesting results.

Bose de Vèze proposes in conclusion for electroculture:

- I. Modes that use "atmospheric" electricity
- II. " " " " "dynamic"
- III. " " " " of " static machines "

In mode I, the Abbé Bertholon de Saint-Lazare built an ELECTROVEGETOMETER which consisted of a wooden post supporting at its top a glass sleeve in the middle of which was fixed a vertical rod (retal?), the point of which supported a broom, with a hedgehog of gilded copper wires. A chain of the same metal connected this rod to another, horizontal, also isolated from the post; this horizontal rod carried at its two extremities brooms or wire brushes facing the ground. These attempts (1783) were not convincing and remained isolated.

Later the Russian SPECHNEFF will resume tests of the same kind by - reparting in a uniform way from isolated posts surmounted by metal crowns with golden copper tips... (the latitude of the tests, nor the nature of the terrain is not specified). Spechneff says he obtained an overproduction of 62% for oats, 56% for wheat, 34% for flax, 55% for barley, 11% for potato, 28% for rye. (The origin of the seeds and plants are not specified).



Experience of the Belgian Lagrange: galvanized iron rods pointed between each plant. Rods 15 cm long.

Paulin device

Broom of copper wires mounted on metal rod

Wooden post on which are connected and stowed iron wires fixed to the metal support of the copper broom on the one hand and on the other hand fixed in the ground asymmetrically like the roots of trees.

Such was Paulin's Geomagnétifer.

Some featured right-hand spiral windings for the wires of iron around the post as well as a sheaf winding on the left for the copper wires of the aerial broom on a series of posts alternated with another series whose windings were reversed.

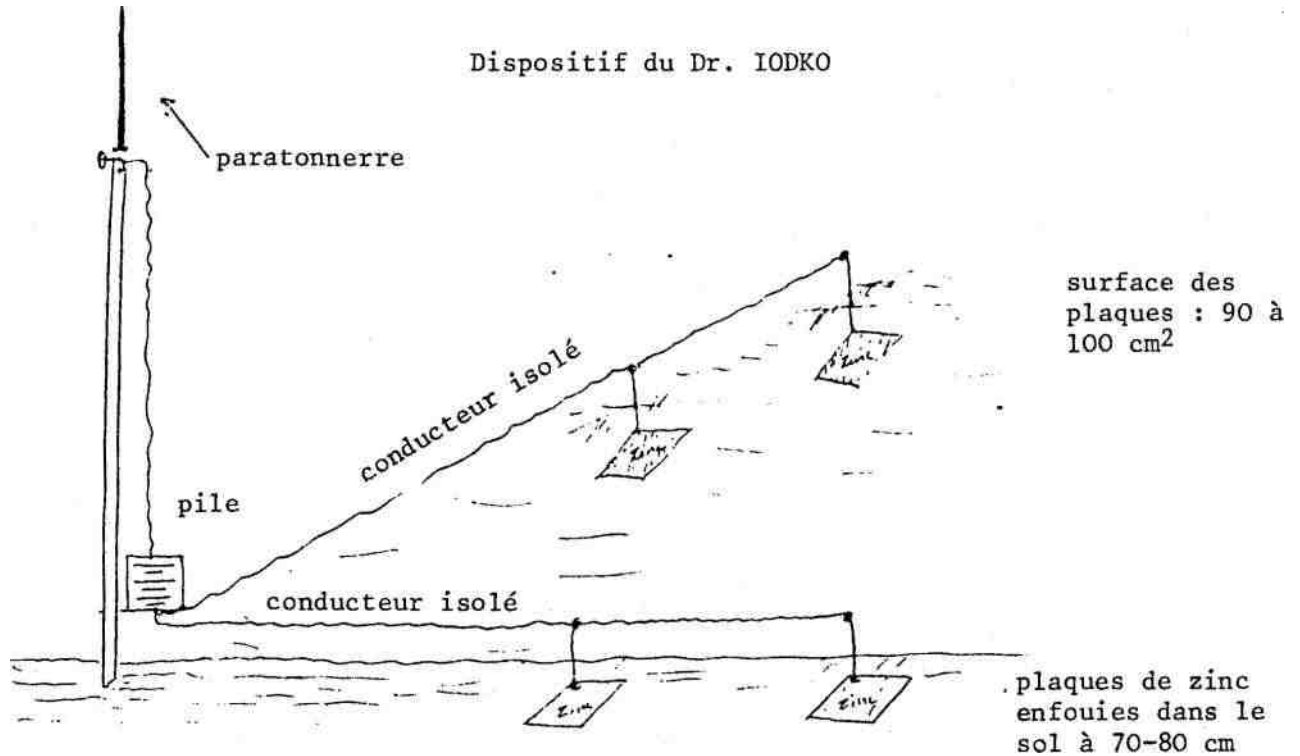
The pitch of the windings, or pitch of these solenoids.

Well after Spechneff a Belgian: Lagrange, professor at the Military School of Brussels made simple and practical experiments; he drove in 15 cm small rods of galvanized and pointed iron whose aerial part did not exceed approximately 0m50 . The yield results were reportedly remarkable. Each stem was placed between the plants.

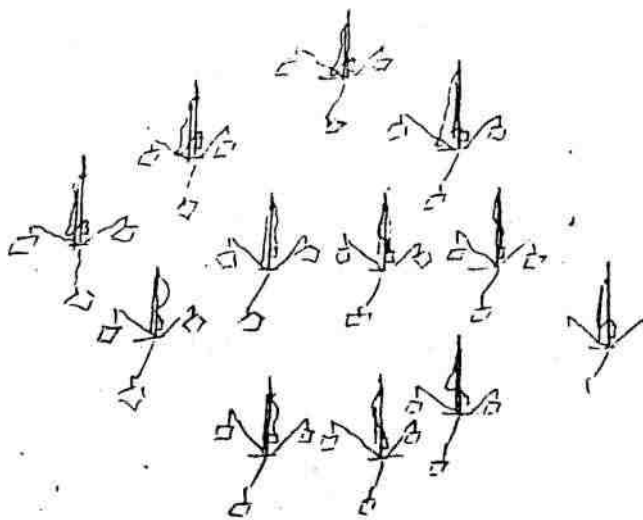
Paulin, another experimenter of the same time invented around the same time the GEOMAGNETIFERE composed of a pole or post, terminated by a metal rod armed with a broom of copper wires; wires start from the stem and branch out into the ground like the roots of a tree. Paulin advised a minimum of four devices per hectare. According to various experimenters both in Belgium and in France, this device provides remarkable results , particularly in viticulture (it is stipulated that the device does not replace amendments or rotations on other crops). The grapes were richer in sugar and must (exp., from 1891). In 1892 in Vais (Ardèche) a spinach crop gave a production of 24,330 grams on 5.40 square meters against 19,700 grams on the same type of land and surface. . . . 2 identical. In Merlien 32 m of potatoes produced 90 kg of potatoes against 59-60 kg on other fields of the same nature and of the same extent

without device.

A farmer from Clifton (County of Gloucester, England) Mr. Pinot published the results with the Paulin device. He claims that potatoes come three weeks earlier and provide up to 50% excess. At Clifton, market gardening produces a 40% surplus of beans and peas. Certain fruits: figs and grapes ripen despite the climate.



A dozen such devices are needed per hectare with a certain geometric arrangement.



Approximate view of a set of lodko battery devices on an area of approximately one hectare

Note: lodko was interested in systems for printing photographic plates as opposed to hands, using high voltage and high frequency generators with the aim of fixing the effluvia emanating from living beings in the image. It is a precursor of Kirlian that the authors never cite.

Works_from_dr._lodko of the Minsk government and in the Uzda zone to Sanatorium_of_Nad-Niemen, with the supervision of professors Baranowsky, Sowietoff, _Voyeikoff of St. Petersburg University .

Period from 1885 to 1897, work recorded in the reports of the Ste d'Agriculture de Saint-Petersburg, with description of the devices.

Dr. Lodko used electroculture devices for cultivation that were based on the same principles that underlie electrographic and electrotherapeutic methods.

It uses an electric battery of a few elements which must necessarily be of very constant flow in direct current.

One of the poles of this pile is connected to a lightning conductor (characteristics ? height, material, shape?) ... connected to the ground (mass?)

The other pole communicates with conductors insulated on zinc plates 90 to 100 cm square in area, which plates are buried in the ground 75-80 cm deep (horizontally? vertically?) [This is an induced polarization].

A dozen of these devices were placed per hectare. He obtained surprising results from the point of view of the speed and vigor with which germinate, develop and bear fruit the seeds and plants subjected to this "telluric" electrification, variable effects of the rest according to the fluctuations of atmospheric electricity. Lodko had also developed a geohygrometer or hysimeter giving precise ground humidity up to three meters deep, a storm detector up to 400 km (See AFMAG) and atmospheric electricity measuring devices (?) (See Mathias)

Methods using dynamic electricity (by E. Bosc de Vèze)

- They use (according to the author of Universal Magnetization). The current produced by plates of zinc or copper buried more or less deeply and connected to each other by conductors, insulated or not.

(As for the physical effect, it is first of all a spontaneous polarization *= wet earth metal contact - to which is added a channeling of natural electro-telluric currents whose frequency and intensity are linked to the activity of the sun on the chromosphere, the photosphere and the corona) [see Mathias, Treatise on Atmospheric and Telluric Electricity, Paris 1924, PUF, French Committee for Geodesy and Geophysics, 6th Section preface by MD Berthelot]

- Brief description of the Lagrange device following the work of Spechneff with additional use for the wires of porcelain insulators (?) which, moreover, did not give good results...

A method that seems good consists in placing plates in the ground connected to a network of metal wires arranged above the plants, this network is traversed by electric currents (??? continuous, one supposes) coming from a source of electrical energy (unspecified).

Bose then evokes an underground connection of wires ... presenting no guarantee of duration and more troublesome for the culture.

(The problem of the orientation, the diameters, the metals, the shape of the network or the shape of the plates does not even touch upon the authors of the work or the author of the book. It is not mentioned that the method copper wires buried is figured in detail in the Chinese TCHOU -LI eight hundred years before our era with a description of the "oriens" (Sic) to be respected for the teeth and the claws of the Dragon).

Methods using static machines (after Bosc de Vèze)

Lemstroem Selim of the University of Helsingfors carried out electroculture tests with an electrostatic machine as well as other tests with high-frequency Hertzian waves around 1885. The experimenter designed an electrostatic machine with nested cylinders 30 cm from 0 by 50 cm in length at high rotation speed, the output powers of which made it possible to supply a large-area metal network made up of a network of bare aerial conductor wires supported by posts with porcelain insulators. The wires were spaced between them by about 1m and provided every 50cm with a metal rod directed towards the ground (material? shape?). The network was fed from 7 a.m. to 11 a.m. and 4 p.m. to 8 p.m. The entire metal network was arranged around the fields to be exploited and consisted of a wire of two

millimeters from $\emptyset$ placed on insulating supports. This assembly is connected to the electrostatic machine. A second 0.5 mm wire on the second circuit was connected to the other pole of the machine. On this second interior circuit were hung the rods directed towards the ground and of approximately 12.5 cm in length each.

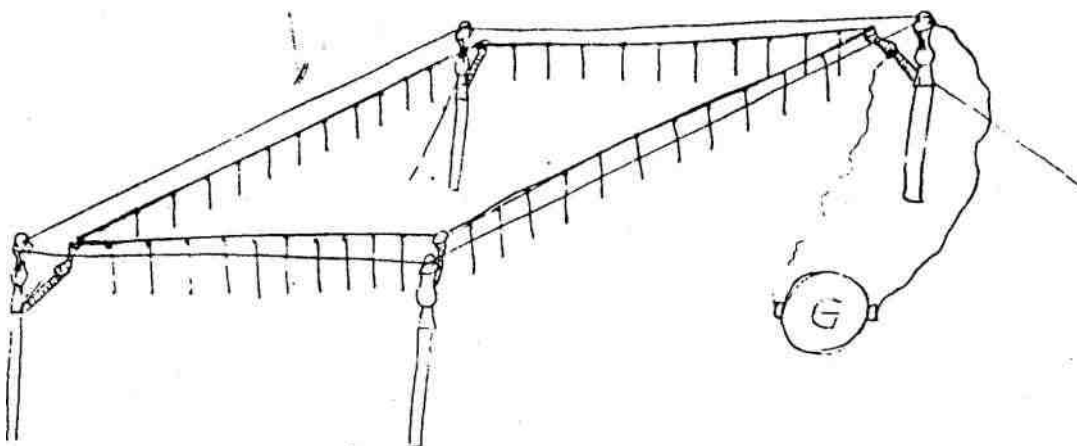
(See "Power of the points" charge carried by a body, Coulomb's theorem, dvssvmétrie between positive and negative charges, electric turnstile).

(Note on the distribution of charges developed by influence on a conductor J. Rocard, Electricité, p.41).

(Bosc does not indicate the high frequency tests, type of generator, means implemented, results etc...) whereas quoted... Lemstroem

- 1°) The proportion of plant growth is about 45% due to the fertility of the soil.
- 2°) Certain plants only tolerate electrical treatment when heavily watered .
- 3°) Electrical treatment is rather harmful during strong solar heat.

Outer iron wire $\emptyset$ 2mm mounted on porcelain insulators



$\wedge G$) = electrostatic machine of Lemstroem Selim. The inner wire is 1m from the outer wire; on this inner wire are fixed the metal rods directed towards the ground. The inner wire diameter is 5mm. Each circuit is connected to each pole of the machine. This network is placed around the cultivated fields.

Lemstroem's experimental work focused both on the electrification of seeds in a humid environment, then of young plants, and of adult plants.

In relation to the vertical electric gradient and the potential difference, see the work of Mathias on telluric and atmospheric electricity as well as the text by Alfred Lambert and Doctor Pierre Creuzé: *Etudes sur les Influences Cosmiques*, Paris 1947.

Numerous remarks on the nature of atmospheric events from 1500 to the present day with the variations attributable to electricity, magnetism and solar activity on the behavior of living organisms in general are all small indications that can be used in agriculture and electroculture. Numerous bibliographical references complete the concrete examples.

Author index: Electrostatic Machines (Bruhat: Electricity) (Y. Rocard: Electricity)

Friction machines : RAMSDEN Jesse (English) Yorkshire 1735 - Brighton 1800.

[previously experiments of Abbé NOLLET Jean Antoine 1700-1770, inventor of the metal leaf electroscope 1747]

Influence machines : The Greek Electrophore

- : TOEPLER (-) with movable glass disc and tin sheets :
- WIMSHURST (1883) with two coaxial glass plates
- : FELICI (1947) with driver carrier
- : Van de Graaf (1932) with "belt" conveyor
- : PAUTHENIER (1939) with insulating dust conveyor in air.
and Moreau-Hatot.
- : FELICI-GARTNER 1950 / FELICI-MOREL in Grenoble 1952-53
with cylindrical dielectric rotating in an enclosure
without pressure (yield currently the best).

Obtaining High Voltages by non-electrostatic processes: Pulse generator initially proposed by GREINACHER.

NB The Lemstroem Selim machine is not cited by the French authors. Its construction gave the following results in its experiments in electro culture:

1898 = 164 hours of treatment gives an excess of 38% + for tobacco,
11.2% for beets; 8.7% for carrots and 11.2% for beans.

1899 = Oats 28%, Barley 23%, Carrots 37%, Potatoes 47% excess; by
against peas showed a deficit of 17.5% and 19% for cabbage.

1900 = Barley 26%, Peas 55%, Potatoes 17%, Strawberries 88%, Beets 42%,
Carrots 92%, Beans 33%.

(There is no climatological indication to support these results).

[But in the beginning of Bosc's text, it is about better results in the polar regions =
Lapland, northern Norway, Russia]

Electroculture by direct influence according to Bos de Vèze 1910 .

Based on the effects that arc lamp light has on vegetation . As early as 1861
Hervé Mangon recognized that the arch helps the formation of chlorophyll in plants.

In 1868-69 Prellieu believed that the absorption and decomposition of carbon
dioxide for plants was done under the light of the Voltaic arc (which varieties?)

In 1880, C. Siemens resumed the work of Prellieu more completely but noticed
that the too intense UV caused the plants to suffer. He interposes thick panes of glass
(filtering) and this interposition eliminates etiolation and plant dieback.

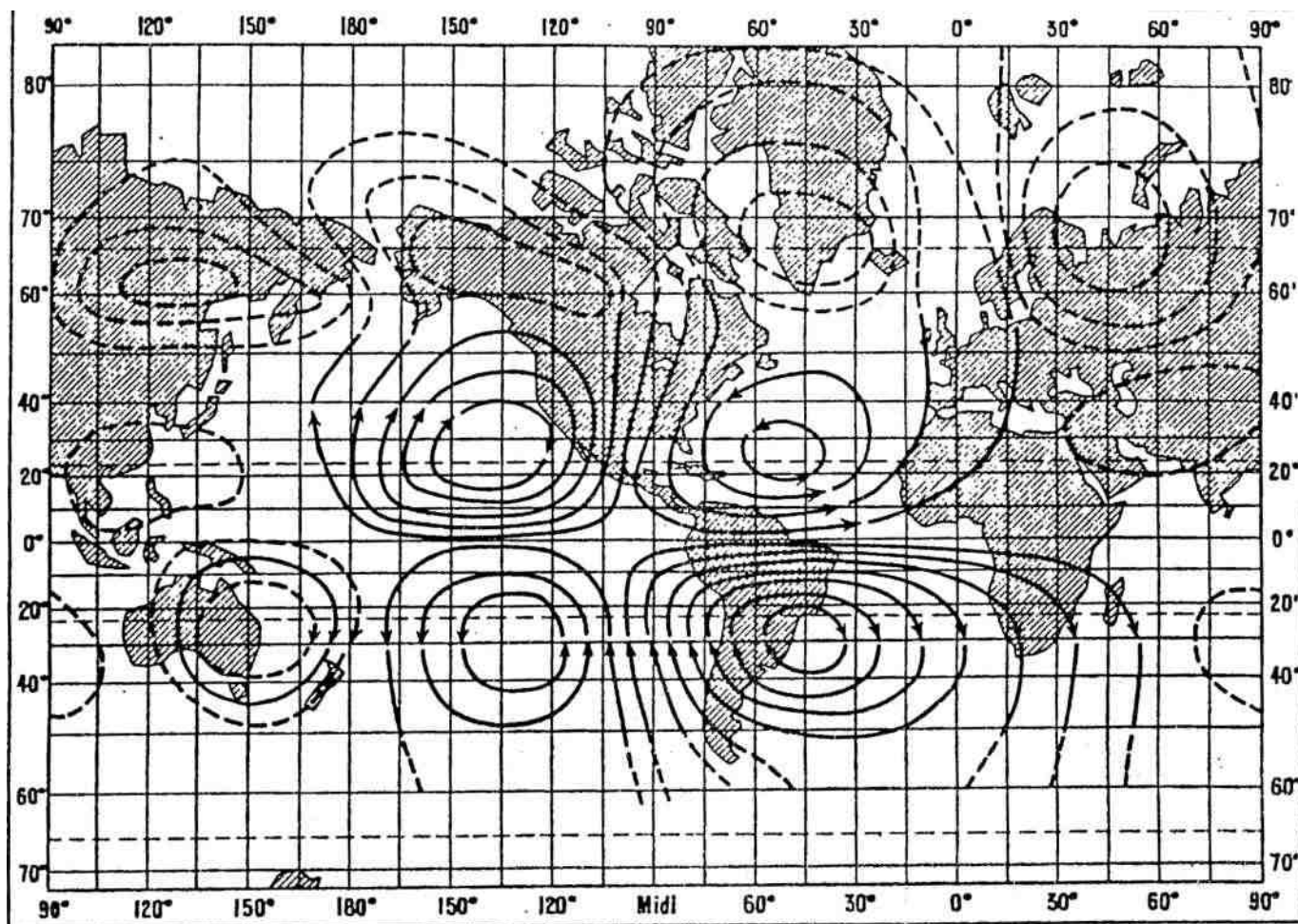
(It remains to be determined under what conditions the trials were carried out. Open
air? Greenhouses? Which varieties?) Hygrometric degree?

Around 1881 Sheraier and LH Bailey of Cornell University performed experiments
which confirmed previous observations (Where?); they specify that the light emission
from the coal arch activates fruiting and that the aerial parts seem to develop to the
detriment of the roots (?).

(No information on the spectral concept and the filters colored in sunlight - water
treatment , seeds , plants by choice monochromatism. Yet the technique of displaying
water in colored bottles is known in India and used for a thousand years. Bosc cites in
his booklet the experiments of Yoghis, Hindu Fakirs causing the rapid growth of
seeds...).

From 1892 to 1893 G. Bonnier studies the action of the light of the arc of

X.

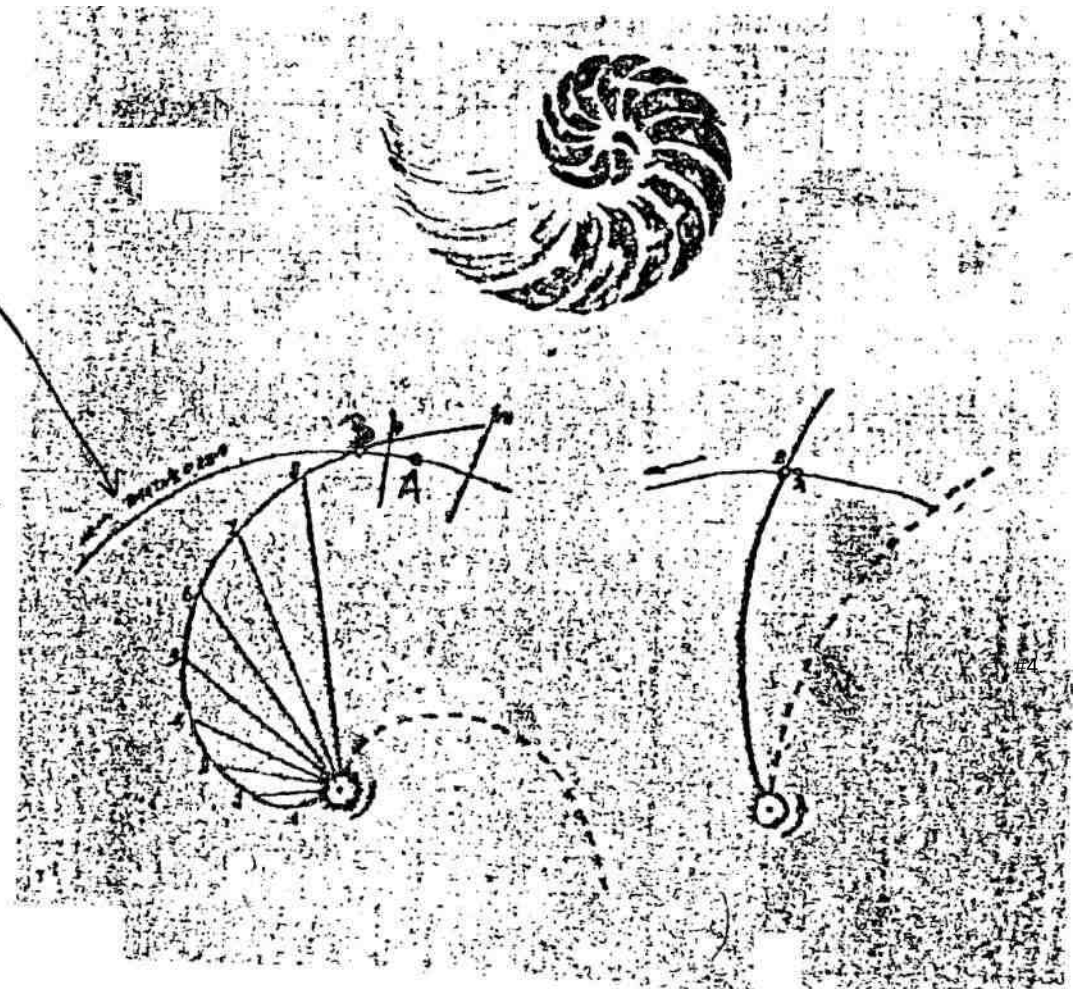


Telluric currents flow at rpidi, Chicago time (according to Rooney).

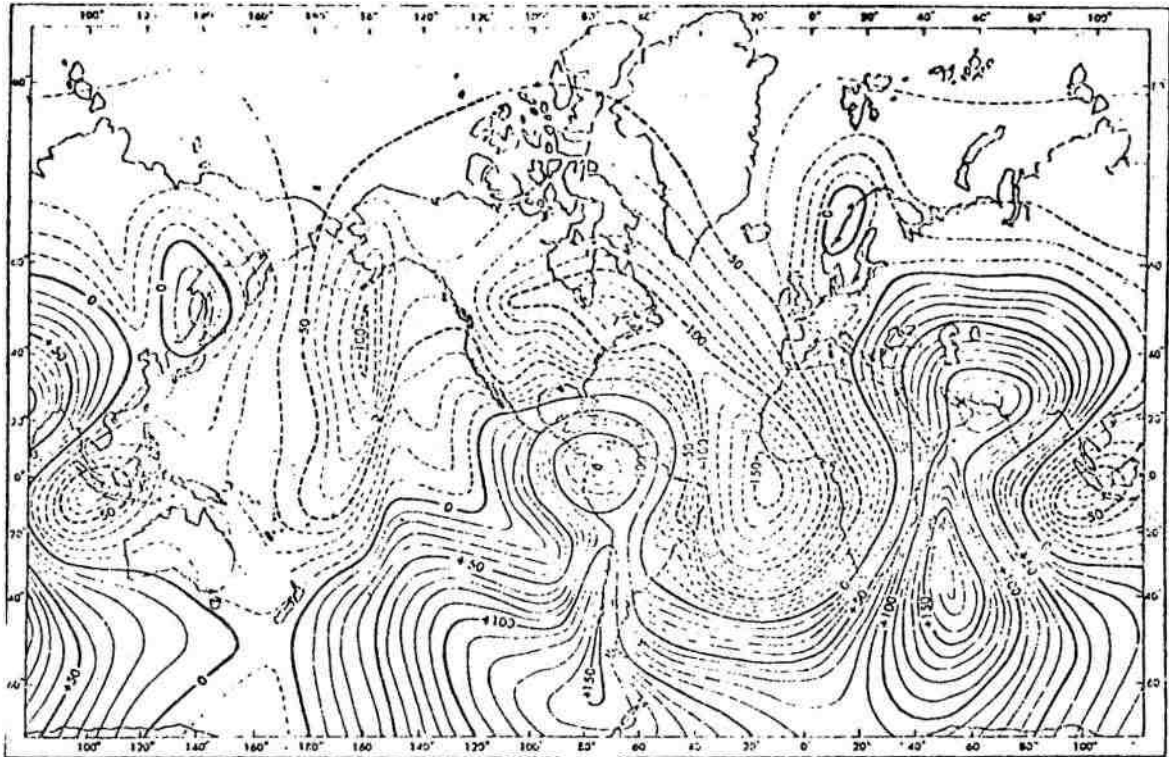
Rotation of
the earth.

solar
wind which is
reported here hour
by hour (we
must not forget
that the sun
turns on itself)

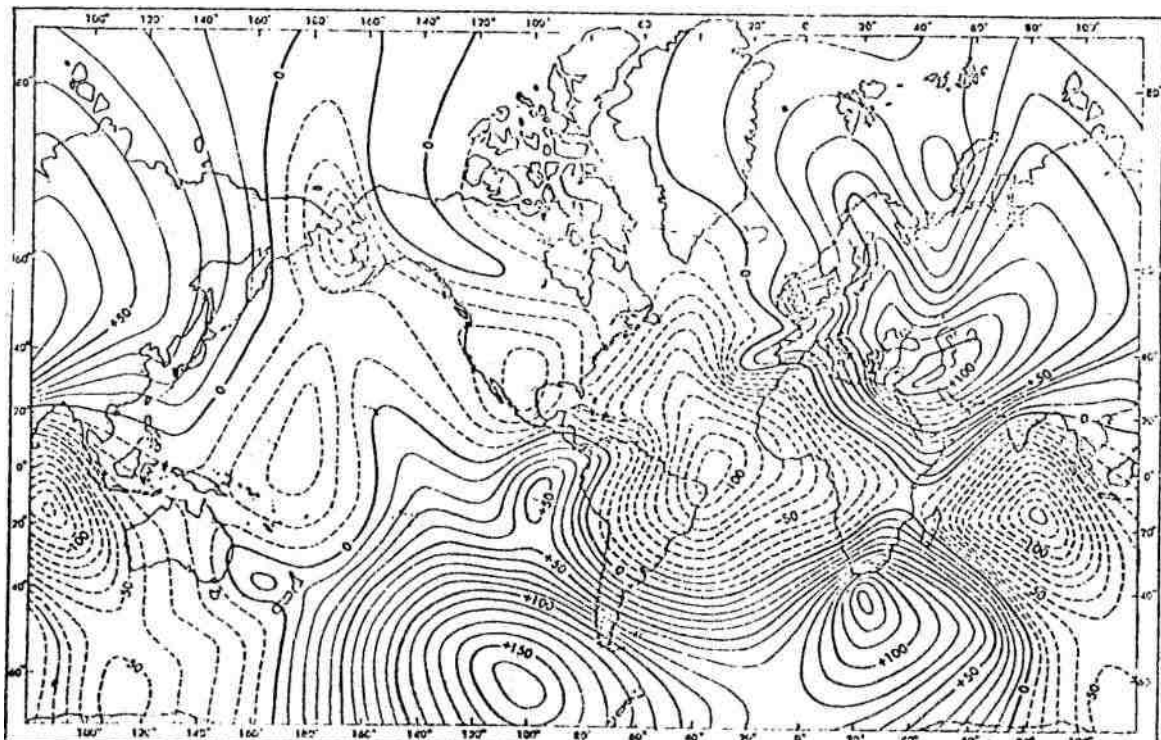
The rotation of the
sun on itself is
about 27 days.



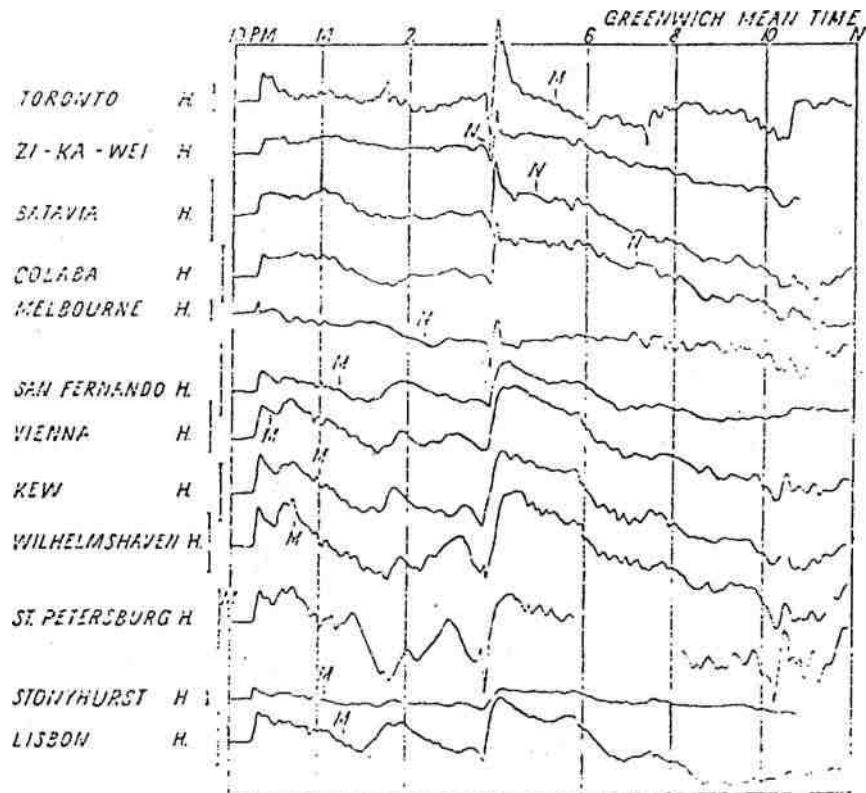
Form-Relation between the winding of
the earth and the sun in of a seashell and moves it-
interstellar space.



Geomagnetic secular change in gammas per year, vertical intensity, epoch 1912.5 (after Vestine). (From Walter M. Elsasser, *The earth's interior and geomagnetism*, Rev. Modern Phys., Vol.22, no.1, pp. 1-35, 1950)



Geomagnetic secular change in gammas per year, vertical intensity, epoch 1942.5 (after Vestine). (From Walter M. Elsasser, *The earth's interior and geomagnetism*, Rev. Modern Phys., vol. 22, no. 1, pp. 1-35, 1950).



Records of the variation of the horizontal force during the the magnetic stonn of 1885 June 24, 25 (After W.G. Adams). The scales are indicated by the lengths of the vertical lines marked H , which correspond to 100, or 0.001 c.g.s. Two changes, at 10.32 p.m. and 3.48 a.m. were simultaneous over a great part of the earth. Local noon and midnight are indicated by N and M.

Extrait de Chapman et Bartells : GEOMAGNETISM.

Mesures de variations du champ magnétique terrestre en plusieurs points du globe = simultanéité du phénomène : Quelle influence possible sur les cellules vivantes ?

continuously and discontinuously (12 hours over 24 hours) on the modification of ligneous plants and herbaceous plants. Bonnier observes that the arc by its light causes ligneous plants to undergo considerable structural modifications in the young stems and in the leaves; on the other hand, little change is brought about in the roots.

Subjected to constant lighting, the structure of plants (cells) has become simpler, has regressed; subjected to discontinuous illumination, the cellular structure remains normal.

On herbaceous plants, intense arc lighting for six or seven months under a glass globe resulted in development and intense greenness or viridity.

The species that have thrived the most under continuous lighting are bulbs, flowering bulbs, certain grasses, tree plants, and especially aquatic plants .

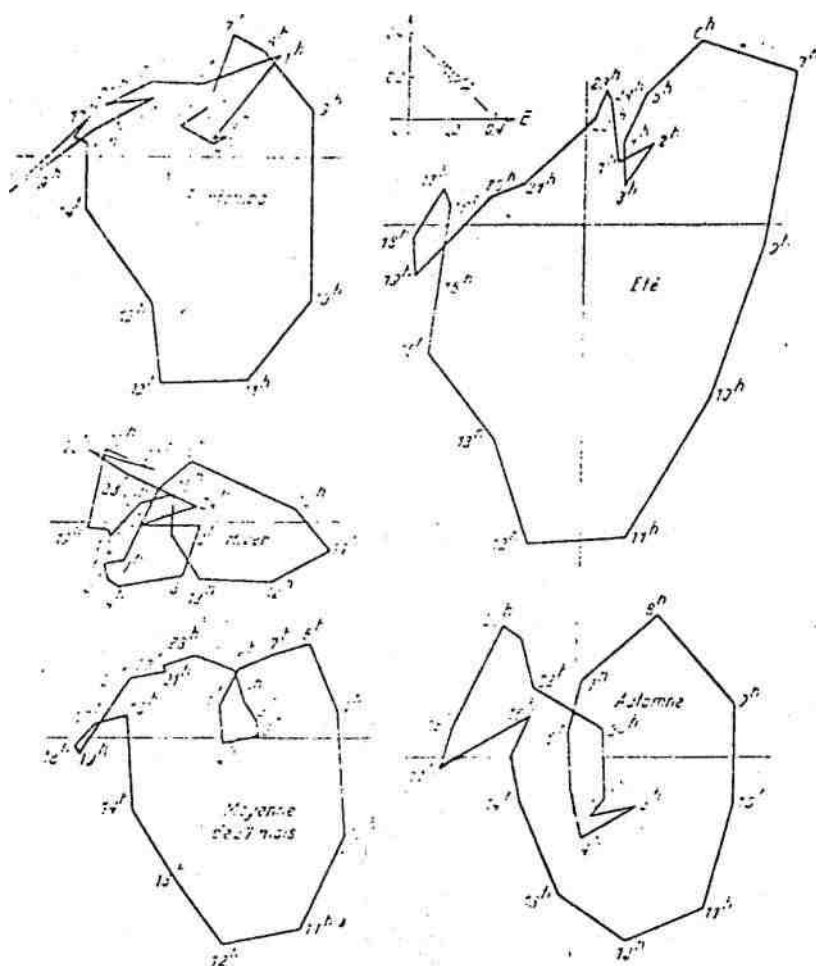
In floriculture , electric light gives more brilliance, more brilliance to certain flowers, provided that the plants are sprayed frequently.

Bosc also cites the work of Georges WILLE of the Paris Museum who was carrying out laboratory work to try to capture ammonia in the ammonia... and thinks that electricity must stimulate the respiration and perspiration of plants: can -be allow plants to fix carbon and ammonia from the air. Bosc then deals with parahail, the formation of hail in relation to atmospheric electricity (work of Dr. Clément) finally he concludes in a general way that the techniques of electroculture are to be created from scratch with machines which have yet to be found.

Observations from 1926 to 1930 of Labergerie at the National School of Agriculture in Montpellier (Hérault). According to the theories of Georges Lakhovsky on the Cellular Oscillator (excerpts) . /

Preamble on the Vine (old observations)

- In connection with certain magnetic storms: sudden lengthening of vine shoots...
- Relative immunity of vine shoots against downy mildew dominated by copper or brass wires (copper and zinc alloy) with reduced growth of the grapes under this last metal alloy!



(*)
Polar diagrams of diurnal variation in Paris (Parc Saint Maur). According to P. Rougerie.

L. Cagniard: Telluric Electricity

(*) of the electro-telluric field at various times of the day.

- Reduction in the production of rows of vines equipped with iron wires intended to capture and transport the electrical energies of storms, an experiment that lasted five years and was canceled when, during certain storms, reductions in vegetation and also destructive accidents were perceived of vines by lightning strikes.

I. Action of the oscillating circuits according to the nature of the metal and the orientation .

The oscillating circuits act when their device is oriented so that the open ends are directed towards the north: this rule has an exception, this activity can be moved in the event of prevailing winds , the opening must be along the axis of the line of flight of the winds, the extremities always turned towards the earth. The circuits have optimum activity being inclined relative to the horizontal at an angle close to 24° . The horizontal circuits have little effect and those straightened with their ends raised upwards, with an orientation of the opening and the angle with respect to the horizontal not respecting the optimum rule, have a paralyzing effect on the vegetative activities of certain plants.

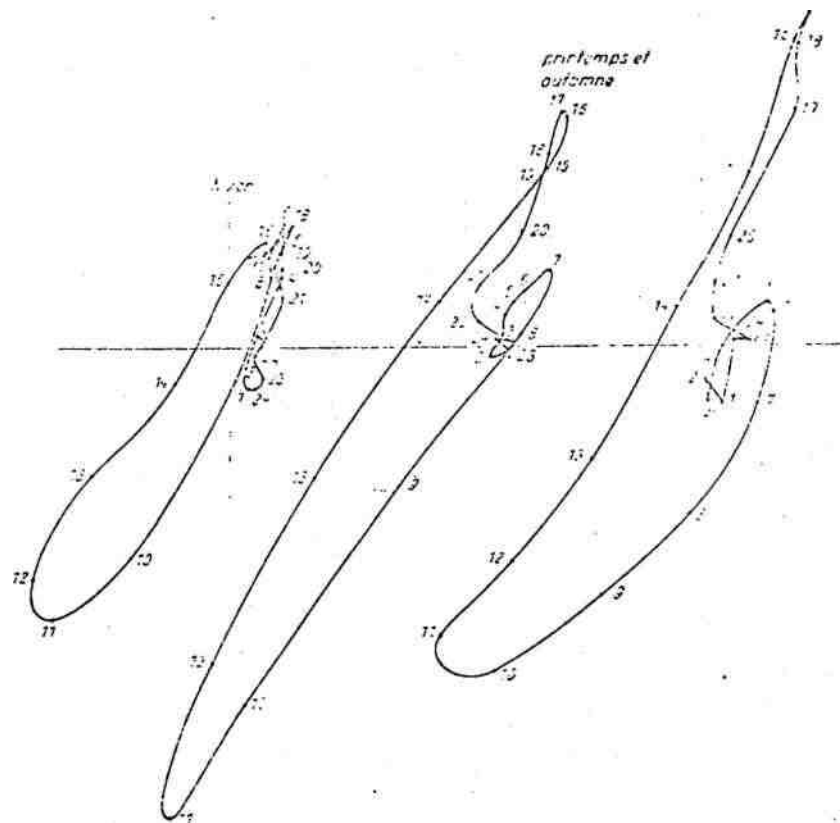
Note : Labergerie emphasizes: "The actions of the 'waves' always act downward as if driven by gravity ".

On this subject and with regard to the observation, it would be appropriate to reconsider in theoretical and practical electricity: a) the Rowland effect, b) Coulomb's law on the curvature of a conductor, c) the effect of the points, d) the experimental work of Mathias on atmospheric and telluric electricity as well as the work of Louis Boutard and Armand Hatinguais.

Labergerie resumes: Blades of soft copper (annealed), striated lengthwise, were more active, those striated across were partially inactive and those striated obliquely were active by obliquely deflecting the plants and their activities in the direction of this obliquity. Silver and lead slides gave results comparable to those of copper.

Note : Consult the work of Enel on emissions and absorptions due to materials having a chosen shape and a given orientation in space associated with a given color and a chosen material.

Labergerie continues: the effect of the circuits is less intense when they are placed vertically; this action moves in the horizontal direction, following the direction of the wind. Verification was made with oscillating circuits of the same horizontal section but of different length. Effect Activities



Polar diagrams of diurnal variation in Stockholm. According to D. Stenquist (*)

(*) Natural electro-telluric currents.

varied considerably with variations in length and distance from the ground.

Note : see work by Mathias; electricity from Y. Rocard and electricity from Bruhat. Comment on resonance phenomena between solar radiation and the immediate environment: Relier (HV) and Thomas (GE), Max Planck Inst. Aeronomy Kallenburg-Lindau 3411, DEU). Have published in "Astr. and Astrophys., DEU (1979, 80, n°2, 227-233, bibl. 20 ref.) "Multiple scattering of solar resonance radiation in the nearly interstellar medium".

The authors have obtained approximate solutions of the radiative transfer problem which represent a highly idealized description of the multiple scattering of solar radiation by resonance in the near interstellar medium. The first and second order contributions are calculated with precision and the "Eddington approximation" is used to estimate the higher order components of the radiation field.

Furthermore Kurth (WS) from the University of Iowa has published in J. Geophys. Res., USA (1979), 84, No. A8, 4145-4164 detailed studies of electrostatic disturbances due to intense plasma waves near the upper resonance frequency in the region outside the plasmapause. the most intense in the Earth's magnetosphere, their polarization is such that the electric field vector of the wave is oriented perpendicular to the geomagnetic field. Electrostatic disturbances show that they can occur at any time and at magnetic latitudes oscillating and varying between the equator and 50° north or south latitude.

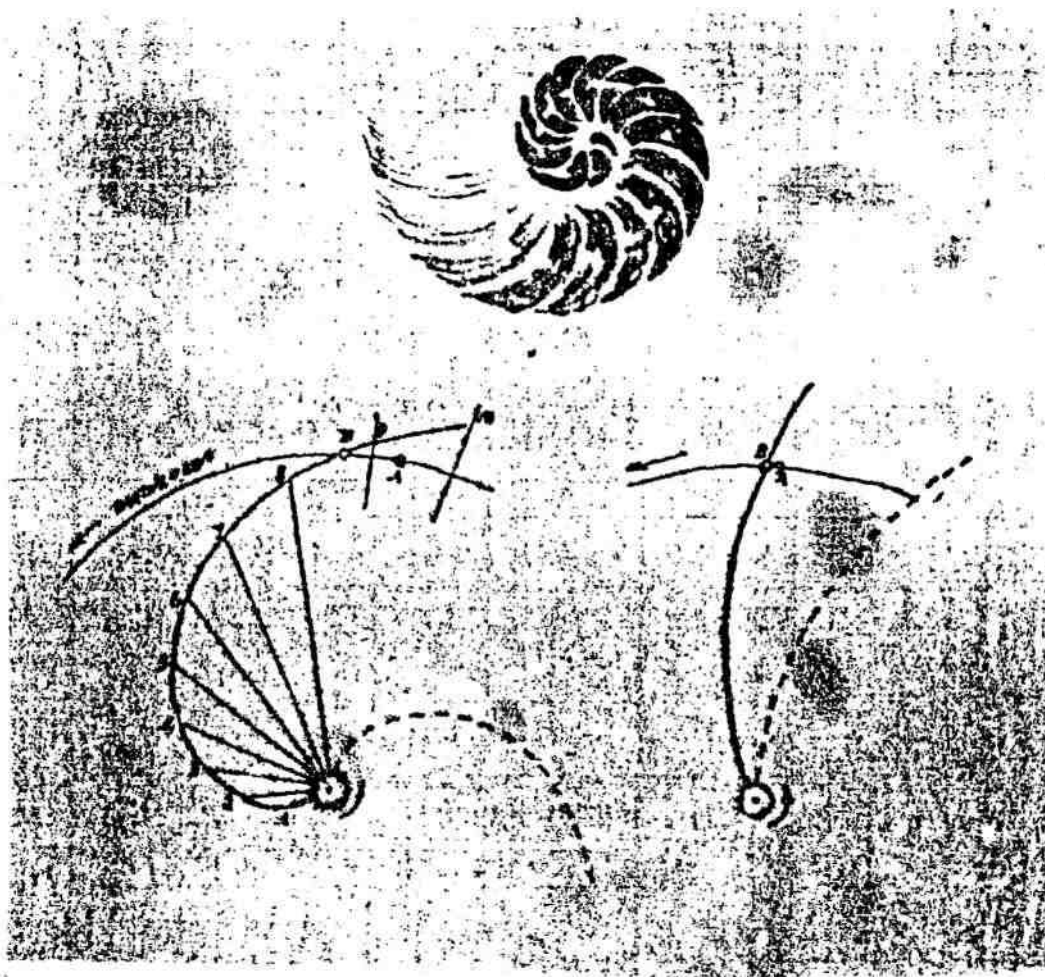
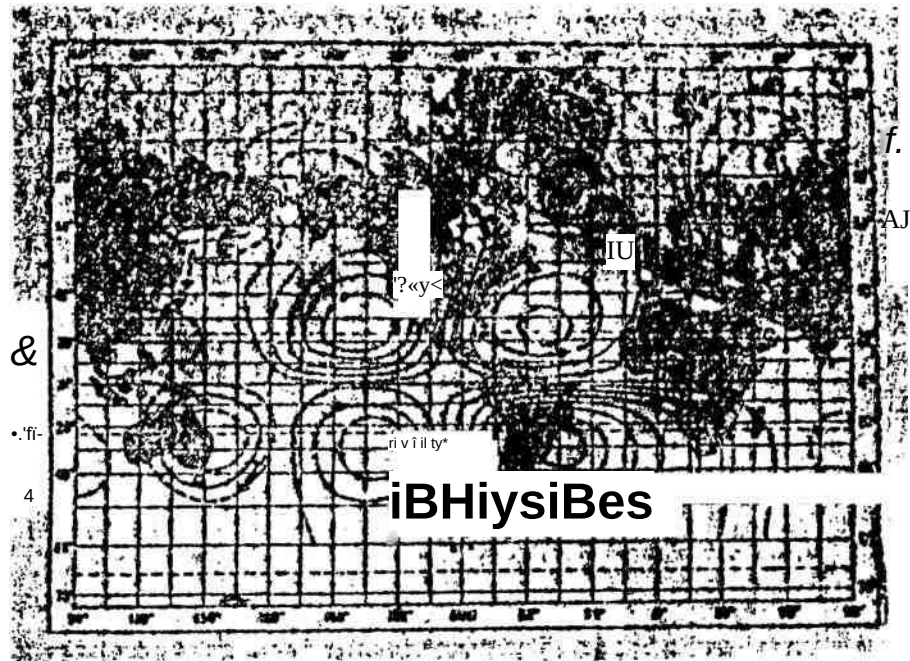
II. Effect of the material isolating the oscillating circuits and of the color (Albedo of a body) .

Waxed or varnished insulating circuits with lacquer range showed a more considerable oscillating power. (See to understand the work of Louis Boutard and Armand Hatinguais). Laberge has applied various products from the lacquer range to the circuits, incorporating these various substances:

White : Starch, alumina silicate: overactivity Red : Various metal oxides: less activity Black : Carbon black, anthracite dust: act. very retarded

The tests were repeated with papers of the same color; the results were similar to above (papers of various compositions).

According to
Rooney. World map
of the direction
of electro -
telluric currents
at mid: for the



Form-Relation between the coiling of a shell and the movement of the earth and the sun in interstellar space.

Note : Consult NASA work on Albedo of bodies. Enel: use of colored filters on water, the light source being the sun (bottles of colored glass in the mass). [Defined in physics: Radiation from a blackbody]

Other tests involved powders of various colors under the devices with light gauzes on cardboard tubes. The results confirmed the effect of absorption or reflections as above (Albedo).

Note : The Boutard-Hatinguais work on the Dorian Gnosis explains all these observations relatively clearly with regard to the ancient theory of

(*)
1 Aether.

With regard to color, apart from the theories of Goethe and Newton, the questions of "colour reflections" and "colours of radiance" dealt with by Rudolf Steiner must be taken into consideration, and in particular the phenomena linked to the Polarization of Light (Nicols-Filters in Herapatitis - double hydroquinine sulphate)

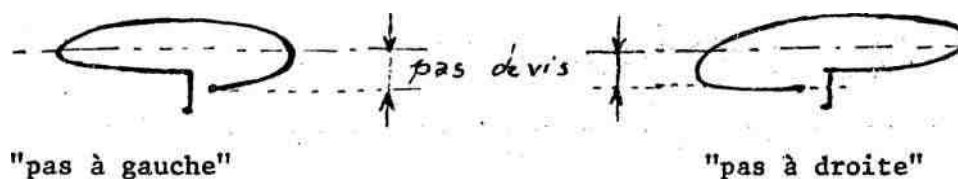
III. Different effects of oscillating circuits according to the nature of the metals used .

Circuits made of "hard" metals have a very weak influence on vegetation; their activities are reduced compared to the activity of "soft" metals especially annealed; and this in a considerable proportion. (See on this subject the work of Sir Jagadis Chunder Bose, Calcutta 1929).

Devices used by Laberge in Montpellier:

A) Oscillating circuits placed vertically or inclined with respect to the horizon - tality, the aperture being placed differently (like a time dial of 01h at 12 o'clock) and the orientation plan having been directed according to the cardinal and inter-cardinal points. (The lengths and θ being variable for each circuit).

B) Oscillating circuits characterized by one of the ends initiating an internal spiral (like a thread), the other end being folded over to form a descent of 3 to 4cm in length (angle of the wire = 90°).



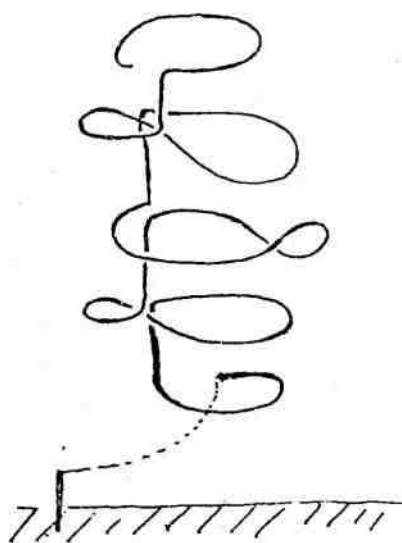
(*) See in this book all that relates to this work.

This very effective device has the particularity of exerting a variable activity with the direction of the winding according to the direction of the wind.

[To bring together the dextral and sinister effects on living organisms, the effects of Routard self-generators and the dextral or sinister Moebius strips as well as the polymetallic "eight" circuits used by Dr. Ja. Fontaine] Consider problems of symmetry in crystals, windings of gastropods etc... The directions of the fields of force in relation to the spiral winding directions called "right" or "left" can be approached by projective geometry, symmetry effects result.

Apart from the problem of electroculture and in relation to the use of "figure eight" of oscillating circuits, in 1930, a veterinarian specializing in small animals, MG Mercurin de Cannes, ing. agricultural EAM treated a large suppurating skin lesion on a four-year-old fox (dorsal region) by placing a Colysa GL collar permanently "in 8" on the neck and on the chest behind the shoulders After 15 days an improvement is noticeable note the practitioner, two months later, the hair has grown back and there is no trace of the old lesion.

On certain shrubs and on orange trees, Dr. Peyré in 1938-39 carried out experiments with a certain Mr. Raynaud from Boulouris in the Var, then on diseased oaks; Oscillating circuits in 8 open superimposed, with right or left winding were used successfully for the treatment of trees. One of the devices used had the following form: A thin set of tubes



bent copper tube began with a "straight" Labergerie-type shape, then continued with three "eights" of asymmetrical loops forming three pseudo Moebius, the last and fifth turn of the copper tube around the trunk of the tree (not shown in the drawing) of the "left" winding was connected to the ground by

.

- C) Metal strips made up of links made of different metals: copper, brass, iron, nickel etc... and suspended between two supports by silk threads. Variable orientations .

Straight or twisted metallic ribbons with pendants forming small oscillating circuits (small.crescents)

- D) Twisted strands of metal forming by spreading out at each end a sort of widened brush (vertical or horizontal)



here too the windings of the "right-step" and "left-step" twists have been retained.

According to Lakhovsky and Labergerie the latest devices have made it possible to check, with intensity, the action of the direction of the wind, and especially the favorable or unfavorable effect of certain metals under certain plants.

- E) Small circuits with descending spikes at various lengths surrounding, as the case may be, the parts of the plants to be stimulated or paralyzed.



The supports are always made of insulating material, either by pinching the circuit in a fork, or by a tri-wire silk support. All of these devices have been very effective; each device has a different action depending on the intended purpose.

In 1924 the French Committee of Geodesy and Geophysics published a Treatise on Atmospheric and Telluric Electricity under the direction of T. Mathias, then director of the Puy de Dôme Observatory. The book begins with a study by MR Dongier devoted to the electric field of the atmosphere and its measurement involving the use of three distinct organs:

lunar influence and you are %éomaa ,nitiqûe,

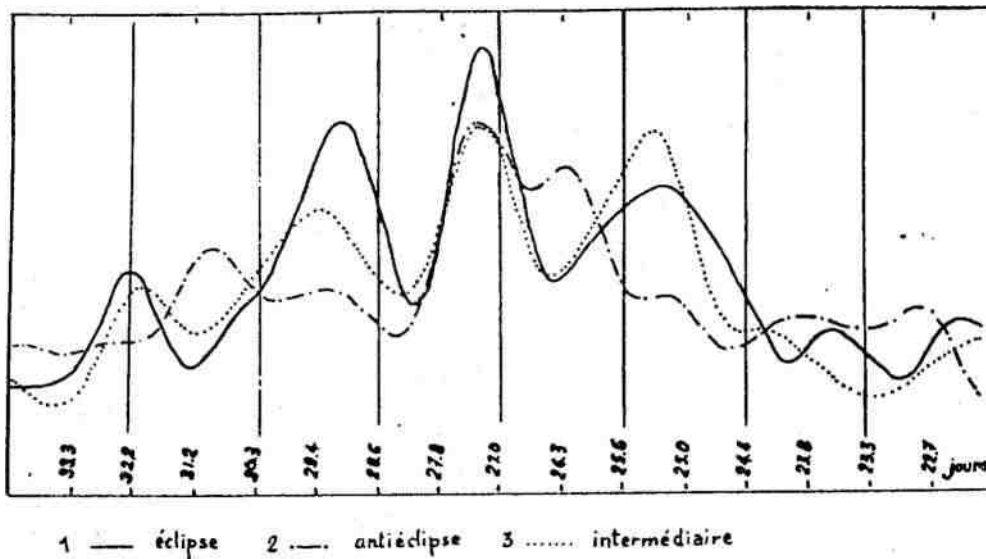
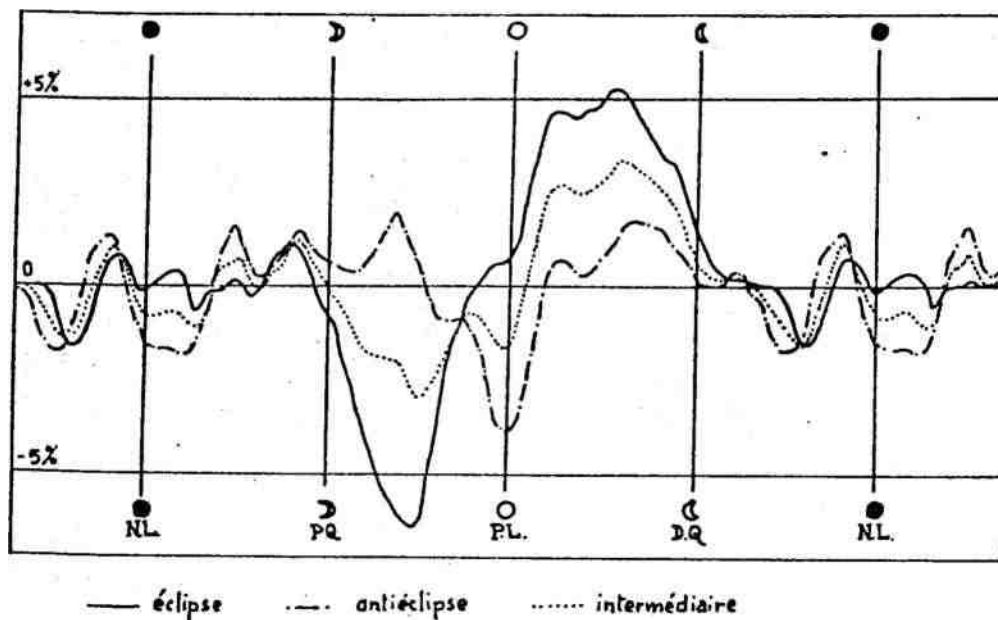


Fig- 1

Autocorrelation functions of the spectral energy densities of the three series:

- a) - strong energy band centered on the 27.3 day period (rot.©)
- b) - neighboring line of $2^{1/5}$ days, duration of a lunation.

Lunar impulse tur geomagnetic, terrestrial activity.



Existence of lunar modulation and dependence of this action on the angle of the Earth-Moon and Sun-Earth straight lines.

Measurements made by the Institut de Physique du Globe in France and by the Ebro geomagnetism observatory in Spain

- 1°) The Collector or potential take-off automatically and at all times puts itself at the same potential as a determined point in the atmosphere.
- 2°) The Electrometer whose role is to make known the difference in potential that a determined point of the atmosphere can have in relation to the ground, or also the ddp between two points located at a known distance from one of the other.
- 3°) The Recorder of the indications of the electrometer.

In a second part, Mr. Ch. Maurin, director of the IPG ' y of Paris exposes the electrical loss, the conductivity, and the atmospheric ionization; subjects supplemented by the study of atmospheric convection currents.

Mr. E. Mathias then discusses the mobility of ions in air, electrical meteors, and the electrical study of precipitation.

Air and water radioactivity is studied by Dr. Loisel and telluric currents by Mr. Basler of the Marseille Observatory.

Note: The understanding of the effects of the oscillating circuits of Lakhovsky and the work of Labergerie can be clarified by a better knowledge of the aforementioned works.

In the experimental remarks of Labergerie it was noted a case of coupling of the different metals that it is the higher circuit which prevails on the lower circuit, in the same way in associations in twists, it is the external metal whose l action dominates over the outer wires. This observation is electrically consistent with the definitions of the distribution of the electric field of the atmosphere constituted by all the points in space where the electric action is felt, in particular in insulating or dielectric media and on the surface of bodies. drivers. Actions are governed by Coulomb's elementary law.

Moreover, the oscillating circuits behave like an insulated conductor introduced into an electric field, electrifications of opposite sign appear on its surface, the purely electrical aspect can make CO assimilate to a collector with a point of the old models of equalizers of potential used with gold foil electrometers. In relation to the aspect of influence phenomena, the power of points has long been studied by electricians (see treatises by Bruhat or Y. Rocard). The load carried by a body (*) Institut de Physique du Globe.

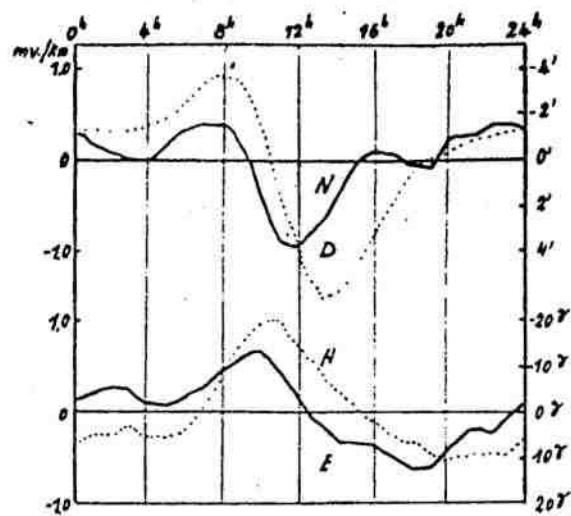


Fig. 3

— variation diurne des composantes telluriques *N* et *E* au Parc St Maur (Paris)
 *H* et *D* représentent les variations de la composante horizontale
 et de la déclinaison magnétique.
 Moyenne : 37 mois

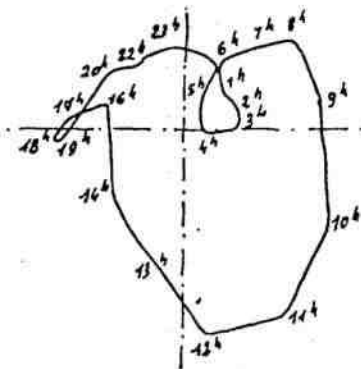
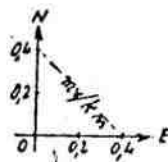


Diagramme polaire de la variation diurne à Paris
 Moyenne : 37 mois



Magnétisme ; Géomagnétisme. (en mV par km)

tends to cluster where the curvature is greatest; according to Coulomb's theorem the electric field will have its highest values there, the points are discharged by ionizing the air. On this first effect is superimposed another effect due to the terrestrial magnetic flux (total field) and more particularly to the vertical or inclination field. If the flux varies proportionally to time, a potential difference arises between the two ends of the circuit. This effect is known as the distribution of charges developed by influence on a conductor. (See "Electricity", Y. Rocard, 2nd ed. 1956, p.41) (See Treatise on Atmospheric and Telluric Electricity, Mathias, 1924, p.5 and 9) .

Action of the circuits on the vine precisely as a superior plant: summary of observations in the open air and in enclosed premises made in Montpellier by Labergerie . (Excerpt from Cellular Oscillation).

Copper, alone, has shown itself to be very stimulating to the vegetation and the productivity of the vine. The early maturity was confirmed by the much sweeter flavors of the grapes, the increase in verified mustimetric richness of which corresponded to two degrees of alcohol compared to the control vines (rows interspersed between the rows activated by copper).

Compared to circuits of nickel and aluminum or brass (Cu+Zn), the differences have been even more accentuated, these metals having acted very favorably on the feet subjected to their activating or retarding influence.

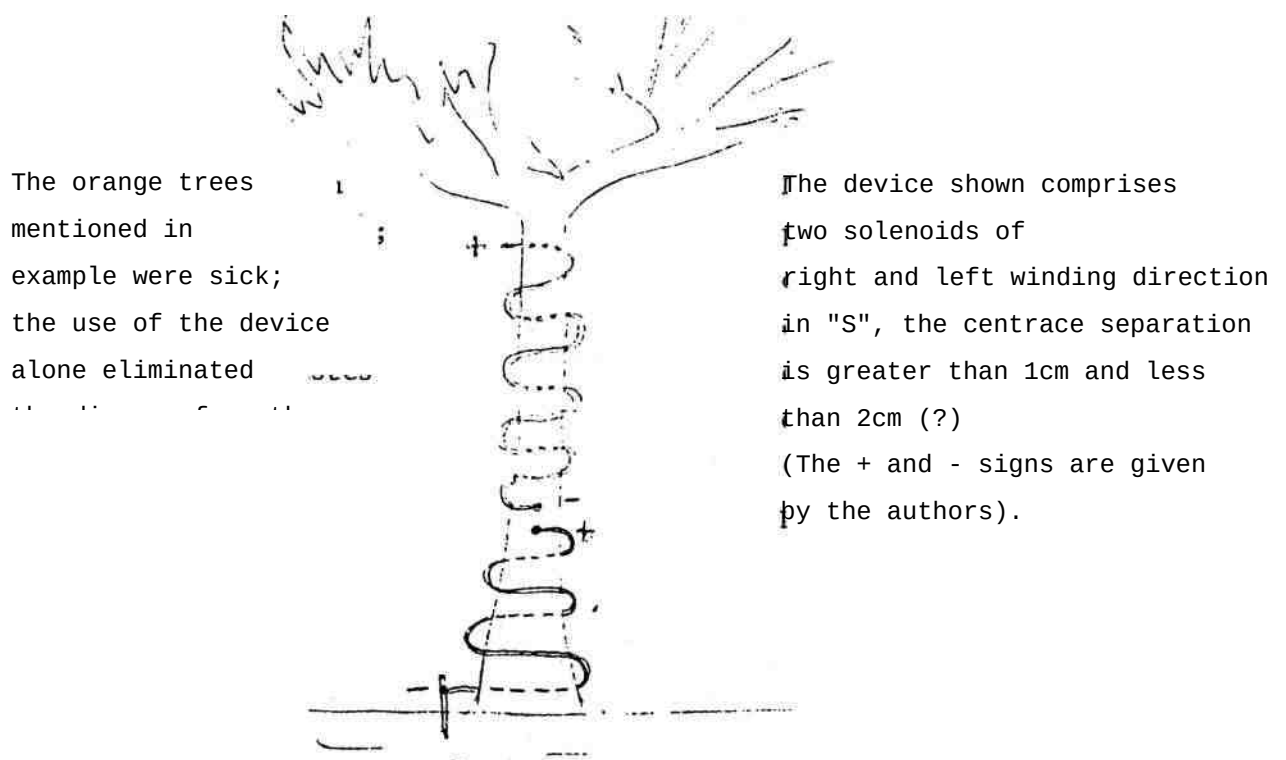
Magnetic iron on vines stopped all fruiting in 1929 and 1930. (Consider the question of magnetic rigidity) .

Outdoors, attention naturally focused on the actions of perenospora (vine blight). From 1927 to 1930 copper circuits left around the vines showed very marked vegetation activity compared to the same control vine without circuit. The dark green shade of the foliage has been more accentuated. The productions were better.

Note : On the action of atmospheric phenomena is superimposed outside the regime and outside the origin of atmospheric ionization all the influence of the radioactivity of the ground, alpha, beta, gamma, neutron rays. In addition, radon, thoron and actinon influence air ionization through their presence. Madame Curie established a theory of the activation of negatively charged wires.

clothes. The influence of wind on the activation of a negatively charged wire has been studied by Kinoshita and Ono (Proc. Tokyo Math. Phys. Soc. t.6, p.92, 1911).

^ The medical doctor Peyré of Bagnoles de l'Orne and contemporary of G. Lakhovsky published in 1947 a text of experimental relations entitled: "Cosmotelluric Radiations". Their topography over the entire planet, their possible relationship with human, animal and plant pathology and in particular with cancer. The quoted remarks cover the period 1936-1943 among others in gardens. Dr. Peyré talks a bit about agricultural and forestry electrocultures. He cites a device used by Mr. Raynaud from Boulauris (Var) on Orangers at Cap d'Antibes, at the villa "Dilecta" then belonging to Mr. Laurens



The author cites the case of treatment of old withering oaks which would have regained vigor with, however, different devices.



Right and left windings and
" left and right superimposed.

Laberge's observations relate to phenomena of correlation between plants and atmospheric phenomena and go in the direction of the theory of cellular oscillation developed by Georges Lakhovsky who sought to highlight the relationships between the field variations of cosmic phenomena and the vital phenomena whatever they are .

Plants have been able, thanks to their sensitivity, to be considered as useful integrators whose variations can be followed every day, or even from hour to hour for very fast-growing plants such as watercress, • (*) . Alenois. The influence with date and time of the following events was significant: meteor showers, remote earthquakes (Italy, Greece, Asia, Japan, America) [See about the variation of the terrestrial electric potential near the sea, the works of Louis Claude Vincent: inversion of the direction of the current before, during and after a movement of a tectonic plate] ... canic eruptions of all countries, falling shooting stars, sudden variations of the barometer and sometimes also number of sunspots .

[Reconsider the work of Kolysco on "capillary rise ", the examinations of flocculations of various metallic salts in suspension in liquids, the behavior of solutions of gold and silver salts, including the chromatographic examination on filter paper follows a particular law of distribution presenting statistically very good correlations with the positions of the moon and the sun : more than 100,000 measurements on the five continents between 1928 and 1933]

JA

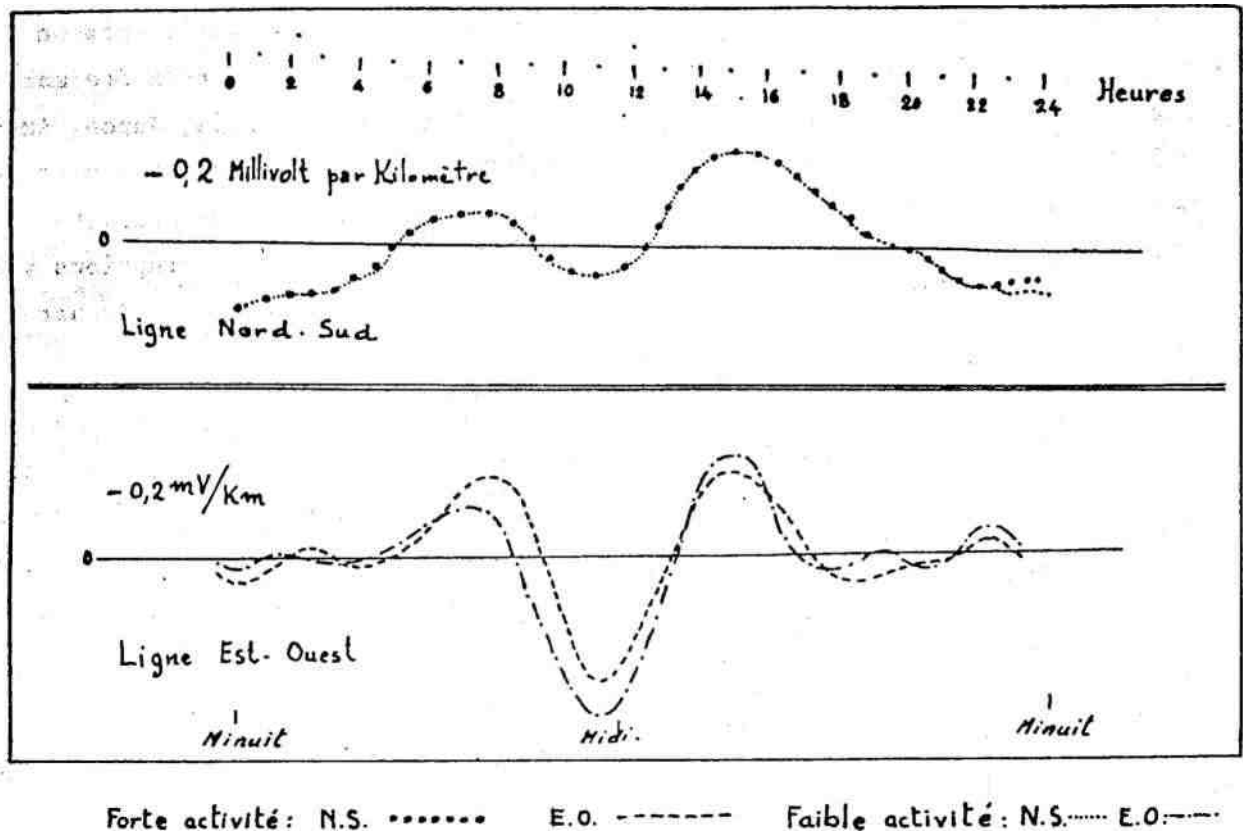
following the work which is statistically significant, also review the experiments of Professor Piccardi of Milan and the measurements of Emile Allais on the paraconic pendulum. (period 1947-1962) .

On the subject of ionization which seems to play a certain role (1924), E. Mathias indicates: The ionization is strong in the air near the thermal springs, on the eruptive grounds... in certain alpine valleys... in the caves... and in a general way in the layers of air near of the ground. Temperature and humidity also play a major role (ionization growth from 1. to 2.2 for a temperature interval of 30° C. between -17° 5 and + 12°5). When the hygrometry increases , the conductivity decreases, the number of small ions and their mobility both decrease when the hygrometric state increases. The diurnal and nocturnal variations give roughly a maximum at the beginning of the day and a

(*) See work done at the Goetheanum in Dornach, c/o R. Steiner: infl. of the day of the full moon.

Fig. 4

Solar activity and variation of the current» t«Uuri<^u«»



Daily averages of the variation of the electro - telluric currents
for the south of Europe (values on dan.: 65-66)

NB The device for measuring electro-telluric currents comprises a mirror galvanometer to the terminals of which are connected two insulated conductor wires at the end of which are placed in the earth electrodes of a suitable nature to give them non-polarizable characteristics.

minimum around midday or at certain places in the evening, which **is** roughly the inverse of the most general variation of the electric field... The various influences involved in the formation of the large ions disturb the conductivity... depend on the weather circumstances of the wind direction ... during ordinary rains there is an increase in conductivity and the increase can become very great during stormy rains... in experiments made in 1923 at Cambridge by CTR Wilson, it was found that a grassy field towards which we direct an artificial electric field we begin to cause discharges of negative electricity by the tips of the blades of grass from a field of 15,000 volts per meter; when the field was in the opposite direction, the positive discharge did not begin until around 20,000 volts...

During solar eclipses before totality and 20' after the electric field decreases by about 20%; the conductivity and the content of ions on the ground increase by about 20 to 25% compared to an average value. In relation to the properties of storm water on plants, and the triggering of lightning, two facts are to be linked with hailstones (water frozen - then cracking) and the ionic emissivity of ice under the influence ultraviolet radiation (dry ice loses negative charges under the influence of UV rays: see Brillouin-Buissau work). Two researchers from the Kirov Institute: Dyachuk and Muchnik have experimentally verified a hypothesis concerning the mechanism of lightning initiation. The study of the particularities of the corona effect of wet hailstones splitting in an air current allows, by measuring the intensity of the corona discharge around the particles of frozen water in suspension, to consider this effect as a basic mechanism. of the initiation of lightning (1979). Dokl. Akad. nauk SSSR, ISSN 0002-3264, SUN, 79, 248, 60-63, gidrometeorol. inst. kyiv.

They also consider that only the ice does not have the privilege of creating this mechanism but also that the currents of precipitation have an influence on the role of electrical generator of the storm. Freyer (GD) from the Univ. of Minnesota, Minneapolis, USA published it in J. atmosph. SCI. AUS (1979), 36, No. 10, 1967-75. Finally in relation to storm water and plants by considering the effects of the geomagnetic field in the lower atmosphere, Hay (PB) and Roble (RG) from Univ. of Michigan published a model in the J. Geophys. Res. USA 1979, 84 no. A7, 3291-3305, considering thunderstorms as pairs of points

positive-negative constituting sources of currents distributed in several storm zones around the earth. This model includes the effects of terrestrial orography and electrical coupling along geomagnetic field lines in the ionosphere and magnetosphere. Furthermore, Weidman (CD) and Krider (EP) from the Univ. of Arizona have discussed in JGR (1979) the physical implications of radiation field shapes and polarities in cloud interior lightning discharge processes. [See on this subject "pulsation on plants", work by Sir Jagadis Chunder Bose, notion of cell irritability]

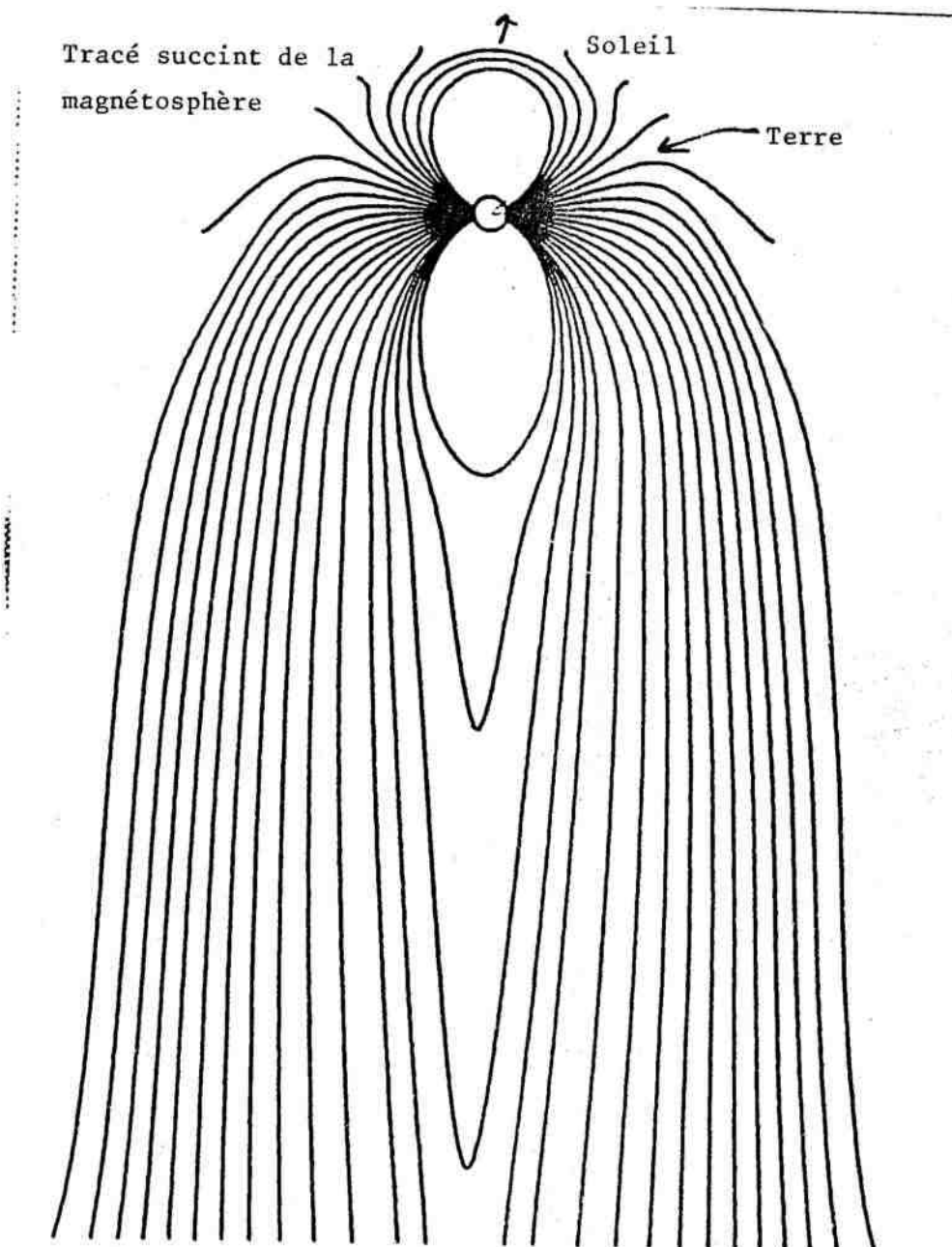
Linked to the problem of solar activity Georges Lakhovsky studied the wine statistics of Bordeaux and Burgundy wines; it shows a set of correlative facts connected to each other by the set of solar and cosmic radiation.

The years of good wine coincide with the maximum activity of the sunspots: graphs covering the period 1845-1914 show a quite striking parallelism of the curves between the activity of the spots of the chromosphere, the magnetic disturbances, the polar auroras and the taste appreciation of great wines.

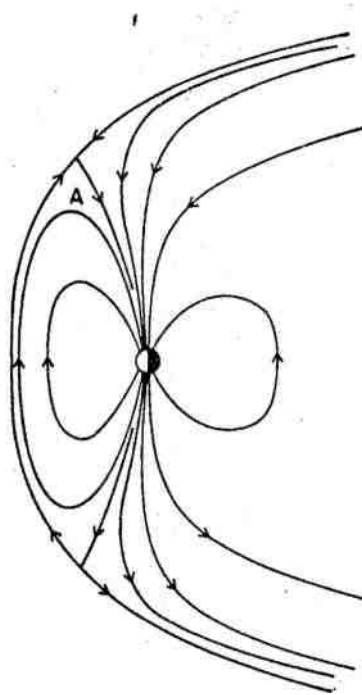
Based on the theory of "Cellular Oscillation", a set of experimental research on plants and other living beings was carried out in 1925. The discovery of the remarkable properties of oscillating circuits was the starting point for a multitude of works biological, both on the therapeutic effects and in the development of plants and living germs. Besides curing experimental cancer *Pelargonium* inoculated with *Bacterium tumefaciens*. [CR Acad, des Sciences, t.186, p.1019, session of April 11, 1928], Professor Vincenzo Rivera of the Observatory of Plant Pathology in Rome undertook a considerable series of experiments on the germination of seeds and seeds of all kinds. Then G. Mezzadvoli and E. Vareton (1929) examined the action of oscillating circuits on the germination of seeds and on their catalastic power.

The increase in germinations and the germination power on barley, beans, maize give growth figures of 120% to 200%. The techniques used were presented to the Italian academies in 1929 and are described in the work of G. Lakhovsky: Cellular Oscillation.

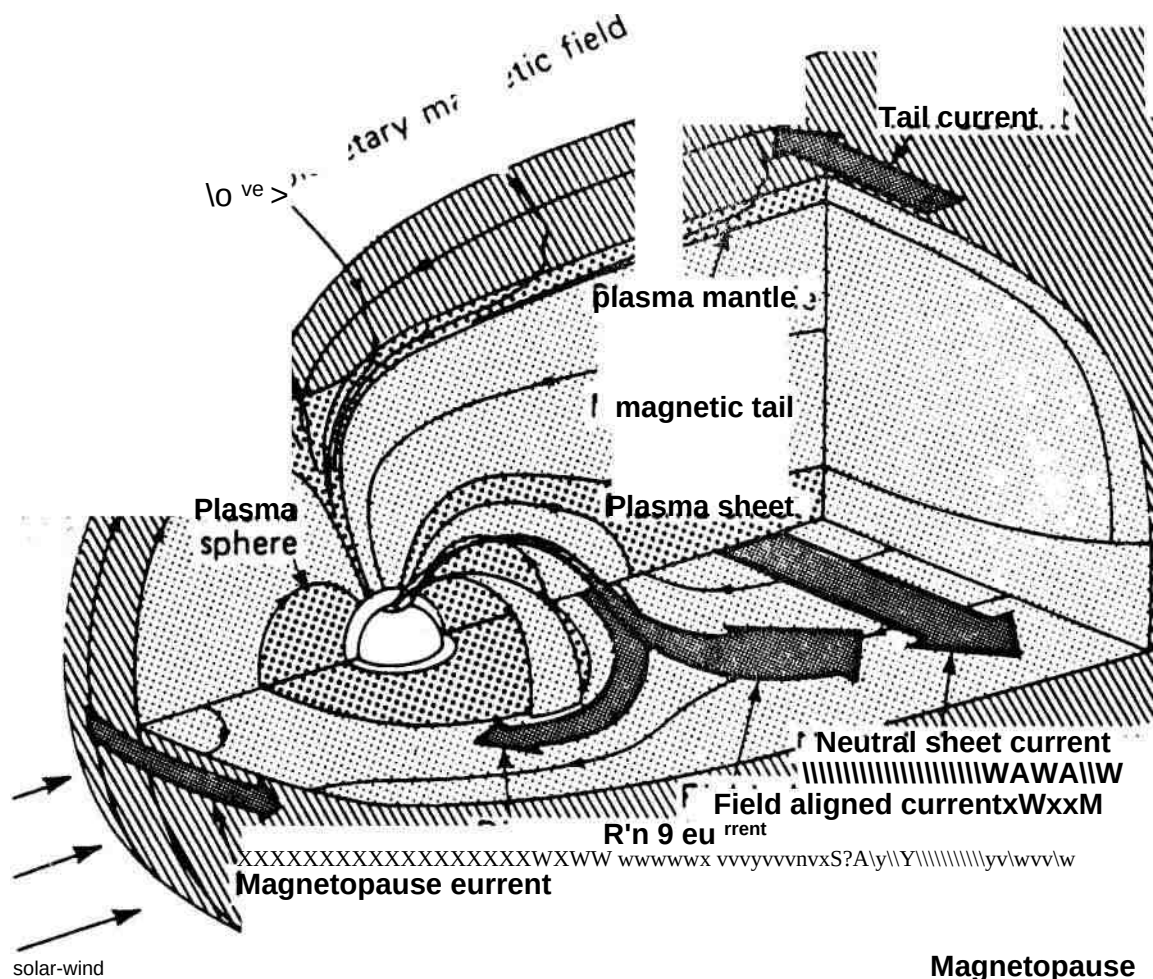
Experimenters from various universities concluded from experiments with the radio-cellulo-oscillator and with the Lakhovsky oscillating circuit that:



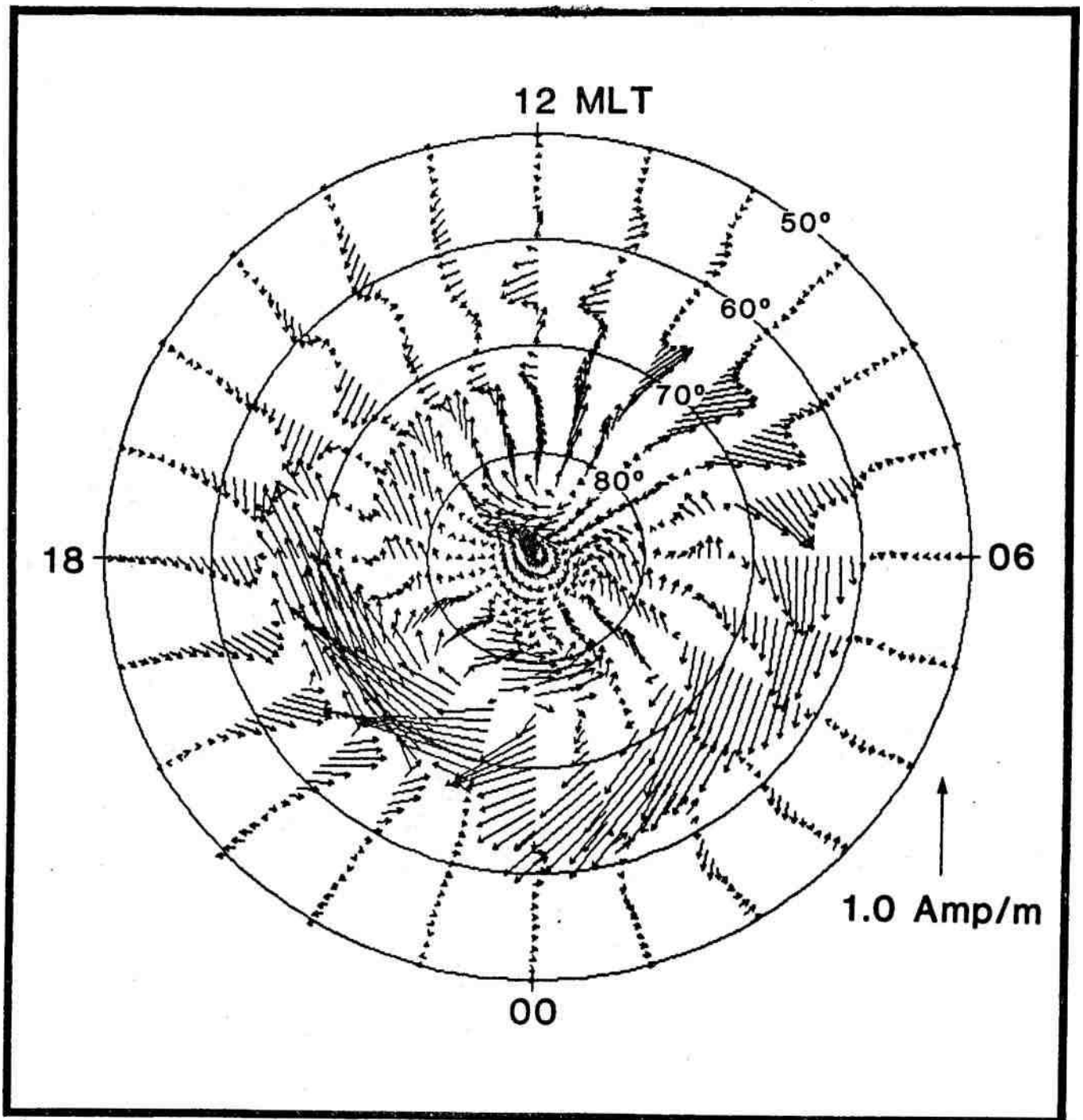
Brief plot of the magnetosphere
where we see the opening in the
field lines



The magnetic field
external to the
magnetosphere



In more detail, we obtain this diagram taken from EOS Transacting American Geophysical Union, Vol.75, n°5, January 31, 1984.



At the pole, over 24 hours, we obtain the figure above. From EOS Transactions, American Geophysical Union, Vol.64, No.10, March 8, 1983.

All of the explanations on the earth plane are given in (4) -

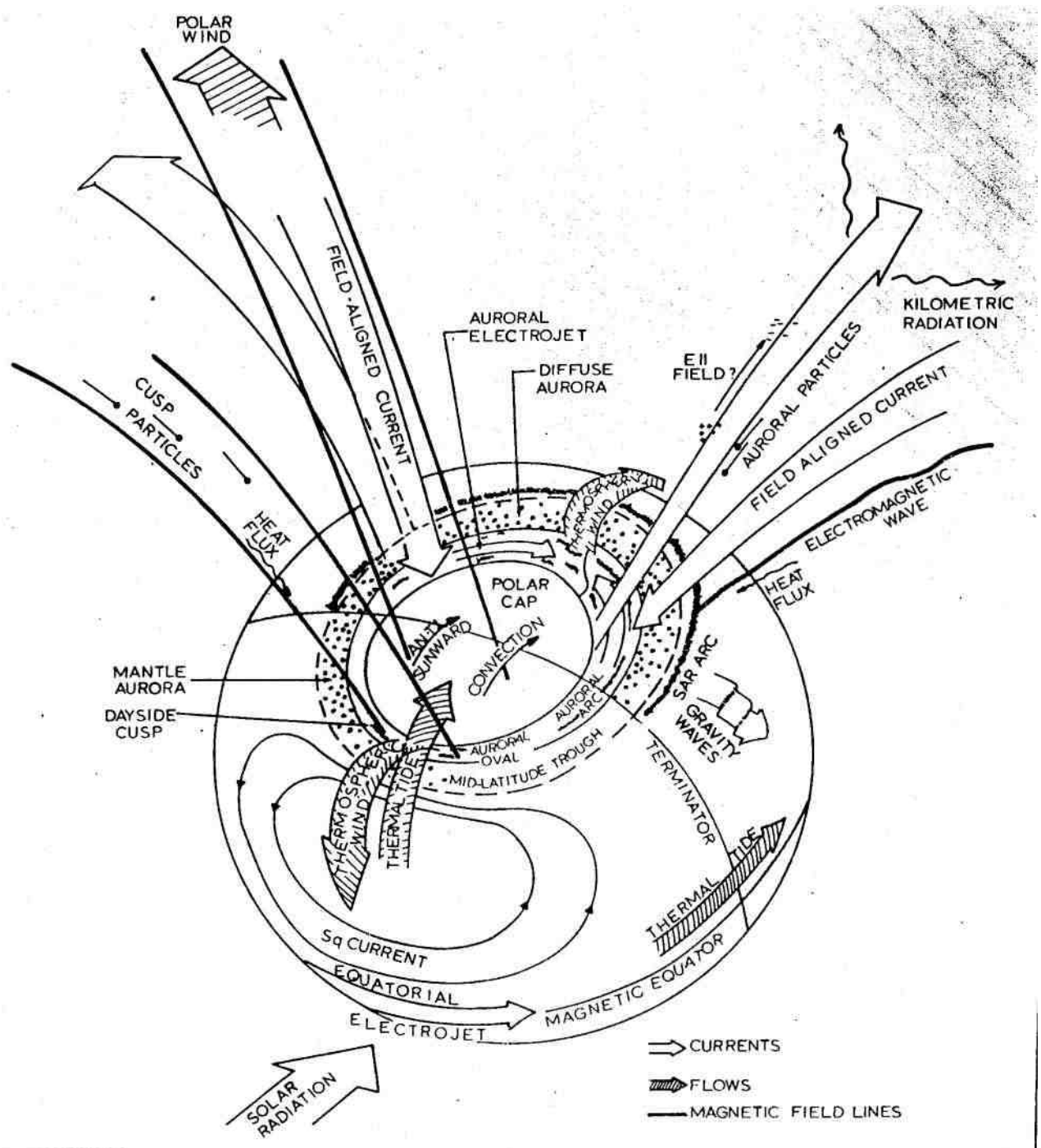
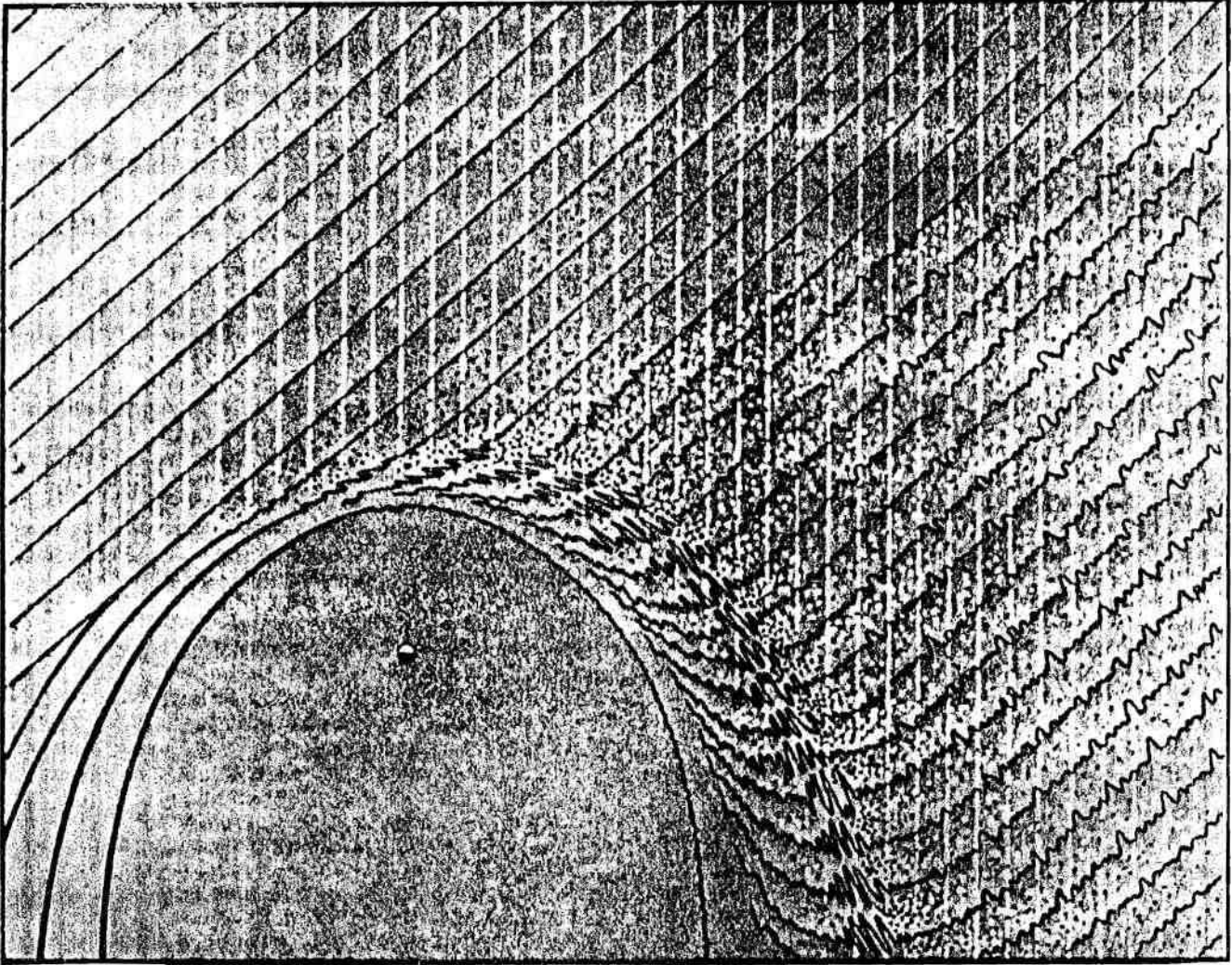
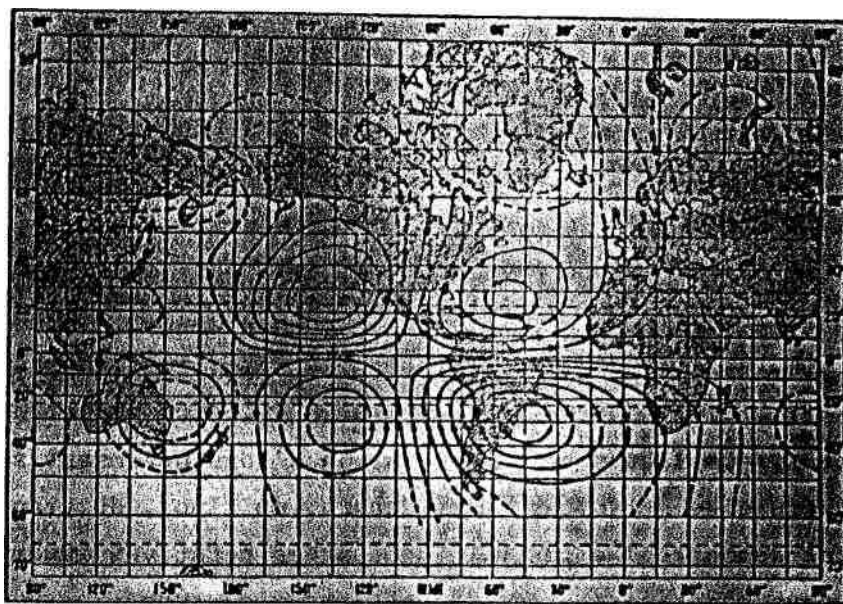


Illustration of many phenomena associated with the ionosphere-magnetosphere coupling. Here we have all the explanations on the terrestrial plane of what has been discussed in the previous boards.



A view of energetic particles and waves confronting the earth's magnetosphere and this in the plane of the ecliptic. The solar wind plasma comes from the sun to the earth (down from the top of the figure) and indicated by white vertical lines. Due to the rotation of the sun, the lines of force of the magnetic field of the solar wind are twisted into Archimedean spirals which; make an angle of 45° with respect to the line joining the earth to the sun. The electrons (black dots) and protons (white dots) are forced to circulate in the region preceding the magnetosphere and guided by the lines of force of the interplanetary magnetic fields, from which we obtain the asymmetrical distribution of the particles, magnetic anomalies, and plasma wave instability. The solar wind is made up of electrons, *a particles* , g particles, y radiation, etc.

Fig. 3



Mean direction of telluric currents
at noon for the 1945 era according to Rooney
(1958)

1°) Electromagnetic waves with a length of 2 to 3 meters exert a favorable action on the germination of seeds and on the growth of plants.

2°) There is an analogy between the action exerted by these waves and that of the Lakhovsky oscillating circuit.

3°) The action of ultra-short waves of 2 to 3 meters in wavelength is greater than that of the oscillating circuit, the favorable effect being a function of the intensity of the waves themselves.

4°) The action of the radio-cellulo-oscillator ($\lambda = 2$ to 3m) is more constant than that of the Lakhovsky oscillating circuit.

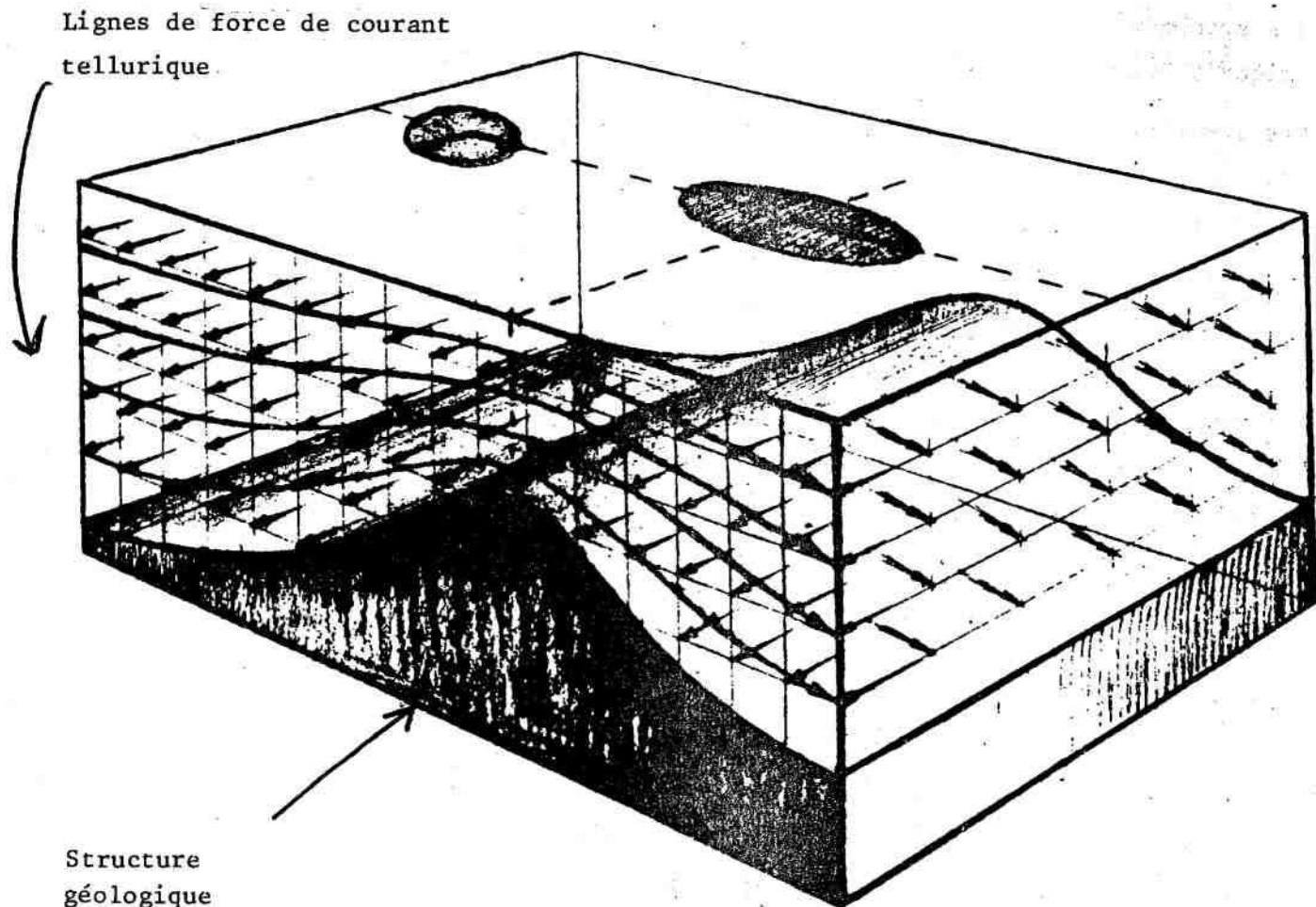
Note : The authors of this work did not examine ionization by light and ultraviolet radiation. Hertz had discovered that explosive distances were altered by producing effluvia in the vicinity. Hallwachs demonstrated the emission of electrons when solar radiation strikes solid bodies. [W. Hallwachs-Wiedem., Annalen, t.33, p.310, 1888]. This phenomenon was observed at about the same time by Righi. Lenard discovered the ionization of air by short wave radiation [Ph. Lenard, Annalen der Physik, t1, p.485, 1900]. Among the solid bodies struck by solar radiation, Elster and Geitel demonstrated photo-electric effects in the air, complicated by the formation of ions, in particular on fresh breakage, the powderiness of spar, feldspar, fluorine, granite. Brillouin and H. Buisson demonstrated the emissivity of large negative charges of blocks of ice under the action of ultraviolet radiation.

Applications of variable and high frequency currents to the biology and treatment of plants from seed to adult plant .

Adam and Givelet, two collaborators of G. Lakhovsky have published a complete summary of the work carried out in Europe and in the USA from 1924 to 1935. Michel Adam indicates in "The high frequency and its multiple applications" (Librairie de la Radio, 101 rue Réamur) block diagrams of continuous or short damped wave generators such as those of Saidman, Cahen, Schereschewsky, Esau, Hallberg, Kohl, Schliephake, Kotzareff.

The following effects and actions are examined with high frequency:

– The nature of the metal constituting oscillating circuits and diffusers have a - decisive influence on the modification of the refractive index of alcohol



Regional transformation of the density of the lines of force of the telluric current: The circle becomes an ellipse on the geological structure. See demonstration on figure 3.

To properly consider the effect, it is also necessary to take into account the local magnetism and the variations of the horizontal component and the declination.

All the variations will be followed in detail hour by hour for magnetism and telluric currents, taking into account on the NS and EW axes the solar and lunar influence (see Fig. 1 and 2).

The polar diagram of the place for the diurnal variation will give the orientation to be taken.

ethyl and acetic acid.

- The systems ethyl alcohol + acetic alcohol, anethyl alcohol + ac. form follow a new law of etherification.
- Organic liquids: olive and peanut oils, wines, vinegars, beers have their degree of acidity lowered. Early wine aging. (Tests made at the wine hall without the control of the wine administration).
- The resistivity of the electrolytes decreases (it can be seen at the Wheatstone bridge that in 1/2 hour the resistivity of a solution of sodium chloride at 1Z, at 21.5° C. falls from 48.312 to 47.339)
- The action of intermediate waves of 100 to 1,500 m in wavelength (frequency 3000 to 200 Kilohertz) on seeds of maize, barley, wheat and rice accelerate germination and promote the development of seeds during the first weeks of their growth. existence. (E. Benedetti, CR Acad. Roy. Lincei, Rome, 13-9-26).
- Action on the fermentation of brewer's yeasts.
- Action on aquatic plants: Elodea Canadensis with an appropriate oscillating circuit of 0 30cm in pure copper of 6m/m of 0 increases the length of a plant by 14 times in 130 days^*).

In another work by Michel Adam and Armand Givelet [The work of GL] "La vie et les ondes" (E. Chiron, Paris, 40 rue de Seine, 1937). All the effects of the oscillating circuits are taken up in detail and particularly the physico-chemical actions, the biological actions on plant cells and on animal cells. All references to authors, subjects and articles appear in the bibliography of this book. The nature of the problems studied, the implementation and the quantification of the results are given. The problem of frequencies and exposure times are also mentioned and specified for each test. The texts of Adam and Givelet complement the works of Georges Lakhovsky. In the continuation of work related to electroculture, a contemporary experimenter carried out "Attempts to update the Lakhovsky process at multiple wavelengths : its implications on the mutability of some plant species". The author, Jean Paul Laurens published the results obtained in volume III, May 1979 of "Communications on Emotional Vital Energy" edited by: LOGBP 23, F 75923 Paris Cedex. The author has made a multios'ci binding HF generator and has experimented with the effects? on Hops• .>; • -/ '1 • ♦ from Japan, hemp sycamore, conclusions and a discussion on the topicality of the Lakhovsky process is presented at the end of the article on trai

(*) compared to the witnesses who grew only 04cm.

ments of tomato tumors. The plan of the apparatus, the times and modes of operation are described. A reminder of the basic works of GL is made as well as an analysis of the theory of the universe-ion in parallel with the bions of W. Reichs. Currently it is the only recent text emphasizing that the conditions of experimentation with regard to the environment have changed since the 1920s.

The electrotelluric currents, electrical activity around mineral deposits, do mineralized faults have effects on plants, roots; born, stems, leaves; are they affected by this form of displacement of exchange whose the regime with an oscillatory character is known for certain periods of oscillation?

One of the first relationships made between the location of plants in nature and the composition of the subsoil is found in the Chinese work written around minus 800 of the modern era called the Tchou-Li. The text indicates close correlations between the frequency on the ground, or the concentration of certain plants and the substances sought by the miner such as precious jade, . . . z (*) . . .
 . . . z
 metal ores, their fluxes, etc. The same work also indicates that hammered copper wires joined together and surrounded in the ground in certain directions promote the growth of certain crops.

In addition to the chemical considerations linked to the variation in composition. soils which are the subject of permanent studies, the electrical behavior of the soil and of the nearby subsoil as well as the variations of the earth's magnetic field in its various components seem to have received no significant practical application except at the level of the prospecting for raw materials useful to human industries.

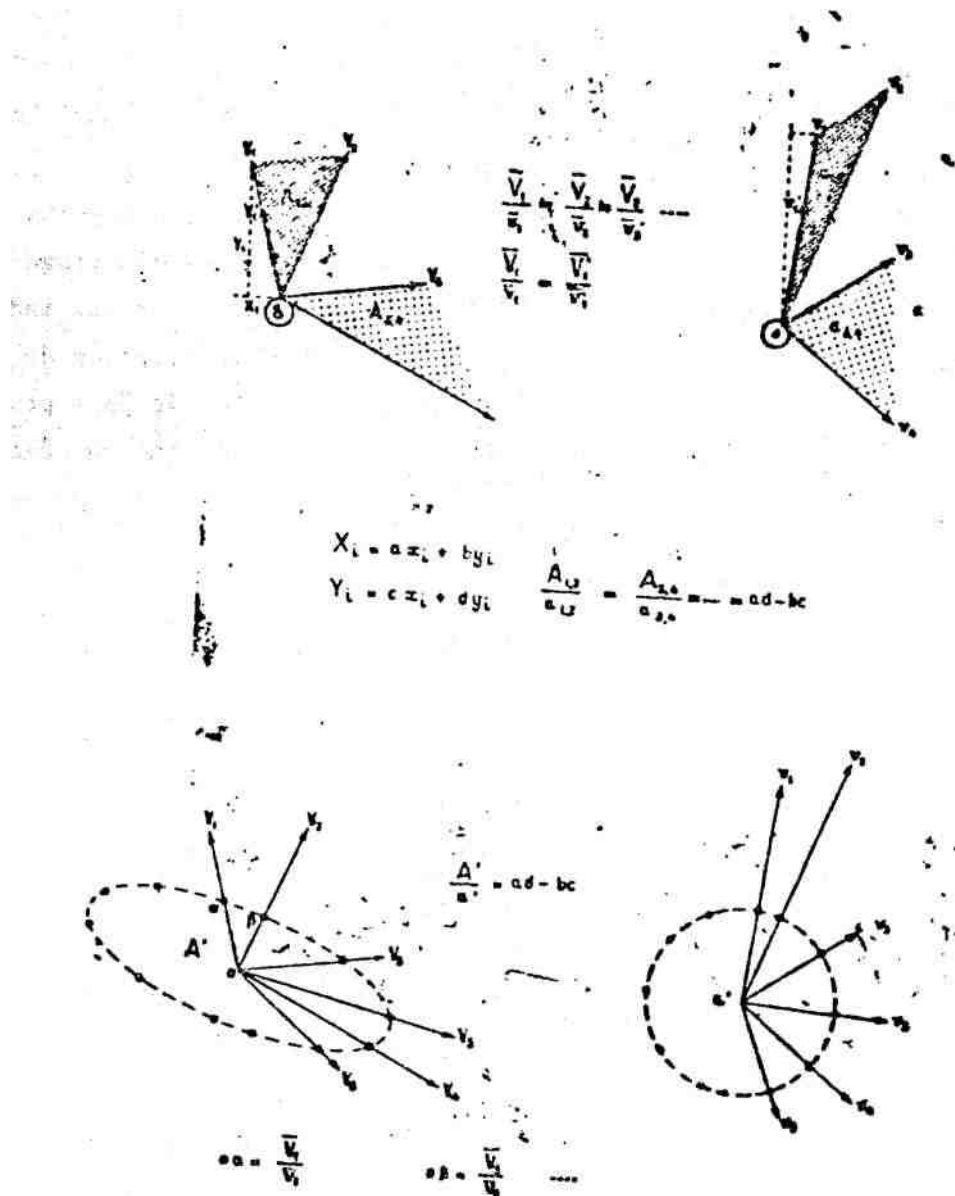
In 1914, the Department of the Interior of the USGS, Washington published in Bulletin 548 a study by Roger C. Wells on the electrical activity associated with mining deposits. The first modern observation is said to have been made by Fox, RW (On the electromagnetic properties of metalliferous veins in the mines of Comwall). The original text includes a table showing the electrical conductivity variations of about twenty minerals, ores and metallic combinations. The author's conclusion was that there were phenomena of an electrical nature in the mineralized veins.

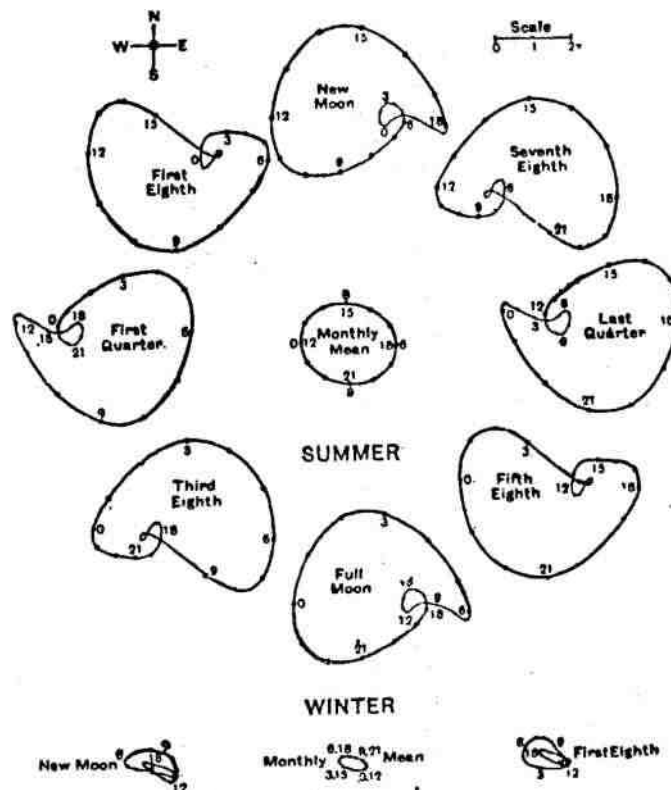
From 1830 to 1844 various authors commented on these activities

(*) A metallurgical flux such as silica, fluorite, borax are products which, when mixed with the ore during fusion, considerably lower the liquefaction joint of the metal.

Density Transformation
 natural electro-telluric

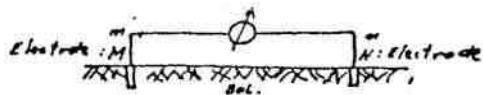
lines of force of the current
 see fig. 2, action of an anticline.





Yes. 3. Vector diagrams for the lunar daily variation of the horizontal magnetic force components at Pavlovsk, in summer (above) and winter (below): for the mean of a number of whole lunations (in the centre), and also at various particular epochs in the lunation, the day-light part of the latter curves being drawn in thicker lines. Scale for 2 y in upper right corner

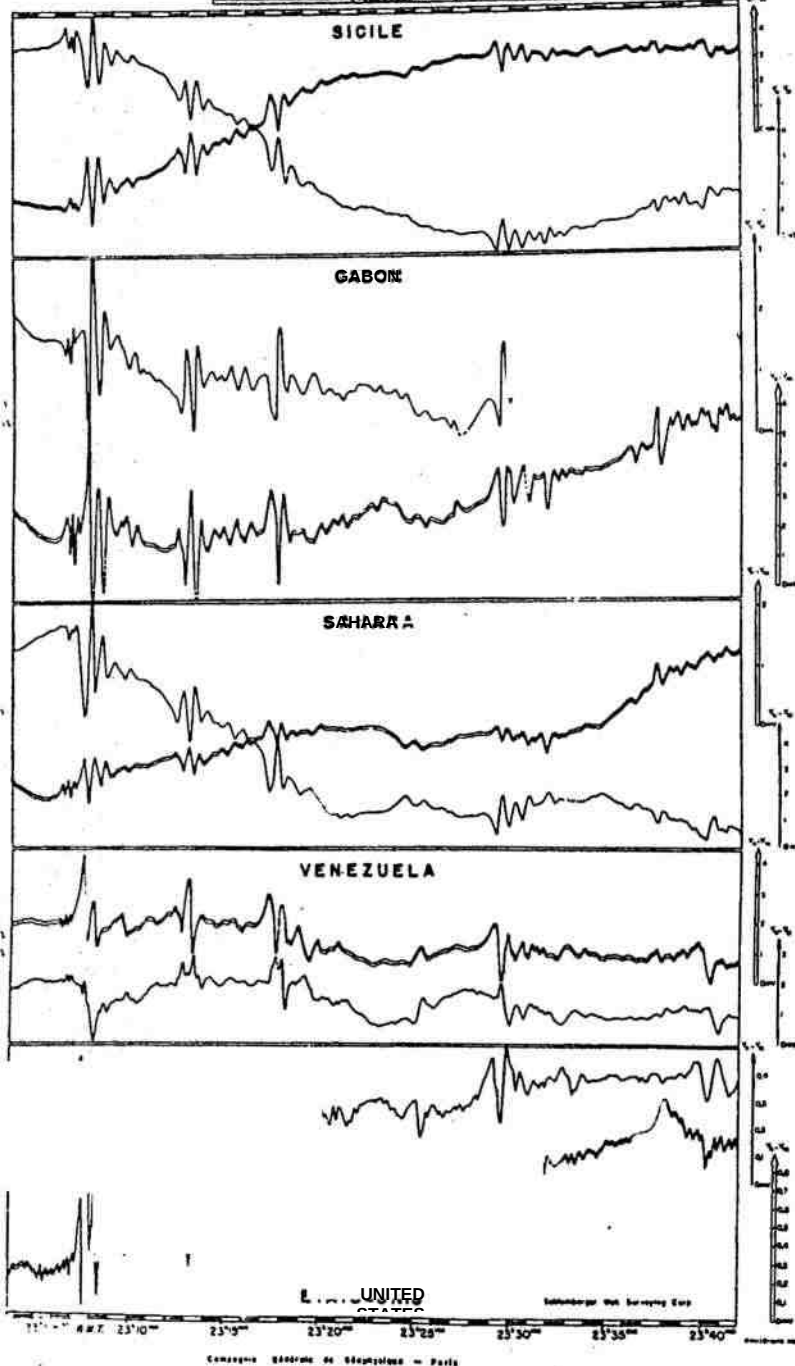
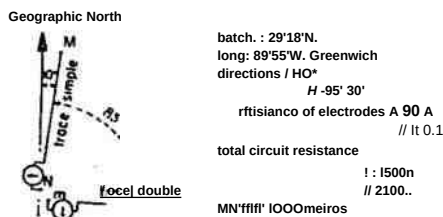
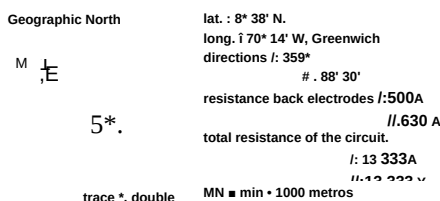
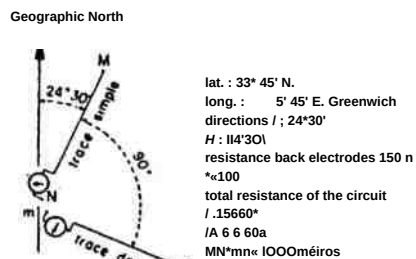
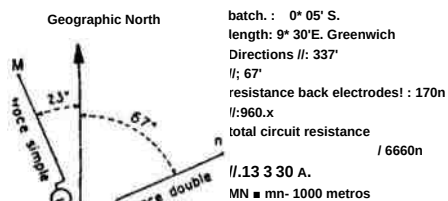
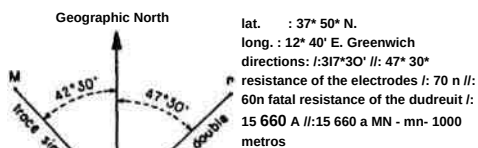
In France, the development of the method for measuring electro-telluric currents was made under the impetus of the scientist Léon Migaux as well as brothers Marcel and Conrad Schlumberger.



Principe du dispositif de mesure du C.T.
pour les traces des bleus et simples.
l'angle droit de déroulage est toujours
le meilleur.

SIMULTANEOUS RAPID VARIATIONS EARTH CURRENTS

FEBRUARY 24, 1952 FROM 23ⁿ 04 TO 23M2GMT



Comparison of global measures. At any moment the correlations are always real and controllable in their variations.

See analysis for averages.

Results published in Erdoel-Zeitschrift, N° 9, Sept. 1958

- Grundlagen und Anwendung der tellurischen Prospektionsmethode Von Kunetz-Urban-Verlag, GMBH, Wien-Hamburg.

Let us mention studies relating to terrestrial electric currents linked to ores:

- Henwood, WJ, On the electric currents observed in Cornish filans, Annals of Mines, 3d ser., vol.11, p.585, 1837.
- Fox, RW, Account of some experiments on the electricity of the Huel Jewel Mine, British Assoc. Adv. Science. Rept., vol.3, p.572, 1834; Report on some experiments on the electricity of metallic veins, etc...; Same, vol.6, p.133, 1837; Saine experimnts on subterranean electricity, made at Pennance mine, near Falmouth: Phisos. Mag., 3d ser., Vol.23, pp.457, 491, 1843.
- Strombeck, A. van, Über die von Herm Fox angestellten Untersuchungen in Bezug auf die electromagnetischen Ausserungen der Metallgange, Karsten's Archiv, vol. 6, p.431, 1833.
- Reich, F., Notiz über electrische Ströme auf Erzgangen, Poggedorff's Annalen, vol.48; p.287, 1839; Versuche über die Aufsuchung von Erzen mittelst des Schweiger'schen Multipliers: Berg un Hüttenmann'sche Zeitung, vol.3, pp.342, 386, 1844.

These remarks on the subject of electric currents in the earth were - subsequently reconsidered by Skey, W. in the laboratory: On the electromativ^ power of metallic sulphides, New Zealand Inst. Trans. and Proc., vol.3, pp.232-236, 1971.

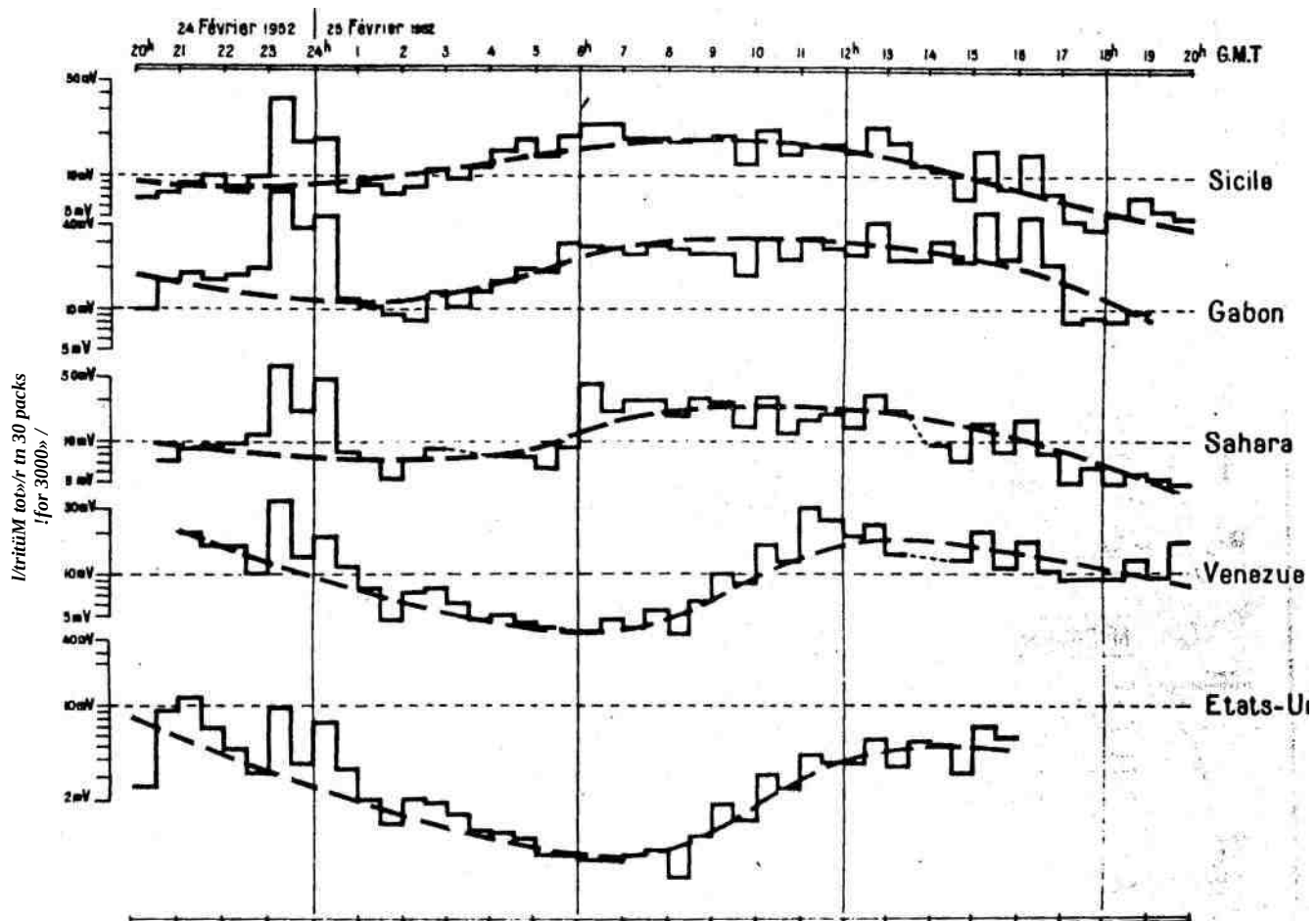
All of the work on the subject from 1830 to 1913 appears in the text by Roger C. Wells cited in the first reference with examples for many ores and natural mineral salts.

In the above and in relation to the polarization of mineralized threads in the soil there has been no attempt to correlate with plants ... it is more particularly geochemical practitioners who have established the close association the vertical of certain mineralizations with certain plant species. We will see more in detail these technical points of prospecting for minerals in the part looking for minerals from plants and soils. However from now on one can notice the separation in pure or applied research of the fundamental notions of the unity of thought that it would have been desirable to develop in electroculture.

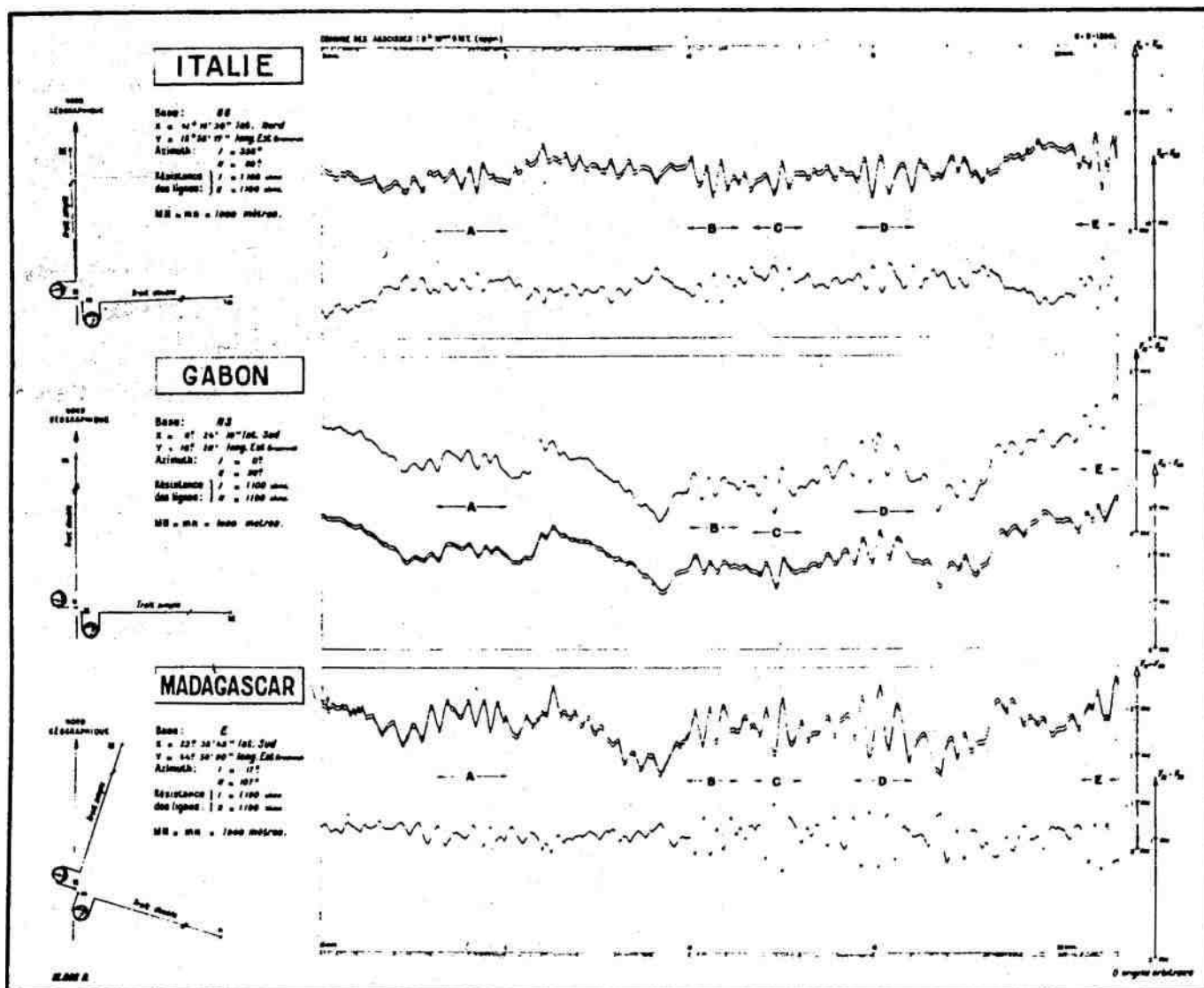
. Apart from the spontaneous polarization of ores in situ, another very important electrical activity exists in the ground and the subsoil in the form of electro-telluric currents whose now well-known oscillatory regime is related to the activity of the chromosphere. solar. The first

Average level of telluric agitation

(correlations in GMT)

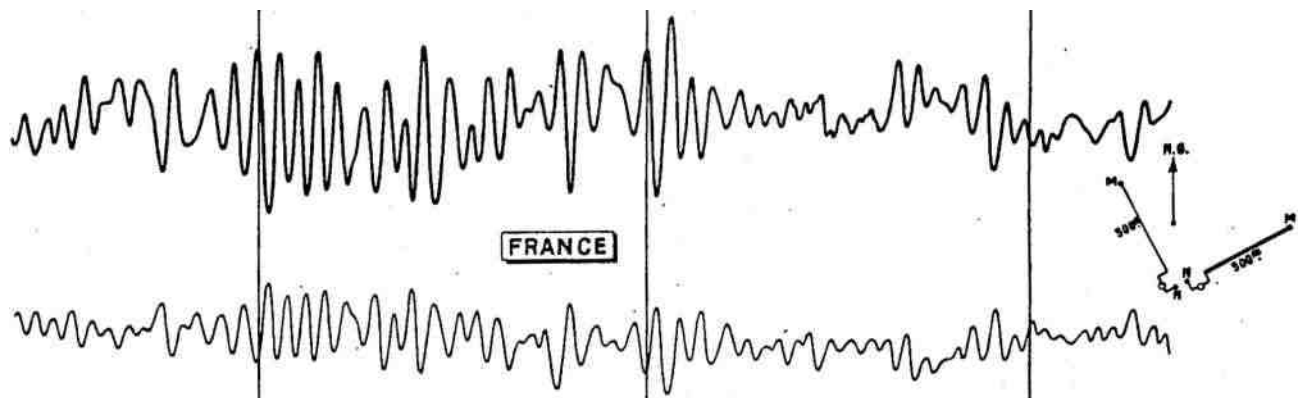
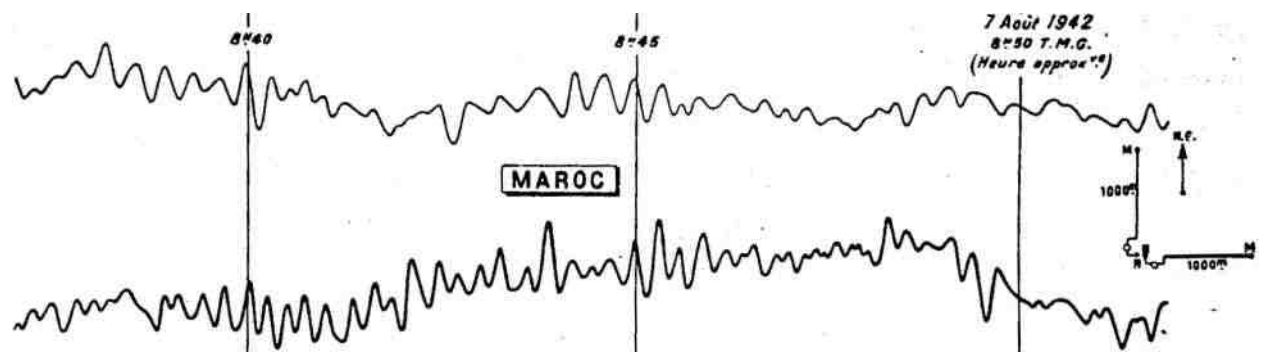


Breakdown of oscillatory telluric pulse groups: average curves. Plate following the correlations.



Telluric currents: variable regime with superposition of oscillatory regime; the strongest variations ABCD make a perfect correlation. These three recordings are free of spurious signals.

Doc. Cie Gle Geophysics



Abscissa : Tampa
Ech aHa :
Ordonnées : Différence de potentiel
 Echelle ~ 0,1 mV/cm

trsa

Telluric currents: variable regime with superposition of
 oscillatory regime

In 1849, Barlow^{v 7} showed that periods of trouble in land and sea telegraph lines were always accompanied by storms. magnetic. The problem of the relations Terrestrial Magnetism with Telluric Electricity was posed; Airy devoted many remarkable works to him. In the general appearance of the phenomenon of electro-telluric currents, JC Maxwell in his treatise on Electricity and Magnetism, t.2, p.322 describes them as a sheet of current enveloping the entire globe, a sheet whose lines of current , orthogonal trajectories of the equipotential curves, have a more or less regular distribution, linked to the geological structure of each region.

Paul Rougerie of the IPG in 1942 attempted a synthesis of work on marine currents and coastal land currents. Interesting correlations were established with the tides and the phases of the moon. A harmonic analysis due to Labrousse highlights the delay effect of 2 hours on average of certain aqueous tides with the direction, the intensity and the direction of the currents etc. Currently we recognize a coastal geomagnetic effect. Two contemporary researchers: Parkinson (WD) and Jones (FW) of Tasmania University, dept. geol. Hobart Tasmania 7005, Aus./USA published in 1979 (vol.17,n°8, p.1999-2015) in Rev. Geophys. Space Physics. USA an analytical study measuring the effect of coasts; it appears that the vertical component of the geomagnetic field is abnormally wide and positively connected with the horizontal component of the internal geomagnetic field of the globe. This phenomenon normally and generally observed near the lines of the coasts, however, offers exceptions. The few coastal locations for which this effect is absent are tectonically remarkable anomalous zones. It would be interesting to examine the wild vegetation on these coastal points as well as the marine flora on the one hand and on the other hand the examination of the cultures on these particular points should be rich in consequences in the hypothesis of the variation of the fields. electrical, telluric, magnetic on the surface of the biosphere-lithosphere contact with stimulation on plants.

Experiments made with submarine cables in 1871 by J. Graves over lengths of 1850 nautical miles and 1820 nautical miles revealed_oscillations of the electromotive force of telluric currents whose (*)

(*) Philosophical Magazine, t.34, p.344 and Philosophical Transactions of the London Roval Ste. n.6:.. 1849.

fluctuations seemed to coincide with that of the tide. (Journal of the Society of Telegraph Engineers, t.2, p.102, 1873).

The order of magnitude of the telluric currents and their directions both those measured at the Ebro Observatory and in itinerant measurements range from a few millivolts to 1 volt per kilometer in the event of measurement during certain magnetic waves and on average.

E. Mascart in his treatise on terrestrial magnetism published in 1900 indicates that during solar flares accompanied by aurora borealis, from August 29 to September 3, 1859, on telegraph lines of 500 to 600 km, electromotive forces of 700 to 800 volts. On July 16, 1892 during another magnetic storm, an emf of 210 volts was observed on the New York - New Jersey line. At Elisabethtown the line length being 33km which corresponds to approximately 9 volts per km.

The first measuring device taking into account a length of measuring line compared to a galvanometer on which the intensity of the telluric currents appears roughly proportional to the distance of the two earth connections " ' i ' " was carried out by EE Blavier and published by the Imprimerie Nationale in 1884 " ' i j ' " under the title "Etude sur les courants telluriques" .

Some researchers have carried out a number of experiments aimed at linking the immediate influences of certain physical agents with telluric currents.

The temperature would have, during calm days, a very clear apparent role. The Memoirs of the Academy of Sciences of Saint-Petersburg (3rd series, t.12, p.44, [3], 1901) report Bachmetjew's numerous researches, highlighting so much a parallelism of paces continuing in the details between the curves of the diurnal variation of the temperature above the ground and the "regime" of the natural electro-telluric currents. It is reported in "Terrestrial Magnetism", t.23, p.78, 1918 the experiments of P. Mauchly which are decisive in this respect: by burying two earth electrodes at a vertical distance of 1m80 and surrounding the whole from a vertical insulating cylinder, he recorded potential differences whose path follows exactly that of the temperature difference between the two earth electrodes: 1° centigrade of temperature would roughly correspond to 1 millivolt.

Humidity plays a big role in increasing the amplitude of the regular variations of the oscillation. Indeed currents due to the polarization pro

evoked by superimposing on the spontaneous polarization appear at the ground connections. Prospectors using current measurements and have been led to eliminate this inconvenience by using "impolarizable" electrodes either of the Conrad Schlumberger type or of various types Brander turns the hollow wooden electrodes, stuffed with copper sulphate, and surrounded by a bare copper wire with contiguous turns, the whole being sheathed in strong braided fabrics themselves impregnated with copper sulphate. For these experiments see the work of M. and C. Schlumberger published in 1920 by G. Villars in Paris and titled "Study on the electrical prospecting of the basement".

One of the aspects of global tectonics, of the general movement of the continents and of the rapid magnetic variations of rocks (relation to be established with electro-tellurics and magneto-tellurics) has been tackled in J. Geomagn. Geoelectric, JPN (1978) 30, n°5 - 585-592 by Nulman, Shapiro, Ivanov, Carmichael (USSR acad. sci. inst. geophys. Sverdlovsk, SUN) in an article: "Magnetic susceptibility of magnetism under hydrostatic pressure, and implications for tectonomagnetism". The authors have studied the behavior of ores, rocks, magnetic crystals and various synthetic samples with regard to variations in hydrostatic mechanical pressure. For a particular crystal, the susceptibility to saturation remanence decreases with pre-oxidation by approximately 25% for 2.5 kilobars (Kb), and is found reversibly with relaxation of the pressure. For rocks and ores, the magnetic susceptibility at low remanence increases slightly up to 0.2-0.8 Kb, then decreases for higher pressures of about 10% Kb. She is found after relaxation.

If the hydrostatic pressure variation varies according to the porosity of the rocks in place with the water table, we can therefore expect variations in the propagation of the lateral velocity waves in the various levels of the layers of the ground, going hand in hand with local magnetic variations. . It is conceivable to thus predict certain constraints of the crust or movement of the plates, distant earthquakes, etc., which would confirm the observations of Laberge on the "influence of meteors" at distances far from his laboratory in Montpellier. , on "plant sensitive integrators".

Note : See also on the points analyzed: The autographs of the Plants and their revelations by Sir Jagadis Chunder Bose, Gauthier Villars, Paris 1934. (*)

(*) CRA Sc. t49, p.460, 1859 and t58, p.942, 1864 (Mattenei-Brander).

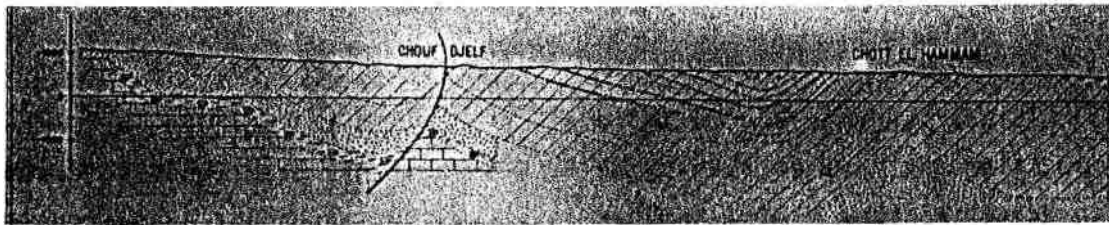
The general study of telluric current variations began in 1884 in Germany with the work of B. Weinstein on the Berlin-Dresden (120 km) and Berlin-Thom (262 km) telegraph lines corresponding to the NS/EW directions. Ebro Observatory in Cortosa continues the study of these variations and their relationship with the diurnal variation of the Earth's magnetic field in its various components.

As early as 1922, the publication of vector diagrams from 41° North latitude to 67°, analyzed statistically from 1882 to 1921, indicated on the closed curves obtained the progress of the phenomenon of telluric currents with magnetic fluctuations of all kinds. Thus diurnal variation and magneto-telluric disturbances are much more marked in the high latitudes than in the low ones and especially in the tropical zone. However, latitude does not explain everything, so at Sodankyla (Finland) on 5km lines, Lemstrom observed, for example, telluric currents 75 times more intense than at Pawlosk (Russia), 7° further south (Terrestrial Magnetism, t.4, p.65, 1889).

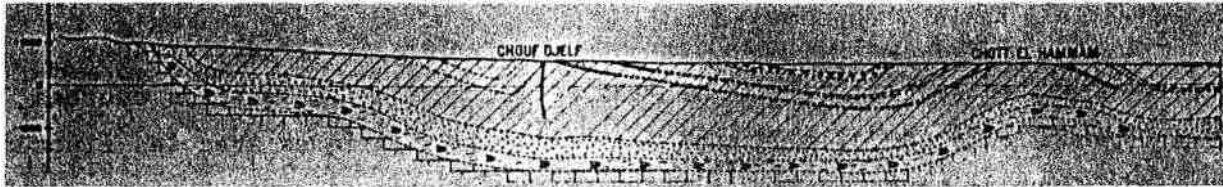
Currently, thanks to measurements by satellites and rockets, it is easier to know the nature of the activity of the sun, in the corona, on the photosphere and the chromosphere. The nature of the particle fields and the behavior of the solar wind around the earth allow a better understanding of the electromagnetic phenomena resulting from the powerful source of solar emission. Certain resonance and oscillation phenomena can be approached with statistical measurement equipment which the precursors could not have. This long presentation on the subjects of electrical and magnetic activities allows us to glimpse the difficulty of controlling certain parameters of Electroculture.

The best references concerning this vast subject are, one, a study by Paul Rougerie (Dr. es sciences) who published in 1942 at the PUF "Contribution to the study of Telluric Currents" text still current and note n °17 (Sept. 66) by Hugo Fournier published by Knowledge Magneto-Telluriques". The author gives 1300 titles and references for the magneto-telluric bibliography. All the problems dealt with are very complex. between solar and terrestrial phenomena, and it seemed to us necessarily to have a relationship with the life of the planet although this aspect is not addressed in the texts cited in reference, which remains in accordance with the principle of scientific non-interference of physics towards biology and vice versa.

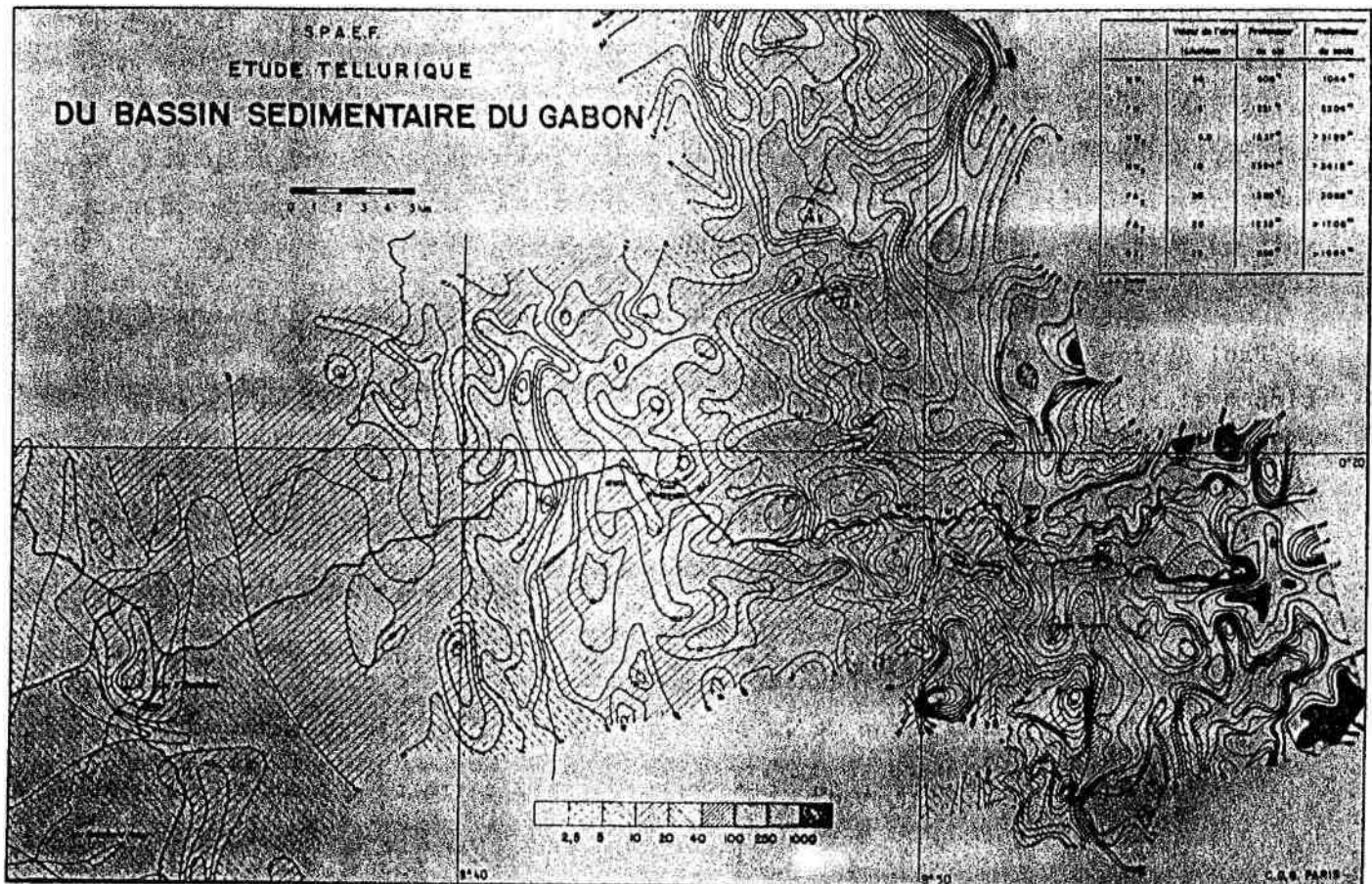
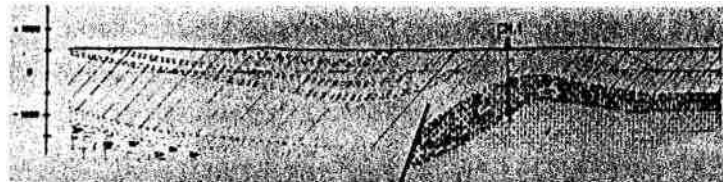
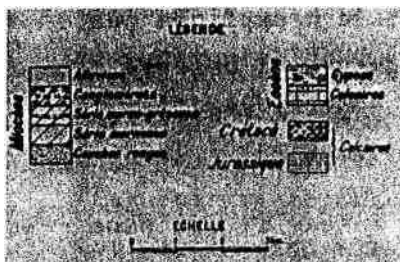
Hugo Fournier from the "conductor" earth treated his subject in application

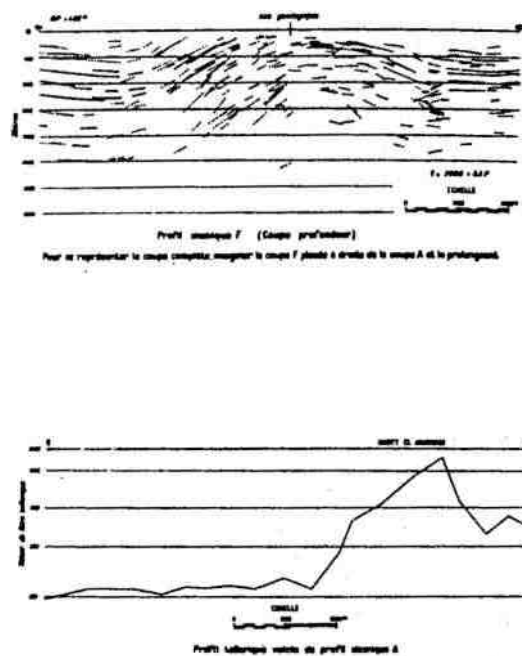
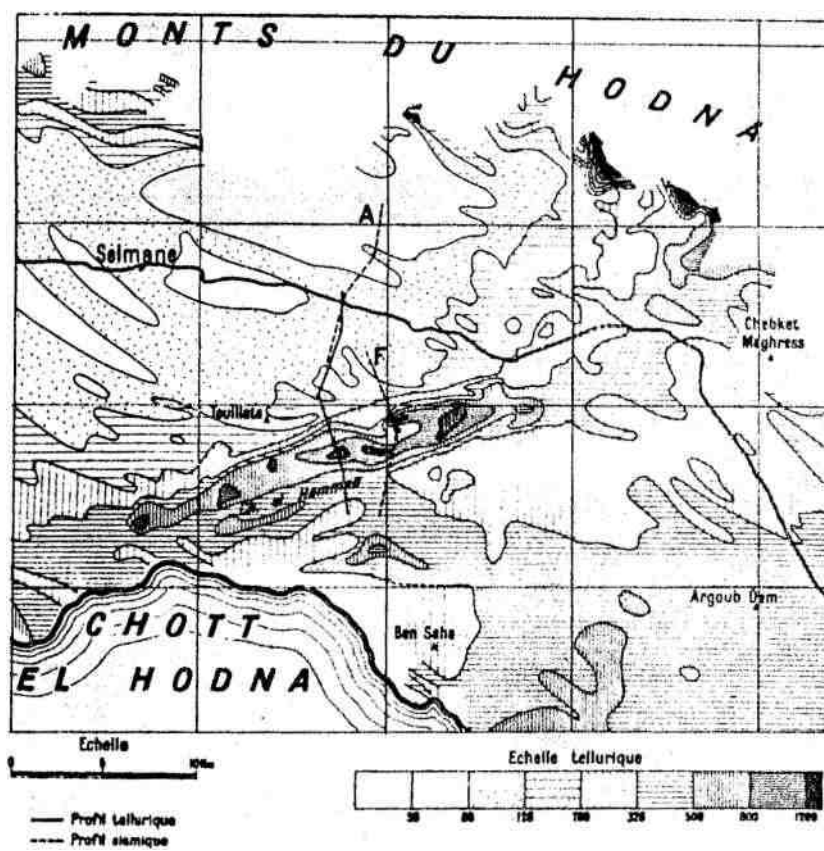
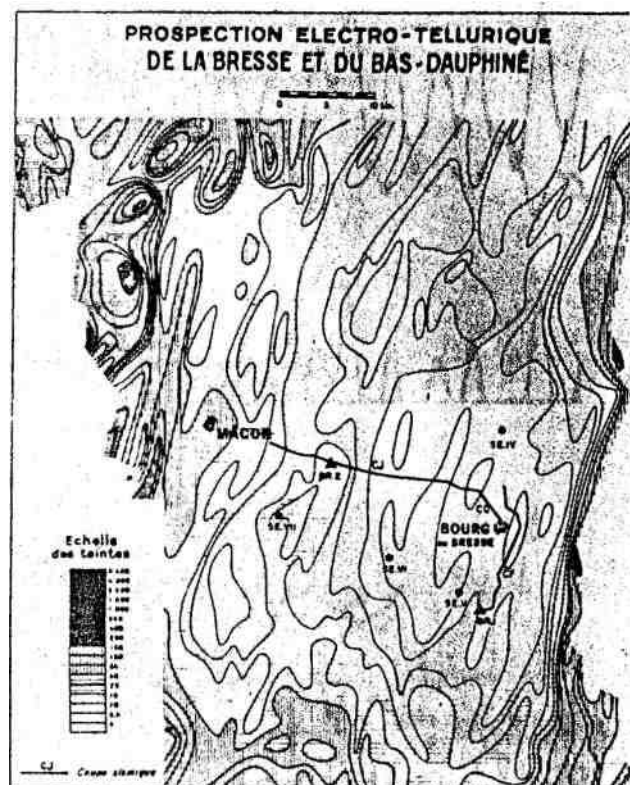
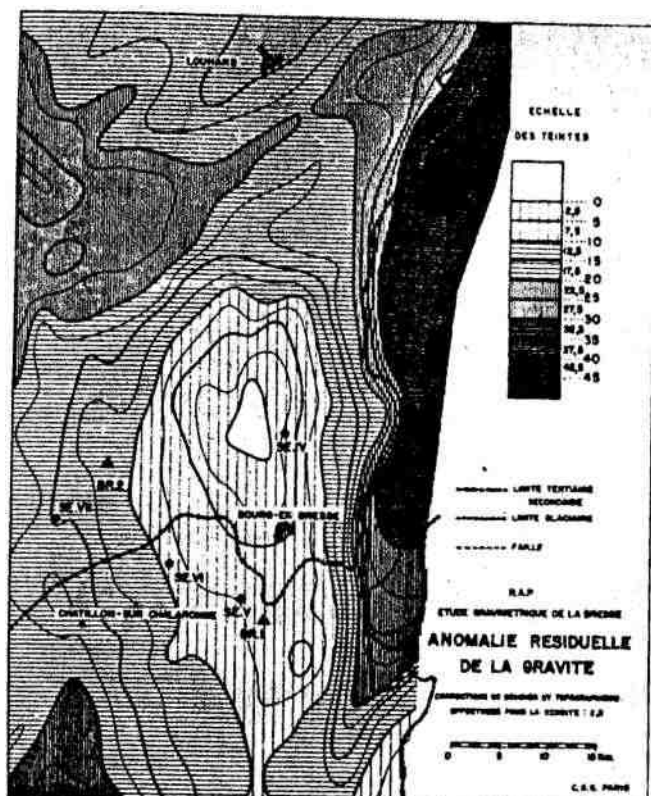


Hodna Basin Cup
< SHReral ISM >

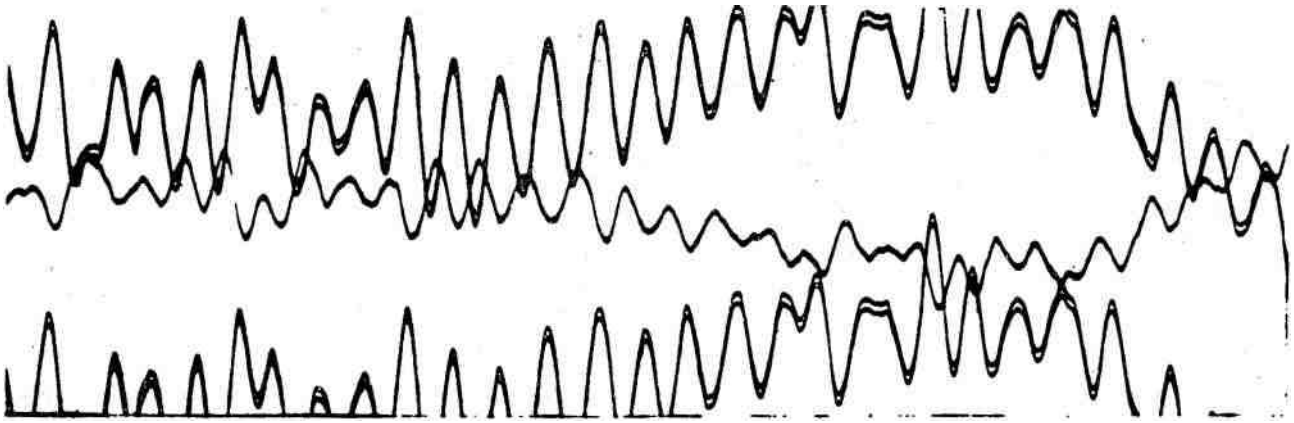


**Revised section of the Hodns Basin after a seismic study
(SNREPAI isso)**

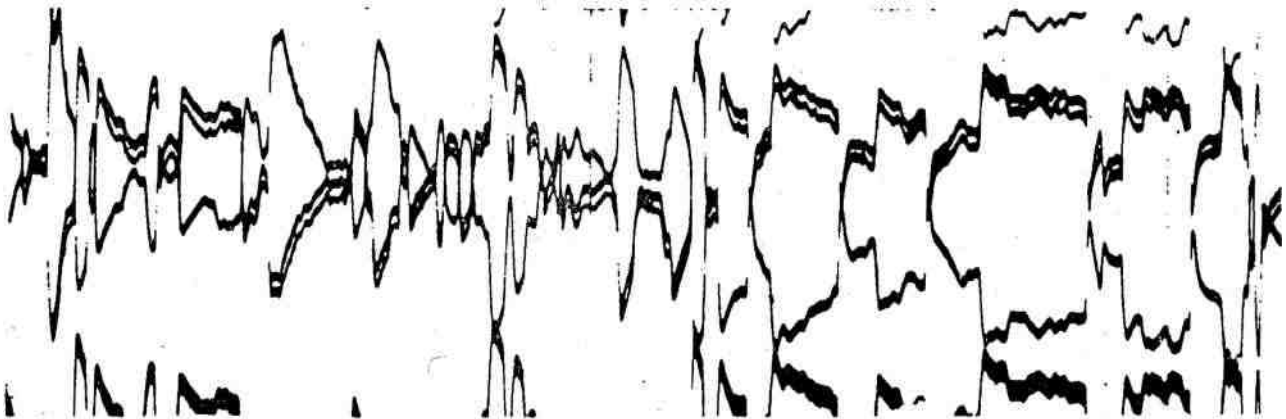




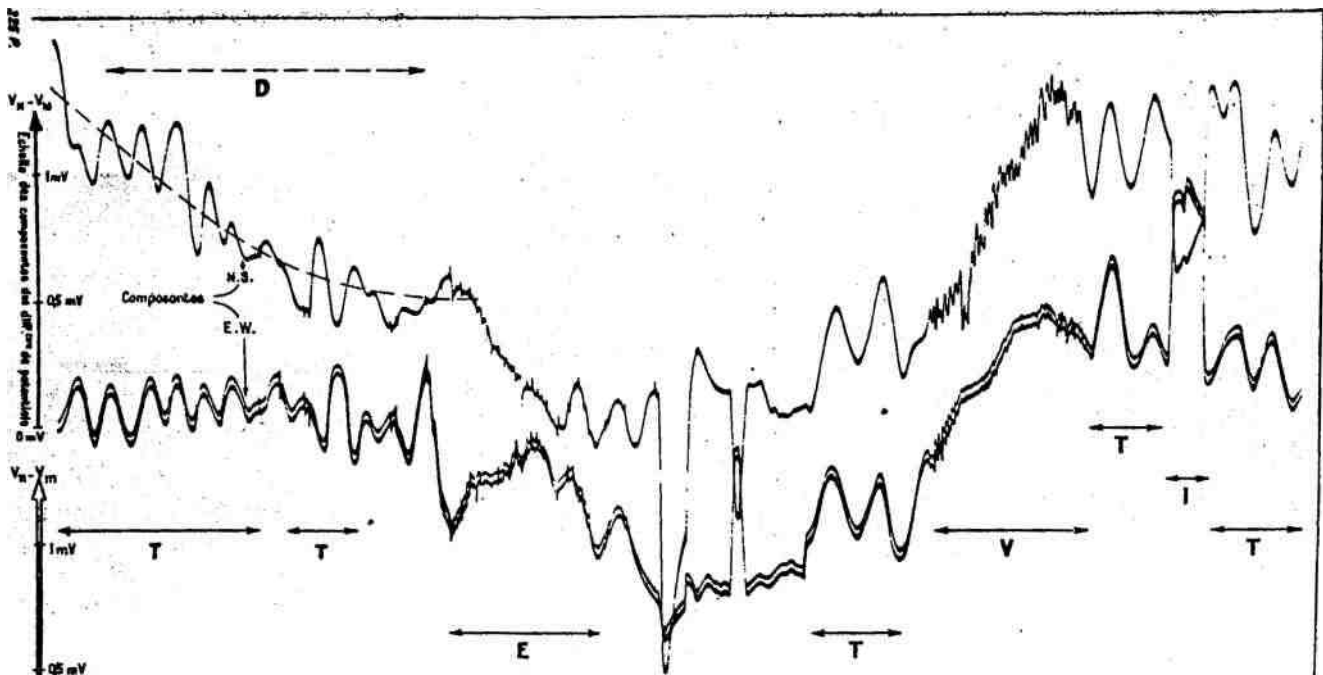
Field variations of telluric currents



Variations in the field of industrial currents



ONE MINUTE



T: Pure telluric
e: Atmospheric electricity.
i: Yagobonde industrial

G: D: wind effect
Derived polarization of the
electrodes)

1

Qmin. I
EcMito toa»

1

3

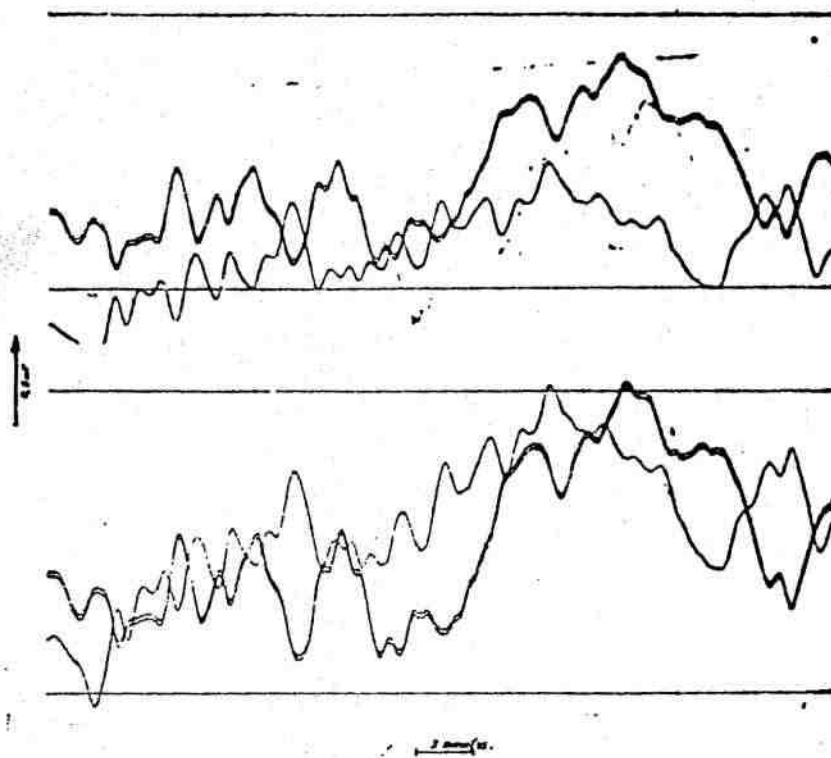
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1

10 minutes. He

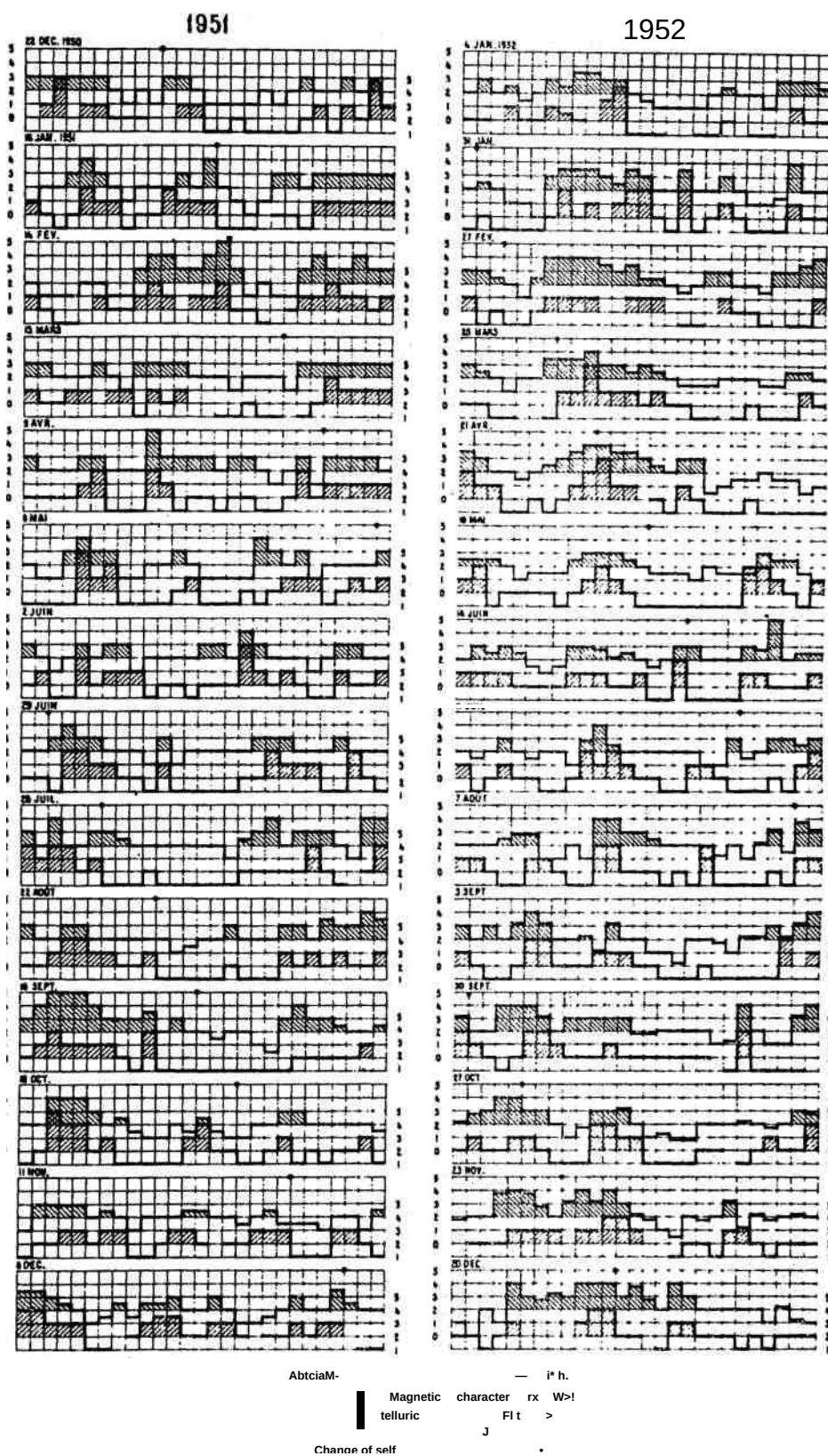
12



Telluric currents: single trace: North-South: MN double
trace: East-West: mn

The measuring device (p. 61) consists of two earth connections (impolarizable electrodes) approximately 1000 m apart, for the MN line and 2 sockets of the same type for mn. These sockets are each connected to a mirror galvanometer via an insulated cable. The slightest variation in potential between the electrodes causes the galvanometric frame to vary, the mirror of which sends back a luminous "spot" onto a photographic paper undergoing continuous unrolling. The voltages vary between 0.1 mv and 3 volts per km (storm weather). Doc. Cie Gle Geophysics.

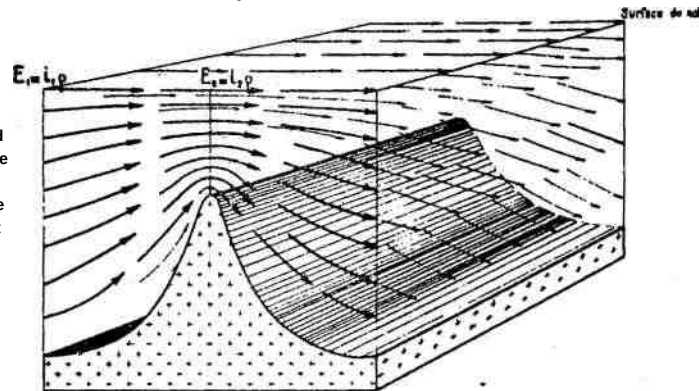
Correlation and recurrence of variations in telluric currents ■
and the magnetic field.



CYLINDRICAL ANOMALY

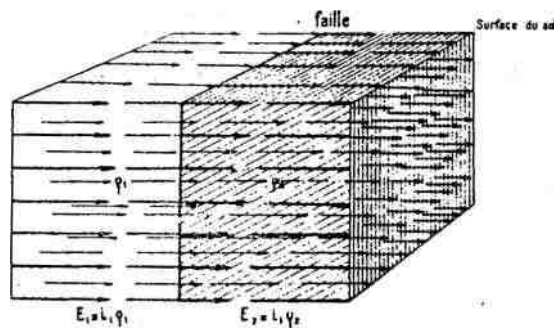
a) Anticline

All the current passes between the obstacle and the surface of the earth. The variation of the field is proportional to that of the current density (constant resistivity).



b) Vertical fault

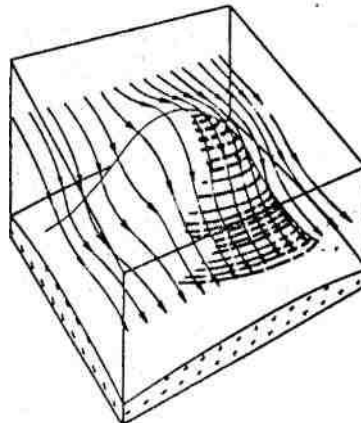
All the current penetrates the resistive medium. The variation of the field is proportional to that of the resistivity (constant current density).



NON-CYLINDRICAL

c) Dôme résistant

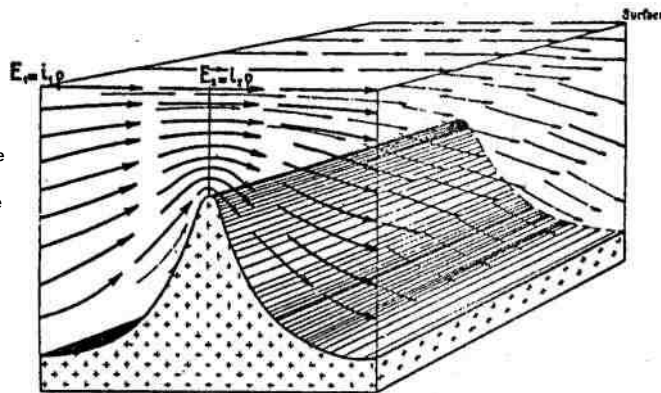
Part of the current bypasses the obstacle.



CYLINDRICAL ANOMALY

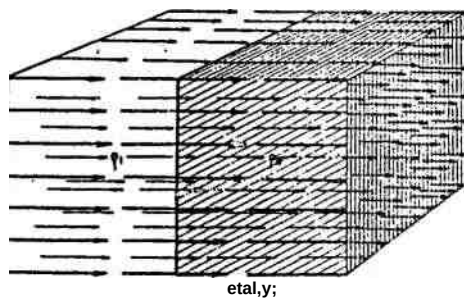
a) Anticline

Any current through enters the obstacle and the surface of the aoli The variation of the field is proportional to that of the current density (constant resistivity).



b) Vertical fault

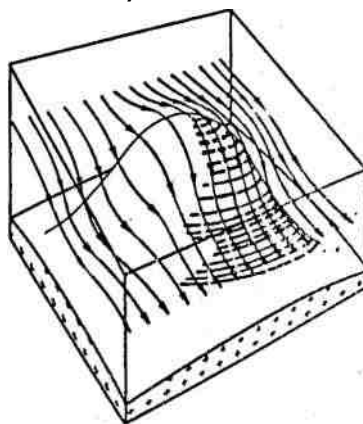
All the current enters the resistive medium. *
The variation of the field is proportional to that of the resistivity (density of the constant current).



NON-CYLINDRICAL ANOMALY

c) Dome resistant

Part of the current bypasses the obstacle.



as for the equations of Maxwell (1865), he arrived at a classification in five methods whose classification is based on various criteria

- a) Knowledge of oscillations
- b) The comparison of oscillations (criterion of in-depth investigation).

In a) we will have the classification: Magnetic sounding, telluric prospecting, magneto-telluric sounding, electric sounding, electromagnetic prospecting at shallow depth using relatively high frequencies. In b) the magneto-telluric classification will be placed at the beginning.

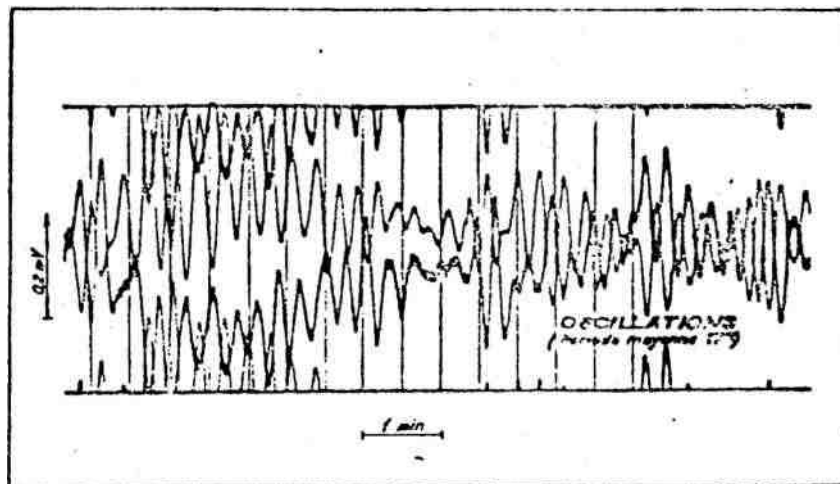
The analogy with the phenomena of "earth mechanics" as oscillatory phenomena does not come within the subject of the author; neither is the coupling of mechanical phenomena with electromagnetic phenomena (magnetic field, marine currents, electrofiltration, earthquakes, explosions, etc.).

Biogeochemistry, Geobotany

Apart from Chinese observations recorded for two thousand years, the relationships between plants and minerals (without mentioning the aspect of the latter's electrical activity or electroculture) have been known in Europe since the 15th century. Observers cite: as characteristic plants of copper content : Polycarpea spirestylis, Viscaria Alpina, Melandrium dioecum, Bouteloua Rothrockii, Eschscholtzia Mexicana, Quercus turbinella. Those known for their zinc content : Talaspi calaminaria, Talaspi alpestra, Viola calaminaria, Viola Tricolor, Thomantheria auriculata. For lead : Amorpha Canescens. Selenium (around the middle of the last century): Stanleya Onopsis, Xylorrhiza, several genera of Astragalus and for Uranium Vanadium contents, several genera of Astragalus as well.

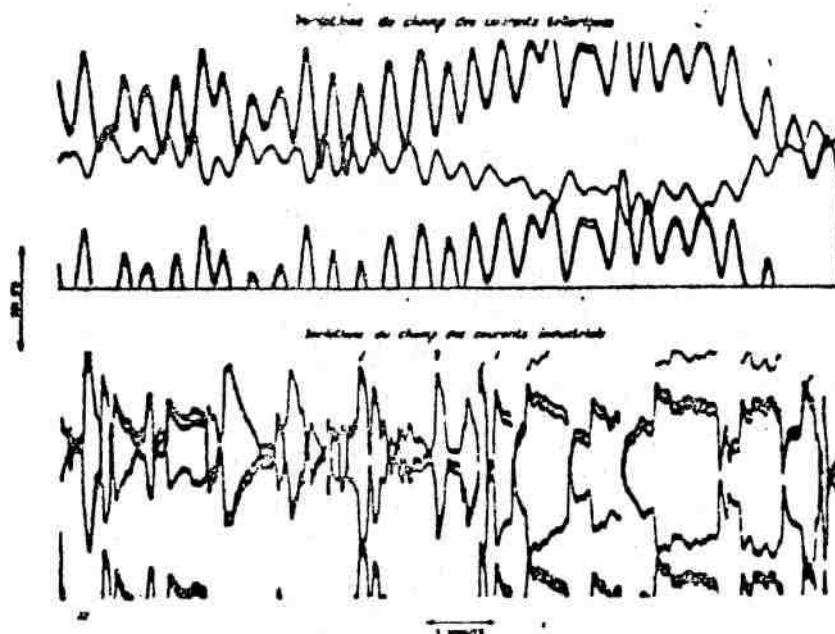
Around the middle of the last century, much work was devoted to the relationship between plants and calcium and magnesium. It was prospecting in an uncultivated area that first led to these remarks, based on the principle that each soil originates from the alteration of the minerals constituting the rocks and forms a limit of the lithosphere with the atmosphere and the biosphere. It is a dynamic system with complicated physical, chemical and biological processes.

130 years ago, Wegmann and Polstorff established the elements that are normally or naturally found in plant organisms. They were:



osci LlfToiRt PUB rtnuiiçtie.

The average period of a pulse lasts 17 sec.



Comparison of two recordings, one of which is parasitized by industrial currents (leaks). The purity of the oscillatory is 'chopped' by artificial discharges completely blurring the meaning of the signals.

[eg On the Bordeaux-Bayonne electric railway and over a distance of 20 km on each side of the track all the recordings are completely polluted.]

[Around cities and industrial installations the troubled zone extends over several km: eg recordings made west of Vienna (Austria) show the parasitic jolts made by the trams when the wattmann passes the pads of the potentiometer of 'acceleration']

O, Fe, Ca, K, Mg, H, C, P, S, N. Alongside were found in plant ashes varying amounts of Na, Si and Cl.

More in-depth research on the ashes, especially in spectral analysis, has proven the trace of many more numerous elements. The composition of the ashes depends on the species, the climate, the nature of the soil, the Ph, the nature of the water. In the stock of salts in solution in the soil resulting from the decomposition of the minerals in place, the plant absorbs a certain quantity , , , . ' '>' d elements that will end up in the ashes.

>'■

■ t-? k ~4v ku

One finds on clay complexes, an absorption of elements from which no soluble part is excluded.

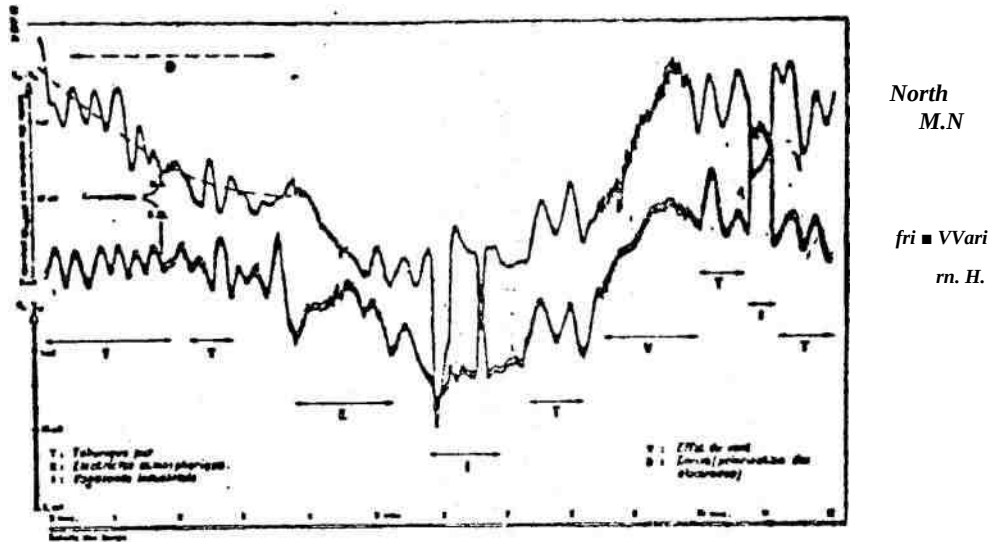
According to F. Leutwein and L. Pfeiffer (Ergebnisse und Anwendungsmöglichkeiten geochemischer Prospektionsmethoden auf hydrosilikatischer Nickelerze, Geology III-6/7, 1954, pp.950-1008), there are complexes between the colloids of the plant and the cations absorbed into the soil a dynamic equilibrium. In the domain of the "rhizosphere" roots, the minerals are decomposed by the carbon dioxide produced by the roots in conjunction with the water in the soil.

In another phase the plant can supply the soil with individual elements in large quantities. The statistics give an enrichment of the humus in Be, Co, Ni, Zn, Ge, As, Cd, Pb, Ti, Ag, Au.

On the content of trace elements in plants and soils, there is a very rich but scattered bibliography. The important texts by the nature of their data are those of Scharrer, Biochemistry of Trace Elements, Berlin 1944 and AP Winogradow, Geochemistry of Rare and Dispersed Chemical Elements in Soil, Moscow 1950.

In uncultivated areas, geochemical/geo-botanical prospecting techniques can be used to establish trace element density maps. If in the same area electrical soundings, electromagnetic measurements, "grounding", induced polarization are carried out, the anomalies, the equipotential curves generally correspond well by superimposing the results maps. In the 1950s and 1960s, the mining and chemical company St. Gobain had experimental measurements of this kind carried out near Valtorte in the Rhône with the aim of locating an ore subsequently recognized by drilling. (Pyrrhotin). In some regions of the Creuse, deep seams (10 to 15m) are indicated on the surface by rows of wild broom. The region of

Twelve minute recording



Superposition of signals on a recording of telluric currents.

T: pure telluric: perfect harmonious curve. .

E: Atmospheric electricity: electrostatic tension. The signal is hatched.

' . I: Industrial stray current: shear of the telluric oscillation.

V: Wind effect: The power supply line has moved over ^^^^a few meters. The movement of the metallic conductor cable by breaking the earth's magnetic field causes a current. hatched parasite coming and going which scrambles the signal.

D: On the N.South line: single spot MN The electrode, although impolarizable, has caused a "drift" (The entire NS is no longer parallel to the double mn)

Quillan in the south of France presents on aerial photographic surveys areas of causses covered with deciduous trees corresponding to geochemical aureoles of sulphides. Above and vertically to the deposits, the atmosphere in calm weather presents a density of metal ions present in the subsoil and on the ground sufficiently characteristic to have given rise to an aerial prospecting method used in the USSR.

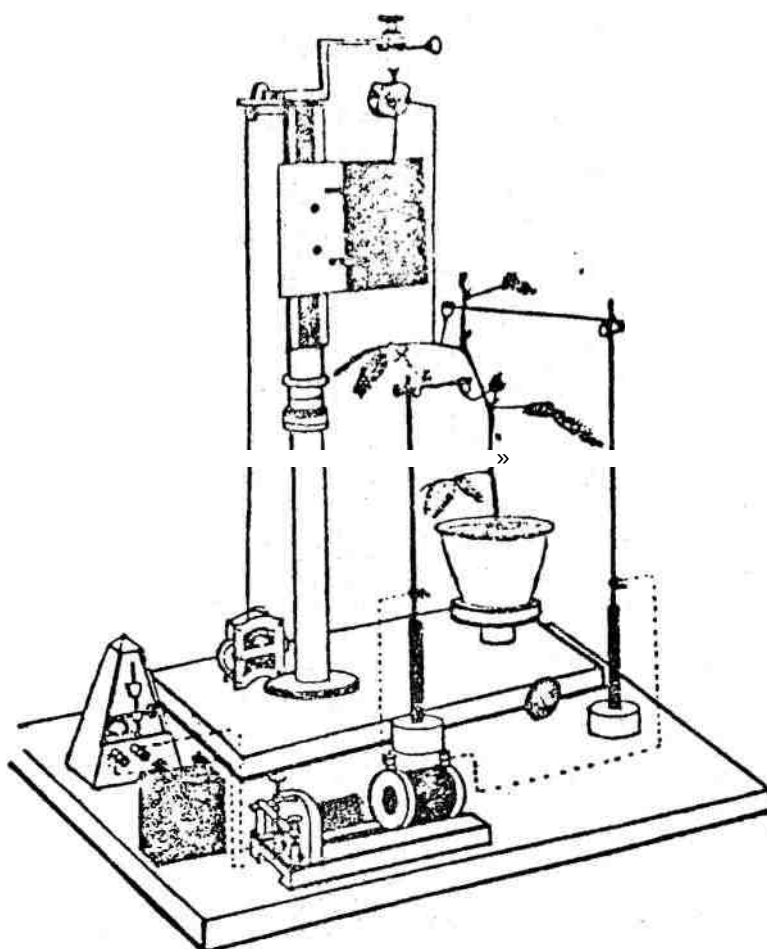
Concentrations made up of water, soil and plants made it possible to put in evidence of the presence in aerial plants of all the chemical elements found there in excess and thereby indicating significant mineral deposits in all types of soil encountered on earth. On this subject, see a 1956 thesis by P. Maljuga, published in Moscow under the title: Geochemical principles for the search for minerals according to plants and soils. Also consult the work of a symposium organized in Royaumont in 1959 on Geobotany and spectrography.

However, research work for minerals by geobotany seems to have stopped only at the chemical aspect of the question, which is very complex. The reversible problem in electroculture seems to have interested only the ancient Chinese who sowed in certain fields metallic bodies of small dimensions surrounded by various insulating rubbers or not. (Needles, hammered son , spherules obtained by projection of molten metal in rainwater kept in jade vases) .

D'Arsonal in his time thought that "every living being was a reversible electrical machine".

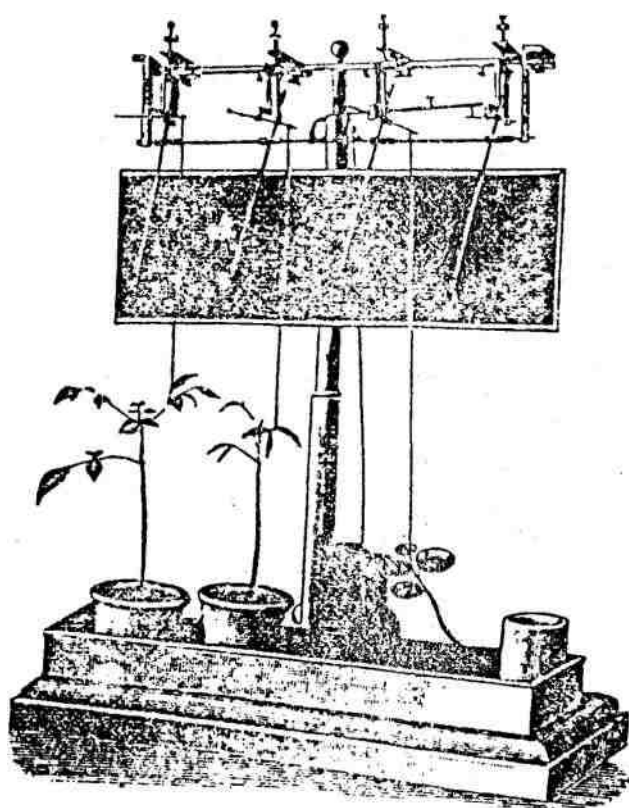
About plants a remarkable set of experimental research was done in Calcutta, at the Bose Institute from 1917. In October 1926 Sir Jagadis Chunder Bose published a text highlighting the electrical discharges of plants each time their tissues are excited. A summary of the research of the Bose Institute was published in 1934 translated from English by E. and G. Monod-Herzen under the title: The autographs of the Plants and their revelations , exposes and at the same time popularizes the research made at the institute. one of the aims of which is to combine the development of knowledge with the concern for its greater dissemination.

In the preface of 1934 Bose expresses his wonder at having seen the boundaries drawn between "living matter" and "non-living matter" vanish. It proves by the graphs obtained from the reactions of the plant it



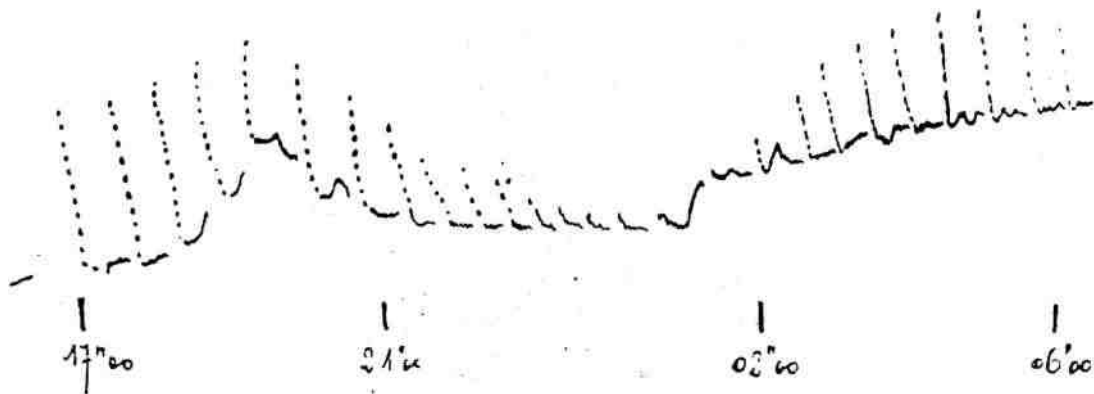
-- .1 />artillagr complete for the automatic recording
of .Mimosa's actions.

automatic recorder



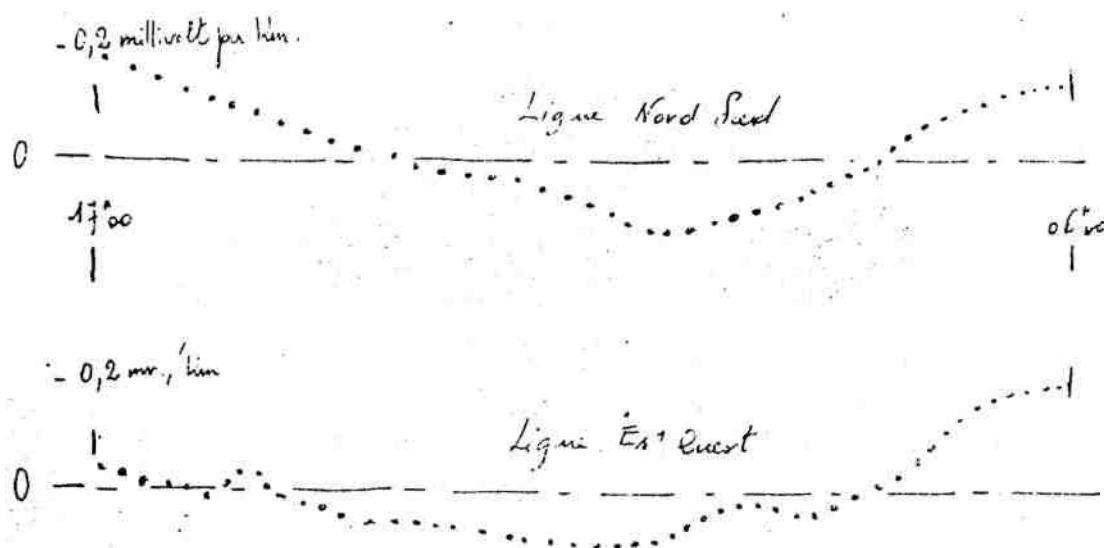
Lur, cislreur du fnouvciuent des leuilhs.

even that there is not a single vital reaction, even in the most superior animal, which is not sketched in the life of the plant. The measurement and recording of the effects studied necessitated the invention of numerous measuring devices by Mr. Bose.

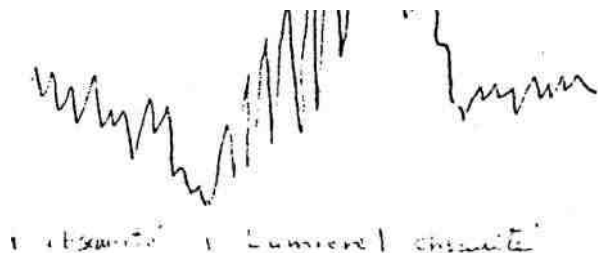


Continuous recording for several hours showing the variations of the sensitivity of the Mimosa with time.

Compare this curve with the hourly variation of electro -telluric currents following a North-South measurement and an East-West measurement (annual average of recordings over 24 hours at the Ebro observatory.)



Mean values for southern Europe and central India (Roaney)



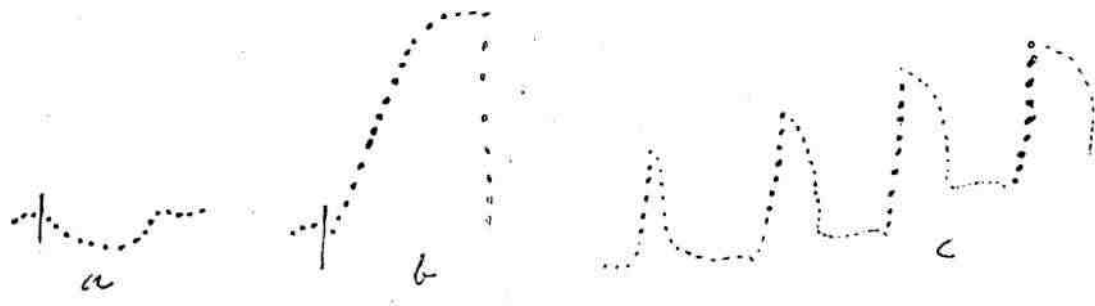
Stimulating effect of light
on a hypotonic subject in
darkness, weak pulsations,
illumination: increase.

The author endeavored to record the motor reactions, and the latency times produced by a stimulus. The following effects have shown various reactions recorded on diagrams by appropriate devices: Effect of the passage of a cloud, intense cold, carbon dioxide, hydrogen sulphide, ether, chloroform, alcohol, strychnine, cobra venom, effect of watering and drought on reaction pulsations, effect of mechanical pressure, twisting, scratching...; ' effect of electric exciters, electric shock, light, ultra violet, Hertzian waves.

Reactions to stimulation were studied on the following species: Mimosa, Carrot, Phoenix sylvestris (recording of diurnal variations of the "praying palm" at Faridpore), Desmodium gyrans, Auerrhoa, Tropaeolum (nasturtium), Nymphéa, Chrysanthemum, Brassica, Mimosa pudica, Pithecolobium (or diurnal pressure variations in the Rain Tree) Mango tree etc...

All the devices intended for recording the movements of plants consist of mechanical optical amplification of motor reactions and plant development. The simplest device consists of a fine silk thread attached to one end of a weighted pivoting lever with a very fine stylus, the point of which scratches a glass plate coated with lampblack. This plate descends vertically by its weight at a speed regulated by a clockwork movement. The free end of the silk thread is attached to the leaves, stems or part of the plant whose reaction is expected. The use of a double coil with electric excitation, allows without contact with the plant to obtain by induction stimulations identical in force, direction, intensity and also variable. The various parts of the plant to which the silk thread has been bound react and mechanically drag the thread.

Reaction of a plant to Hertzian waves (Bose Inst.)



(a) reaction to weak excitation (b) reaction to a

EM Secondary induction:

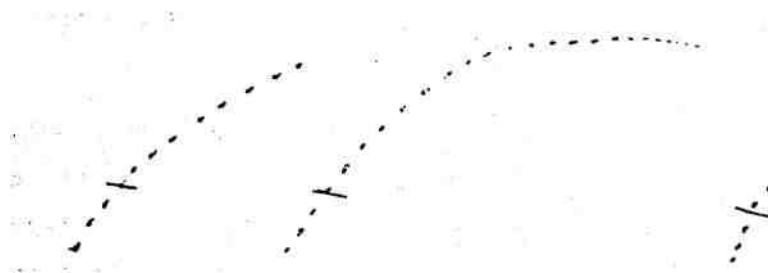
strong excitation:

Growth acceleration Growth

retardation

(c) reactions to average intensity excitations: delays followed by returns to the normal state.

Recording with the Bose oroscograph with magnetic amplification .



Effect of increasingly strong electrical excitations: (0.5, 1 and 3 units) on the elongation due to growth.

Reaction to an electric shock.

Successive dots are separated by 4 seconds.

Two other types of recorders were built by Bose, one is a resonance model for self-recording; the other with amplification by optical lever (the Sphygmograph). The amplification of all signals depending on the device chosen varies from a few hundred times to a million times. Appropriate electrical probes have made it possible to locate nervous tissue in the petiole of certain plants.

The recording diagrams of electrical excitations by induction coil on the plant generally show that a very slight excitation increases the speed of growth, a medium excitation retards it. A vigorous excitation results in a momentary contraction.

Tests carried out at various temperatures show that at 60° centigrade, the death spasm occurs which puts an end to all growth.

Recordings made with the balanced compensated Crescograph showing variations in the growth rate of a plant on the order of a billionth of an inch per second prove the effect of a flash of light produced by a single electric spark whose short excitation translates into the reaction of the plant itself.

Subjected to Hertzian waves, the recording diagrams show that ultraviolet retards growth, that this retardation decreases as it moves towards yellow and red, and that wavelengths from 0.6 cm to several tens of kilometers produce variations in growth depending on the intensity of the stimulus. Weak waves accelerate growth, intense waves retard it, the effect continuing long after the excitation has ceased. Medium intensity waves produce a delay followed by a rapid return to normal. [These are the only indications of the public text... Lakhovsky and Laberge had made the same statistical remarks at the same time].

In relation to the phenomena of influences, the ancient Chinese retained two so-called "ritual" instruments, one represented the "channel of influences from heaven to earth"; it was the "Pi" jade disc with a central perforation of 3/5th of the apparent 0; the other represented the "Tsung" or chimney of earthly influences in the sky, natural influences of course. Nowadays an influence of an electromagnetic nature has been demonstrated at the Stanford laboratory in the USA: It is shown that the starting frequency of "chorus" emissions is strongly linked to the harmonics of the frequencies of high voltage power lines. These results are interpreted by a flow of radiation from high voltage lines

in the magnetosphere where it stimulates the generation of very low frequency choruses . The "Chorus" is one of the most commonly observed electromagnetic (EM) waves in the magnetosphere, which leads to the conclusion that radiation from power lines has significant effects on the level of wave intensity in the magnetosphere , which affects the orbits and lifetimes of energetic particles trapped through wave-particle interactions.

[See on this subject: "The Control of the magnetosphere by power line radiation". Lurette (JP), Park (CG), Heliwell (RA) Stanford univ. radiosci. Lab. USA, J. geophys. Res., USA (1979), 84, no A6, 2657-2660, bibl, 10 ref.]

This for the "Tsung" of ancient China, when to the "Pi" analogically, the radar center of the same university indicates that the gravitational field of the Sun acts as a spherical lens vis-à-vis a source of semi-infinite EM radiation . The gravitational focusing of electromagnetic radiation that could be used for directional observations and communications over interstellar distances. The Solar Corona severely limits the lensing effect at radio frequencies . See: "Science ", USA (1979), 205, n°4411, 1133-1185 bibl 9 ref. (Eshleman (VR) "Gravitational Lens of the sun: its potential for observations and communications over interstellar distances".

Reactions of mineral matter

Around 1896, Bose had noticed a property of radio wave detectors that under continuous stimulation lost the usual sensitivity, as well as non-activation for a long time. Subsequently he realized that certain substances made the receiver extraordinarily sensitive and allowed to record extremely weak signals. Other substances diminished sensibility or abolished it entirely.

Having obtained indications reminiscent of one of the sensitivities of living tissues, Bose carried out stimulations on metal wires made of tin, copper, platinum by employing the same method of electrical reaction used for research on animal tissues. The results of the diagrams show, according to the periodic oscillations recorded, either by successive points, or by continuous line, an astonishing similarity of outline with the reactions of the nerves or the muscles. Uniform reactions are obtained following identical excitations, provided that suitable intervals of rest separate them. A weak excitation produces a weak reaction and the return to the initial state

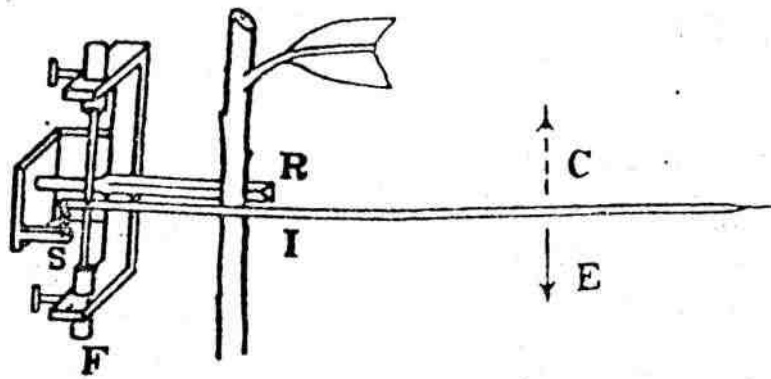


Diagram of the Sphygmograph

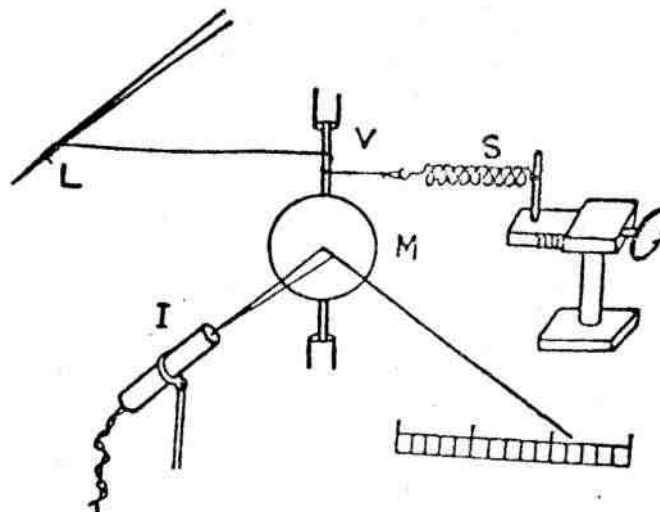
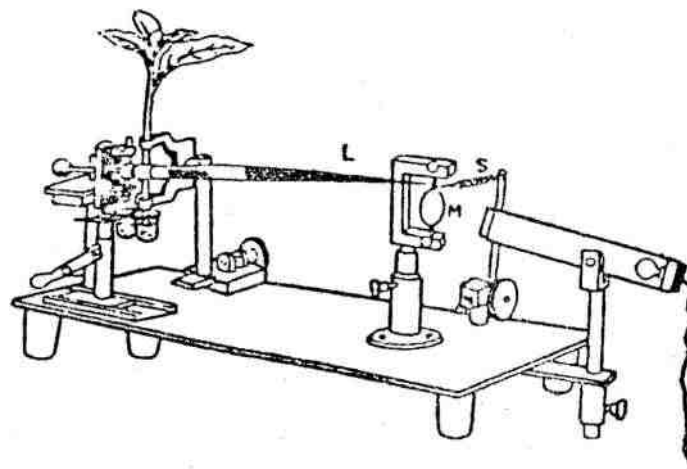


Diagram of the assembly of the Sphygmograph, with its optical apparatus

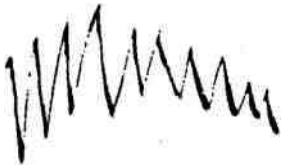


The Complete Optical Sphygmograph

produced quickly.

A stronger excitation gives rise to an ample reaction, the time necessary for the return to the initial state is increased. A weak inactive excitation in the isolated state, becomes effective by repetition... this remark is interesting and draws attention to the manufacture of copper cauldrons hammered by the Chalderash specialized in the work of metals starting from the ingot mould. . The long-hammered copper utensils with a tenon-mortise assembled bottom had the reputation of making mold difficult on the jams cooked in them... the other simply rolled or turned utensils did not have the same property, although also in copper ...

The metal subjected to long repeated excitations shows fatigue on the diagrams of Bose recordings. An identical form of signature registers on the excesses of stimulation for a plant.



Here is the reproduction of a signature obtained by Bose for the "fatigue of a metal".

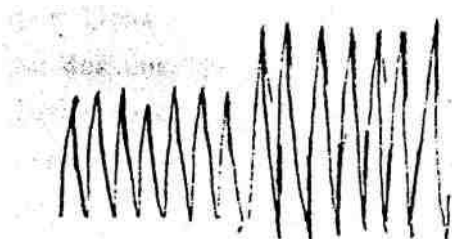
There is no specified scale in relation to the constraints imposed.

As for plant stimulants, after recording the reactions normals of the metal to successive identical excitations it is touched with a stimulating; there is an increase in the electrical reaction. The signature opposite is produced by a solution of sodium carbonate . Other stimulants produce further increases

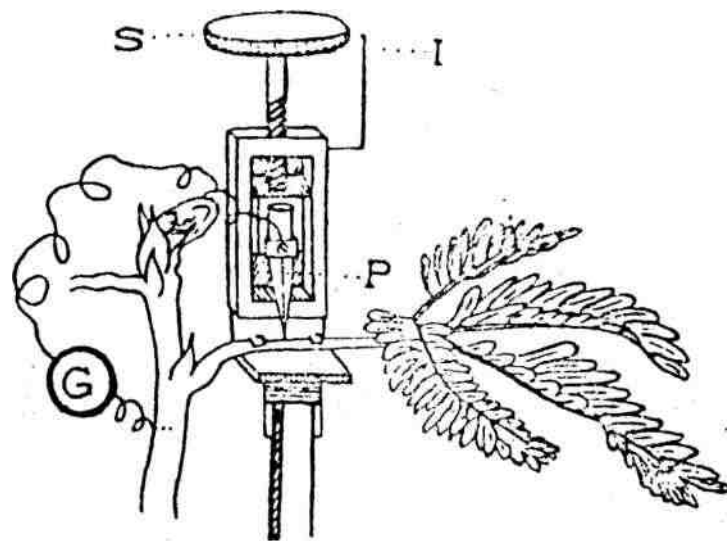
most amazing of irritability just like plants.

In the examination of the curves resulting in a total immobilization which characterizes the death of a plant by poisoning for example, it remains to confirm the unit of reaction with a metal and a poison, final test characterizing the stopping of the life for physiologists.

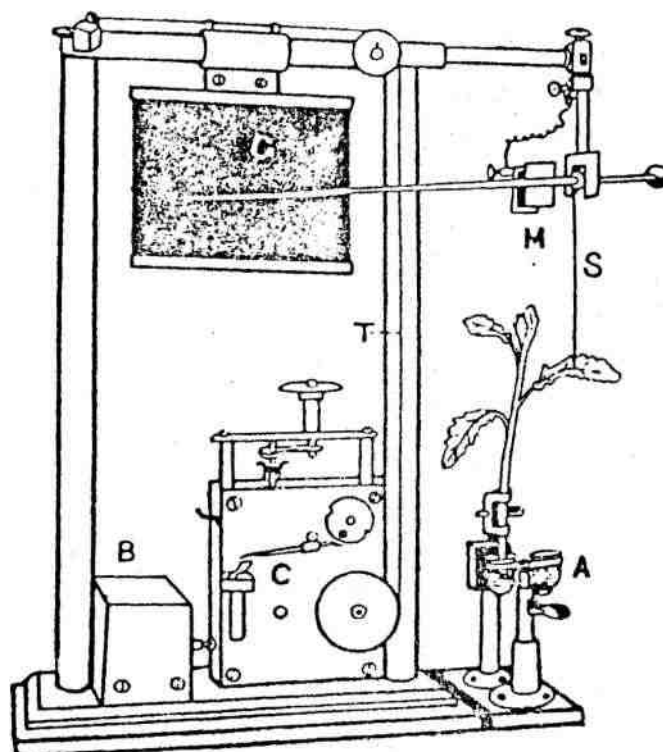
Strange as it may seem, the electrical reaction is intensified



by a small dose of poison, while it disappears entirely by applying a large dose to the metal. (Known technique of the ancient Chinese for the manufacture of poisoned needles)



Electrical probe used to locate nerve tissue
in a mimosa petiole



The electric phytograph

The nature of the "poison" for the metal is not given in the text.

In the presence of these facts about the similarity of reactions and curves Bose posed in 1920 the question: "How could we say: Here end physical processes and here begins physiology?" He added: the distinction between external and internal stimuli comes only from a difference of point of view, because in fact they play the same role. Everything inside a living organism has come to it from outside. By localization in the surrounding universe energy and something else flowed into the first living granule. To understand the activity of cells and living forms something more than electrical or mechanical solution is needed. The cellular "machine" (and the complex structures) can only vibrate with life when it is placed in constant communion with all the energies of the universe which surround ^{it}. The conclusions of the 1920s are echoed in Lakhovsky's theory of the Universe-Ion, as well as in Wilhelm Reich's proposal for Bions. ?

Bose also indicated that after his experiences he better understood a message proclaimed thirty centuries ago on the banks of the Ganges: "Those who, in the multiple changing appearances of this universe, see Unity, possess eternal Truth, and no other, no other".

The excessive specialization of current science has the danger of losing sight of this fundamental fact, that there is only one truth, only one science, comprising all forms of knowledge. And yet how chaotic

* appear natural events! Between theory and practice, between the study of the inorganic world and that of organized life in its multiple manifestations, growth, vibrations, oscillations, resonances, movement, even sensation. Thus converge and meet Physics, Physiology, Psychology. It is in this figure that those who, among the multiplicity, will know how to perceive unity.

Note on phenomena with periodic variations: (April 8) - General Sabatier. "Technical and Scientific Information Bulletin of the Technical Section of the Army" .

Periodic variations are considered in three categories:

1) Vibrations of matter affecting position (pendulum, vibrating string, tuning fork) limited by mechanical inertia. Variations in shape (contractions,

flexions, dilatations) reach the electronic device of the bodies, resulting in electrical phenomena of the same frequency and vice versa. These variations- , . . .

4

Oscillatory periodic frequencies are between 0 and 10 referred to as "Low frequencies" or "acoustic range". Beyond 10 the material vibrations are excluded (at least defined CCid'). The capacitor - choke system 12 provides oscillations of up to 10 (TSF) periods per second which 12 associate with corresponding variations of the Elec fields. Mag. After 10 electrical oscillations are excluded; light, x-rays, rays 24 gamma, the penetrating cosmic rays give frequencies of order 10 Beyond that, General Sabatier supposed possible vibratory variations and 40 oscillator which would go up to 10 and which would be of the Biological domain.

These indications will serve as support for Marcel Violet in his experiments from water and living bodies for the expression used in his writings and called "Biological Waves" improper terms just like the expression used by other researchers working on localization and delocalization and which use three-dimensional objects of variable shapes and materials oriented in space according to certain parameters; expression known as: "Waves of Forms" (now called eifs).

There is a question of semantics here, the words under their anthropomorphic expression change the meaning of the lost word which it hides from the sole creator verb.

.Beyond these conceptions, let's see what convincing facts have been executed experimentally by Marcel Violet.

Experimental tests on water and plants of

Marcel Violet

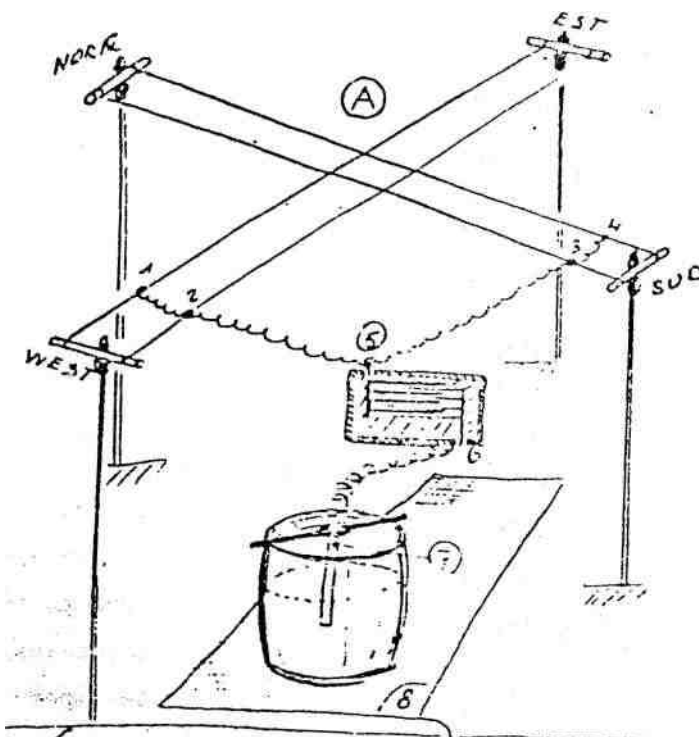
From 1938 to the last 70 years an engineer AM Violet from remarks, laboratory observations, adaptation of indication or tra- (*) values carried out by researchers such as S. Bignand, has carried out or had carried out a large number of experimental manipulations on water subjected to various radiation fields in which the vibrations of electromagnetic waves and photons as well as ultra penetrating radiation seem play an important role . In the fields of health, animal husbandry, the chemical industry and

(*) See study in this book.

Agriculture, the results are convincing. From the effects obtained, Marcel Violet will try to identify theories and put forward hypotheses. He tried an industrial experiment with so-called "electro-vibrated" water and stuck to "the facts".

Between 1960 and 1965 experimental work on water subjected to variable frequency fields (from 50 hertz to 1000 hertz) was carried out in the USSR. This water mixed with cements to make concrete gave the finished products a hardness 1.5 times greater in all cases. samples compared. We found no Soviet publication on agricultural trials; on the other hand, the descriptive bulletin of the CNRS in the analyzes of publications in Russian language in 1962-1963 gives a "theory" of the displacement of the spectral lines of water subjected to treatments of the "electro-vibratory" type (in the terminology of (M. Violet) Russian analysts bring in quantum theory and wave mechanics, and we relate these insights in relation to the agricultural implications of Violet's work.

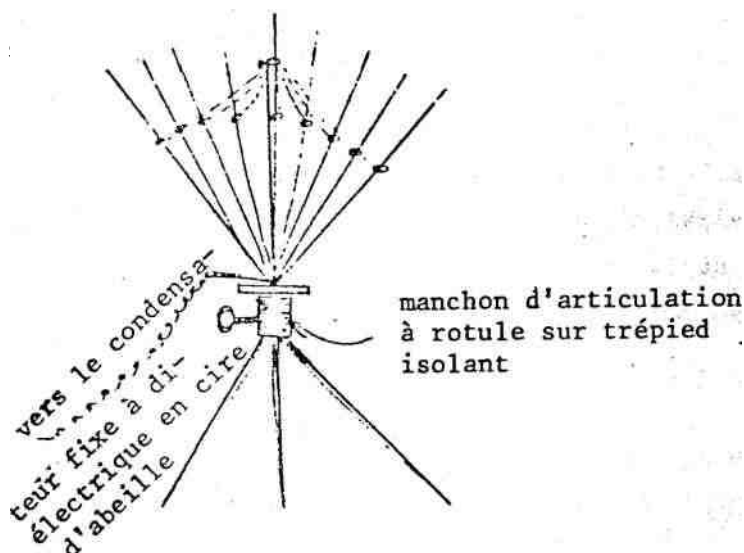
First treatment devices tested by M. Violet on water. Dynamization by "biological waves" (an improper term used by the author). (See this term attributed to General Sabatier).



The device (A) comprises two times 2 antenna wires oriented NS and EW in copper. The wires are carefully insulated on rods and porcelains. Connections on these antennas, 1,2, 3,4 come in (5) on the armature of a capacitor whose dielectric is beeswax (proc. Bignon). The outlet (6) of the armature is connected to a carbon electrode (7) immersed in rain or spring water contained in a tub of non-metallic material (wood,

glass, plastic). A zinc plate (8) placed on the ground under the water container is earthed at (9). After eight days the water is ready to be used: immersion of seedlings or seeds, watering, washing of leaves etc... Subsequently the antenna (A) was replaced for other tests by "umbrella" antennas in copper whose opening cone of the nine lerins could

open from 10° to 160° . The ball joint of the articulation : possible to orient the cone of the antenna rods at various points in space.



Marcel Violet has published certain results of experimental research in the book "Le Secret des Patriarches". A fifth edition in May 79: 'Cosmic energy in the service of health' was published in the Courrier du Livre. (A few extracts):

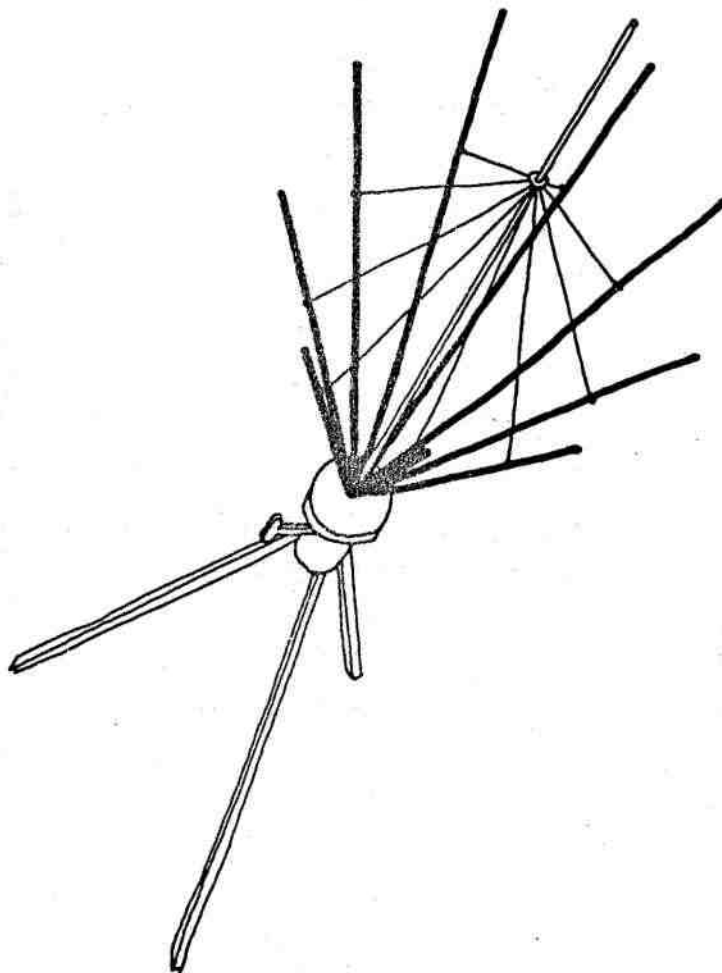
At the request of the Ministry of Agriculture, systematic tests were carried out on the influence of "biological waves" on the growth and composition of plants. Various processes are used, including:

- a) Soaking seeds in treated water for 15 minutes before planting (comparison with controls soaked in untreated water).
- b) Watering plants with dilute treated water at mid-vegetation (various percentages and various periods) [Major choice $10^\wedge$]
- c) Treatment of plants with gases charged with "biological waves" (see General Sabatier's note)

The first results of tests on the plant have been recorded in the

- Carrots from Croissy: after 8 weeks of vegetation, 80 grams, controls 60 grams.

Vue en perspective de l'antenne "parapluie"



beets sugar bowls: 1 sugar point more than the controls: 9, against 8, ⁸

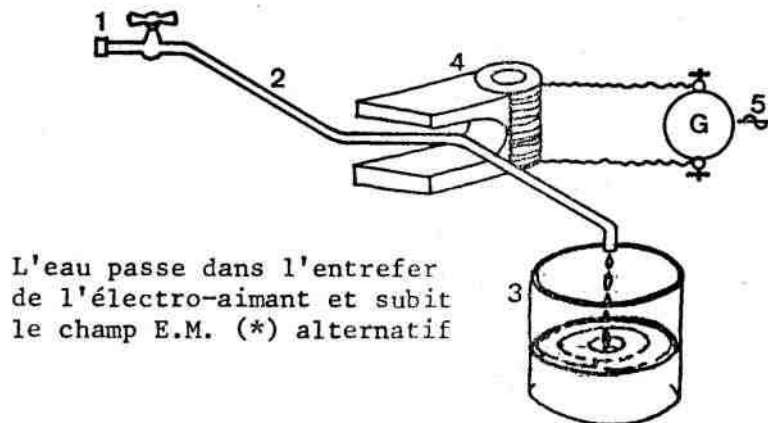
- Fodder beets average weight: 2,300 kg, controls 2 kg; plus watering with treated water at mid-vegetation 2.4 kg with some 8 kg.

The taste of the vegetables is not affected, the chemical compositions for these tests (wheat and barley included) do not reveal any new bodies. The richness in sugar of the beets makes one think of a possible treatment of the vine in the years when the sunshine is low.

- Potatoes: 1st experiment: soaked for a few minutes in treated water and placed on shelves stacked under a waterproof tarpaulin. Inside this cabin, oxygen, ozone and nitrogen are sent pneumatically, treated for 24 hours. One plants, the output is higher from 6 to 10[^]; the percentage of starch also increases compared to the controls.

Second water treatment devices for agriculture and livestock developed by Marcel Violet on various suggestions from those around him: radiesthetist friends, Mr. Bignon, Mrs. Tricot etc...

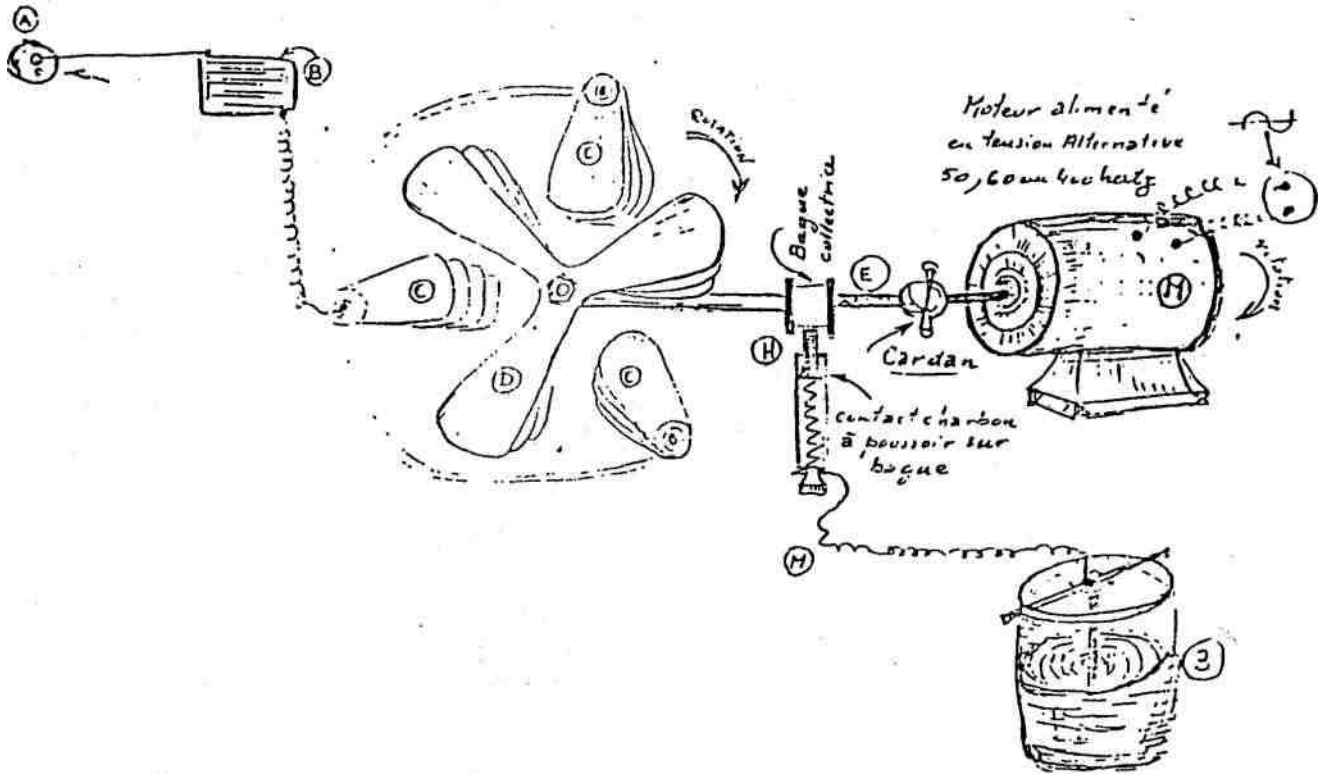
Electro-vibrated water



First treatment : the filtered rainwater (1) distilled from a drinking source without dripping, flows through a plastic pipe (2) into a glass or plastic container (3). Between upstream and downstream the plastic pipe passes (and guides the water) inside the magnetic field of an electromagnet (4) powered by an alternating voltage (5) 110/220 volts, 50 hertz at 24 volts, 400 hertz (voltage commonly used in aeronautics) (*)

(*) EM for Electromagnetic.

Second_treatment



Alternating source (A) 50, 60, 400 hertz with "Phase", is connected to the armature of the fixed beeswax capacitor (dielectric) (B). The other part of this CF with beeswax dielectric is connected to a fixed armature of a mobile capacitor (C). The three-pole mobile part (D) is driven by a shaft connected by cord to the axis of a motor supplied with alternating 50, 60 or 400 hertz. (M). The shaft section (E) is insulated from the motor, it is a teflon or hard rubber shaft segment. (Mechanical supports on bearings are not shown). A carbon ring collector (H) is connected to an electrode either of carbon or of various metals or alloys (mono or poly-metallic). This electrode simply soaks in the water of the container (3) which has received the first treatment in the air gap EM of the device drawn above. The water treatment time in the alternating EM field lasts a few seconds (this is the first treatment). The average treatment time of this water with immersion of the carbon or metal electrode should be six hours; this is the second treatment before use.

Second experiment on potatoes (Bringing together the Labergerie Lakhovsky work and the communication to the Academy of Sciences on potato degeneration)

Cultivation in two boxes filled with sawdust (absence of nutrient salts) of plants (with controls) of treated and untreated potato. Watering with ordinary water. Result: 3 pdt give 10 tubers with a total weight of 80g for untreated plants; 3 pdt give 7 tubers weighing 205 grams for treated plants.

Third experiment on potatoes : two calibrated sorted batches germinated in boxes. The treated batch is sprinkled with treated water in a treated atmosphere; after 48 hours the germs of the potatoes watered with ordinary water show no change visible to the eye. The treated plants presented during the same 48 h time a hair of numerous roots reaching several centimeters.

Note : Following these tests, a chemical company attempted to accelerate plants using a multimetallic electrode following the design of the second Violet device, taking as a succession of metals those of the Lakhovsky oscillating collars on the one hand and on the other hand by misting water treated according to the first Violet devices; this water being pulsed by a gaseous composition also treated with "Biological Waves".

The fourth experiment on potatoes focused on reducing the minimum growth time (90 days) to be able to sprout and reproduce. On treated plants a gain of 10 to 12 days is observed.

Note: In general, each technique taken in isolation in all electroculture systems each gives a minimum increase either in time or in overproduction of 10%, at least in aerial cultures [In aquatic Labergerie went from 4cm to 40cm for an adult plant] .

In 1967 in market gardening in Aisne, under greenhouses, plants were watered with electro-vibrated water in a 10% dilution twice, at 3-week intervals after sowing.

Result : Melons, larger size, larger leaves on treated plant.

Cucumbers: Increased vigor, even grading, disease free. Salads: increased speed of growth. Nice features.

Celery: huge percentage increase in germination.

Tomatoes: considerable width development with 2cm wide trunk in $\emptyset$, extraordinary abundance of fruit clusters with at the same time flowers at the top.

Flowers: Seedlings of nasturtiums and sages emerge prematurely and give vigorous plants.

During the tests in the chemical field, sodium chlorate was put in a bottle (1/4 of the capacity). We filled with treated water. The bottle was sealed and half buried vertically in grass. The grass was destroyed in a few days in a circle, the center of which was marked by the bottle for a few decimetres.

General Sabatier in his articles of the 1948s pointed out (he is not the only one...) that man tries to probe phenomena with known devices; and add i l can not do otherwise!

René Barthélémy, from the institute, undertook systematic research to try to identify the wavelengths (?) that we try to attribute to the effects obtained by the devices built at Mr. Violet's or that he had built . Barthélémy concluded: 1°) You are indisputably in the presence of oscillatory phenomena, 2°) My systematic research, going from $\emptyset$ to $10^{\wedge}$ did not allow me to detect them. You are beyond... $\wedge$ $\wedge^{\circ}$) Sabatier's hypothesis is probably correct and the effects on living organisms confirm it.

With regard to Agriculture, the tests consist of soaking the seeds in electro-vibrated water without any contact with any metal whatsoever taking place. The duration of immersion varies from 1/4 h to 4 hours depending on the size of the seeds. The water can be used again for new grains and seeds: beans, peas, etc...

For comparative tests, alternate the rows treated or not. Impregnation time: Small seeds : depending on the size from 1/4 hour to 4 hours. In principle, peas, beans, broad beans: 3 to 4 hours. Allow to dry before planting. Potatoes: 1/2 hour. Leave to germinate: before planting, soak for 2 to 3 minutes.

Transplanted plants : leeks, tomatoes, cabbage, lettuce: soak for 1/2 hour to 3/4 hour.

Electro-vibrated water electrodes in agriculture have mainly focused on them Copper-Magnesium combinations. (*)

(*) It is actually EF and C. of T. which are detectable in the vicinity of these assemblies.

Other associations have also allowed a SUPER concentration

| | | |
|-------------------------|-------------------------------|----------------------------------|
| <u>trace elements</u> : | - Silver + Copper + Gold | - Zinc+ Copper |
| | - Manganese + Copper + Cobalt | - Zinc+ Copper + Cobalt + Nickel |
| | - Nickel+Zinc | - Gold Copper + Silver |
| | - Cobalt + Vitamin B12 | + - Zinc+ Copper + Cobalt |
| | - Ferro-Vanadium + Clay | - Versatile 6 metals: |
| | | Ag,Cu,Mg,Mn,Ni,Zn+ |
| | | Phosphorus + Iodine + Lithium. |

All Marcel Violet processes obviously apply to animal and human food and to the treatment of certain ailments.

About the nature of dielectrics in capacitors, see CRAS 17 Jul. 57 - Note JATAR-SRARMA: Silent electric discharge in air with the use of insulating electrodes (Study of the Joshi effect at low frequency with beeswax electrodes). The discharge current to the oscillograph consists of many high frequency pulses superimposed on a sinusoid. Ebonite, bakelite, mica, porcelain and other waxes.

j.- have been studied.

Stanislas BIGNAND

by P. SMIRNE

(Paris)

In applications to the stimulation of seeds, seeds, plants in general, the radiesthesist experimenter Stanislas Bignand carried out from 1941 to 1949 in France, Switzerland and Belgium a remarkable series of experiments whose spectacular character could have interest, if not large manufacturers, at least small and medium- sized enterprises.

Stany, as his intimates called him, had at first made a large number of observations in the field of the emission or absorption of "-OSCILLATING ENERGY" by various food substances with a view to improving of their properties or of the development in these substances of new properties.

This researcher filed a number of patents in which he described, based on his experiments, that all bodies absorb or "admit" a certain wavelength and return or emit part of the energy as well. received by them.

determined and suitably chosen wavelength .

The device included a rotating pole variable capacitor. The mobile part comprising 12 blades of cruciferous shape, each mobile blade being offset with respect to the previous one such that the 12th blade has its outer edge at about 30° from the edge of the first blade along the generatrix of an ideal cylinder. The stator or the intermediate fixed blades of this capacitor enveloped the moving part by 180° and was supposed to store radiation trailing in its field, increased by a high-frequency induced current due to friction in the air, to the static currents inside the rotating masses and to the vibration of the electric field.

In addition, 12 capacitors combined with oscillating electric circuits produced by thermionic lamps allowed by their adjustment

and their coupling to four fixed bimetallic capacitors (copper-zinc) whose dielectric is an "active" beeswax to obtain a considerable activation of the vegetation either by watering the crops with treated water, which according to Bignand was tantamount to emitting waves directly on said cultures, either by germinating grains or growing plants in this water. Tests prove that the time of immersion in water plays a considerable role. Potatoes on 1 ha treated could reach per plant from 2 kg to 2 kg 1/2 per potato, each plant giving 5 to 6 tubers. The treated barley gave in normal soil without fertilizer a yield of 64% more than on the control surfaces. In the Strasbourg region, diseased whole fir stands have been successfully treated by Bignand and Bostrich.

In the Bordeaux region, the vine was treated by spraying this water subjected to rhythmic high-frequency oscillations in which distilled water or rainwater had been placed. In some cases Bignand added various products to the water being treated.

For example, a decoction of oak leaves and bramble leaves added in very small quantities and coffee substitute develops the aroma of coffee in the infusion to the point that this infusion can be confused with pure coffee. To prepare such a product the author began by measuring the wave admitted or absorbed by the coffee substitute, and by the same means the various combinations capable of enhancing the aromatic properties of said coffee.

Stanislas Bignand filed some of his patents with Marcel VIOLET and Raymond PAIRIER de WOŁODKOWICZ. According to the researches of "Stany" they thought that each body, each gas, each molecule, could be "activated" by means of the treatment of one or more specific wavelengths. Bignand considered in his devices that the 4-pole and 12-blade rotor acted as an energy modulator on the characteristics of the circuit in which it was placed and which included the 12 adjustable capacitors and the 4 fixed bi-metallic beeswax capacitors treated with high frequency emissions during the cooling of the dielectric of which we have already spoken.

On the device described above, each capacitor had its capacitance adjusted on demand and for a specific purpose. Acting alone, the whole apparatus was capable of capturing by transfer certain energies of the ambient space conditioned by "waves" or "other energies" of which one of the aspects known is of so-called electrostatic or electromagnetic origin^ the link whose source (*)

(*) These are mainly EF and C. de T.. Many of these devices have an ext.

primary is extremely distant, i.e. galactic. We are, in reality, in front of the apparatus where relocation intervenes.

Bignand according to his conception, by combining specific capacitors, chokes, various resistors, forming a discharge circuit in which - by acting like a resonant filter - this discharge was modified according to a rhythm imposed by the action of the variable capacitor at rotor. The resulting energy action can be amplified and controlled at will by an appropriate electronic assembly.

Bignand explained his oscillating assembly, the adjustments to be made and its operation as follows: On the terrestrial sphere, taking into account the latitude, the longitude and the altitude, we receive through the intermediary of the moon playing the role of an energy distributor a differential flow of galactic energy emitted by 88 listed constellations. The various states of matter, masses and radiation that we know on earth also exist as plasma on our star, the sun. During the relative movements of our satellite and the mobile planets, all the movements of the bodies of the solar system bring about a phenomenon of oscillation and resonance of the galactic energies that we receive on earth at the point where our device is located, provided that it is tune by energetic adaptation.

Bignand's experience and findings show that out of the 88 listed constellations we can initially isolate twelve characteristics on the equatorial plane, twelve energetic characteristics on the tropical terrestrial planes (N. and S.) twelve circumpolar energetic characteristics on the circles poles, 1 energy characteristic on the (*) polar group, 1 energy characteristic on the constellation of the Boreal Dragon, 1 on the Southern Dragon, 1 on the Southern Cross.

The apparent place, from a geocentric point of view, of the sun passing over the zodiacal constellations brings for each body, each element, each mineral, vegetable, animal, human association, every thirty degrees of a circle a modification of resonance. and oscillating said terrestrial body at the point where it is on our planet.

The moon by its complex movements and its age during the lunar month

(*) Big Dipper and Little Dipper.

causes in its rapid circulation a distribution and modification of our differential of galactic and solar energies which requires a monthly adjustment of the devices in operation at the place where they are.

Bignand considered his rotor or rotating variable capacitor as the rhythmic oscillator of the galactic field modified in harmonic, electrostatic and electromagnetic mode. The number of poles, their shape, the inclination^{on} the generatrix of the cylinder of the leading edge of each pole as a modulatev njfa e this "energy" aether.

The four fixed bi-metallic capacitors with wax activated during the installation of the dielectric as assimilated to four of the states of matter-radiation on earth.

The 12 adjustable capacitors like 12 discharge circuits act like a resonator with the luni-solar positions and whose adjustment required monthly adjustment according to the age of the moon and the position of the sun in each sign.

Two other tuning capacitors allowed fine tuning of the resonance of the grouped constellations Big and Little Dipper as well as the Boreal Dragon.

chokes , resistors then *made* it possible to amplify, to control, energy transfers for one of the various purposes for which the device could be used.

Bignand could thus improve, exalt certain properties of the bodies or give them entirely new properties after having determined the characteristics of the energetic radiation absorbed or admitted by the body or bodies to be treated, then subjecting them to complementary radiation of character. determined teristics whose primary initial energy was captured - according to him - in the field of cosmic and telluric radiation with all the desirable selection.

The results obtained by Bignand exceeded, so to speak, the . witnesses of the time. We have cited the activation of plant states and l[^] Wt of all this information should be of interest to us for 1 'Electroculture, but we also cite the modification of the raw materials of certain dielectric or insulating products, considerably improving the properties of these dielectric materials at inside which can be incorporated into colloids

(*) When we talk about aether, there is no question of talking about energy because everything is delocalized.

other materials making it possible to obtain new effects at the level of the radiation of the usual physical fields.

Furthermore, the selective power of the apparatus allowed the colloidal dissociation of metals and various other substances in water or other products in the liquid state at ordinary temperature. A series of experiments was carried out under the supervision of M. Doladilhe at the physics laboratory of the Faculty of Sciences of the University of Dijon.

Checks carried out by Georges Carrère, an agricultural engineer, have highlighted not only the optimization of legumes already mentioned but also the neutralization of harmful effects on animals and humans from telluric and cosmic radiation. Bignand considered this use in dwellings and stables as a possible air conditioning of premises whose characteristics to be neutralized were determined in advance.

This aspect of the "air conditioning" of the premises found its application in agriculture at the level of the storage of the plans as well as the establishment of particularly stimulating greenhouses for the shoots requiring an operation of $r_{gffi} \wedge age$. Competitive achievements were made in Belgium and were known to the public under the name of "Boites à Bignand",

The application of the method required a transmitter device such as that described above, suitably adjusted for the particular application of ambience harmonization. The purpose of this setting was to load the air conditioning unit in question. This apparatus comprised a tank containing pure beeswax into which were incorporated various micronized substances chosen according to the particular result sought: colloidal metals, colloidal substances, finely dissociated particles, of metals, of metalloids, alloys, solutions or homeopathic products etc... etc...

In this set, in this medium were immersed two electrodes, one for  vant e one plate of zinc, the other a plate of copper, the forms of which may vary. Metals could also vary in their nature.

To achieve air conditioning, rebalancing, harmonization

of a room, Bignand operated in the following way: after having "charged with waves" the air-conditioning device by means of the transmitter generator described above, Stany placed in the electrical circuit of the alternating network by a two-wire connection the two electrodes of the air conditioning unit inside which was the dielectric containing the suspended matter previously chosen for

their characteristics. This small air-conditioning box is placed in the room to be treated. Stany considered that a polarization appeared on the plates creating a "field" serving as a support for the radiations stored in "1 ' eWjlfeyte" although dielectric of this air-conditioning device, which after the avoîjS" , • -, (*) primed, maintained their oscillating and resonating vibratory energies.

Depending on the nature, the dimensions of the place to be air-conditioned, also the materials of the building, the composition, of the ground of the basement, the cover of the construction and according to the particular conditions of "air conditioning" or atmosphere to be achieved, finally according to the results of the preliminary study of the elements of the environment, Bignand manufactured either a simple tank made up of the parts and materials already described, or a tank combined with more or less important oscillating circuits.

The observation of the effectiveness of Stany's devices, in addition to numerous users, was made by veterinarians and agronomists in the case of certain diseases, epidemics, epizootics or various places and circumstances. George Carrère agricultural engineer signed an article relating to some of the results obtained by Bignand in the International Review of Dowsing No. 19.

Another application of Bignand processes was in 48-49, a little before his death, the exaltation of the perfume of certain products, or of certain flowers such as the carnation. An experiment carried out in Geneva for Coco Chanel gave impressive results on the fixation of the natural essence of oeillets in wax, although this operation is, according to certain experts, not feasible. This realization, taking into account, because it must be said, the position of the planet mercury in space (transit on the sun) proves once again the validity of the theoretical and... practical ideas of Stany.

Articles and references relating to Stanislas Bignand . Patents.

- International Journal of Dowsing and AC, N°19. Article by Georges Carrère, Agronomist on the optimization of the growth of legumes and cereals stimulated by S. Bignand.
- International Journal of R. and SC, N°17. Article signed M. Doladilhe, Master of Conf. Adj. of Physics at the Faculty of Sciences of the University of Dijon. Title: Nature of Atmospheric Energy discovered by the dowser

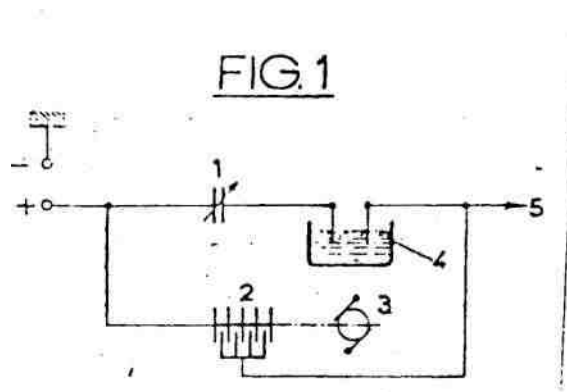
(*) Some of Bignand's solutions were electuaries.

says Bignand. Articles published shortly after Stany's death.

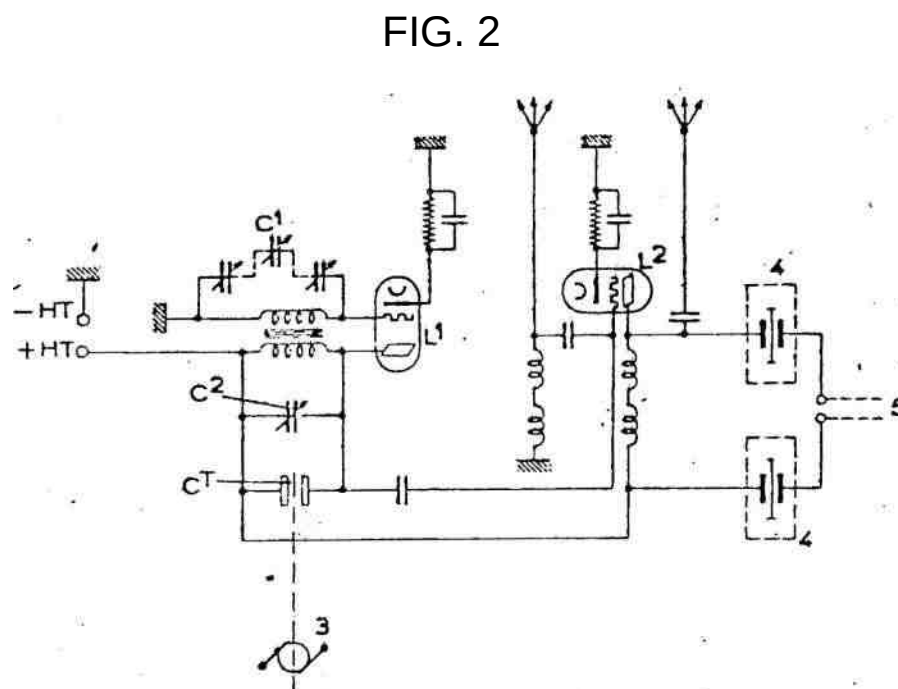
University of S. Bignand patents ;

- Process for treating various substances to improve their properties or to give them new properties and products obtained: dem. 2 aë uf| | ^_941, pub. January 5, 1943, Gr 14.Cl 8, No. 877858.
- Process for developing the qualities of different gases: dem. October 26. 1945, pub. April 28, 1947, Gr,5, Cl 8, N° 921.109,
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(**)F. Dufourg had also developed a process for exalting the perfume of violets, thanks to certain information obtained in a state of trance from his wife. The paths to discovery are multiple and astonishing! !



1. Common type variable, adjustable capacitor.
2. Special variable capacitor consisting of a rotating element with 12 blades staggered so that the gap between the first and the last is 30° .
3. Rotary electric motor of the universal type supplied with 110 or 220 volts 50 hertz for driving the rotor above.
4. Dielectric to be treated: waxy or other materials, various electuaries containing certain bodies in suspension. An electrolyte possibly.
5. Usage network.



Appareil plus complet pour principe ...

[1]

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Georges BERGE

par R, SMIRNE

(Paris)

Alongside the work of S. Bignand, an experimental researcher living in Seine et Oise, Georges Auguste Albert Bergé used between 1960 and 1944 various devices intended to increase the vitality of living beings and especially seeds. He had constructed a device made up of several metal rings arranged concentrically and openings of various sizes. The section of the rings was variable, or more exactly adapted to each type of seed, seeds, bulbs, tubers to be treated. To the action observed on various plants in layers, greenhouses, he had adapted a form intended for the stimulation of poultry and eggs.

Under the open rings placed in boxes of harmonious dimensions, (*) his particulars (use of the golden ratio), Bergé had placed colored screens in the order of the solar spectrum or the rainbow. The screens had an adjustable horizontal offset, the distribution of colors being made by a lamp whose rays were projected by an adjustable parabolic mirror and an optical condenser.

A variant of the color projection was achieved by a superposition and a shift of 7 transparent rhodoïd bands folded in 8 or by Moebius band placed in front of the progressive openings of the open metal rings. (The light beam directed through these bands hitting a selected area.)

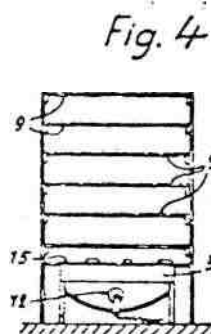
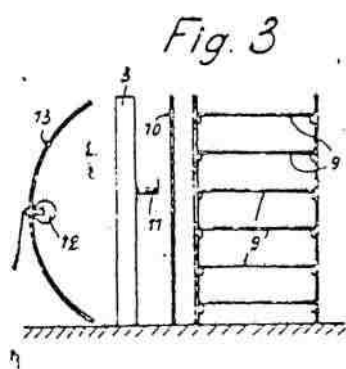
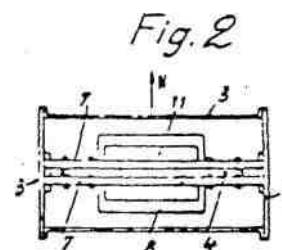
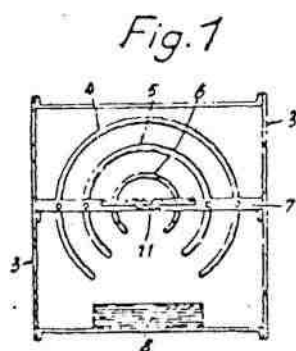
In some models was associated with the rings a sheaf of soft iron wires tightened by a copper wire connected to the ground,

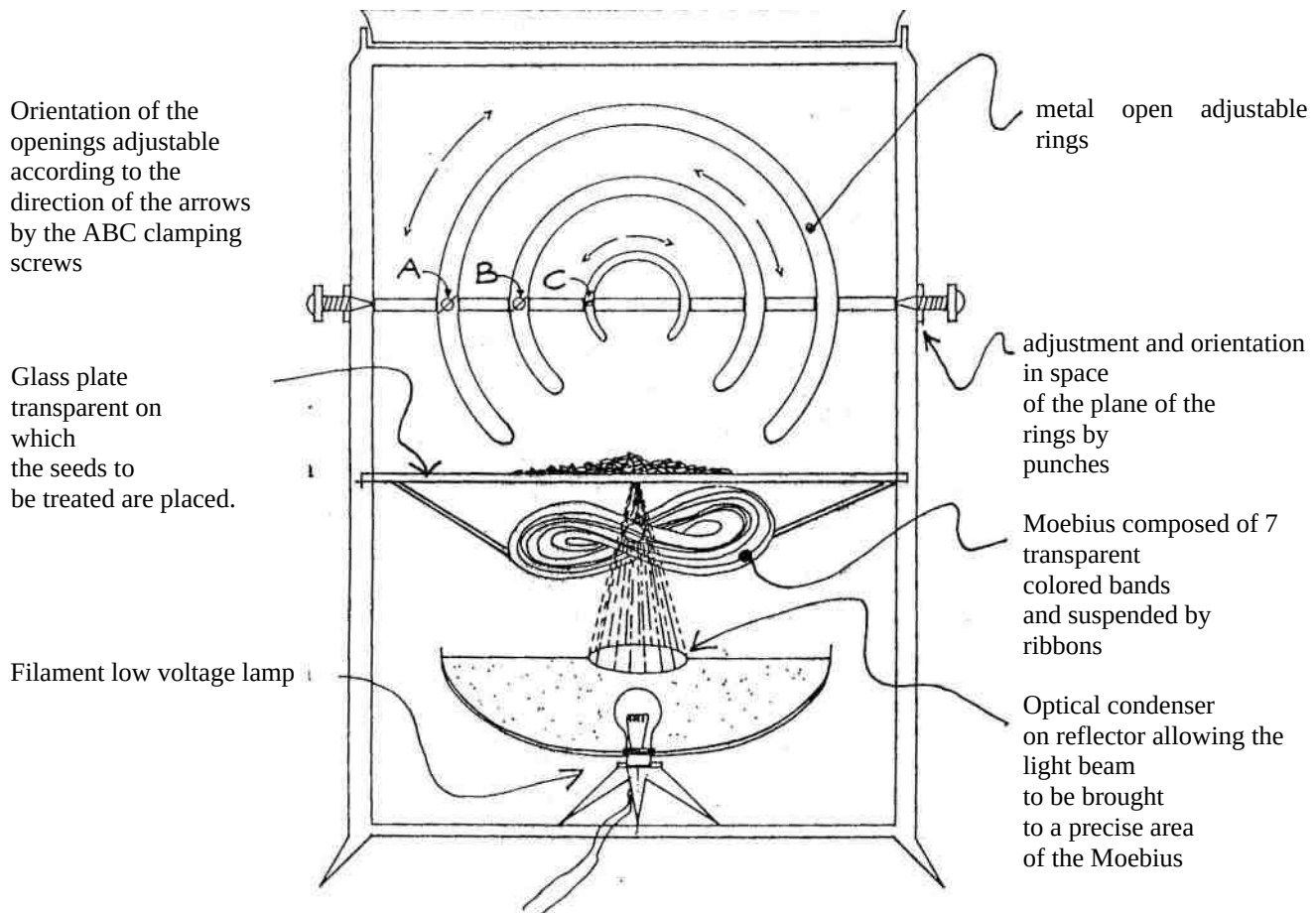
In other models the device and the colored screens were enclosed in insulating boxes whose internal walls were also colored in mono, bi, tri-chromy.

At the top of the device, a receptacle S intended to contain fragments of various metals or metalloids as well as various bodies could also be placed in various orientations by two perforated cords.

(*) It would be preferable to use the radiant number of G, Cobbler,

putting a precise positioning in height of the small container. Finally the number of rings changed according to the desired effects as well as the distance of action.





In the devices of Bergé, this researcher and his entourage, could note the following effects: 1°) for each variety of seed an appropriate adjustment had to be made; this "adjustment", this ad-hoc adaptation, consisted in orienting the openings of the rings according to the magnetic meridian of the place. 2°) to vary the plane, or the planes of each open ring from the horizontal position to the vertical position by the rotation either of one or more of them. 3°) One or more colors had to be projected by filters 1 on seeds, grains, bulbs or tubers. On this subject, other devices projected the color outside on superimposed shelves. 4°) All of the adjustments were made simultaneously a) empirically, b) by dowsing, with a pendulum and a rod.

Subject to these conditions, the experiments proved an increase in yield, depending on the variety of vegetables, of 15% to 40%. Let's do the re

marks that the Bergé devices are a variant of the work of Labergerie at the Montpellier School of Agriculture and whose conclusions appeared in "l'oscillation, ^CELLULAIRE" by G. Lakhovsky, that is to say that: the shape of the open rings or oscillating circuits, the section, the diameter, the inclination, the orientation in space, the nature of the metal(s), the coupling of the circuits, their colors, play a precise role for each species of plant, however Bergé mainly treated the seeds.

Note that Bignand, Bergé, following all the official or unofficial researchers who have approached through their work the elements affecting the living cell through plants have been led to note an extension of the effects on animal cells but our subject being mainly electroculture, we will not develop veterinary, breeding, or medicine and pharmacy subjects. However, let us point out the work of a therapist from Marseilles, who from 1950 to 1968 modified Lakhovsky's devices with a few collaborators, taking inspiration from traditional sources, Eastern and Western, as well as certain more recent discoveries.

Edmond Vernet and some of his collaborators aimed to act on the Deoxyribo-Nycleic acid present in the nucleus of each living cell and which, according to them, has ferroelectric properties; also emphasizing that the planets of the solar system as taught by traditional Chinese medicine act on our organs: thus the lungs are under the influence of Venus, the liver depends on Jupiter, etc. At the same time, Vernet and his friends had linked to these elements the remarks of physics, astronomy and meteorology that the famous botanists Linné (to whom we owe the Taxonomy) and de Candolle had been led to make while studying the blooming of flowers and who left their names to this " time-measuring-device" that are floral clocks.

Thus, with the aim of acting on certain properties of the nucleus at the level of deoxyribonucleic acid, Vernet had produced an oriented antenna composed of nine Hertz resonators or open oscillating circuits. Each of these resonators was divided into seventeen sectors. This antenna-diffuser had (A) κ 153 squares (17 x 9) or sum of all numbers from 1 to 17 inclusive. Each box represented for Edmond Vernet the location of an organ according to the diagrams of iridologist doctors. The diameter of the circles was a transposition of Bode's law, their calculation having been made according to the ratios of the distances of the planets to the sun in the order: Mercury, Venus, Earth, Mars, Jupiter, (*)

(*) It is also the number of fish taken from the net of the disciples of Christ at Lake Tiberias (Evang. according to St John)

Saturn, Uranus, Neptune, Pluto. Each open circuit represented the orbit in which each planet travels around the sun. Vemet does not speak of the various so-called correspondence metals attributed to the planets. He used the positive disc in the following way: a "crocodile" type clamp used in electricity is fixed on the circle representing the earth in the sector indicated by the iridologists, according to the organ on which one wants to act. Then another "crocodile" clamp is placed in the orbit of the planet which, according to Chinese tradition, influences said organ; the cell regeneration effect is obtained when the two clamps form a certain angle.

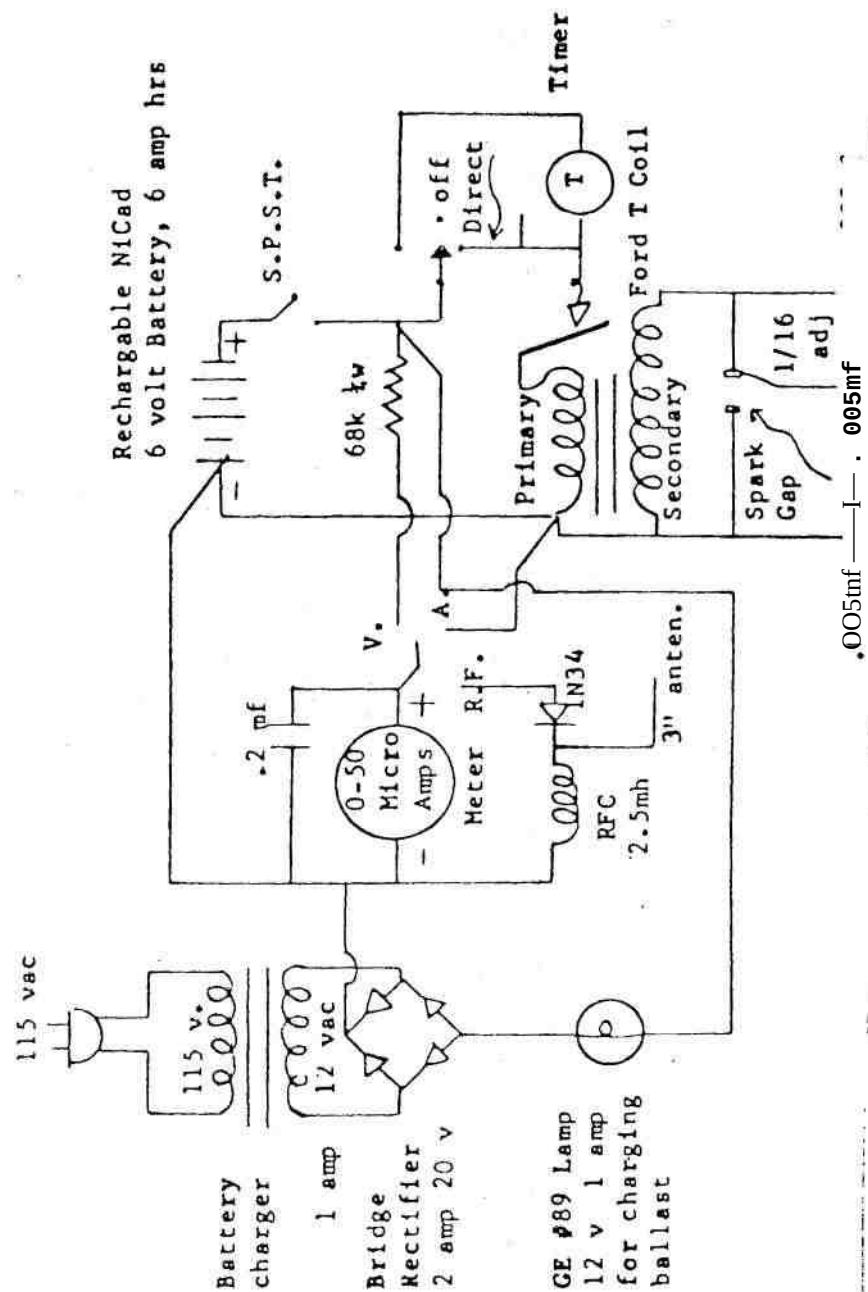
This "astral energy" captured in the cosmos by this antenna with 9 circuits and 153 boxes was then transferred to a high frequency generator whose principle is the Lakhovsky multiple wavelength oscillator and whose oscillating circuits had been modified and calculated so that each of them can act on an acupuncture meridian.

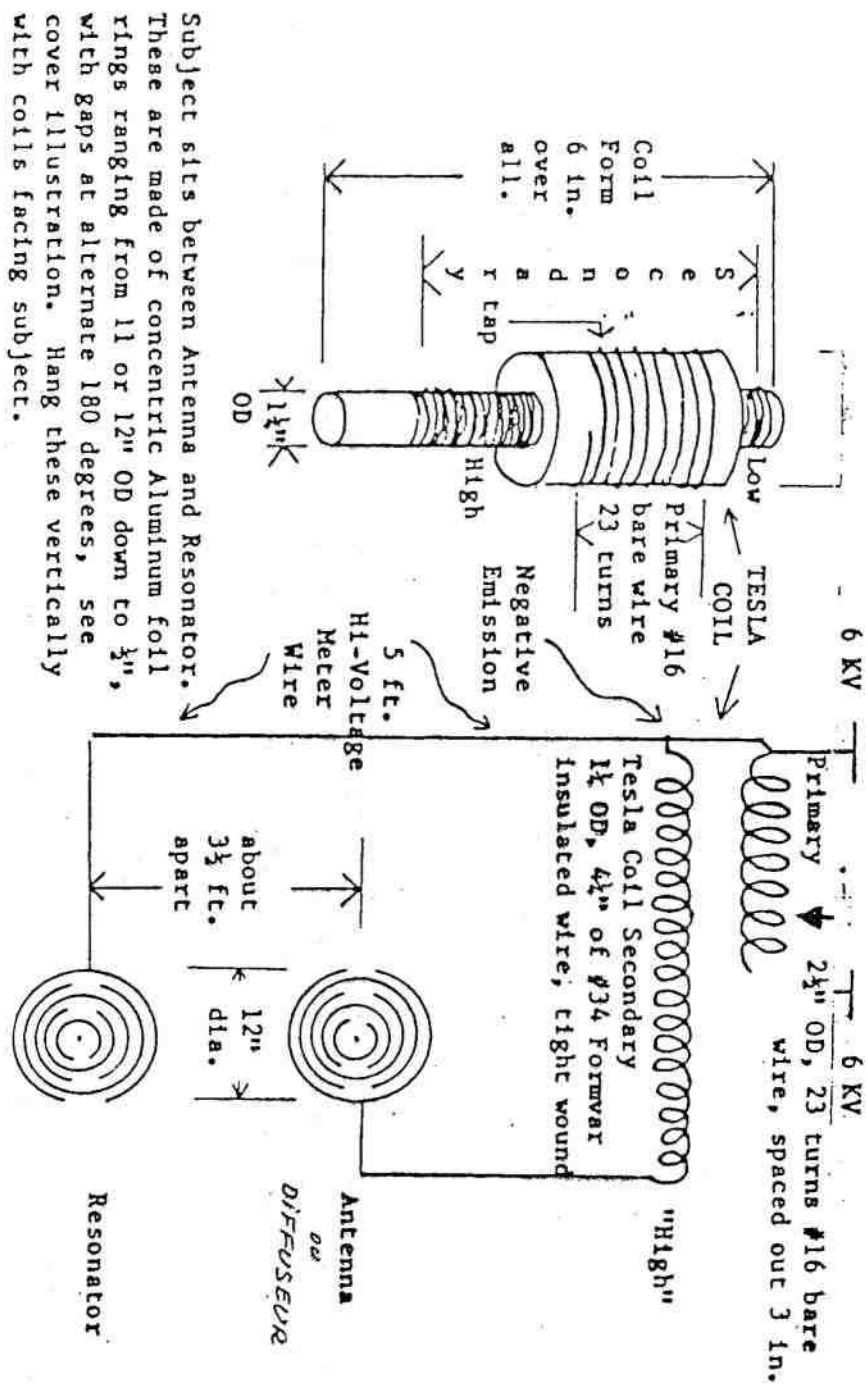
The two diffusers of this high frequency generator comprised twelve concentric open circuits ending in metal spheres welded to a threaded rod acting as a micrometric screw to adjust the spacing of the opening of the circuit. Depending on the cases to be treated, E. Vemet thought he could calculate the choice of wavelengths to be projected into higher harmonics by the least common multiple (?).

In other cases of treatment Vemet coupled to its cellular regenerator with multiple waves instead of the oriented antenna with 9 resonators, a metal statue on which the acupuncture points of the meridians were pierced. He inserted plugs at the points indicated after examination of the Chinese pulses and connected to the oscillating circuits with 12 rings the conductor coming from the point of the statue. According to Vemet, he thus achieved an "electronic acupuncture" without contact. His tests led him to also couple by adopting them to each open metal circle rare gas electrodes similar to those of the Fluvita or Holo-electron devices.

But another application had to be made in relation to our subject of electroculture with regard to the floral clocks of Linnaeus and de Candolle

Edmond Vemet and his collaborators indicated in the press of the time and in patents that they wanted to act on the cell nucleus and on deoxyribo-nucleic acid. According to their hypothesis close to that of Bignand, they considered that the nascent states in the organization of living cells





Principle of the multi-wavelength oscillator.

The diffuser and the resonator may include circuits whose number and metals vary with the nature of the applications.

The characteristics and genetic states are intimately linked to a local atmosphere on the earth, which is conditioned by a distant atmosphere such as the position of the planets in the solar system and the position of the Sun-Moon couple in the equatorial zodiac . Their observations to affirm such a thing came on the one hand from the reality of the circulation of energy in the acupuncture meridians (12 bilateral and 8 extraordinary meridians) and from the hourly cycle of activity of the said meridians, and from on the other hand, observations made by gardeners, phytotherapists and botanists on the blossoming of the bud which, in order to become a flower, follows time rules linked to the position of the sun according to its right ascension, to the air humidity and latitude, longitude and altitude.

The Swedish naturalist Cari Von Linné (1707-1778) establishes a classification of plants into 24 classes, based on the characters drawn from the number and arrangement of the stamens then establishes a subdivision of the classes according to characters drawn from the number and the arrangement of the carpels forming the pistil. His observations also led him to give certain plants the name "meteoric" because of the observation of their sensitivity to atmospheric variations : their flowers open differently depending on the shade, the humidity, the dryness of the air. . He designated others by the name of "tropics" because they advance or delay their rising according to the length or the shortness of the day. Finally he considered other plants as equinoctial because they open or close at well-determined times.

With all these plants which bloom at different times but always the same for the same species, Linnaeus imagined composing a floral clock by marking, as on a dial, by the blooming of a different flower, each different hour. , every hour of the day and night. Here's what he got:

| | |
|----------|-----------------------|
| 1 hour | • Lapland sowthistle |
| 2 hours | • yellow salsify |
| 3 hours | • Great picridia |
| 4 hours | • Bindweed |
| 5 hours | • Crepid of the roofs |
| 6 hours | ◆ Scorzonera |
| 7 hours | • Nenuphar |
| 8 hours | Chickweed |
| 9 hours | ■ field marigold |
| 10 hours | • Ficoid |
| 11 hours | • eleven o'clock lady |
| Midday | • freezing |

13 hours: Carnation
 14 hours: Crepid red
 15 hours ; Barthausic
 16 Hours: Alysse
 17 Hours: Belle de Nuit
 18 hours: Geranium
 19 Hours: Beautiful Day
 20 hours: Nocturnal ficoid
 21 Hours: Nyctanthe
 22 Hours: Purple Bindweed
 23 hours: Night-flowering catchfly
 Midnight: Large-flowered cactus.

After Linnaeus, de Candolle was struck by these phenomena and noticed that the Lady of eleven o'clock opens at eleven o'clock to close at 3 o'clock, that the night-flowering Ficoide blooms at 7 o'clock in the evening to close around 6 o'clock in the morning and that the Large-flowered candle blooms at 7 p.m. to close at midnight.

He noticed, moreover, that certain flowers undergo the influence of the heat which acts on their blooming and on their duration and that these two phenomena vary according to the latitude of the country and, for the same country, according to the season.

He therefore modified Linnaeus' Clock accordingly and erected a
autre qui avance sur celle dressée par Linné

| | |
|-----------------------|-----------------------|
| Between 3 and 4 hours | hedge bindweed |
| at 5 a.m. | Poppy and chicory |
| at 6 a.m. | solaneas |
| at 7 a.m. | water lilies |
| at 8 a.m. | chickweed |
| at 9 a.m. | icy |
| at 11 a.m. | purslane |
| at noon | ficoids |
| at 2 p.m. | squills |
| at 5 p.m. | catchflies |
| at 6 p.m. | la Belle de Nuit |
| at 8 p.m. | large-flowered candle |
| at 10 p.m. | convolvulus |

Moreover, Linnaeus was struck by the fact that certain flowers possess the property to announce the good weather or the rain.

These are the ones he called "meteorical" whose development is modified by the atmospheric state. Such is the case of certain Chicory plants which do not open in the morning when it is going to rain, as well as the rain marigold which closes when the weather is rainy. The day will be pure and beautiful if this flower

blooms early but expect rain if it stays closed after 7am.

However, these plants cannot predict storms; they remain open during stormy rains which surprise and deceive them. Only clovers and wood sorrels know how to fold their leaves a few moments before storms, like the springtime log drive which bends its little head at the approach of the rain as it does at nightfall.

Facts of the same order, rather difficult to observe and from which it is difficult to draw exact conclusions, have nevertheless served to draw up a veritable barometer, called by Linnaeus and other botanists, a floral hygrometer.

On the other hand, the time when flowering begins varies according to the species, the latitude, the climates, which did not prevent Linnaeus from having the rather poetic idea of drawing up a Floral Calendar, which Van Helmont called in his time the "Balsamic Time".

Here is one that is usually accepted for the most part in France:

January: violet
February: anemone
March : primrose
April: small periwinkle
May : lily of the valley
June : blueberry
July: carnation
August: magnolia
September ; colchicum
October *. chrysanthemum
November: verbena
December: Christmas rose

The Japanese, whose country is in bloom all year round, have not failed to establish a floral calendar, where each month is designated by the flower that embellishes it or best perfumes it.

Thus the second month is that of Orchids, the third that of Paulonias, the fifth that of Magnolias, the ninth that of Chrysanthemums and the twelfth that of Camellias.

We have underlined observations of electrical, climatic and other variations on plants in the work of Sir Jagadis Chunder Bose, there is an obvious relationship, in the observations of Linnaeus, his successors and Bose.

E. Vernet thought that ancient and traditional observations, numbers, shapes, appropriate geometries, have for the observer of nature the key to understanding the secrets of the structure of the universe. Also in an attempt to integrate sparse data, Vernet just like Bignand but perhaps with the advantage of additional knowledge that he did not have (*) neither did Lakhovsky think that the incipient and varying state of the constituents of the nucleus of the living cell was the seat of phenomena presenting microcosmically remarkable similarities with the incipient states of the macrocosm.

Armed with these assumptions, the friends and collaborators of E. Vernet carried out series of measurements on plants in correlation with Linnaeus' remarks. It should be noted that the beginning of Véret's work had focused on the treatment of cancers, just like G. Lakhovsky and his experiments in healing *Pelargonium immaculatum* with *Bacterium Tumefaciens*, thus continuing series of measurements carried out by dowsing for most researchers. endeavored to discover "wavelengths" carrying unknown energies coming from the cosmos, for the stimulation of plants from their cells.

We notice in Vernet's design for his antenna intended to capture the astral energies in the cosmos that

- 1°) the area of the telescopic planets is not really represented
- 2°) The small planets GERES, PALLAS, JUNO and VESTA are not either.
- 3°) By using concentric and regular circles where always hangs a harmonic.

What is unclear on the other hand in Vernet's antenna devices is the connection, or the branching of said 9-circle and 153-box antenna with the 12-circle resonator diffusers of the multiple wavelength oscillator .

Still, the users of the seeds, seeds, bulbs, tubers treated by the device as well as the rainwater treated by these positive devices allowed Rhone experimenters to obtain exceptional vegetable crops as well as splendid varieties of flowers by choosing the time of action on the plants according to their biogenetic biological clock.

(*) Deoxyribo-nucleic acid.

N° 1.482.874

M. Vernet

PL unique

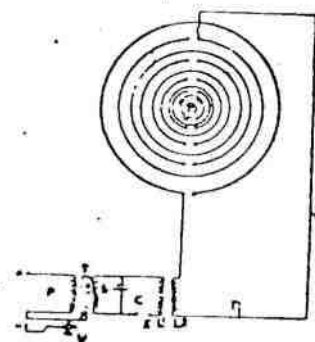


Fig. 1.

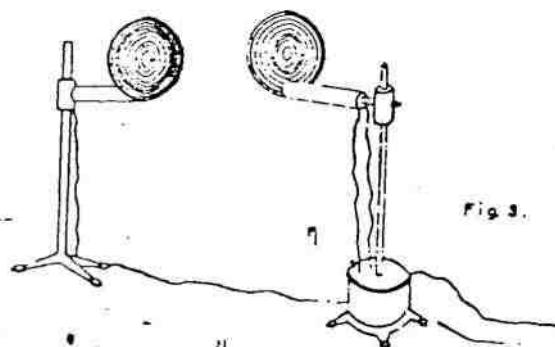
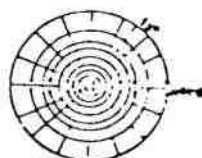


Fig. 3.

VERNET branch

The concentric circles were divided into 17 spokes
giving 153 squares.

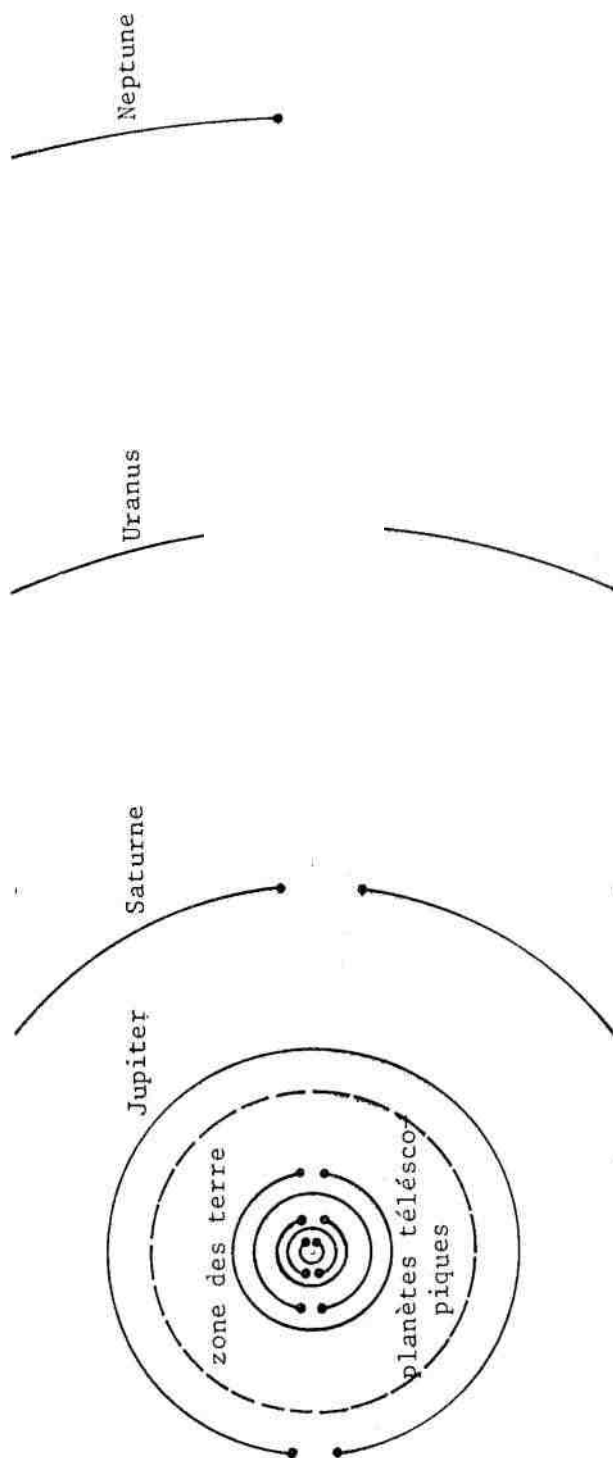


Diagram of the solar system showing the Sun at the center, with concentric orbits for the planets. The innermost orbit is labeled 'zone des terre' and the outermost is labeled 'planètes télescopiques'. The planets are labeled: Jupiter, Saturne, Uranus, and Neptune. The Sun is labeled 'Soleil des planètes, celle de'.

Diagram of the solar system showing the Sun at the center, with concentric orbits for the planets. The innermost orbit is labeled 'zone des terre' and the outermost is labeled 'planètes télescopiques'. The planets are labeled: Jupiter, Saturne, Uranus, and Neptune. The Sun is labeled 'Soleil des planètes, celle de'.

Mercure : 0,387 5,203

Venus : 0,723 9,539

Earth: 1, Uranus: 19,18

Mars: 1.524 Neptune: 30,07

Unfortunately, apart from a few scattered articles of these results, there has been no synthesis of the results and at the present time perhaps at the Ministry of Agriculture one could find in the archives the results of the experiments of this mine. of researchers whose names are now almost forgotten

[1] .

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[1] P. Smirne, ARK'ALL Commun., Vol.8, fasc.1, 139-152 (1983).

GNOSTIC DOCTORS AND THEIR SYSTEMS

by

P. SMIRNE

(France)

This text is supplemented for understanding the work of

L. Boutard and A. Hatinguais, but the presentation is made to relate a little to the mode of expression and the representation of the occultists. According to Simon of Samaria, at the summit of entities, that of fire, infinite power of all the aethers and states of heat. This power to act goes into action. To express its thought, intelligence named it, and having named it, it was able to unite its different ideas and form a whole. From this all were drawn six beings or emanations of infinite power. These beings were obtained by syzygies: two by two active principle, passive principle. Simon also had the six aeons of the superior.

- | | |
|--|---------------------|
| + We (mind) | - Epinoia (thought) |
| + Phone (the voice) | - Anoma (the name) |
| + Lagismos (reasoning) - Enthumesis (reflection) | |

In each of these six aeons or primitive beings (nixaïs), the infinite power was found entirely in "potency" but not in "act". To pass in "act", it is necessary to make an appeal to an image which is in conformity with what one awaits infinite power by an "image". For this the aeons themselves became principles of emanation and six intermediate aeons were created. Finally in the terrestrial world took place the material creation itself according to the law of the similarities which says that all the beings of a particular world develop in a way in conformity with that in which developed the higher world and that all the beings in each world are themselves eager to instigate what their superiors in the emanation hierarchy are doing. The desire carried out leads to the principle of the fall.

Energetically, the principle of Gnostic generation is always the

same: the active power enters into communication with the passive power, deposits therein a germ which is only a spark; the germ grows and develops according to image and likeness. The doctrine of Basilides of Egypt completes that of Simon. At the top, the god who is not, the nothingness that exists "o wx ontheos" (which can become something), passes through internal action in God called Giotés, of triple essence and triple activity; under its action, there is a first emission (Katabolé) which is, which becomes the act. The first rises to the pleroma; the second (carried on the wings of the Holy Spirit) only reaches its limits; the third remains in the germ of the world: for it redemption is necessary know. Thus the upper world is created. The intermediate world was then created by the union of this germ of the world and the treasure of all the seeds: the great Arkôn was born enthroned following the upper world. The nature background of God is Love, and Love cannot do without an object to love. This Arkôn generates a son. The link where these mysteries were realized is the agdoade which extends to the moon where reside the principalities (Archai), the powers (Dunameis), the dominations (Exausiai) and three hundred and sixty five gods governed by the great Abrasax (corruption of the great Hebrew blessing (the Barecha (from Berech: to bless)). Abrasax can be given either the etymology: Al Braxas (the blessing stone), or "abra-q-ad-habra" (meaning : "Send your lightning until death"). From this world of the Ogdoad emanates a whole other world which reproduces the first according to the principle of similarity. There was thus a second Arkôn dominating the space where we live where the potentiality of all the possible germs is. But the beings emanating from a world cannot conceive of a better world than their own. The second Arkôn was unaware of the first and wanted to dominate in turn the geniuses. These resisted him. and it was a terrible struggle.

Valentine of Alexandria humanized the teachings of various Gnostic schools. Emanation is the basis of his system. In the intermediate world the prince is the "Demiurge", "The Ancient of Days". The material essence has another prince: Diabolos; the spiritual essence has as its prince Beelzebub; the Demiurge is foolish and "coarse"; while the divine wisdom Sophia acted and created, the Demiurge believed that everything came from his hands and imagined himself to be God.

Valentine definitively establishes the Gnostic science of the pantacles; destiny therefore depends on the powers that preside over acts; as for the acts themselves, they have no influence, they only indicate the strength of the dominating powers. For Valentine seven planets enter the twelve signs of the zodiac; the rising of these planets is directed by the powers. They indicate

the movement of matter. There is often a struggle between the powers.

In gnostic pantacular science, in addition to the "name" (the name is the thing), source of the power of the pantacle, appear the respective representations of the god, the angel, the daimon evoked as well as the symbolism of planetary time.

The Gnostics modified what seemed to be usages and brought very precise, new and important operative elements. Tables were set up in a way that seemed strange by the proximity of the names: Yahvé, Adon, Sabaoth, Michel, Osiris, Isis, Anubis...

The DLXXIV papyri from the National Library - Mimant 2319 from the Louvre, the demotic gnostic papyrus from Leyden, the Greek papyrus XLVI from the British Museum... show this work of synthesis on the problem of the psalmation of "energies". The Cabinet of Antiquities of the National Library contains a large number of Gnostic amulets including the ABRAXAS, which according to the Greek numeration system gives:

$$\begin{array}{l} A = 1 \\ B = 2 \\ R = 100 \\ A = 1 \\ S = 200 \\ A = 1 \\ X = 60 \end{array} \quad \text{i.e. a total of } \overline{365}$$

These considerations are profane; they circulated among the people. They are not initiatory. Gnostics wore abraxas and according to their rank they had to be able to explain to which device was related such or such abraxa except some which were known only to higher grade gnostics.

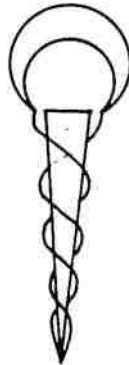
Gnostics have related it to Mithra whose Greek name meitras has the same number

M=40-, E=5, I=10, T=9, R=100, A=1, S=200.

The Abbé de Montfaucon divides the abraxas into seven classes:

- 1) Those with the head of a rooster
- 2) Those with the lion's head
- 3) Those at the head of evapis
- 4) Those with the heads of Anubis, snail, Sphinx, snake, monkey
- 5) Those with human heads with or without wings
- 6) Those with Hebrew inscription
- 7) Others more bizarre.

The divine nomen (definition of what is to come) is known as Abraxas for Abrasan, a substitution which is explained by the Latin scribes of 3rd and 4th centuries. Subsequently, the use of the angelic name par excellence has given the general name of the Gnostic stones found. The name of Abraxàs was associated with figures of which the most frequent was Anubis carrying



The inscription of Miletus in Asia Minor, published by Le Bois in 1847, gives the correspondence between all the Greek vowels and an "energetic" sphere of action linked to seven planets, satellite (the moon) and star (sun). The planets were:

Saturn
Jupiter Mars Sun Venus Mercury Moon

The 2nd century Gnostic musicographer, Nicomachus of Gerasius says: "The sounds of each of the 7 spheres produce a certain noise, the first making the first sound, and to these sounds have been given the name of vowels... The Theurgists invoke the divine by lip-clapping, whistling, inarticulate sounds without a consonant. It is the cerebral configuration of the human being which creates the relationship between the sun, the moon and the 5 planets and the sounds. The human being, by his thought, arranges the correspondences in a given representation.

The seven-stringed Greek lyre (heptachord) is quoted in the papyrus W of Leyden and in the "Papyrus Mimaut". One establishes the correlation by psalmody of seven vowels with seven virtues or powers. (*) (**)

(*) published by C. Leeman, Papiri graeci

(**) 9319 from the Louvre.

| | | | | | |
|----------|-------------|--------------------|------------|------------|---|
| <i>n</i> | Mid | hypact meson | Chronos | Lead | |
| <i>O</i> | Fa | parhypate meson | Zeus | Tin-Zinc | |
| <i>t</i> | Floo | lichano meson | Ares | | |
| 0 | The | my e | Helios | | 0 |
| I | whet | tritrus synemmenon | Sphraodite | red copper | |
| H | her | paramet synemmenon | Hermes | Mercury | |
| E | C | | Silenated | | |
| H | D | netes ynemmeno n | | | |

Seven metallic principles preside over it with 4 x 7 moonlights in seven intonation sounds. The Gnostic incantatory formula will be:

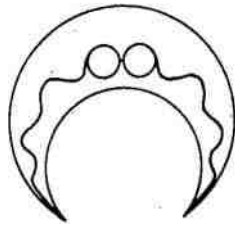
Pronouncing towards the West, the right hand extended towards the left and likewise the left towards what is on the left, you will say A; towards the North, the extended right fist, of the say E; extending both hands towards the south of the say H; then towards the South, the hands turned towards the stomach _z you will say I; spitting on the ground and touching the extremities of the feet, you will say 0; looking in the air, the hand on the heart, you will say there; looking at the sky and both hands on the head, you will say 0 .

The magical papyri that have come down to us indicate constant employment
Greek vowels both as voice notation and as sacred syllable,
but also as an "energetic" indication.

Remark. It would be hasty to identify the Boutard atoms with the atomic nucleus and the electricities with the electronic cloud: one cannot reasonably confuse the arrangement process of the little serpents with that of the N,O,P,Q electron shells. ; moreover for iron and copper, we only have the N layer.

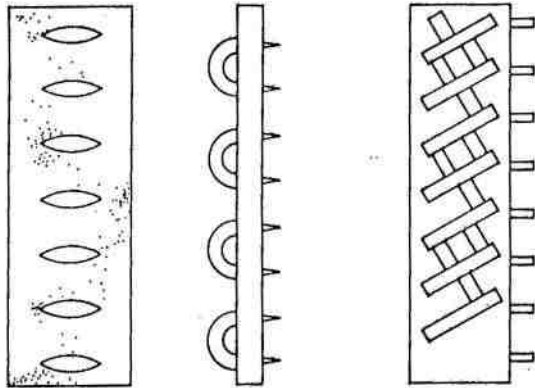
All one can do, if one follows the Gnostic theory, is to note that the world of matter (of the fall) is the imitation of a process of which only prototype matter is the primary form (arkon).

Amylifying generator : While the truck represents the element
fundamental of the automatic generator, the steel crescent is that of the amplifying generator ' ^ u: *- before reading the
abraxas, the truck
carries a small snake whose head rests on the sphere of the truck and the
doubled snake is worn in double winding on the crescent. The heads meet towards the center, at the belly of the crescent,
the tails being directed towards



peaks; the croissants are fixed on a strip of baked goods, the bellies on one side, the tips on the other.

Electricity is collected by 3 strips of red copper very thin and perforated in regular diagonals

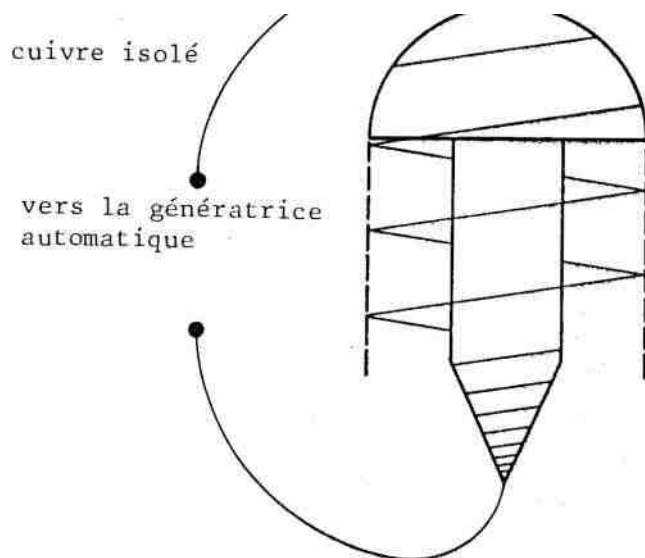


The perforated red copper band equipped with lead wires is sewn on a cotton case and applied on the stomach side of the croissants on the leather band. The red copper band equipped with brass threads is sewn into a cotton sheath and applied side to side to the leather band (carrier). Both collectors and together. The band of red copper equines three more.

the wearer are all three sewn sword of red threads is wrapped around

red copper strip winding (*)

(*) There are 3 such strips: one whose collectors are lead wires (black blue), one whose collectors are brass wires (yellow), one whose collectors are copper wires (red).



wooden barrel its shape

The sausage is rolled up on the wooden mandrel, The nature of the wood of the mandrel is not specified.

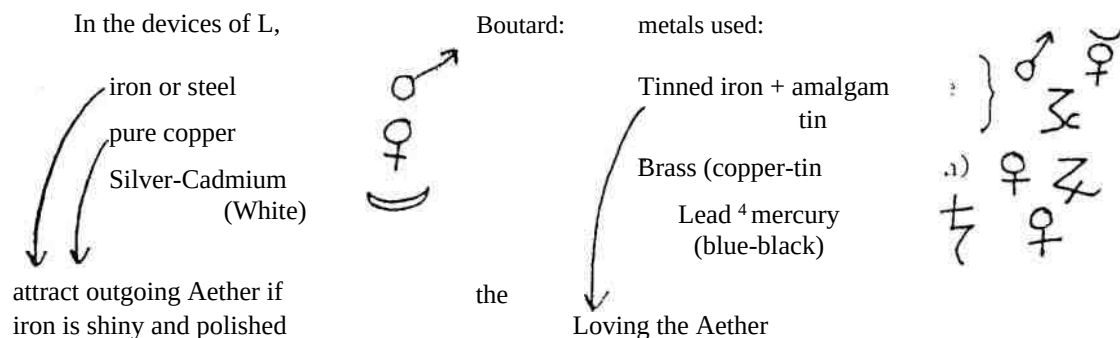
The conductor of the inductor current consists of a strong insulated copper wire which, after having been wound on the head, enters the calote through an opening provided for this purpose to receive the coil already wound as well as,,

Outlets are provided for the lead wires,

brass copper.

All of the 4 strips above are wound on the cylindrical part of the wooden mandrel with a mushroom. The base of the cylinder ends in a cone. The set is about 25 cm high.

The choice of wood is important. The height is not.



Copper collects red atoms

Brass " " " yellow

Lead " " " blue-black

Mercury promotes the disintegration of the Aether by attracting the blue-black atom.

The goal used for Boutard is to dissociate the Aether and to draw from it the elements from which one expects an energy by collecting the male atoms and the female electricity, using appropriate conductors.

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P. Smirne, ARK'ALL Commun., Vol.8, Fasc.2, 113-119 (1983).

TH MORAY'S "RADIANT ENERGY"

by J. RAVATIN

(Paris)

T. Henry Moray began his work at the age of 9 in a small laboratory he had set up. That was in 1901. He began experimenting with what he called "Radiant Energy" in the summer of 1909 and 15 months later he was confident enough to have a small device that worked.

It corresponded to his idea of drawing energy from the earth. He worked as an electrician at the "intermountain Electric Company" of Salt Lake City and in 1910, he was the main shareholder of the "Independent Electric Company", still in Salt Lake City, Utah. He obtained a doctorate at the University of Uppsala in Sweden, and in 1914 started electrical installations and this for a few years during which he became a project engineer for the "Aarastad Construction - Company". Later he continued his trade in a larger firm thanks to a contract with the "Phoenix Construction Company" associated with the "Utah Power and Light Company".

After his military service, Moray worked as a civil engineer. In 1926 he founded the "Research Institute" and then became chief consulting engineer in the "Western Division of the Rural Electrification Administration". From 1940 Moray continued his work as a consultant and also his own research in the laboratories of the "Research Institute" in Salt Lake City.

To consolidate his ideas, he relied on the remark of Nikolas Tesla.^Through space, there is energy. Is this energy static or dynamic? If it is static, our hopes are in vain, if it is dynamic, and I take it for granted, then it is only a matter of time before man manages to harness his technology to the very engine of the (*)

(*) His mother was Swedish and his father Irish; both emigrated to the United States where TH Moray was born.

Radiant energy, in Moray's mind, means energy from the cosmos, coming from one point and spreading in a straight line in all directions.

Moray said "I do not claim that all my theories are proven, but I claim and prove that the device works". With his device, he lit an array of lighting lamps of 100 Watts each. There were 20 to 30 electric irons or hair irons; he could also run engines for hours on end without loss of power. When he lit lamps with his device, the witnesses observed a whiter and softer light than if they were under the tension of a usual electrical source.

On June 24, 1925 Moray discovered that by combining germanium with various substances, he could make a valve for a radio receiver capable of operating without batteries and other accessories necessary at the time for radio receivers. This germanium alloy was successfully employed for the detection and amplification of radio signals without the use of any extraneous source of electricity. He had discovered what came to be called after semiconductors but unfortunately Moray, their inventor, was completely forgotten.

On the other hand, the criticism of the patent office observer was very biting and in order to consider the invention null and void, he used the arguments of the time.

In 1911, Moray had the idea that the energy (that's what he called it milk) that he seemed to be using was oscillatory in nature. Rather, he was thinking of oscillating bursts of energy coming either from earth or from space.

Moray's efforts continued and he discovered that his radiant energy device was capable, in modified form, of detecting radio waves. The latter was ready in 1927,

It was in 1931 that in a further improved version, he redesigned his device with a very purified germanium crystal and after purification, he included some impurities.

Moray thereafter continued his experiments in order to convince skeptical and recalcitrant observers and made them check all his apparatus and their vicinity so that they were persuaded that there was no hidden source of energy. Experts were summoned to also examine

electrical equipment before start-up. They were amazed that there was no change in the setup when Moray turned on a 50 Watt lamp or a 300 Watt lamp; everything happened as if the current was adapted to the device. On the other hand, there was no sound coming out of the device in operation (it had no moving parts). The capacitors in his assembly seemed to charge slowly, the wires of the circuits remained cold whatever the time of use and the power expended.

The scientists who examined the device said that the current coming out of it was high frequency. For Moray, his device pumped electricity somewhere to give it back elsewhere.

On the morning of September 3, 1926, he operated successfully with a sound-detecting device. After months of work, in the laboratory, it was possible to capture a conversation and music several kilometers from the place of experimentation. An interesting test was conducted in the presence of the lawyer RL Judd and his son Thomas and EG Jensen. In October 1929 a Russian scientist, AA Yakalov, came to his laboratory and checked the equipment and experiments of the various devices based on the principle of radiant energy. He was able to follow in particular the listening to conversations held about 10 km from the laboratory, that is to say at the station of the Denver and Rio Railroad, in Salt Lake City.

By 1936, the sound-detecting device had been improved and it could be used by radio personnel. The signals were received loud and clear.

An interesting test involving both the Radiant Energy equipment and the sound detecting apparatus was made by General Shinkle of the US Army in December 1936. (*)

(*) The general said he was very impressed with the results and asked under what conditions the sound-amplifying device could be acquired by the government. Moray replied to the general that he was giving up all right to his apparatus for the benefit of the government, but he demanded only one condition; is that the government is naming the device Moray. The general said he would report to his superiors. Moray and his associates were informed some time later that the official report had been directed to Washington by the general, but that the government did not agree to accept the offer. (This notification was given by the United States Representative, Thomas Amlic, of Wisconsin). [1].

From 1936 to 1938, several tests were carried out with a large number of researchers from Moray's group and applied to radio. They received direct broadcasts from Antarctica where Admiral Byrd and his party were. These broadcasts came in clear and distinct, even when the regular radio stations admitted that the interference was such that they could not receive Admiral Byrd's broadcasts, due to poor weather conditions.

If we test the lit lamps, we find high Taofel fields. We are in the presence of Moray devices, Phenist objects. A localization is done.

For more details on the work of Moray, one can refer to the work of TH Moray [2]. This text is taken from [3],

Bibliography.

- [1] Dowsing Magazine.
- [2] T. Henry Moray: The Sea of Energy in which the Earth floats, (Edit. Cosray) Salt Lake City, Utah, 1960.
- [3] J. Ravatin, ARK'ALL Commun., Vol.8, fasc.2, 121-148 (1983).

X

Department of Commerce
United States Patent Office
Washington

Please find below one of this examiner communication
reviewer application

Roman E. Robertson
Commissioner of Patents

MORAY Products

Applicant: Mr. Moray

Claims are dismissed as being derived from an apparently ineffective device. It can be seen that the apparatus does not produce any sensible current, since there is no means of supplying to agitate and heat the electrons of the metal of the cathodes by thermal agitation. The current produced in the cathodes by the antenna will not heat the cathodes to a temperature at which an appreciable number of electrons per second are ejected by the thermal agitation of the surface of the cathode. The applicant mentions the use of radioactive substances to make the cathode-anode space conductive. But this will not produce a current in space in the absence of electric potentials on the plates (cathode-anode).

Translation by J. Jelodin

914 Continental Bank Block Knows Lake City,
Utah

November 17, 1926

Mr. AC Cooley New House Building Sait Lake City, UTAH

Dear Mr Cooley,

This letter is sent to you for your information and in order to make a record of an electrical experiment made October 29, 1926, by the invention TH Moray for MM. JWM Knight, RL, Judd and myself.

As decided October 27, 1926, I met Messrs. Moray and Judd at Mr. Moray's residence, 2484 South 5th Street East-Salt Lake City, Utah, at about 8:00 a.m. October 29, 1926. We charged electrical equipment in my car and left Mr. Judd's car at Mr. Moray's home, as only three of us were to make the trip from Sait Lake, We had planned to take Messrs. AW Iving, CW Nidley, DO McKay and also James E. Ellison, but none of these gentlemen could come. We arrived at Orem, on Provo Bench, at about 9:00 am and just a few minutes before MJVM. Knight arrived. After transferring the electrical equipment, lunch, and rubber mat from the chair to Mr. Knight's car, we searched for a suitable place for the demonstration in the vicinity of Strawberry Lake, the object being to keep a distance respectful of high voltage electric lines, in order to avoid any induction of these.

I remembered that the odometer registered 19 miles in Charleston and 26 miles as we left the last power line near the entrance to Daniels Canyon; just as the meter showed 52 miles where we stopped to experiment, a distance of 26 miles from the nearest power line.

As we entered Daniels Canyon a light snow began to fall, we met a car about half way up Daniels Canyon, in which there were two men and two women. The driver was Mr. NJ Keffer who owns the Internation Brake Co. He says he chased the deer, but came back empty-handed. We asked him about the road conditions far out in the Canyon and Strawberry Valley and he advised us that a lot of snow had fallen but the roads were passable. There was about an inch of snow on the radiator cowl, covering the top and roof of his car. We decided to continue, hoping that the weather would clear up, which it did. As we approached the summit the sun showed through the clouds as desired and as we descended into Strawberry Valley the weather improved, so Mr. Moray quickly agreed to give a demonstration at any place we could choose. His only reservation seems to be that he didn't like experimenting in a thunderstorm because of the danger involved.

Mr. Moray requested that we choose a place near a torrent of water, so that the metal tube would be sunk into the bed of it to be more effective. We stopped at a place about 10 miles southeast of the top of Daniels-Strawberry and 200 yards west of the main road leading to Duchesne, this location being nearly east of that which we Mr. Knight called Haystack Mountain and maybe 3 miles east of Strawberry Lake, over a small stream whose course zigzagged through a gentle grassy slope, to the lake.

The weather was bitterly cold and we sat in the car before unloading and mounting the equipment, and each of us had two cups of hot soup, which was brought by Mrs. Knight, in a thermos flask. After that, we ate a few sandwiches and then proceeded to assemble the equipment. The antenna wire was attached without any help or instructions from Mr. Moray, except that he suggested that the wire be pulled tight, to prevent any weakening of the stiffness of the wire in the center. This was done and the wire appeared a good distance above the ground, about 7 or 8 feet at its lowest point.

The metal tube establishing the ground contact was made of a half inch section (12mm5) of a pipe made up of two parts. The lower one was sharpened into a point at its end, in order to be able to make it penetrate more easily into the bed of the brook. Its length was 6 feet and after being sunk about 5 feet into the bed, the second part, which was about 4 feet, was screwed in with a Stillson wrench and the pipe pushed deeper until it met a hard object and began to bend near its top. I judged that there must be 7 feet of pipe in the ground.

The antenna wire was insulated from the poles (tubes) by two glass insulators about 6 inches long with holes drilled at both ends. A piece of wire about 2 feet long connected each insulator with the pole. The main wire was attached to the antenna at a point about 10 or 15 feet east of the pole. I helped Mr. Moray solder the connection where the main wire was attached to the antenna wire and also helped him solder the ground wire to his pipe. I counted the distance between the antenna pole pipes and estimated it to be 87 feet since I took 29 steps of 3 feet each.

Mr. Moray's equipment, apart from the antenna and ground wires, consisted of a box about the size of a butter dish, another box, unpainted, slightly smaller, a fiberboard box about 6" x 4" x 4" (inches) to which Mr. Moray referred: like the one containing the tubes, and a metal base plate about 14" x 4" x 1" (inches) containing what appeared to me to be a magnet at one end, a switch near the middle, and a receptacle for an electric light globe at the other end. There were also several poles supporting the wires connected to the plated metal base.

Mr. Moray took this electrical equipment out of his automobile and placed it on the running board of his car. Two dry boards, which Mr. Knight

brought, were placed on the ground and a rubber mat, usually placed under his desk chair, was placed on the board for Mr. Moray's protection against electric shock when he had to stand on the board. The running board was barely big enough for the equipment, so we took the front seat cushion and placed it on the mat and Mr. Moray transferred the equipment to the seat cushion and connected it there.

Very light flakes of snow fell occasionally, and a tarpaulin was hung above the upper part of the car doors, when these were opened, in order to protect the equipment from dampness. When all the wire connections were made and everything was ready, Mr. Moray began the adjustment. It was just at 1:05 p.m. on my watch. Before starting the adjustment, he placed the key on the post, then he said that she could make the ignition while the light was burning but no light appeared. The adjustment was to strike the end of a magnet through two projections of metal coming out of what I referred to as magnets. After adjusting for slightly over 10 minutes the key was put on the post and the light came on immediately, It was shortly after 1:15 at my watch. Mr. Moray having put the key 2 or 3 times during the adjustment in operation but no light appeared. We allowed the light to burn for 15 minutes, until about 1:30 p.m. The brightness of the light, a 100 watt globe supplied by Mr. Knight, was 75% as bright as a 100 watts plugged into a regular household outlet, in my opinion. It was an even light without fluctuation of any kind. '

As the light burned, Mr. Moray disconnected the main wire going to the antenna of the device and the light went out. He connected it again and the light appeared. He also disconnected the ground wire and the light went out. He then reconnected it and the light reappeared.

Mr. Moray seemed confident that he was able to get light and conducted his work in a skilful, unhesitating manner. He says he knew he could do the same thing in the Sahara Desert or in the deepest mine. When the experiment was over we congratulated Mr. Moray and I felt confident that he had a real invention and that no hoaxes were committed.

Mr. Moray disconnected the wires from the equipment and placed it in the car, while we dismantled the aerial and poles and prepared them for the return trip. We left the earth pipe embedded in the edge of the creek, as a landmark. When we reached the main road, we wondered if the underground concrete conduits provided for small streams and drainage were counted, thus, I examined one about 3/4 of a mile, - hence the test was done but found no marks at either end. The underground conduits, however, were made of approximately 36 inches of galvanized iron pipe. There were two pipes in the underground conduit, just above where the test was done.

A mile or two from the summit, on the side of the Strawberry, we passed a flock of lambs, which John Roundy had purchased from the Investment Company at Duchesne. Mr. Roundy was on horseback leading his lambs; we spoke with

I

him for a few minutes and we continued on our way.

When we reached the first power line at the entrance to Daniels Canyon, we checked the meter again and verified the accuracy of its first reading, which showed that there were 26 miles to the line! electric where the test was done. Two-thirds of the way down the Daniels

■

Canyon, we passed a herd of sheep. This herd belonged to Mr. Cummings, of Heber City, whom we talked to for some time. Warren Strong, his brother-in-law, was with him and I shook hands with them and spoke to Mr. Warren.

The pistol I borrowed from MW Woolley was not taken out of its box, but was left in my car at Orem. The rifle that Mr. Knight took was not used, nor even taken out of its leather case. After the demonstration, Mr. Moray fired his pistol five or six times into the stream.

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il

A copy of this letter was mailed to Mr. J. WM Knight, RL Judd and a copy was held for Mr. Arnold Herman, who typed this letter and myself.

Sincerely yours, I

(signed) EG Jensen

EGJ/AH

Translation of j, Jelodin

To the interested person(s).

On March 16, 1929, I witnessed the demonstration of an apparatus invented by Dr. TH Moray by means of which he produces electrical energy. A report of this visit has already been made.

On December 13, 1930, I was present at a second demonstration of the same - ensemble; only slight changes in the device were noticed although a great improvement was made in the performance. Capacitors, chokes and other parts from the previous test were now inserted into a small box approximately 10" x 10" x 30". To start working the operation of the assembly, instead of striking with a magnet, as before, now two discs - one of hard rubber, the other of iron - were rubbed against each other and produced an electric charge; the electric charge being discharged into the whole, began the operation of the whole (device) and immediately it was operational, and it continued to recharge.

The panel, which formerly contained 6 x 100 watt lamps, is now replaced by a larger panel, which contains 32 x 80 watt lamps, 3 x 1000 watt lamps and a receptacle for an extension cord with plug. At the other end of this cord was a 575 watt electric iron. During this test, all the electric lights were on for a while and were brighter than the other lights in the room, which were powered by the house circuit. When the lights shone, the electric iron was connected. There was no noticeable fluttering or sudden dimming of the lights, as usually occurs in lighting systems when an electric iron is plugged in. The lighting was as bright as before the iron connection. The iron heated up quite quickly and in about two and a half minutes it was hot enough to do the ironing.

I have made a careful examination of the complete system to see if it was possible for the electricity to have been brought into the system from another source. Every part of the system was inspected - the antenna, the main connecting wire, the switch; the switch was removed and examined; the wires from the switch to the box containing the capacitors and chokes; the box itself was lifted and examined; the wires from the box to the lamp panel; the lamp panel was lifted and examined and the ground wire from the switch to the water pipe by which it was connected to the ground; not a space was left where it would be possible to introduce a metal which could carry energy and I am prepared to say that no connection existed from which electrical energy could be obtained from another system.

A Radio crystal was fixed on the wire-director going in the apparatus, and coming from the antenna, on one side and going to the ground wire on the other side, and a perfect reception was obtained. If there had been energy from other sources, on the antenna or the incoming leader wire, when connecting the radio to the ground wire, reception would not have been possible but a loud (humming) sound would occur. would be produced. This was demonstrated by connecting the radio set to the lighting system of the house. When the system worked, the lamps lit up brilliantly. The terminal wire attached to the antenna was disconnected. A vigorous arc appeared and the sparks shot through the air, across the space of at least six inches. This could indicate that the energy was of a high voltage and the nature of the sparks indicated a high frequency.

I did not understand the principle by which Dr. Moray produces electrical energy. Wire capacitors and chokes were ordinary. The two cylinders called "oscillators" and the small conical envelope called the "detector"

are the only things that are not well known, but the system works well. It produces an abundance of electrical power and does everything Dr. Moray says it does.

I own no stock or interest in the company that promotes this endeavor and my sole purpose in issuing this statement is in the interest of the advancement of science. I consider this development as a great advance in the field of the production of electrical energy for science.

Very sincerely yours

In witness to the above statement, I, below, sign my
name

State of Utah
Salt Lake County

Thomas H. Yates EEME

Thomas. J. Yates having first duly sworn, deposed and declared: That he .
has read the previous declaration and acknowledges that he wrote and signed the same
as that set out above.

(seal)

Subscribed and sworn before me this 18th day of December 1930.

Notary - residing in Salt-Lake City, Utah

Translation by J. Jelodin

December 22, 1956

To those concerned :

During the spring of 1938, I was invited by the Utah State Safety Commissioner to accompany Mr. Cull to the Laboratories of Dr. Henry Moray for the inspection of the Moray Radiante Energy Equipment, My father, M, JC Jensen, an engineer, was also present. He saw a lot of testing done on the Moray equipment. This invitation was given to me because I am an engineer and a battery expert.

After seeing Moray's assembly lit, 30 luminous globes of 150 watts, a 1000 watt heated iron and a special high frequency motor operated for a period exceeding an hour, I placed my measuring instruments on the connections of the antenna and ground (ground). They were electrically . inactive (dead). He was checked whether there were any hidden wires;- there were none! The box containing the equipment was opened by Commissar Jull and myself. Each piece of equipment was taken out of the box, weighed and tested. These tests were made with very sensitive voltmeters and ammeters to determine if the box could contain batteries or if certain parts of the equipment had some sort of battery action in them. They didn't. (Implicitly, search for weak electric currents which can be provided by concealed cells or batteries. Nothing was detected).

There were no concealed electrical wires and no battery current or other power source (power supply) could be detected other than that claimed by the inventor.

The commissioner then asked permission to punch holes in the box containing the equipment, in order to be satisfied. That every part had been removed from the box and there were no hidden compartments. All this was done to the complete satisfaction of those present.

I had been asked for my comments regarding the tests and demonstrations of Dr. Moray's "radiant energy" equipment, and the above is a review of the test and demonstration which I testified. I affirm that this apparatus is that represented by Mr. Moray,

AB Jensen 426 Felt Building Sait Lake City, UTAH

State of Utah
County of Sait Lake

AB Jensen, having first duly sworn, deposed and said; that he has read the above declaration and acknowledges that he wrote and signed the same as that set forth above.

Subscribed and sworn before me on January 31, 1963

KK Steffensen - Resident Public Notary in Sait Lake
City, UTAH

Translation by J, Jelodin

923 E. Grogson Ave.
Knows Lake City 6, Utah

December 28, 1962

Dr. T. Henry Moray Knows Lake City, Utah

You asked me recently if I remember a test done in 1938 on Moray's Radiant Energy Apparatus. To the best of my recollection, the test was as follows:

I accompanied a group of people consisting of Mr. JC Jensen, Mr. Greely Snyder, Dr. Moray, and myself (I remember another person, who was also with us, but I will not give his name, because I'm not sure I'm not wrong). We were driving past St. Johns in Toocle County, Utah, and then 25 to 30 miles southwest of Johnson Pass. At that time, there were no facilities in Dugway, no telephones or power lines in the vicinity - simply, empty land.

Dr. Moray had packed up his equipment and the tools needed to erect it. We selected a location 50 to 75 feet away from our automobile. We helped him drive a 6-foot iron rod into the ground. With a torch the wires were welded to the steel rod. We erected two tent poles and attached an antenna made of - *fh* 10 multi-strand copper wire.

The antenna wire was attached to the box containing the Moray assembly. and the ground wire which had been welded to the steel rod was also attached.

Dr. Moray had brought a board with about 33 150 watt lamps mounted on it, a high frequency motor, a 1000 watt iron, and I believe, a heater or fan.

After connecting the ground and antenna to the box, Dr. Moray plugged in the lamps, iron, motor, etc. The energy from the box lit the lamps, heated the iron, ran the motor, etc.

We all examined the apparatus and determined that there were no batteries, no hidden connections to power lines, as we were miles from high voltage power lines .

Mr. JC Jensen drove the automobile and chose a place for the ex-? experience, so he could not have been pre-prepared for the test by Dr. Moray. Although the demonstration took place 25 years ago, I still vividly remember the reaction of everyone present. Thus we had the firm conviction that the Radiant Energy was indeed what Moray had - presented.

Very sincerely yours

AB Jensen

AB Jensen, having first duly sworn, deposes and declares: that he has read the foregoing declaration and acknowledges that he wrote and signed the same as set forth above.

AB Jensen

Subscribed and sworn before me this day, January 31, 1963

KK Steffensen

KK Steffensen - Notary Public residing in Salt-Lake City, Utah

Translation by J, Jelodin

CERTIFICATE

UTAH STATE

Salt Lake County

David I. Gardner, having duly sworn, depose and declare that I am a licensed Civil Engineer of the State of Utah, License No. 427, and have obtained a BS degree in Civil Engineering from State University from Utah.

I was a practicing Civil Engineer for 37 years as District Engineer for the Salt Lake County Water Conservancy District, District Engineer for the Granger-Hunter Improvement District and various other positions .

I am also an inventor having filed a patent on an apparatus for laying Line Pipes in a ditch, and also have a patent application pending on a dredging system , which works with full success. It was during the year 1936 that I was invited to visit the Laboratories of Dr. T. Henry Moray in Salt Lake City, Utah. When I arrived, there were several people present to see Dr. Moray's experience demonstration. During the evening Dr. Moray demonstrated what he called Radiant Energy powering a set of electric motors, irons, electric fans connected to a panel on which are mounted outlets for each globe and electrical appliance. The demonstration used a thin copper wire of very small diameter stretched across the room. Connected to each end of the wire were two porcelain insulators small enough to easily fit in one hand, in order to tension the copper wire tightly. I was given the insulator to hold by its end and the second insulator was given to another person at the other end of the room.

Dr. Moray asked me to pull the copper wire tight, which I did. I asked him what could happen if I touched the wire. He says nothing could happen except to just cause the lights to go out. I asked him if it could give me an electric shock? and he replied "No". So we pulled the wire tight and Dr. Moray turned the switch or did some other thing that I don't remember now; but the lamps lit up at their maximum power and the electrical appliances functioned marvelously. Having an inquisitive mind, I wondered if he was actually telling us the truth. So I let my hand carefully touch the copper wire, and the lamps went out just as Dr. Moray had told us. Dr. Moray said someone touched the wire. I removed my hand from the wire and the battery of lamps, as well as the devices worked again. How it worked, I can't say, but I'm sure it all worked.

There was, of course, much discussion among the crowd regarding the phenomenon I had witnessed. The gentleman who invited me to the demonstration - mentioned an ingenious device which Dr. Moray had also invented (and with which he can pick up sounds, without these being broadcast by a broadcasting station. He says that he heard a demonstration of this machine. I asked him if I could hear it too. He called Dr. Moray and asked if I could see a demonstration of the "sound pickup device" and I was granted that privilege.

There was an old radio box of a conventional design with two headphones plugged into the box, Pr, Moray tuned the device by turning a knob on the front. He passed me a pair of headphones and another pair to my nephew Mr. Clark Gardner,

He asked three people to go out for a walk in the street and strike up a conversation, which we were accustomed to. One of those people who chose to walk was my brother Archibald Gardner; the others were Louis Gardner who invited me to see the demonstration and Mr. Albert Dimond whom I knew well and whose voice I could determine. I was sure that I listened to these three men. I distinctly remember that it was raining at that time and I could hear the rain pounding on the pavement as they walked. I remember them saying they had better get home quickly if they wanted to avoid getting completely drenched.

As they returned I wondered if I could hear other people talking and despite Dr. Moray's advice not to fool around with the adjuster, I moved the knob slowly and heard very loudly. distinctly the sounds of a station: the train whistle, a porter or the station master saying "All aboard". From the nearest station to the receiving device there was a minimum of 5 miles. I don't know which station I picked up, or which railway line, but people were talking in English.

I know people walking down the street didn't have a transmitter or other instrument with them. They didn't want to believe me when I told them what I had heard, but they said I repeated their conversation.

David I. Garner

Subscribed and sworn before me today, September 14, 1963.

Translation by J, Jelodin

RESEARCH
Knows Lake City, Utah

850 Three Fountains Drive Unit No. 198
Morrays, Utah 84107
March 19, 1971

To those concerned,

During the latter part of July, 1938, I witnessed an experimental test done by Dr. T. Henry Moray at his laboratory, which was then located at 2484 South 5th East, Salt Lake City, Utah. This was only one of several tests; experiments I was confronted with with different groups of men present.

At this experiment there was a panel of lights consisting of 35 electric luminous globes, twenty of which were 150 watt globes and 15 of which were 100 watt globes. There was also a 600 watt heater in full action, a standard 575 watt iron and a special motor, which was designed by Dr. Moray. A photograph of this experience was taken in my presence, showing the people in the picture. I would like to draw your attention to the electric globe which did not light up and which was in series with the device when it was in full operation. It will be noticed in this picture, which really shows all that has been taken from this instrument in operation, the whiteness of the light, which came from these standard commercial luminous globes. •

I have been presented with many tests, which have been more or less similar. These tests were made in order to prove the successive improvements made to the apparatus from time to time.

I also read many letters from those who also witnessed the various tests of the equipment in operation.

Samuel E. Bringham

UTAH STATE
Saltlake County

Subscribed and sworn before me this March 19, 1971

Com. E.g.: 11.1.71 Jacques

B. Bell

Notary Public residing in Salt-Lake City, UTAH

Translation by J. Jelodin

2508 South 5th East Salt Lake City, UTAH March 19,
1971

To those concerned,

On a Thursday evening in mid-February 1937, then I was a student; - teacher at the College of the South, I called my uncle TH Moray to authorize me to bring my professor Mr. Vivian Decker, and my dentist Dr. Fred K. Bischoff, in order to testify at a demonstration of his apparatus producing UEnetgy--Radiant • Accordingly, the three of us had this opportunity at 2484 South 5th East, Salt Lake City, Utah. The apparatus was housed in a wooden box, something measuring 12" X 18", with an antenna and ground wire going into the ground. The wires coming out of the box led to a panel of 40 luminous globes of 100 watts each and also to an iron. My uncle touched a switch on the top of the box with a plate of electrostatic material in his hand and all the globes lit up brilliantly. We all then noticed that the. bulbs burned "cold" except that each had a very hot spot about the size of a dime on top, offset slightly to the side from the center. I also remember that I could turn the lights on or off by influencing the equipment by approaching it and moving away from it, either with my whole body or with my hand. If my memory is clear, the machine had to be regulated with a graduated dial whose needle had to be placed in the right position to bring about the condition required for its operation.

I vividly remember leaving them with the firm conviction that there was a free source of energy and an alarm system against burglars.

Chester Tood

State of Utah Salt Lake County

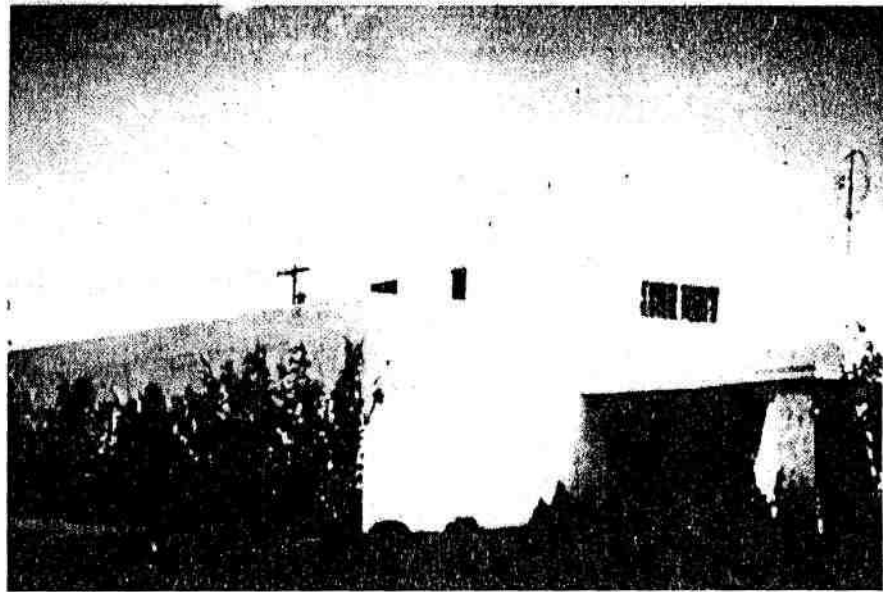
Subscribed and sworn before me, this day, March 19, 1971

Jack B. Bell

Com.Ex. 11.1.71

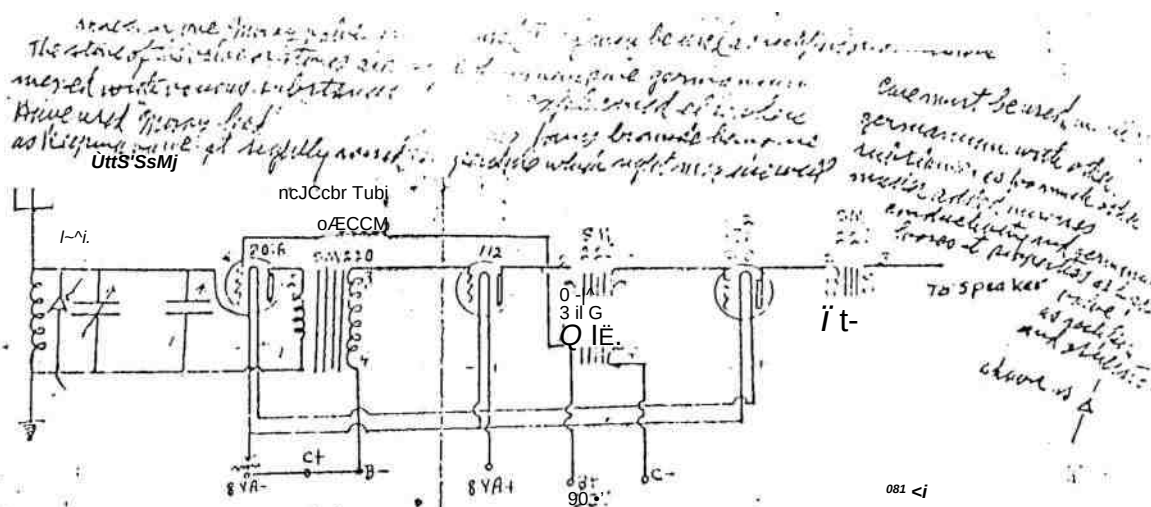
Resident in Salt Lake City, Utah

Translation by J. Jelodin



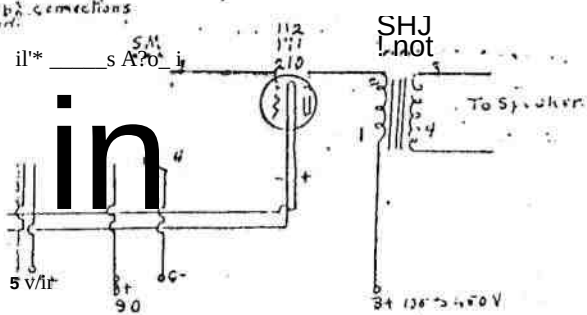
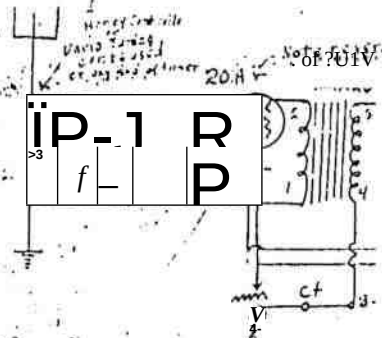
Exterior of the laboratory

TH Moray



To whom it may concern
 I have personally appeared
 before me and submitted a drawing
 this 14 day of November 1987

Notary Public



ray

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 in 80pF if desired

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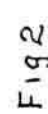
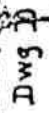
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Schéma de montage de T.H. Moray

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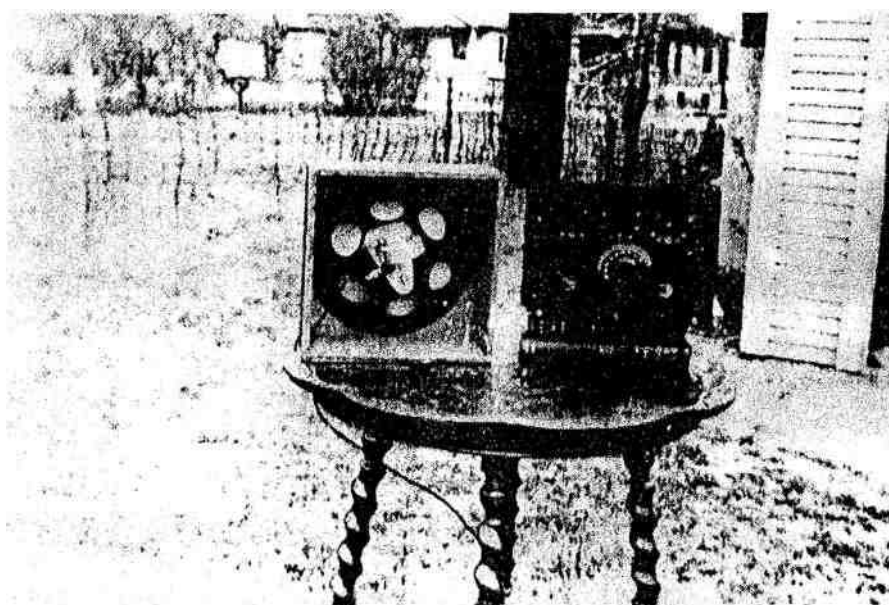
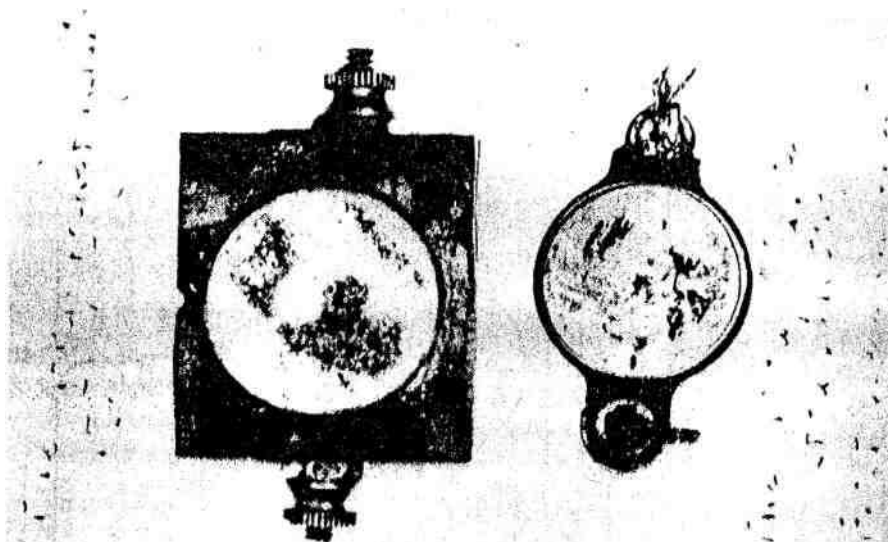


MV = means
 Money
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 something more
 (rare)
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 pebble of Money
 mixture of gemstones
 see written instructions
 on how to make
 mixture

| | | | |
|----------------|---------------|-----------------|-----------------|
| D. Thomson | L.A. Hansen | Rf Schubert | E. Fiedler made |
| March 21, 1931 | April 2, 1931 | Mark! 21. 1931. | May 20 and 19. |
| | | B.L. Garmworthy | |
| | | MW 2-28-1931 | |

Appendix I

- 4A. Picture of the Moray Valve radio itself, no batteries, referred to by Moray as pocket radio; operated full-sized speaker.



Photographie d'un générateur à

"Energie Radiante"



FIG. 6.



FIG 7

1/2 Circle based on moving

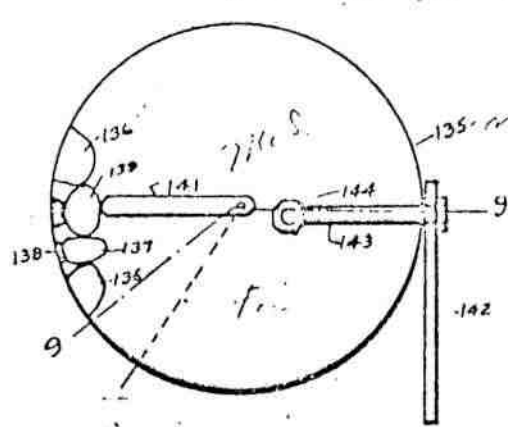


FIG. 8

Murray valve

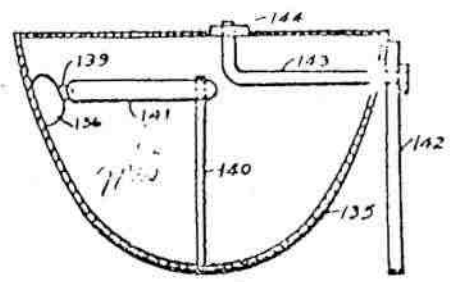


FIG. 9

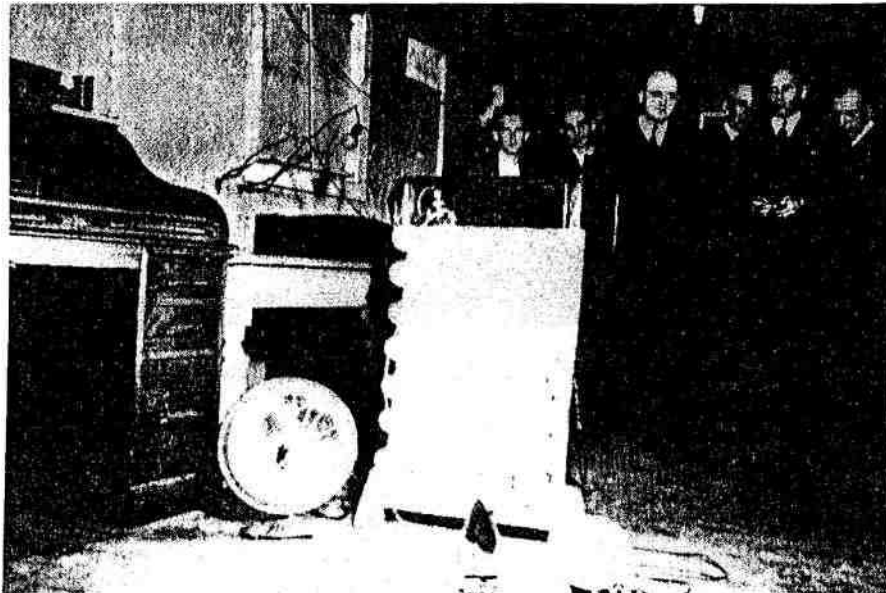
INVENTOR

7717 NOAH Y.

BY

Murray O. Hayes
ATTORNEY

S-ÆJ



Expérience de T.H. Moray

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Feb 19

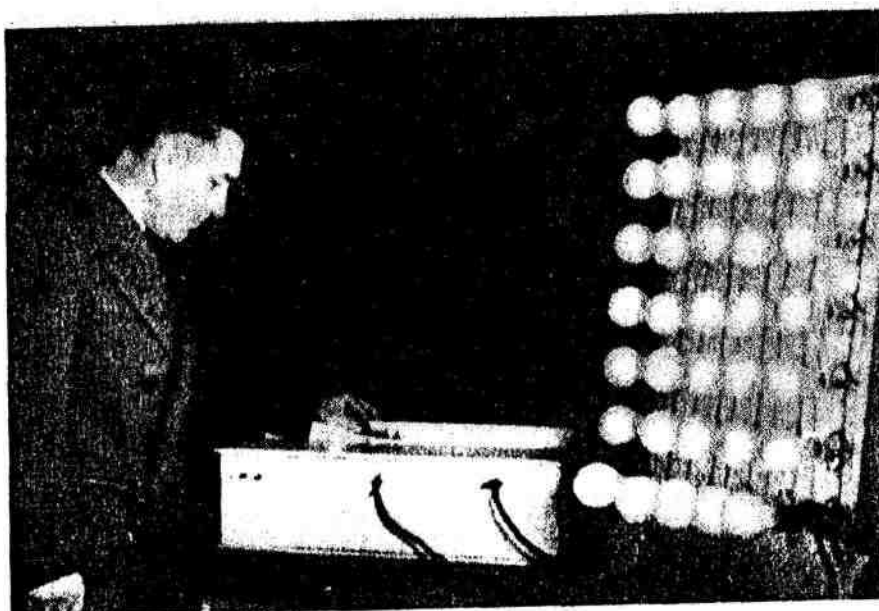
Charles E. Cole

State of Utah) ss
County of Salt Lake)
Subscribed and sworn to
before me, a Notary Public, this
19th day of March, 1971.

Russ H. Petersen

Com. Ex. 7-12-1973

Residing: Salt Lake City, Utah



TH Moray in front of his generator
Bench of lit lamps

Frederic DUFOURG

by P. SMIRNE

(Paris)

Frédéric Dufourg is mentioned for the first time by Octave Uzanne in [1] as an old man from Toulouse living in rue Saint-Léon in Toulouse. F. Dufourg pursued work alone since 1892, work which he titled: "Studies on the Brute Forces of matter of the three kingdoms on all phenomena and on all phenomena of the Universe by Intelligent Forces" .

There was a first work which dealt with "Brute Forces", as for "Intelligent Forces", Frédéric Dufourg was to publish them later but the document was lost; he reduced his study to that of the "prodigies" of the initiates of (*) of Antiquity. He had to cross at times the results of Louis Boutard

Gaston Bouvier, member of the institute, professor at the Sorbonne, witnessed Dufourg's experiments in 1915 at the home of Mr. François Gayral, president of the Cooperative Society for the Production of Violets in Toulouse at Lalande (in the suburbs of Toulouse), Dufourg worked with the help of his wife who was a medium. She has also done a lot of research for lost and stolen objects, missing persons, fugitives for the police.

Dufourg called himself an alchemist and he really was, but he had nothing of the attitude of the usual alchemists.

His language, like all those researchers who came out of the usual C. de C., put off the rationalist people and he quickly found himself isolated, misunderstood . Moreover, he had a rather brittle character; his mail with chiefs

(*) F. Dufourg writes "I did not deviate from the method adopted by the philosophers of antiquity, of which the Ionians (by their Gnosis) were above all the phyricians".

.(**) An American journalist Jack Wins-thon reported in July 1914 in a New York daily newspaper on the validity of the observations made by him of the experiments and demonstrations on agriculture carried out by F. Dufourg.

of state shows it; its mordant was not enough to make it recognized.

Let's take a look at how Dufourg expressed himself:

"I produce 'water' or 'earth' by transmutation of the imponderable . All my science relates to the discovery, nature, condensation, transmutation of the multiple energies of Nature and the return of universal matter to the imponderable".

Even this should not have put off scientists. On the contrary, they should have felt the sign of another approach, without reducing his work to rational patterns.

It is necessary: 1) to put in order his "model" built from the words of his personal vocabulary

2) compare the results obtained with certain texts encrypted by the methods of Johannes Trithème, even if the texts of Dufourg are not coded

3) establish a parallel between his work, that of Rudolf Steiner in agriculture and that of Louis Boutard. (Dorian Gnosis).

According to the language adopted by Dufourg, this man considers the Universe composed of "Brute Forces" and "Intelligent Forces" whose phenomenon of manifestation is "Universal Life".

Dufourg's notion of force is an analogy that can be expressed as follows: There are visible, ponderable forces and imponderable "invisible forces" which he calls "fluids".

Dufourg insists on the fact that "the brute forces of the three kingdoms" can only be fully assimilated on the condition of associating the " Intelligent Principle" with them.

Dufourg's experiments and results only have a chance of being realized and brought to fruition by following his vocabulary. We can then interpret everything in the C. de C. that we are developing.

Let's continue with another example of his language, this time relating to an experience:

"Plunged into the earth up to half its own length, a steel brooch takes on the properties of a magnet after a certain time. Suspended by its center from an untwisted silk thread, this brooch undergoes a

horizontal attraction and a vertical attraction" (declination and inclination that Dufourg considers to be caused by the earth and its magnetic field). This is in agreement with physics. But he goes further. For him, the properties of any attracting body are said "magnetic" (he considers wires, pins, rods, tubes of various materials as magnetic attractors). This is "the magnetine attraction" of the ancients. Dufourg considers cold as a force, as well as heat (he never does not define this word). Cold aggregates the elements of the masses, heat disintegrates them. Cold materializes, heat dematerializes. The earth as a planet *and* for it a purely attracting body. The "heat force" is the cause of the "magnetic attractive force" (this is the central heat of the earth).

We will not continue further in this presentation. No matter the language, it's the results that matter. Dufourg actually arrived at al chemistry. His concepts, which constituted only a coherence for himself, served him to foresee and connect his experiences. His mistake was to ape the universally used terminology. He also used in his approaches a kind of analogy. Its pseudo-scientific portrayal had deep roots.

Dufourg's atoms are closer to Boutard's little serpents than to the atoms of physics.

To molecules he associates colours, to atoms degrees of heat and force. He arrived at what he called the liquid materializations of the "subterranean and atmospheric energies", this by burying bottles of colorless glass containing salt (sodium chloride) and from which came out metal rods or elder arches.

In 1915 he took care of perfumes (violet); he amplified the intensity and range of the scent of violets in enormous ratios. Then in Finland, he *energized* fuels. The list of applications of his work is long. He too opened a royal road!

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- [1] Octave Uzanne: "La Dépêche de Toulouse", September 27, 1929.
- [2] Pierre Smirne, ARK'ALL Commun., Vol. 8, fasc. 4, 147-149 (1984).

IS IN THE

par

P. SMIRNE

CRAVE

France

Enel has practiced all his life through an initiatory chain that he simply calls "the Brotherhood" a teaching lightened of the thick veils that have covered the "Great Tradition" for centuries. His friends, predecessors and contemporaries, and Enel himself, designated under the term "Gnomology" this experimental way which reveals to man the key to the constructive forces of nature. "Gnomology", a word of Greek origin composed of: Gnoma: proof-experience, Gnomé: knowledge-knowledge and Logos: creative Word, develops in the follower, through a reasoned and intelligent practice of various manipulations and reflection- meditation, his spiritual faculties to bring the follower to a balanced harmony in relation to the forces directing the Universe. The pragmatic part of the oral teaching of gnomology covers in this tradition of transmissible powers various degrees of subtlety whose concrete reality is directly linked to the possibilities of access of the aspirant to the supersensible world. A tripartition of time was offered to students whose training in the initiatory practice included prayers, work, rest. The nature of the prayers was based on the right word, at the right time. Some prayers were in Latin, others in Hebrew, others in Sanskrit or ancient Egyptian. Prayers having the value of mantrams were designed for concentration in general; for the restoration of mental and physical balance; to come to the aid of a third party afflicted. The first condition of admission to "La Confrérie"

- of which Enel was a representative as having received authorization to disclose
- is to apply a double principle: need to serve, from the teaching given by "Gnomology", all those who would need help while not ignoring the need to avoid "throwing pearls to swine". The part concerning the "work" included elements specific to semantics, the roots of languages, grammar as well as the bases of the various astrolo-

gies that combined with both the Hebrew Kabbalah and the Egyptian Kabbalah provided an understanding of how and why the prayers were structured. The Written Trace of the Revealable Elements of this Science was published in 1928 by Enel under the title: Trilogie de la Rota. The "Gnomology" in the practical teaching for what concerns the "Work" gave the elements intended for (*)

the formation of an "Egregore of the servants of Good" intended to help man in his spiritual ascent. The methods employed being the same in the Brotherhood as those which have guided the elite of humanity during the millennia of its existence. By regularly and scrupulously carrying out the indications given, the adept then became a participant in the creation of an extremely powerful magnetic chain whose results can only be directed towards a goal of good. Some supports were designed, made and used as transmitter-receiver devices for sympathetic currents. Draw into these supports various materials, mineral, vegetable, organic whose geometric shapes and their combination in volumes made it possible by orientation in three-dimensional space to obtain from a distance certain effects of modification of the states of living matter and particularly cells or beings. These supports, used with the appropriate prayer derive their origin from various devices that Egypt bequeathed to us with the "texts of the pyramids". Enel used and had his students and friends use, from 1909 in Serbia, in particular with Kazimir Stabrovsky, Director of the Academy of Fine Arts in Warsaw, an oscillating resonator allowing them to emit and receive effects according to their orientation. due to a form acting at a distance through the physical contact of the operator and the concentration of his thought. The originals of these early devices consisted of a strong slab of green rock of a basic nature or of metamorphic micaceous schist of a square shape in which were deeply engraved two concentric circles centered on a Khejra (Egyptian scarabs) mobile orientable and removable staff; in the first circle was engraved an equilateral triangle inscribed in the circle and of which two of the vertices were prolonged by two other smaller triangles, included between the central circle and the second concentric circle, In all the thickness of the engraved circles were placed two blades of money opened in one seal each. The position of each opening becoming mobile, an "adjustment" of each circle was used in relation to the general orientation of the apparatus on the terrestrial sphere according to the position of our planet compared to the sphere of the mobiles (cortege of planets) and the sphere of the fixed ones (zodiac). The principle diagram

(*) P. Smime has retained certain words used in occultism, as they were commonly used by Enel; he, indeed, felt what he manipulated

of this device and three prayers were presented in a very simplified way forty years after its first applications in 1959 in the French edition "Message from the Sphinx" published in English in 1936. From the beginning of the century, several people from various ethnic origins will place themselves under the authority of Enel for the specific purpose of working towards this. "egregore of good" mentioned in his Gnomology. The technique of the work is based on the daily exercise of mantrams, the choice of which is based on the so-called Babylonian hours with their master of the day. At the solstices and equinoxes, carefully prepared theurgic operations are performed. The use of the third meaning of hieroglyphs is retained for this particular work. The principles, Enel le.s indicates under firm allusions borrowed from other authors; citing them as an example in some of his writings: D. Martinès de Pasqually de la Tour, Louis Claude de St Martin in "Le Crocodile", Ossendowsky in Bêtes, Hommes et Dieux (about the king of the world), R. Guénon qu he had moreover met in Cairo on several occasions before the Second World War. This definitive commitment on the Way is taken by Enel following a meeting which he described in his text "The Initiation". This man, his predecessor, gives him a glimpse of the physiology of the Cosmos and gives him tangible proof of the powers of man over nature by explaining to him the maneuvering of the energies implemented during what some might take for "miracles". This man, distinguished in the Order of Noble Travelers, leaves Enel a cryptographic key which, provided that it appears in the Old Testament, finds its echo in the Vedic texts, can be commented on in the light of the Gospels to explain passages obscure of the Buddha's Doctrine but also is situated in the Future through the symbiosis of the book of Revelation with the book of nature and eternal life. This true man, in the sense that Ouspensky, another friend of Enel, was looking for him around the same time, this Man sent from a "hearth of light" will therefore choose Enel to entrust him with a work which will find a partial realization much later in the world of energy structures whose forms are an expression in the sensible world. On the same practical level, Enel was able ^{to} confront certain actions of the word just at the right moment in Egyptian theurgy through the operative contacts he had had with Count Vincent! (PV Piobb) between 1902 and 1930. Both among others with various people of different ethnic groups co-opted, on various levels will combine the keys of the verb by the thorough knowledge of the vibratory forces of the sound and energy of the gesture which hieroglyphically structured make it possible to associate as geometrically at will the forces

of the Astral. The Platonic Idea which presided over these works associated knowledge and faith in a pragmatic gnosis. Enel denounced the empirical recipes and proved that the Initiatic Teacher is not to be found in any grimoire or any course but through the daily exercise of the intensive development of the will towards the single goal of evolution. He stressed that true initiation could only be transmitted orally by a master, an initiate controlling the novice by graduated tests, a severe and indispensable disciple, sanctioned by the "oath which seals the mouths". Seen from angle c; Practical Mysticism, Enel creates for its students a safe and convenient method of dowsing eliminating the personal factor. He demonstrates through the practice of their curative application the symbolic meaning of hieroglyphs, the square Hebrew of Esdras, the hexagrams of Fo-hi, the Indian quipus... which, coupled with various materials and forms assembled, oriented in the The space will allow, among other things, certain remote therapeutic actions from which many sick beneficiaries owe him a total cure. Under his pseudonym, notebooks taken by his students will appear between 1949 and 1959. First steps in therapeutic dowsing; Radiation forms and cancer; Remote Radiation Treatments. He cites in his practical notes the people with whom he *worked* such as Jacqueline Chante-reine, Doctor Camille Savoie, Dr. Naret, Engineer Capes, Pierre Beasse, Dr. Régault, MM. Chaumery and de Bélizal, Georges Lakhovsky, the engineer J. Cartwright and whom he himself partly trained on certain points. As for the other texts of Enel, behind the allusions, the practical remarks, the "devices", another quest, another orientation, another essential of an occult order appears and which he specified from 1928 in his Trilogy of the Rota dedicated to a lifelong friend the CHC ... Colonel Henri Chardon Thus what has been called radiation of forms, emission in absorption of forms by their assembly oriented in space contains in essence the numbered combination formative forces which then come into action and are adequately applied to the appropriate supports to create the desired phenomenon either on the nascent oscillatory principles within the living cells or on the protoplasmic supports of the various waters in the universe. The magnetization being created by part of the devices, it is a question of condensing, multiplying and projecting the active phenomenology of vibrations, oscillations of which physical and chemical science ignores the magic essence captured and used with fruit [1].

NOTES.

Enel : May 6, 1883, St. Petersburg - November 16, 1963, Gliou/Montreux (Switzerland).

Journal articles from 1928 to 1938 ; Annals of Hermetism - Psyche - Astrosophy -
The seer - The Papers of the Marvelous - Initiation and Science.

Works with publication dates : Trilogy of the Rota 1928 - Magic Cures in the 20th
century 1929 - Post Mortem 1929 - The Sacred Language 1934 - The Origins of Genesis
and the teaching of the Temples of ancient Egypt 1935 - Message from the Sphinx
1936 - First steps in therapeutic dowsing 1949 - Radiation of Forms and cancer 1951
- Remote treatment by Radiation 1959 - Gnomology 1959 - The mystery of Life and
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A MUSICAL INSTRUMENT - THE VIOLIN

by J. RAVATIN

(Paris)

Let's say a few words about a musical instrument which is the violin. We have already seen that a violin has an ext., some have a parent; these are often the most famous. On the other hand, the violin as an abstract entity has a progenitor. Many violins respond to KShPh and the shin is found right above the soundboard, roughly vertical to the sound; if we remove the soul, (* (**)) the shin disappears and the KShPh also if any time we had detected this state On a violin we also find a vital field.

When the instrument is played, there are phenomena of overlapping which can occur between the instrument, the instrumentalist as well as transfers with the piece played, or a part of this piece. If it is an orchestra , transfers can also occur with instrumentalists, more rarely with other instruments. If it is a soloist, transfers are often made through the conductor. orchestra.

(*) The idea of removing the soul of the violin was put forward by François Saison.

(**) A musical note emits one or more eifs; depending on the instrument, different states can be associated with these eifs. If we test the very first notes of Beethoven's 5th Symphony, we find V^, (it was A. Philippe who had the idea the first time to do this test. You have to be careful that the 'we search :

-or they are the eifs of some notes,

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---------------------------------|
| - | " | " | " | " | " | " | " | of a passage from the symphony, |
| - | " | " | " | " | " | " | " | of a part, |
| - | " | " | " | " | " | " | " | of the entire symphony. |

To this we can add the state of the composer that we can detect in the work, his personal state when we listen to the work (states associated with memories when we listen). Like John Dee's Table, the symphony becomes a surdant (that is to say, it has a double role: containing and supporting). Can we deduce that such and such an eif comes from the state of the composer ? It is more difficult to ensure this, because there can be for example a dynamic duality between the piece and the composer, but the piece can serve as a channel in the sense in which we have posed it, a channel in which we will find different states of the composer.



Tartini's Dream

Guiseppe Tartini was an Italian violinist and composer; he was born (*) on April 8, 1692 in Piran (Istria) and died in Padua on February 24, 1770

What we are going to report here is a story that is now considered a nice legend, but which certainly corresponds to an anecdote experienced by Tartini.

It is said that Tartini saw in a dream the Devil who offered him his services , and who, according to his order, performed before him a sonata on the violin; delighted with what he had heard, he tried to remember it when he woke up, and composed this singular sonata which we know today UNDER the name of the "Devil's Sonata". An engraving was inspired by it; we reproduce it here.

Following this anecdote, the violin should not be seen as a satanic instrument. It's much more subtle than that, The era intervenes; superstition was still very much in place in Italy at that time (superstition is now derided; often it corresponds to sensations attached to attitudes which are placed in the equivalent of rituals in the face of events).

Back to the violin. Each violin is to be studied separately. It would not be necessary, in their case, to deduce a too hasty law on the CV, the eifs, the shin, the states etc... There are also harmonics to its form.

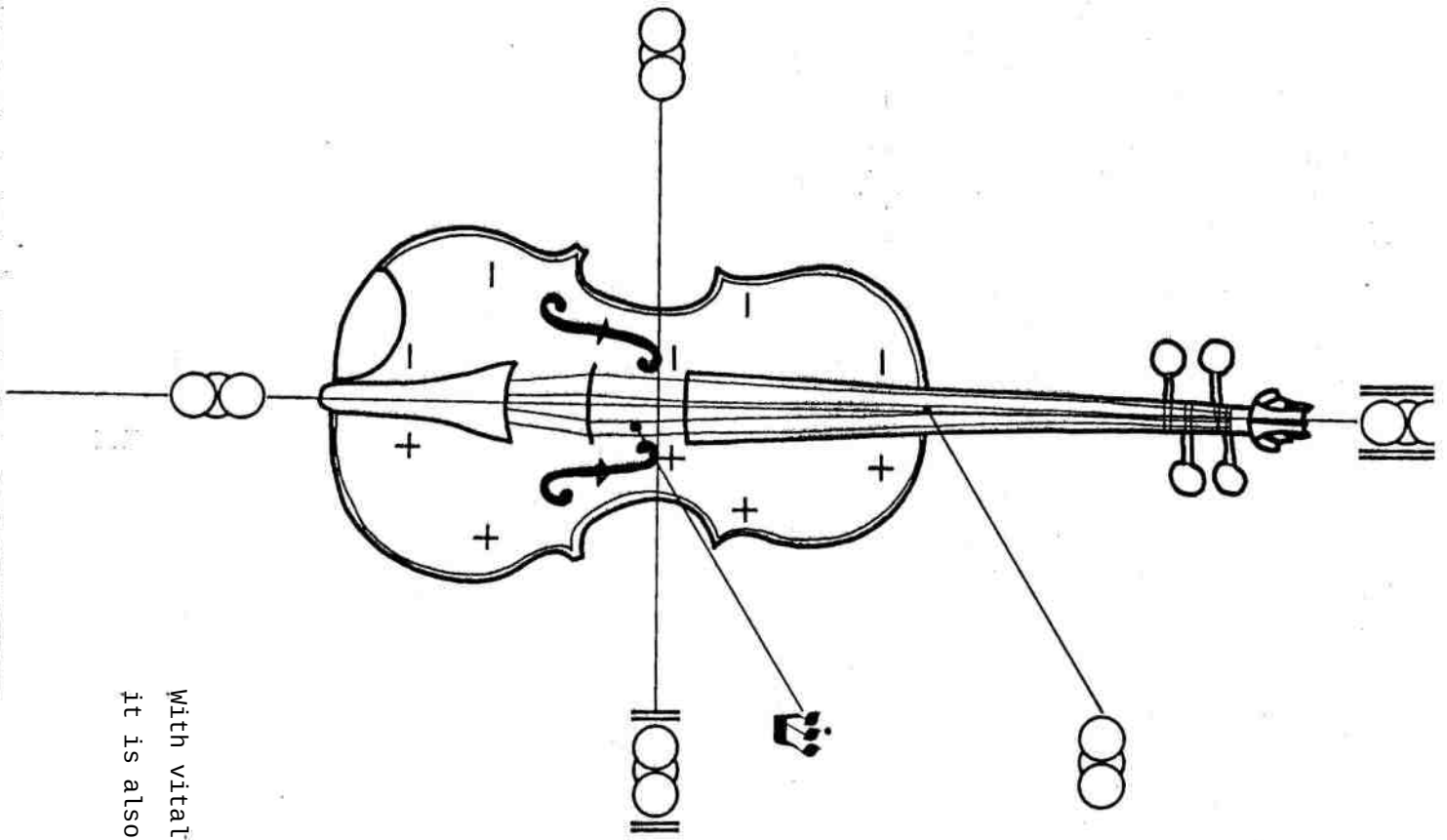
Modern musical instruments can be found that emit at C. of T. high; for example the device designed and assembled by the Bachet brothers called "Le Cristal". It generally emits at very high C, de T. On the other hand, if we play the music of JS, Bach on it, it no longer emits these C. de T. So, if we want to use this device for the quality of its eifs, we must create music for it .

Bibliography

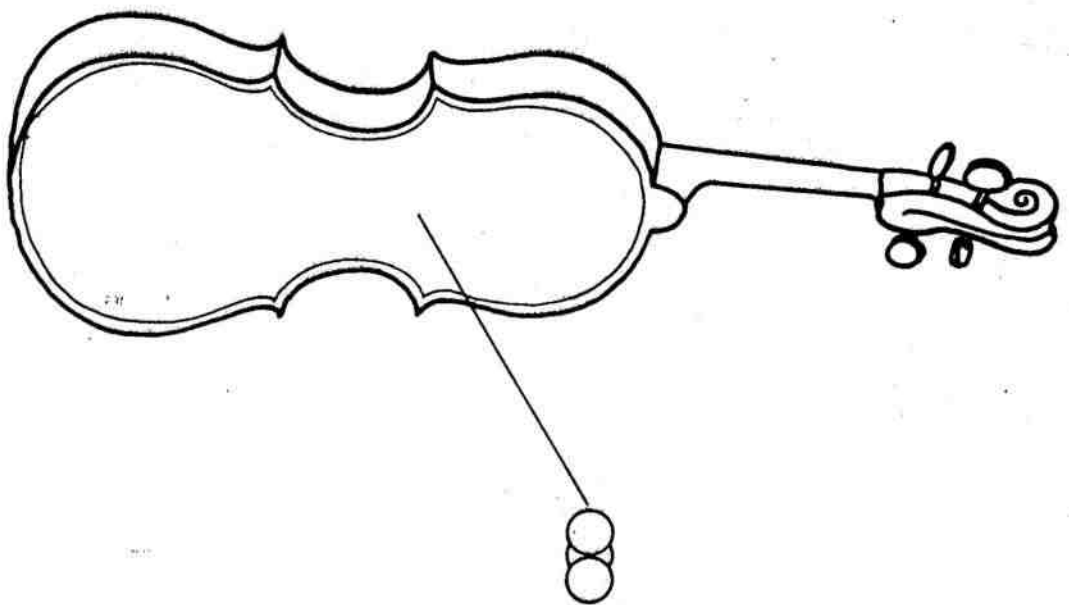
[1]-Dictionary of Authors of all times and all countries, Edit. Robert

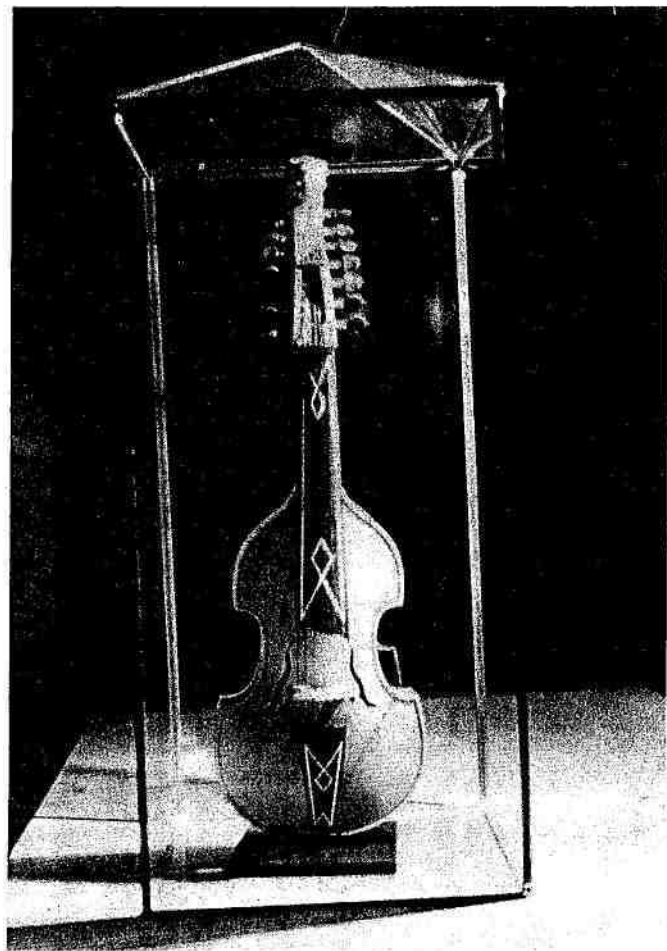
Laffont-Bompiani , Paris (1958),

(*) Nothing is known about his early youth. Tartini was appointed in 1721 first violin of the orchestra of Saint-Antoine in Padua. In 1723 he was the guest of Count Kinsky in Prague. He returned to Padua in 1726 and remained there for almost half a century devoted to fervent musical activity. In 1727 he founded a violin school which became famous and earned him the title of "Master of Nations"; he composed, fulfilled his duties as choirmaster and devoted himself to research on acoustics. But Tartini longed for a peaceful life and henceforth refused any invitation from abroad; he was very generous to his friends in need. His work has an important place in instrumental music. It includes about forty concertos, fifty trios, one hundred and fifty sonatas for violin. Many of his compositions are still unpublished [1].



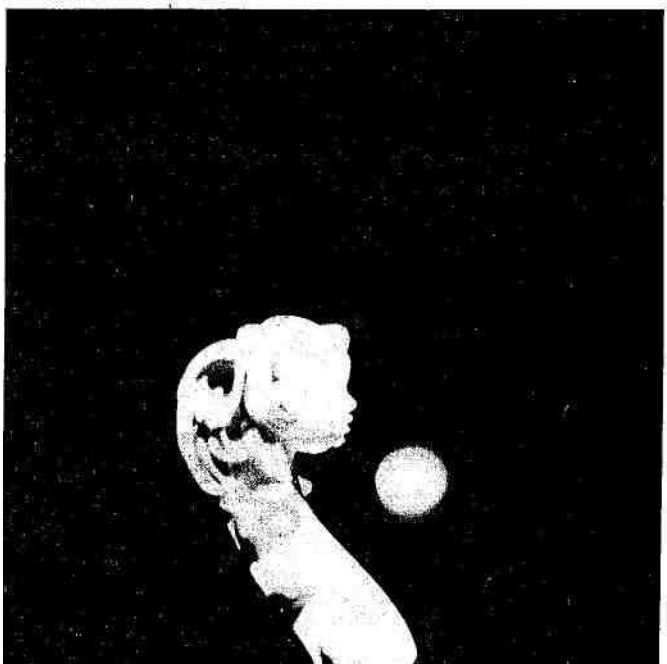
With vital field, right sh.in pointed in the middle,
it is also necessary to add the harmonics of this





Viola d'Amore performed by Samuel
FRANERIC

The right shin is detected vertically
, near the bridge, slightly to the
right (front view). The right shin
also comes out of the angel's mouth,
as well as the life knot. The
instrument has an ext.





Violin made by Samuel Franeric.
special eifs, CV, etc...

Violin making, through the study of eifs
completely new.



On this instrument there are
of the states, presents itself in a day

THE FIELDS AND RAYS OF COHERENCE

by J, RAVATIN

(Paris)

Let us give a square tiling of the plane and let A, B, C, D be four points placed in vertices of squares and being themselves the vertices of a rectangle.

Let O be a point on the line xy which bears AD ; let's place θ in a vertex, of a square and outside the segment AD . Let us consider a family of straight lines stemming from θ , placed to the right of xy and intersecting the rectangle $ABCD$.

Let n and p be the number of squares respectively of which one side is carried by AD and one side carried by CD ; xy will have $n+1$ common points with the vertices of these squares; xy will be said to be $(n+1)$ -dense.

As shown in the figure, there is another straight line which is $(n+1)$ -dense, it is uv which has a slope of 45° with respect to the direction DC . (The line rs is not $(n+1)$ -dense in general). Consider the parallels to AB passing through the vertices of squares having a common side with AD . Taking into account AB and DC there are $n+1$ parallels. Suppose these parallels are not $(p+1)$ -dense and the points can move from one vertex to another of the squares. So there will be occupied vertices and empty ones. These moving points are called wandering points. The plane is assumed to contain wandering points in a finite neighborhood—significantly larger than rectangle $ABCD$. They will only move along these parallels. Thus in some cases xy and uv may not be $(n+1)$ -dense. We thus see that the number of points in $ABCD$ is less than $(n+1) \times (p+1)$.

Consider a line Ot' passing through θ and between xy and OB . Suppose it intersects AB along a wandering point, say R . Let S be the wandering point closest to R on Ot' . Let ab be the parallel to AB going through S . Let R' and S' be the respective closest wandering-points-

-y, lie of R and S on AB and ab . The quadrilateral RR 'S'S will be called unit fundamental of structure.

In the case where xy and uv are (n+1)-dense, the fundamental unit of structure for xy is a square and a parallelogram for uv (if the nearest positive points are at the vertices of squares with one side carried by xy and a diagonal carried by uv).

Neighborhood of a wandering point: these are the wandering points placed on the vertical and the horizontal of the considered point which are at a distance (measured in the plane) less than or equal to that of the side of the squares.

A line from θ may not be (n+1)-dense; it may contain holes or its slope means that it does not pass through n+1 nodes (vertices - squares) and. that the wandering points do not fill it sufficientlyJ We will have straight lines which will be almost (n+1)-dense when they contain only a few holes. This notion of almost (n+1)-dense is on the run. A straight line, despite having holes, may have points located consecutively; it will then be swarms. These swarm lines will be useful to us later.

•you

θ is the Obs; xy the rational C. of C.; the set consisting of all the (n+1)-dense lines, lines of the genus rs (figure 1) (one point out of two or one point out of three), certain lines with swarms having a swarm having n or n-1 points or having several swarms almost filling the line, will be the other C. of C.

. (*)

p te

The straight lines are the Rays of Coherence. We see that the C. of C. usual is confused with its R. of C. and that the other C. of C. is fleeing: flight because of 1) lines with swarms, 2) due to lines of type 1 point out of 2 or 1 point out of 3 horizontal, 3) by wandering points, 4) by shape ; of the unit of structure, 5) by the shape of the neighborhoods of the points of the straight line.

* '

The rights to swarms not retented in the C. of C. preceding will not be not R. de C. but islands of inconsistency. The structural unit generated by an R. of C. therefore has different forms according to the R. and C.. The Obs. θ will fill rectangle ABCD with the structure unit. If it starts from the usual C. of C., i.e. xy , it will be a set of squares (those considered at the start); if if it starts from the R. of C. uv , it will be a parallelogram and it will fill almost another parai 1 el ogr amrn p (***) ABCD; next rs it will be

(*) Radius of Coherence: abbreviation R. de C.

(**) A line and an associated structure unit will be called a

The unit of structure is the polaror seen previously. The filling of ABCD can be done by different polarors, This structure unit makes it possible to describe the echo of the C. of C. which has been launched, in an R. of C. of this C. of C. The rectangle ABCD filled by a unit of structure is the equivalent of the process of spotting an observer in the ET structure. The polarors generated by xy for example or by uv are able to describe localized representations which are very stable in coherence.

That obtained from rs (figure 1) gives a structure of the ET type which is more delicate to approach than that given by xy or even uv because the almost complete filling of the rectangle ABCD is coarser. The rational description is very stable, others will be too or they might be less stable. From straight lines with swarms of the kind islands of incoherence, there will not even be any more generation of a polaror, therefore no coherent description on a human scale.

It may happen that two structural units are simultaneously present **along** a R. de C., in which case the description of reality becomes very laborious and tends towards incoherence, all this due to the appearance of a hole in the R. of C.

Consider the polaror generated by xy and that generated by uv; the first is a square, the second a parallelogram, we will pass in the bands delimited by the horizontal parallels of the squares to the parallelograms by a **cumulative**. The rectangle ABCD thus filled can be considered from several points of view (Figure 3).

1) Two or more flag-like shapes impose a disordered structure in ABCD. The filling of the rectangle is not done from a single polaror which by this process imposes the stability of the markers. An absolutely incoherent world can present itself to the Obs.

2) In the case of FIG. 3, it can be considered that the part to the left of the straight line mn emerges from the left and vice versa; therefore we have a dynamic duality; to this dynamic duality corresponds a ternary. Rectangle ABCD takes another character; it is not known whether landmarks are possible in this case. But that's outside of the inconsistency. It will be a fractant.

If a large part of the (n+1) points of xy pass over an island of inconsistency, they will make it an R, of C.

;CD

We can also have a fractant by dynamic duality of 2 straight lines with swarms, this dynamic duality being caused by a third hold. We no longer know what such a fractant will correspond to.

The deepening of knowledge in the rational C. of C. corresponds to a displacement of points A and B in the direction of x towards y . Rectangle ABCD expands; the line xy can be $(n+q)$ -dense ($q > 1$) . But on this model we see that the other R. de C. are likely to grow at the same time. So deepening in an R. de C. has effects on the other R. de Cs.

If we consider an R. of C., some points by wandering can leave the line constituting this R. of C. which will therefore no longer be $(n+1)$ -dense, but will retain enough points to generate a polaror. The departure or arrival of a wandering point will correspond to what is called a Representation. Thus by moving errant points one can impose its representation, if the mental process is in the model corresponding to the moving of one or more points. For the R. of C. to retain its entity, it is not necessary for several moving points to appear a swarm of holes in the straight line constituting the R. of C.

It has already been pointed out that the C. de C. echo is more or less perfect; the wandering points, the rectangle ABCD and the straight lines constitute what is launched by the human being, the fact that he puts himself in what he launches and the imperfect echo, response to what he has launched.

The Obs. θ can move in the plane, for example on the half-line Dx; then the usual C. of C. is always the same; if it stays in this C; of C. , he will have no awareness of this displacement. On the other hand, if he is in another R. de C., he will become aware of it.

We will assume that the only possible displacements for θ are on Dx .

The displacement of θ on Dx is associated with a power of adaptation of the Obs. It is a certain freedom of development of the Obs. to catch the R. of C. If a straight line becomes impoverished, the Obs by a tiny displacement can find a straight line as rich which gives it a representation quite similar to the

can escape the world he is in since he throws the C. de C. and places himself in it.

We have defined the neighborhood of a point. Let us examine the case of an $(n+1)$ -dense line and another line formed of a swarm of points, these two lines being such that the neighborhoods of one (uv for example) contain the points of l 'swarm of the other for example ($u'v' <$ in figure 2). In the chosen example, the points L',1',P',J',K',L',M',N' are points of the neighborhoods of points of uv which is an R. of C.. Then a representation of θ which will be $u'v'$ will be able to pass from an island of inconsistency to an R, of C. and in this case will be able to contain the coherence of uv . We will thus have another form of fractant on $u'v'$.

Consider rectangle ABCD and all the flags and islands of inconsistency , all of these placed in the chosen plane from the start. At the same time that we have echoes of C. de C. we have islands of incoherence through the possible launches.

The description of L. Boutard corresponds to an R. de C.. The Elders are presented differently depending on the representation.

In the R. of C. of the other C. of C., it is understood that we fabricate the past. xy contains the history of mankind in the usual C. de C. for a
Obs. given. This story can be hung by neighborhoods or by moving errant points on a straight line with swarms. We will then have a new representation
tation.

The position of the Obs. on Qx, the ballad of the wandering points will allow to set up the identification.

P Consider now for a given position of θ , a layout
R. de C. and islands of incoherence, the wandering points: their set goes
om

have a certain configuration. There can occur quite
configurations /

particular ones which will cause localizations to appear in the different R. of C. and islands of incoherence. So c^f is in the plan

] used in this model that will place the most complete localizations.

In the usual C. of C. everything is localized, the concept of delocalization
does not exist. The other C. de C. is impregnated with the escape of landmarks. The R.de C. de p
c this other C. de C. are other possible localizations. Islands of inconsistency are unaffordable locations on a human scale because we cannot build

ou
r

The Global therefore stands above the plane. It has no structure. It is no longer at all on a human scale. The Enels are related to the possible layouts of the flags. The wandering of the points accounts for the fact of the impossibility on a human scale of an absolute mastery of the R. de C. In the launch there **is** the wandering of the points or rather the two are confused. The Global, in **this** model, does appear to be the unknowable, but configurations of flags and islands can act on this Global. They will constitute a role of gigantic ^{ancron} . To the Global one can associate a state: it is the Aîn-Soph. **Say** that one associates a state with the Global, that means: to associate a state with that which... (*) foresees it - by extension, or will say that c is 1 state of the Global.

The Schema of the Sephiroths is presented as a distribution of states i- that the human being takes when he becomes aware of the R. of C. and the figures that they can form.

Malcut corresponds to the usual R. of C.; it is the state associated with a human being who is unaware of the concept of C. de C. a

Yesod, Hod, Netzah, Din, Hesed, Binah. Hochmah, are states which correspond to various manipulations of the R. of C. They are not necessarily distributed along a lattice in the mathematical sense of the term.

Kether is the state of one who has become aware of the possibility of localizations by special arrangements of R. of C. and islands of incoherence ns allowing localizations.

t » We can associate characters with these states by correspondences and one can gradually become impregnated with these states and the imposed correspondences and manage to create generalized dynamic dualities.

One can wonder if the Global could be used directly, without going through the arrangements of the R. de C. (without localizing by these arrangements), n As we have already seen, it is the decalaires and channels which intervene. But according to what we have seen, it still comes down to using these arrangements without knowing it. the • • z >

So one can imagine a human being penetrating the Global by thought; he would really have the Aîn-Soph as a state. This is the last step. She is very delicate. It is good for the adept still in our time to feel and see a little the previous states. At another time, which we do not know, it may be able to work directly with the Global alone.

.ce (*) We can imagine an operability without structure. This would appear in
re our model as erasing any flag and running away from AB and CB. At this
stage manipulations are very delicate. (**)

(**) The Tree of the Sephiroths is what he carries within him; it is a surdans on a human scale. We carry it within ourselves at the same time as we extirpate it to use it, u

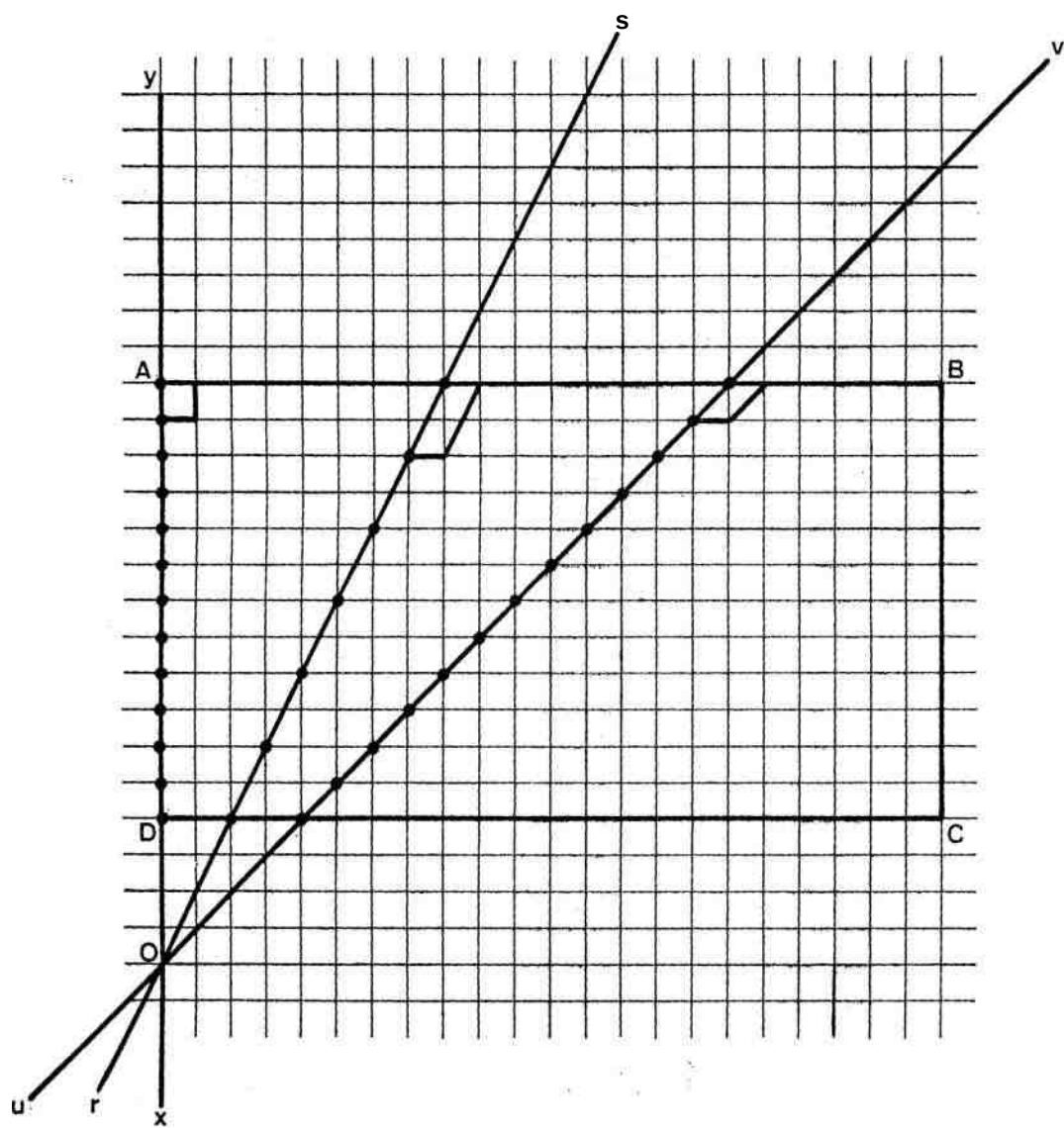


figure 1

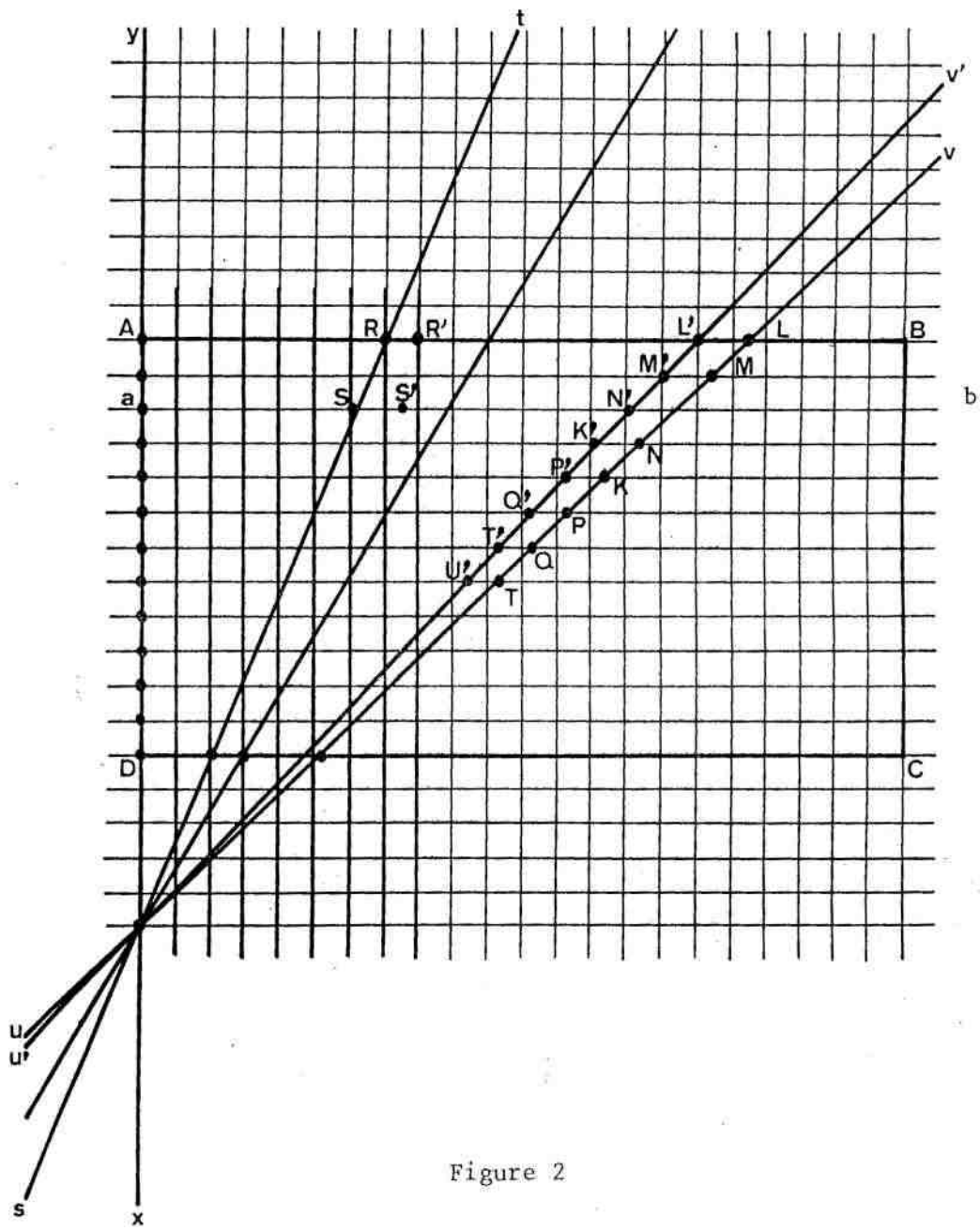


Figure 2

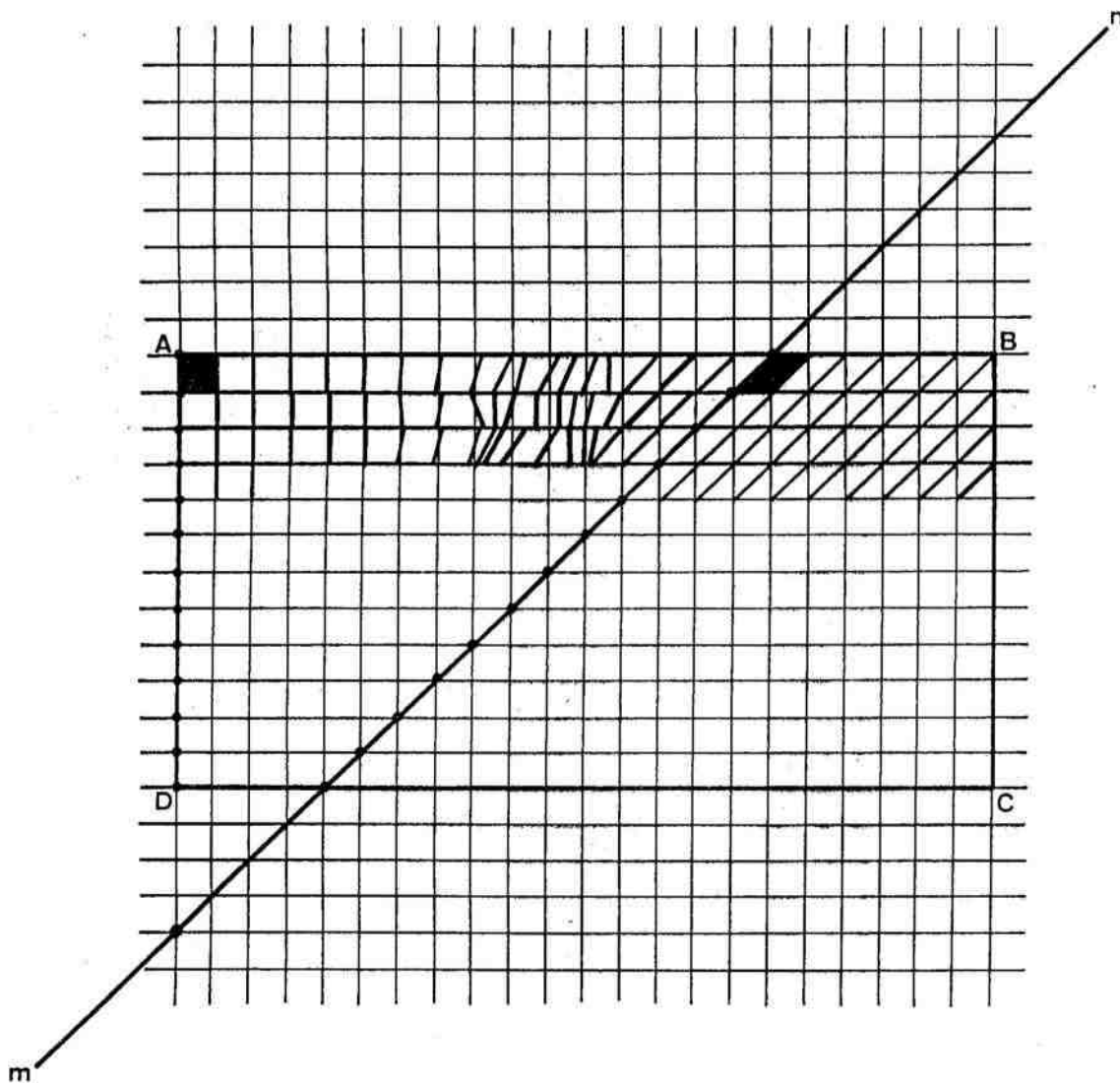


Figure 3

PART I by ROSGNILK

(France)

The human being, because he is alive, participates in universal principles ; let's say rather that he researches them, manufactures them, because it's on his scale; he establishes his understanding.

Let's see this on the following example: around all the capitals that have been built for a long time are "streams"; similarly around major cities, former key capitals. The "currents" form for us, observers, a justification for the establishment of the city in a precise place^

What should rather be seen is that the search for a rational explanation of the implantation of the city in such a place entails the construction of one or more coherences which are placed either in the C. of C. usual for some, or in the other C, of C. for others and leads to the double creation: that of the past, that of the "currents". Coherences are set up and we only have that; the question of whether these coherences correspond to the truth is meaningless, because there *are* only coherences; afterwards we can confuse truth and coherence if we absolutely want to keep the concepts of true, of truer than . .

Creation, because there is creation, is done by building the links that put in place a coherence in a C. de C. We thus arrive at a "sacred geography" which is a representation that we will call "the world". This will depend on the C. de C. chosen and the radius of coherence in the C. de C.

If we take up the texts of the Bible, we must above all not try to understand them according to our rational thought or according to the images associated with this thought. The texts of the Bible must be adapted to the language that one could draw from the dream; let's take an example - the Woman is taken from the rib of Adam. '

It is not a human rib.

It is not a side of man.

(A -kk)

We should rather orient ourselves towards the notion of halo and latticework associated with man: the woman would be like a latticework associated with the man-woman couple (the woman already carries the couple within her). (*) (**) (***)

(*) Pierre Smirne: Private communication. The notion of these currents is not considered here.

(**) it is moreover preferable to refer to the text in Hebrew.

(***) Lattice in the sense of a three-dimensional lattice.

The woman associated with the couple man-woman is entirely to give the halo; hence the expression: "Woman is drawn..." which is not contradictory with this text because there are other concepts. To the couple man-woman, there corresponds a form F; the woman is the part of this form F which builds thanks to F the aureolar of F .

In Alchemy, the Obs. is not separable from his experience, this is often said, but is not at all correctly felt nor by the one who expresses it, nor by those who listen to him. The Obs, when it does its manipulations, places itself differently vis-à-vis what it does. It is a displacement in front of reality which is important and which will cause reality to present itself differently, from which either we assume the form F in the structure ET and we manipulate to make F evolve, or we assume that the manipulation is only a rational view of a displacement of the Obs., then the work on F becomes a transformation (*) of the Obs. We will therefore say that the Obs. has "moved" in relation to the F-form It is this second point of view which must be considered if we want to speak of the 'Obs, in the experiment!' A "displacement" in relation to a system of forms entails

a transformation of this system of forms. So be the Obs. stay on point usual view: he transforms his system $F^\wedge$ into $F?$, Either he becomes aware of the other point of view, which will make him much more operational, he

"moves" relative to the system

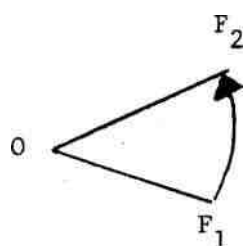
This displacement is abstract

but let's do them

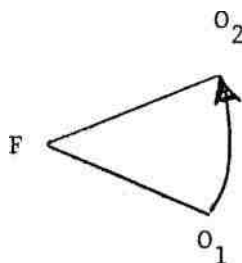
2 following diagrams.

X Obs. before and after $i = 1.2$

$F^\wedge$ Shape before and after $i \gg 1.2$



0 sees F pass in $F_{,,}$. Swap F and 0 , $0^\wedge$ and 0_2
we'll say F sees 0 in



(*) This is what implies reason for which we h preferred to write Obs. instead of observer
an observer's idea in e

and consider $(F \rightarrow F_x, 0; 0 \rightarrow 0_x, F)$ this has an aureolar which we call-

Let's ion the "displacement" of 0 relative to the system. The concept of reference can be either that used in physics, that is to say the data of a unit and an origin, or, and it is the process that the human being uses to place, to recognize, the assembly of systems which are ordered in a coherent way in a C. of C. . It is a more general view than that of the physicist and which seems less precise, but it corresponds to the way in which one fills the space.

To the rational way of thinking, it is good to remember what Agnès Chabrier [1] wrote. "It seems to me that by dint of searching, one hardly goes beyond the instrument of his research, by dint of studying, the limited object of his study, by dint of wanting to create, the narrow limits of his creation". However, there is this sensation vis-à-vis the RBB in the usual C. de C. as soon as one becomes aware of the existence of the other C. de C. Researchers would have to go and search in libraries, libraries of all genres and from all places, they would find something to think about, but above all that they do not reduce what they read to what they usually think, that they use it to escape and after work. Stanislas Lem writes "Lodes of potential discoveries certainly lie in various libraries, but they have not been noticed by competent people" [2] , "We have become accustomed to a clear situation, where what is dark and unknown s spreads before the homogeneous front of science, while what is conquered and understood constitutes its rear. But in the end, it does not matter whether the unknown is found within nature or whether it is formed among the paperwork libraries that no one reads, because data that has not entered the circular domain of science and does not circulate there to generate, in fact does not exist for us... Our culture, in its famous scientific part avant-garde, is a narrow creation, a vision narrowed each time by the observation, historically frozen, of a multitude of factors, among which the combination of circumstances, recognized as being the intangible principles of the methodology, eg can play a leading role"...

The search for sensations is essential, it gradually leads to new concepts and thereby to new models. So let's see again a thought of S. Lem: "The world injects its rules into human language as soon as it has begun to arise; mathematics lies dormant in all language and it is only to be discovered, not to be invented". This Sensation of mathematics which lies dormant in language can be considered as another form of existence such that one only feels it by expressing it in the form of sleep in language.

As has often been pointed out in this work, the human being launches his C. de C., which is equivalent to fixing a representation. The fixation due to the launch contains within it an operativity.

Let us take the example of a marble that someone throws blindfolded to describe what is in a close vicinity, which we can only know by the noises that will appear due to the course of the marble on the terrain (holes, markers, obstacles); By this process, we obtain a description of the room, but the most important thing is the fact of throwing: this is where the operation lies. The throwing of the marble is the shadow of a dynamism, Operability resides in the effort of the throwing; it shadows a dynamism. It is up to the human being to revitalize this shadow. Action is the most important; everything is in the possibility of action. The human being, when he thinks contains his thought in action, this is where operativity takes place. Thus operability is in every human being. This is where we see that everything is in the human being, that is to say that there is the possibility of operativity in throwing ladders. To launch a ladder is to choose an origin and a unit which are not separable.

In the Bible, characters are assigned lifespans of 700 years, for example. This is the clue for the one who reads it, that he must not place a duration of 700 years for the life of such a character, but that he must use this duration to bring about a sensation within himself. Astrology indicates rhythms; these must give birth to other rhythms in the brain and these are the ones that are operative.

What we call rational and the rational description of the world are an extreme case of representation. The OT does not take us back to the heart of this other representation.

The symbol of the serpent biting its own tail is reminiscent of creating oneself^(*), but it is important because it suggests the rational description which, in order to generate itself, desperately accumulates the systems of more

(*) For many terms made by us and presented for the first time in the magazine "ARK'ALL Communications" and then in this work, we have not given definitions. The reason is this: to bring out the concept but not to separate it (which a definition often does) from the intuition of which it is the culmination. By this means, we reinvest intuition. The reader must get used to this approach: it is necessary if he is to go on alone. It's a sure way to develop the deep sensitivity he needs and which has often been erased by excessive rationalism.

(**) There are other interpretations quoted in many works.

more complex, divides, and by this means extends.

It is difficult to understand ancient civilizations and movements. For example, the Bogomils, who were hunted down and exterminated by the Romans, fortunately left a message that Boutard seized. The sculptures of their tombs, sarcophagi, which are still found in Herzegovina, remain very mysterious for archaeologists who consider their descriptions simplistic: for example on the upper frieze of a tomb, we speak of the farandole of "kolo", a folk dance always in the spotlight in the country; on the same tomb, stags, on another monument motif, of a vine rolled up on a cross like a Christian cross, person- z . . . (*) swim surmounted by a crescent moon, many human hands that are found elsewhere in the caves of Dordogne and also as far as Hellas (which is much older than the Bogomil tombs). The latter, some of which recall a dance, are clues for L. Boutard. So the human being could, for the sake of aesthetics, in an unconscious way, attach himself to fundamental notions. And these notions are fundamental because the human being finds them by creating in the arts, games, his daily life, death etc.* then he takes them up, cleans them of their everyday gangue. If we generalize these attitudes, we arrive at the "Know yourself, and you will know the Universe and the gods" of the temple of Delphi. The human being, by its nature, enriches the C. de C.

But there can be deviation by the usual C. of C. which, by its too intense localization imprints in some of its coherences the very expression of the reason for this localization: we are faced with the observation and the causes at a time; observation and causes are in dynamic duality.

One can generalize the dynamic duality to a representation which one will name "dynamic star" where more than two terms intervene. There will be a dynamism and if there are n-1 terms, we will form a divine n-ary, r

If we consider certain centers of our modern world with multiple uses in the form of an enormous building where, for a certain arrangement, plays are performed, where, for another arrangement, operas are performed.

(*) Miroslav Karleja writes [3 1: "The hands of medieval sarcophagi, peaceful and joined in their Thomistic attitude, have survived from Gothic art until our days: these hands of orans are the sign which explains to us and makes us familiar with the human form, and this detail is often more important than all the other parts of the earthly effigy." "Nothing like this for the Yugoslav monoliths... On the Bogomil monoliths, there is nothing these prostrate figures, . . ."

ras, concerts, or even a transformation into a cycle path for runners, or even a ring for boxing, and we can continue in the multiplicity, this center will not be by its multiple possibilities a star but an "anti- star". Moving atmospheres are produced by this process.

If we move on to the UFO minoid, by structure transport, it behaves like an anti-star. A case sometimes mentioned is that of haunted houses. Such a house manifested itself in Belgium a few years ago. Before the haunting phenomena appeared, the house had no exterior; there was a family in the house with a young adolescent boy. The boy had ext and CV. When the haunting phenomena appeared: beds moving, blankets thrown on the ground, marks on the boy's stomach, etc., the house took on an outside, the boy's disappeared. The house and the boy were in equimsey and formed a channel. In the channel, the functionalities are exchanged, mixed. The boy's vital field is gone, the house has taken one. The boy kept his parent. The house is in K^{Sh^Ph} , condition, on the run. The teenager is not in this state. The house and its ext are in dynamic duality. We see that the study of the phenomenon is quite laborious, the phenomenon not healthy. Rebalancing such a place is tricky. It is not certain that all the phenomena of haunted houses can be explained by an identical process ; other cases are variants of these; in particular, cases where there are no adolescents may arise, so it is then possible that the phenomenon begins with a system of the TAG type on which the following is grafted. _____

In this book we have not spoken of an Italian researcher named Prioré. Prioré after the last world war left his homeland to settle in Bordeaux where he remained until his death which occurred in 1983. Prioré was a technician of exceptional skill in the handling of electrical devices and electromagnetic fields. . Stimulated by a personal idea, it seems, and encouragement (he fortunately had some) he built a device of extreme complexity, but for Etienne Wolff of "uncertain logic" [4].

Jean Borsarello in his book [5] devotes a few pages to Prioré, in a chapter entitled: Electromagnetic Field, Cancer and Acupuncture, H describes the relentlessness of Prioré in the realization of a first and then of a second device as well as his successes. medical; Sadly, that's not the case with most people in the 'V' industry."

many doctors did not go into the experiments in depth or wanted to ignore them and made unfounded criticisms of the work of this researcher. Wolff nevertheless agrees that the tumors treated by the Prioré device regress or even disappear. On the other hand, he is surprised that Prioré has systematically opposed the publication of the slightest detail about his camera; it was known that it was a question of repeated exposures to magnetic fields of 300 and 620 gauss. Prioré nevertheless received money from state organizations to install devices in its experimental premises. Wolff therefore criticizes Prioré for his silence about his device; he reproaches him for his "loopholes". It has, it seems, at most authorized the experimenters who took part in these experiments, to speak of a generator of electromagnetic waves and modulated magnetic fields. Prioré also took a patent, so Wolff said that the text was insufficient, that then he wanted (Prioré) to keep the financial benefit he would derive from it (Prioré had spent a lot of personal money for the advancement of his research) . Wolff recognizes certain qualities in Prioré: intuition, relentlessness, devotion to his fellow men in pain (he tried on several occasions to save human beings suffering from irremediable cancer with more or less success it seems?) "Let us also recall the efforts and complete or partial success. sometimes by Georges Lakhovsky who also, with his multi-wavelength oscillator, treated cancer patients, but some who were brought to him were in such a state of progress that he could only let them die without suffering .

It seems that Prioré was irritable, obstinate, secretive and Wolff's conclusion is that it is a total lack of scientific spirit.

That this researcher is irritable, we would be less so, having in front of us a sneering menagerie of so-called RBB researchers or low-class careerists often ready to do anything to stifle and resume the new work of a researcher less in place than them (it's not Wolff of course, but Wolff forgets the environment in which he himself is immersed).

"We become secretive", which is a completely normal attitude, and that in addition he is reproached for his lack of scientific collaboration, we then fall into a vast masquerade. "The inventor of the process lacked method and rigor..." another reproach to Prioré.

"He should never have taken refuge behind a kind of professional secrecy, which made him suspect to everyone.

pleasure and displeasure of competent scientists, the editing and description even master"

(To) tion of a device that he could no longer

But Sprink, Hendershott and the others had great difficulty mastering their devices. Sprink worked from 1923 to 1960-1965 on different devices. He sometimes had to wait a month for the effect to appear, and this after putting on the electrical voltage.

It is unfortunate to see people who claim to be scientists making simplistic judgments, in the name of the scientific method that they

personal sauce, on extremely fine, delicate experiments In addition Prioré, Sprink, Moray, Keely, ... are off the beaten track!?

Finally Wolff thinks he has a clear conscience by adding: "We neither want nor should overwhelm the memory of a man who was of good will" This is to take Prioré for a nice simpleton! That Prioré wanted to make money is completely normal. Besides, if a brilliant inventor earns money, he reinvests part of it in new research; he makes a lot of people work in pure research (on the other hand if he is led by antiscientific attitudes to withdraw into himself, there is no longer any possible opening), and also in the industrial field and this much more than CNRS researchers do. Tesla by his multiple inventions with deep industrial scope, was a model of the genre; he reinvested in his research.

Often we demand evidence at all costs to make hypotheses credible , that's what a so-called good scientist demands. Indeed, the history of science is strewn with cases of researchers who put forward ideas whose verifications turned out to be false. Yet the history of science is also full of cases of those researchers who felt profound discoveries without being able to give the proofs. In the field of eifs, for example, researchers working on the continuation of Sprink's work were asked to prove that the effects due to the device, if we wanted to associate them with eifs, were not due to an electric field, to an oddly shaped magnetic field, electromagnetic wave propagation,... The number of nega

(*) It seems that the last models he built were quickly out of use and did not allow repeating experiments.

(**) Wolff gives this advice "The only possible modification was for example to vary the intensity of the magnetic field. In a well-done experiment it is only allowed to modify one factor"?!! In this field, this is not true!

ments that would have had to be justified to please a few handfuls of RBBs was important enough to spend the time of an average human life there, without being able to prove everything. A researcher feels the way, the necessary concept, he does not build his truth on negations; these can come later.

Two anecdotes linked to Albert Einstein show that he knew his hypothesis true long before it was verified [6]. There were two major expeditions organized in February 1919 to observe a total solar eclipse, from Sobral, to northern Brazil and the island of Principe in the Gulf of Guinea. They both demonstrated that Einstein's calculations were correct and that it was therefore necessary to abandon the conception of the world which had been imposed since Newton. One might think that Einstein was transported with joy by this verification of his calculations. Antonina Valentin relates the memories of Eisa Einstein:

Einstein looked at the photographs with a surprise that changed to joy lighting up his features: "It's marvellous, it's truly marvelous!"

It was entirely believed to be his triumph. When the importance of this confirmation was explained to Mrs. Einstein, she ventured, shy:

"How happy you must be Albert!"

And Einstein, his eyes still riveted on the photograph, replied: "I am delighted"
But he immediately added:

"I would never have believed that photographs could reach such perfection"

So he was talking about the quality of the shots.

His son-in-law, Dima Marianoff also reports: One of Einstein's colleagues from the University of Berlin happened to be there when the photographs were in the researcher's hands. He said to him: "You must be happy, you hold in your hand the proof of the exactitude of your theory".

"Proof?...Proof?" Einstein repeated with that surprised, questioning air that always gives the impression that he is only a visitor on earth. "Proof? They're the ones who needed it! I wasn't looking for it!"

This second anecdote reflects on the first because we see that Einstein could have been delighted by the quality of the photographic prints.

Previously, we talked about magic and music. Western music has lost more and more of its magical character. It was written with more and more precision: the landmarks were accentuated especially in the 17th century. The state of music has evolved. Our society has a kind of landmark fetishism; to manufacture more and more perfected electro-acoustic devices only engenders a local and mechanistic understanding of the musical state; therefore, reproduction quality should not be confused with music condition. The musician is now too caught up in his training, and his training has often been marred by competition. He is rarely himself, the same for those who are called virtuosos. The races of technical difficulty and sonic perfection have meant that in this perfection one senses a localization that will be less and less accepted. Often we no longer feel in front of a human being who places himself at the human level when a known violinist or pianist interprets concertos, sonatas etc... we would prefer to feel the human being; the human being corresponds to a level of comprehension of the world, it is also for that that it is necessary, when one manufactures a form, to think it on the human scale (at the same time level and scale - these two . (*) words are needed both to convey the deeper meaning)

The world of music often becomes unpleasant from those who animate it : a violinist talks for hours about a stroke of the bow which gives such an effect. While we know that in 30 years, this bow stroke will be different and that the present moment will no longer be popular. (We will then invoke progress - what a lack of deep reflection we find at all times!). There is nothing more boring than an assembly of high and mid-rated professional musicians. They establish a "bourgeoisie" in the musical field, and their attitude shows that they are far from music. They pride themselves on sensitivity of intelligence. The same goes for music lovers: they want to talk about everything related to their specialty (music, for them, is not a deep and discreet passion, it's a specialty). Much could be said about these characters; in conclusion, we are sure that they have not felt, and will never even be aware of what the music state is.

(*) The slightly truncated pyramid of Cheops is thus a form which is on a human scale. This does not mean that all pyramids must be truncated!

The tuning fork is a reference in relation to which we will tune those who must together create a musical state. In untempered music (Oriental, for example), various sound scales can be used; you have to agree before you play the music. For example, such a musical state is created among Hindus who play together. Silence is necessary; it brings relief to the sound form which partly emerges in the room. The music state will then be present in silence. In pre-classical music, there are the preludes which were made to tune the instruments; it was to refine the benchmarks that we were going to use. The cue in oriental music is not like in classical music and so the attitude of the instrumentalist is not at all the same: in the first case the instrumentalist is much more creative, in the second he catches up its lack of creativity by a perfection of execution in intense localization. This localization involves a shaking thanks to which others come to be enriched; nothing is completely negative. The rise in the frequency of the note there for a century and a half, it may be, favors a localization of all musical works. But it's only an hypothesis.

In an orchestra it is interesting to take a closer look at the conductor. He gets straddled by the listening crowd, the Enel crowd that the concert hall creates. He can also be overlapped by the symphony entity (if it is a symphony he conducts). Between these two overlaps transfers occur. It also serves as a channel for the orchestra. His vital field can turn and disappear, eifs appear.

Some people try, to think outside the box, to create "open" systems. An open system is never completely defined; to define it, it must be immersed in another. Open systems occur for example in thermodynamics where they have a rather specific meaning.

These attempts were therefore put in place to reduce any system to what is called a formalized system in the sense of Hilbert and from there, thanks to Kurt Goedel (1931) such a system can be entirely arithmetized. We will not insist on the work of Goedel. In an R. de C. one must not understand the prehension of systems as successive interlockings because they are all supposed to be open, therefore each incomplete, therefore complementing each other from content to

(*) We can, for an overview, refer to [7].

container, but you have to see emergences coming together. It is moreover the process that an observer uses to place himself in the C. of C. that he projects. In the rational C. of C., one describes the phenomena of propagation of the sound, by the effect known as Doppler. This has been extended to the electromagnetic field under the name of Doppler-Fizeau effect . is located in zones of emergence or immersion of the ET structure.

René Huyghe in [8] answers a question posed by Simon Monneret about the creation of new terms.

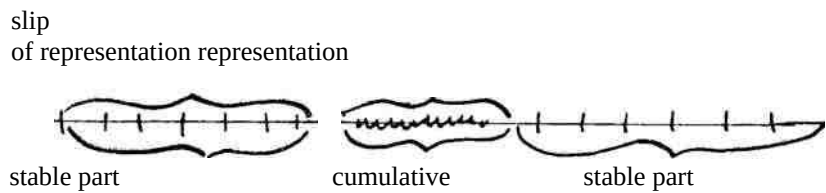
"I am deeply hostile to it. It is a process that is both pedantic and easy; it stems from one of the flaws of our time, where "beautiful spirits" dream, like the cuttlefish, of hiding in a cloud. obscure. Too many contemporary thinkers are enamored of the idea of creating a word that is theirs alone, or at least the prerogative of a restricted group. It will be a retreat for them, a jade cave, at the inside which they will tangerine,,"

Marc Beigbender changed the spelling of certain words to give a new meaning while keeping the pronunciation, which is a link with the term already used. This link is important - the intuition is reinvested - it is a first detachment. If we create new words, and there were a certain number of them in this work, and if in addition we give them without definitions, it is so that people do not get stuck on a word, avoid to a known meaning, avoid reinjecting into a word thanks to which a meaning is evoked, the usual meaning (if it is an already existing one). The person is placed in a sensation corresponding to the surprise which places the new word on the background of the usual knowledge. We have a halo - it is the latter that will bathe the reader. In this technique, the reader gradually and unconsciously detaches himself from the usual C. de C.; the unhooking technique is in the unhooking. Thus (technique of detachment, detachment; intuition, reinjection of intuition) creates an aureolar which will reinforce the previous one by overlapping or equimsey between the two.

We therefore see the usefulness of these concepts introduced here which at the same time justify the criticism leveled at Huyghe's response to Monneret. Moreover Huyghes speaks of the ritual in "dead souls" by the intuitive value of the word "He who gets stuck on a word does not interest me, because we will never be able to

lead far". He further writes "...a certain system of ideas is indispensable in order to communicate itself to others. But he must never forget the dark depths from which he came."

In the examination of dreams, it is interesting to focus on what is often neglected, the deformation of a representation to pass to another, let us say a shift in a representation. Thanks to this shift, we can construct a halo; for that let us consider a cumulo-shift arithmetic; it is composed of discontinuous parts and shifts: cumulatives and a decalar. (Representation slippage, representations; discontinuous (or stable) part of arithmetic, cumulative) has a certain halo. Let us explain the correspondence:



In the dream there would be a sort of photographic effect. It would be the negative of a positive. The halo is not explained here.

If we consider an oracle, we have:

(oracle, rational thought; an emergence, a support). This has a halo and it is he who is the value of the oracle, whence in an oracle one must look for a deep sensation, one will gradually become impregnated with his halo.

We have previously seen interpretations of day (D) and night (N). We can consider N emerges from J and “reciprocally ($J = N$ in duration at the equinoxes , and this is the usual representation of Yin-Yang). We can put them in dynamic duality at the equinoxes; this is the role of certain rituals carried out at these times.

In each civilization a form is projected onto the civilization which serves as a background; a halo is created and reused by this shape. It's a bit like the story of the sphinx in Jean Cocteau's work "La Machine Infernale" [9] (*).____

(*) Let us recall an expression of Cocteau, still in this work: "The time of men is folded eternity".

We haven't talked about Nicolas Tesla or Wilhelm Reich. They were, however, two giants of the same order as Sprink, Boutard, Moray...

Tesla has filed over seven hundred patents. In his writings he relates that it happened to him, when he thought of a machine, to see it almost entirely realized in his brain; then he had only to describe it. He says himself that some have appropriated some of his discoveries [10] and [11].

Wilhelm Reich linked orgasm and bioelectricity. In orgasm, very strong transfers are created between the partners, they pass to high C. of T.. He tackled cancer and got some interesting results. Orgone boxes are associated with eifs as with certain N. Tesla devices.

Parapsychology involves, in order to consider a valid understanding, all the notions developed in [12]. It is useless to experiment with statistics; by this means it is deformed to make it fit into a C. of C, to which it does not belong. Raymond Réant, clairvoyant-sensitive, made predictions of the future for certain people which, in general, came true. Did he predict the near future for each of these people? Did it influence their future? For one or the other of these questions, a break is created which immediately leads back to the rational C. de C., such as the use of the terms objective and subjective.

Let's make a halo thanks to the process used many times (predict x((*) (**)) the future, influence the future; objective, subjective). It is this aureolar which is important. We escape by this trick to the dilemma posed. We also see that the rational C. de C. contains the tricks that make it possible to escape it and this by arrangement of concepts, but this arrangement is not in the C. de C. To obtain the halo, we used objective and subjective. Their order of presentation matters little, which makes the halo unique; it is tinged with an atmosphere. The aureolar-atmosphere is the operation carried out (***) by the clairvoyant-sensitive and it is a form depending on the moment.

(*) As it is presented in the usual language.

(**) Objective and subjective are used as means and no longer as ends.

(***) We see that parapsychology, if we want to approach it correctly, must be thought out with the concepts presented and developed in all these chapters. The usual approach has only distorted it. In a duplication for example, it is good to study the changes of eifs during the evolution of the duplication of the subject, the new states which can appear, his vital field, his psychic field, his possible transfers. It is also necessary for the safety of those who practice it.

J. Ravatin in the courses relating to the eifs that he gave in Paris from 1975 to 1980, pointed out the choice according to civilizations and periods of the approximate value given to certain numbers, in particular that of π . The history attached to π is now considered as a curiosity whereas on this example one can feel the notion of the more or less localized* The importance of the number π is such that we must take up a brief history of it. The Babylonians and the Hebrews admitted that the length of the incenterence is three times the length of its diameter; the Egyptians identified π with $\frac{22}{7}$ (which allowed the value 3.1123). Archimedes affirmed that π is between $3 + \frac{1}{16}$ and $3 + \frac{1}{8}$, that is to say between the two decimal numbers 3.140845 and 3.142857. The Chinese, the Japanese had also found values

decimals; in the 4th century of our era the Hindu scholar Arya-Bhata had found

| | | |
|------------------------------------|---|----------------------------------|
| 3.1416. | In the 16th century the Dutchman Pierre Miétus gave the value | 355 |
| provides | five exact decimal places | $\frac{355}{113}$ ^{qU1} |
| | Frenchman François Viété circa 1593 | calculated |
| π with 10 exact decimal places | in 1610 Ludolph van Ceulen succeeded in obtaining 127 | |
| decimals | In 1873 the Englishman William Shanks obtained 707 | decimals |
| | | In 1882, |

Ferdinand Lindemann demonstrated that π is a transcendent number. Thus is found definitively proven the impossibility of squaring the circle, that is, of a side of a square whose area would be equal to that of a given circle.

In [13] Marcel Boll quoting Lancelot Hoghen, adds: "An American state has tried to restore π to its biblical value (IIth book of Chronicles, IV, 2). tape measure (just measure successively the perimeter and the diameter of the bottom of the pan, to realize that π is greater than 3); but these same people mechanically repeat the famous phrase of the Psalms "They have eyes and will not see", without suspecting in the least that it applies to them; a fine example of this obscurantism, the deplorable effects of which can be seen everywhere".

It is quite surprising to see that the Babylonians, the Hebrews took the value 3 for π ; the error is hardly conceivable. And yet, 22 that's what they took! Let's see it another way. The values $3 - \frac{1}{16}$ - 3.1 - etc... are too coarse to be considered errors. For our C. de C. these are errors, but in a way voluntary errors. Taking 3 for π is to consider a circle already delocalized. Thus (3, π ; delocalized, localized circle) has an aureolar which is a circle a little less localized than the circle thought of in geometry where the localization is complete. They

so thought the halo, that is to say a depth was felt and had to be expressed. The circle drawn on the monument had to be placed where that depth reflected on whoever was looking not at the circle but at the monument on which the circle was drawn. This turn of mind led to paths other than the path used now. Moreover, even if the circle was drawn according to our geometric formulation, the circle was not thought of as a circle alone but on a building, in a place, it could be less localized with the surroundings. This is what we have already signaled by saying that two triangles in a plane, in certain positions, are perhaps no longer quite two triangles, one can partially delocalize the other; looking at both at the same time is not the same as looking at one then the other; in Cartesian thought we confuse two approaches, because we forget the approach, we do not reinvest it. It is necessary to reinvest it if we want to be operative: to be operative is almost to ask permission to start again a second time an act which, for him, had all its value, because it occurred at a given moment. This act was not separable from the moment; to be operative is to take it up alone and place it at another time, even if we know that it is not exactly the same act; it is here that we understand what humility is: it is the expression of the state of an attitude,

“They have eyes and will not see”; Lancelot Hoghen by quoting this extract from the Psalms forgets that it is for him like a “return shock”. You have to feel the delocalization to measure how much the Babylonians and the Hebrews, taking 3 for π , thought of the circle in a totally different way from what it is in rationalism. To consider a delocalized circle therefore amounts to thinking much more than the circle.

One type of shape that we haven't reported is the maze. The word is quite enigmatic. We find drawings of labyrinths all over the world, the Cretan Mediterranean, the New Hebrides, Patagonia, India, Tibet, Scandinavia, among the Zulus. Many hypotheses have been put forward to justify the presence of labyrinths. We will not expand on this subject because the reason for the existence of the labyrinths is certainly other than all that could be supposed.

Many have been found in churches and also in villas made of mosaics. In the Roman villa of Harphar in Yorkshire, in the

Roman villa of Italica near Seville, in a family tomb of the ancient Roman necropolis of Sousse in Tunisia. In Pompeii, we found a "house of the labyrinth" called "house of Diomede". At Calvatone, near Cremona, there is a mosaic floor representing a labyrinth. They are also found in books and in particular in a work by Strabino on labyrinths, published in Nuremberg in 1510.

The Church of Santa Maria in Aquiro in Rome has a circular labyrinth with a diameter of about two meters. The Church of Saint Mary of Trastevere in Rome also has one inlaid with a diameter of about four meters. The Cathedral of Chartres has the largest labyrinth of diameter 13 meters 25 in blue and white stones. The Cathedral of Amiens also has its labyrinth (the latter is a copy of an older one which was destroyed in 1825). Its course is similar to that of Chartres except that it is made by straight segments instead of arcs of circles

There was one in the Cathedral of Reims; it was destroyed on the order of a canon who was bothered by the uproar of the children and visitors who walked through it .

An Easter ritual of dancing and playing pelota in the labyrinth of Auxerre Cathedral was performed by the dean and novices of the town's convent by forming a chain around the labyrinth and holding hands; they danced around passing a ball to each other.

The chapter room of Bayeux Cathedral is decorated with a maze of red, black and ocher tiles; its diameter is 4 m.

Sens Cathedral had a labyrinth which was destroyed in 1769, about 13 m in diameter. The outline formed by hollow lines was filled with lead.

In the Church of Genainville (Val d'Oise) an engraved labyrinth measures 70 cm by 78 cm.

In Norway, Denmark, England, it has also been discovered in churches. In Italy, gardens had the shape of labyrinths [14].

The most worked labyrinths often resemble that of the Cathedral of Chartres. It can be reproduced in 1 m by 1 m format and used like any other shape. We realize that the labyrinth is a very complete form / £ \ because we can make it emit all the eifs we want, even the VC and we can thanks to impose on a place, states and a atmosphere. We even have (*)

(*) VC abbreviation of Creator Verb.

the eifs envelopes. To achieve this, you can use pawns that you place in the cells that are on the perimeter of the labyrinth (this is the case with that of Chartres); then we also have the pawns on the course starting from the entrance and going towards the center. Other methods are possible. Nevertheless, any manipulation on the labyrinth requires great sensitivity; it can be dangerous if the person working is not sufficiently prepared.

The labyrinth of Chartres Cathedral also behaves like a shape analyzer. It can also be used to isolate a shape from a neighborhood of shapes.

The labyrinth also has the property of integrating the model of whoever uses it and this form remains stable in this role; we can check thanks to the labyrinth that a cancer first appears on the ext of the patient and then localizes.

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CONCLUSION AND DIGRESSION - VARIATIONS ON PREVIOUS THEMES

Part II

Designed by

The numbers that occur in documents considered to be ciphered (the Bible is a typical example) are "pseudo-traps" because a researcher can find there a number that will seem fundamental to him; this system of reading which is generated is not fundamental; it gives certainty to those who use it. They create a search path but they are not a search path; they are what we call triggers.

Signs, traced flat shapes depend on the representation; moreover, this representation is very marked by the C. of C. . Let us recall that there is no symmetry between the two C. de C.: the usual C. de C. cannot consider the other; on the other hand, the other C. of C. can consider the usual C. of C..

Let us rethink the expression: The Head (who does not know) created the God of Heaven and Earth. The Head can be considered as the Obs. The God: it is the concept of C. de C.. The Sky: it is the other C. de C. (it contains the concept of flight). The Earth: it is the usual C. of C.. The representations of the human being, as long as there is coherence, are realizable.

In all that precedes, there has only been question of two C. de C. A third and a fourth must not be constructed. That does not make sense. It is good that the person who reads this book redo the intellectual process of the one who felt the need for C. de C. (This is not to redo the history); it will come to sensation. We therefore have the rational C. of C. and we construct another one; this other will give richness to the concept of C. de C. because at that moment all the rational becomes a C. de C.; the extensions of the rational C. of C. are only the rational C. of C.. One of the representations of the Ancients was the Earth at the center of the World. As the Obs launches its C. de C., the representation:

Terre center du Monde, by copying this process, reinvests in this representation . It would be necessary to examine in addition what corresponds to the fact that the Obs. gets into the C. of C.

Consider that for Man, God is a wager: Man makes a wager, he creates God, but God creates Man. We fully find here the C. of C. that we launch and in which we put ourselves. We can think like this: God has Christ. The trace in the room is Christ who became Man.

It is not the breaking of the macron but a trace in the local of this macron. It is from all these concepts that we can begin to approach the Father-Son-Holy Spirit trinity.

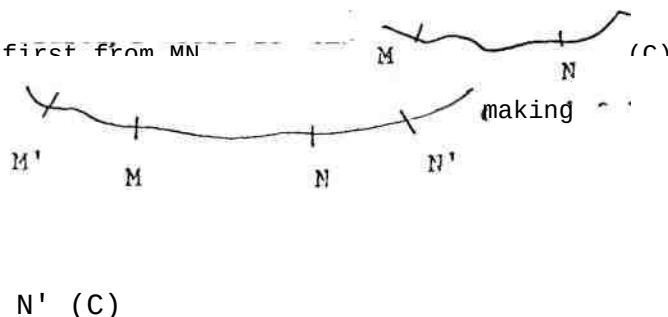
There are often two seemingly opposite meanings of a word in Hebrew. In this case whoever uses the word becomes a third term which will create the dynamic duality between the two meanings. But it's not the Z.z (*) dynamism, it has rather a role of trigger.

We can discover individually or collectively. switch from individual to the collective is to extend to extend is to reinvest the launch of the C. of C? . (**)

In some cases, it is possible to see a shape drawn in a plan, a stone, a branch of a tree in the evening perhaps taken for an animal thanks to the reflections of a fire in the dark; in these cases one stretches oneself;

to extend is to seek another polar: first from MN on a curve (C) we go to M'N'

that we have



(*) M. de Charette in Asturgia uses the concepts of Theme and Version. These are brought into dynamic duality by the Hand. The hand is a trigger.

To seek is to create; we place forms to which we give a meaning: it is to make them a trigger. Christ is a trigger of Man.

(**) In Astrology or in Asturgia impressions are placed on planets. This is actually making a copy of an extension.

The animal is animal (in the Cartesian sense) only by the Cartesian approach which suppresses the extension. Otherwise the animal is other, because in the usual domain we do not communicate because we restrict.

Do not confuse false with absurd; if the false is applied it is absurd, but all that is absurd does not come from such a source. .

The experiments were carried out in the following direction: women sing every day near a pregnant woman: we notice that the child has supple and developed hands. In conclusion: we have a vision of people who observe after the fact; we can see it differently: it is the people (all possible Obs.) who are transformed; it is even much more; it is a representation of the world that is changed.

In a condensed formula, we must say "it is not the beings and the ((*) **) things that change, but the representation of the world

WJ Thomas wrote, "When men regard certain situations as real, they are real in their consequences." It is a very particular case of the creation of the past in the future which fills a coherence. This does not mean that one can posit any situations, assume them to be real and then consider their consequences as real. This is where the notion of coherence comes in, which cannot be defined but is constructed. Coherence is a bit like a surd.

One can also think, taking up the work of Boutard, of the expression of Art from the components of the Aether. Thus Art relies on the $\mathbb{A}$ to express itself and this at a given time, but the $\mathbb{A}$. responds to Art and warps with it; this entails that there may be many constituents of $\mathbb{A}$. .

Let us give some complements to the concept of overlap: overlap $F \cap F'$ of two forms F and F' . F (F by F'), for example obsidian is overlapped by Iceland spar. We also have C. de C. the basic program VC. We can overlap sets of numbers: for example,

(*) You must not say: "but not the child", because this no longer makes sense from the moment you adopt this path.

(**) It is more correct to simply say "the representation of the world changes".

we will write

\ <1^33.34

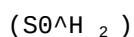
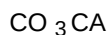
|

which means that 97 and 98 where 97 emerges and 98 is really present are overlapped by 33 which emerges and 34 which is really present.

Let 's see **the case of a** chemical reaction: sulfuric acid which acts on **calcium carbonate**: Write the reaction



is much less interesting than noticing that we have an overlap. In fact, we should write ':



(sulphuric acid, when reacted, is che

covered by calcium carbonate).

This does not mean that **the** chemical reaction **presented** according to the usual writing is false. It is better to consider that we are dealing with a much broader description.

Now let's look at elements of the **Mendelieff table**. Let's keep it find Ca,K,Hg (the case of H potassium and hydrogen and make (K) ; we magnesium overlapped by oxygen). And in (Mg)^ , we also **find Ca,K,Mg** .

If we are in front of a balanced chemical reaction, we can have overlap, but then initial products can be found in the overlapped-overlap system, but not necessarily all of them.

On the other hand, each time there is a chemical reaction, we are not necessarily faced with an overlap.

In the overlap sulfuric acid by calcium carbonate, neither is found. If they are found by chemical analysis, it can be assumed that the reaction is not explained solely by an overlap.

(*) This is K 5 Mg; $^{39}_{20}\text{Ca}$ and $^{24}_9\text{O}$ 44 16

Louis KERVRAN



was one of the few to ensure that biological transmutations were possible. he conducted systematic experiments to confirm observations he had made at a young age. In publications from 1936 to 1939 and after the last world war he presented the details of his results. After the war, he was called to Paris as a scientific delegate to a minister in order to represent him in all the ministerial and interministerial commissions responsible in particular for studying protection against the effects of radioactivity. Although by his functions it was possible for him to ask for an opinion and to summon any scientist (physicist biologist etc...) specialist in a very limited subject, there was no question for him of taking the opinion , about his work, professors, researchers of the CNRS to make a favorable judgment and to have any help. As he wrote, he happened to meet men who were sometimes exceptional, often remarkable, but restrained by a regrettable conformism, trampling on routines, sometimes discouraged, even worried about their future, "while he has so much to do to go from forward, in new ways".

It turns out that the biological transmutations are real; Paul Baranger, a chemist, had also been one of their fierce defendants. L. Kervran also verified that high pressures can bring about transmutations, which geologists had repeatedly pointed out following meteorite falls.

Biological transmutations, refused by physicists because of the low energy they use, are much closer to alchemy than to chemistry; they are localizations by ways related to the Global. (*)

(*) There was also Paul Baranger.

Whatever the **field among** the five cities, since the awarding of the first prizes, we can notice this: a large number of researchers, pen- sisters, etc... really deserved it; others have benefited from it, either because they

■ felt an idea in germ, **and** appropriated it by different means, either for political reasons.

We come to the Lebon-Lebon prize: Nobel upside down, the anti-Nobel. This prize was imagined by David **Franeric**, when he was a student engineer around **1979**. It is reserved for **a** kind of individuals who can be characterized as follows: sure of their abilities, eager for honors, having many of knowledge (libraries-with-legs), competent in certain domains, exceeding in their appreciations the PU their domains of knowledge, having sure judgments on individuals: (**they** possess **absolute** scales of classification of individuals, established according to gender criteria: the school from which the individual leaves, the rank of entry, exit from the school, alertness, the mindset. It's the RBB that reinstalls RBB!!

It must be admitted that we hear far too many researchers lend themselves- dre researchers, scientists claiming to be scientists. Because they are scientists, researchers in laboratories, universities, they allow themselves

in the name of their titles, even more of Science, to speak of what they do not do not know, which is contrary to all scientific ethics; and moreover, i to approach domains which are not the rational C. of C. and by this attitude, to ridicule and reject them. There are very few real researchers who have the soul, very few scientists worthy of bearing this name. We are at a time when titles and denominations waltz; a handyman who repairs his radio calls himself an electronics engineer (this goes even further, when one finds people who in winter having a bad cold announce a flu ' 1 '). It is for all this that another prize had to be created.

While the Nobel Prize is only awarded sparingly, the Lebon Prize will be awarded quite frequently, the people worthy of receiving it being

(*) It would be necessary all the same that these howlings manage to realize that Science does not belong to them, that they do not represent it. Science as an entity is not on the scale of Europe. That's not the same at all!

(**) They treat themselves like having the flu, which makes the drug factories happy. This attitude is an index of mediocrity in a society due to a degeneration that manifests itself in verbal formulation.

quite numerous. It may even happen that a Nobel is Lebon. The reverse is, it seems, much rarer.

We have already pointed out that in our Cartesian universe (that is to say, localization in the usual C. of C.), there is the *operativity of delocalization. This is what the alchemists understood.

Paul Dirac wrote "All matter is created from some imperceptible substrate". One can feel this thought known as the creative motive by the spirit of the search for origins in the infinitely small.

The infinitely small seems to be this imperceptible substrate. If we think that it is the brain which, in this search for origins, creates the infinitely small, then the infirm small is an image of emergence.

A certain number of terms had to be elaborated, as we were able to verify during the reading of the work, to develop the new form of thought. Already, it has been verified that people using these terms heard during seminars, courses, conferences, or read in the magazine "ARK'ALL Communications" and use them, without having understood the meaning, and absolutely distort them by relation to the meanings given by their author. This is really unbearable because the effort of building this new form of thought will quickly be reduced to nothing. In one example: alchemists gave a conference in a small town in central France; during their presentation they used the word global for total, even osteopaths following a conference given by J. Ravatin at their request, made the same mistake on the same term. The words overlap, equimsey are used indiscriminately. People have an uneasy feeling; they want something new; they make an extension of the usual C. de C., that's all they found. They don't understand that in our usual representation, we only see endings and that the important thing is the journey that leads to the ending.

In everything relating to eifs, the structuring of the question asked gives an answer. There is not only true and false, there is also structuring. We have seen the concept of ancron: it completes the cultural vehicle (vehrel). All these developments are left to the reader in exercises.

Let us immediately point out that to any reader requesting further explanations

comments, on all that is treated in this work, it will not be answered to any mail. It should be understood that the researchers of the ARK'ALL Foundation can bring new solutions and by their theoretical concepts and by their experimental achievements; these require investments and these researchers, engineers, researchers from state or private organizations, in order to follow their ideal and their goals, cannot involve their organizations in it and must by their personal means invest in their research, also c there is a pressing need for help in the form of money that they need They do not have time to answer multiple letters asking interesting questions, rich in consequences certainly, but less fundamental for them and for those who want to follow them in this way only small donations which can by their accumulation, allow them to solve problems posed by our technological society in crisis, and ensure their development with multiple openings by the construction and the establishment on the planet Earth of a new C. de C.

Guy Thieux made a presentation at the "Symbolism and Architecture
(*) sacred and Possibilities of Contemporary Orientation", relating to the art of building and adorning buildings, the rules determining a structure which must, in their usefulness for a cycle, a given period, meet material needs, aspirations spiritual and human necessity.

Two ancient texts, the collection of Chou-Li and the ten books of Vitruvius indicate to us the fundamental principles without which the constructions of civilizations of ancient China and those of the Mediterranean world would not have existed, to take only these two examples.

The Asian text compiled around -800 before the modern era, reports extremely precise indications on the use of the "six ritual instruments of the emperor", the Roman text established around the first century before our era although using words of obscure meaning constantly reminds us of the considerations of ancient architects.

A detailed analysis of these two treaties indicates several reflections

(*) which was held at the Pedagogical Unit of Architecture n°6, Paris 19th on March 21, 1983, congress organized by Jacques Famery and Pierre Lefèvre.

inspired by both a solid "naturalist" knowledge of these predecessors and a precise relevance of applying the principles to each case of type of construction, religious, civil, military, land, marine: these reflections - placed in the context of .the time of the constructions prove that the permanent concern of the old builders aimed in the first place for a relationship as balanced as possible, experienced by man on earth, of forces, of energy exchanges , of which the relative apparent discontinuities are a cosmically one.

Human cosmologies have been expressed in apparently very different ways. Their presentations, their poems veil one and the same concern: to constantly preserve a formula, a universal model, under numbers, ratios of proportions, a geometry adapted to each particular ethnic, geographical, climatological case. Keep a formula over time, that is to say the movement of the earth and the solar system in our galaxy. This formula allows by successive anamorphoses metamorphoses, adaptations of the adequations to space, to time and to establish on earth the civilizations whose traces, for the archaeological discoveries, prove the existence to us.

In these discoveries, architecture impresses our minds with more enjoyment, perhaps because it has more durable proportions than painting or plastic art.

The civilization of the dawning era is confronted with a hyper-intellectualized, intensely politicized approach and knowledge of the harmonies of nature experienced by our predecessors and approached by them with another form of reason.

The intense development of industrial techniques has brought about new and considerable facts: man can modify the structure of the atomic nucleus, the information of the genetic message, man can explore materially and by vehicle the extraterrestrial space. The massive use of electricity, radio waves and the development of chemistry have caused pollution over the last 150 years to modify the exchanges of the hydrosphere, the atmosphere, and even our ionosphere.

These considerations imply an adaptation of the "formula" to these new facts: knowing how to adapt, use and harmonize for our time the privileged relations of man with his environment, which the poet of Chou-Li poetically expresses by the marriage of the tiger of Jade with the Azure Dragon or How to Happily Place the Channel of Celestial Influences to Earth Opposite the Channel of Earth to Heaven Influences.

To identify the constants and variables of the problem posed, we need to make a brief interdisciplinary overview of the known physical fields that any manufacturer will now have to take into account, as well as new considerations that we will endeavor to clarify.

The materials used, the materials used by the architect are linked to the states of solid, liquid, gaseous matter, to their interactions on the one hand, and on the other hand to radiant fields, to states of radiation beyond and below the visible optical spectrum.

All of the terrestrial matter, the radiations of terrestrial origin were previously in a state of plasma which still exists on our sun. After the differentiation of these states, there remains a phenomenon of recurrent oscillation that certain working hypotheses relate to quantum physics between the same materials although in a different and more complex state on earth than on the sun.

These relations have a direct consequence on the behavior of living cells whose field of coherence and equilibrium is linked by teleaction to the nature of the soil, the subsoil, its chemical composition, its forms, the materials and to the shapes that make up the building or buildings constructed on the physical envelope of the globe.

Old observations show by the observation of plants, the localization of ores. These ores are sometimes linked to faults, split fractures in the bark; but their nature and their concentration is different depending on whether they are eruptive, sedimentary or metamorphic rocks. Around these faults, the study of which is a matter of geology and tectonics, various possible phenomena occur: soil solifluction, collapse. Magnetic anomalies causing a significant variation in the components of the Earth's magnetic field, variations in field and intensity of sinusoidal electro-telluric currents, the induction of which is linked to the behavior of the upper atmosphere and to the solar plasma by the emission of "wind solar", The gravitational field, the gravity is modified; the natural radioactivity, thermicity, the ionization of the air, the vertical electric field also.

If we are at the edge of the sea, other phenomena will be produced by the tide and a phenomenon of invasion of sea water in the coastal zones made up of sediments with a large grain size will alternate with a reflux of brackish or fresh water of the preheatic water table, which results in a dissolution of the soil prohibiting the balance of a large building,

In this work, it was question of grids. We also need another concept that we won't talk about here, it's that of diving suit . When testing a shape, we have seen that if it is enclosed in too small a volume, its eifs, states can change completely. It can even help him partially relocate (*). This shape will be said to be in a spacesuit By transporting structure, we will test shapes that will not give much because they will be in a spacesuit. If we come back to the gates, people never claim to find any because they consider, and in this they want to put themselves in a position of superiority, that they are strong enough to lift the gates without knowing it or get rid of this concept. This is an attitude that is very often encountered in non-orthodox fields where researchers quickly tend towards very privileged positions (which they create for themselves). They crown themselves. In the case that concerns us, the negation of the grids corresponds only to a perfectly RBB attitude of power and not to the attitude of subtlety which is given by the understanding of the necessity of the grid concept. To deny this concept or not to feel it is above all to show a lack of sensitivity towards what one wants to approach.

Seismology must also play an important role in soil mechanics, in the same way as the detection of cavities. However, these /Logic elements are coupled with very subtle phenomena whose various parameters we will try to define.

Recent work carried out on deoxyribanucleic acid (DNA) shows through Raman-Laser spectroscopy analysis that the place of metals can vary in the nucleus and that this change, caused by various physical or psychological stresses, is a source of disorder in the cells.

These displacements can be easily provoked by various simple devices energetically causing the desired displacements. The known physical parameters are: pulsating variations on the components of the Earth's magnetic field, inversion of the vertical electric field coupled with an increase in radioactivity metallic teleaction by remote induction, prolonged stay in a Faraday cage, prolonged stay in a resonant cavity in oscillating electromagnetic mode contrary to the ambient balances.

(*) it is used in certain African magics where the person who locks himself up in a small hut, for example, creates a TAG system at the same time.

However, all these anomalous variations of magnetic fields, gravitational electric. are both linked to disharmonies in the subsoil and to inadequate forms and materials in the constructions; in some cases, the superimposition of the effects is added without forgetting the electrical pollution of the high voltage lines under which numerous disturbances have been observed. To these nuisances, let us add those of the hydrology of treated and, so to speak, devitalised water, the analyses, of which, although bacteriologically reassuring, do not generally give the real physical parameters for the balance of human health.

Thus, we see the need for a whole branch of "Biological Geophysics" to be created in collaboration for another function of an Architecture of human and cosmic Environment.

The A-logic discussed in the introduction is not at all logic in the usual sense of the term, nor formal logic, nor logic of the kind imagined by AS Yessenin-Volpin, who constructed what he calls » ... (*) shovels ultra-intuitiveness

None of these approaches contains a process equivalent to cue leakage. Yet Yessenin-Volpin went furthest in an escape from the usual approaches; the link which exists between the movement and the interval as it is suggested by Bergson when ^{he} takes up the artifice of Zeno which consists in "recomposing the movement. of Achilles according to an arbitrarily chosen law", is not expressible in any of its approaches.

A-logic is the foremost intuitive approach to any formulation. It is placed before the formulation.

(*) He has so far only developed ultra-intuitionism, but he also speaks, without developing them, of:

- ultra-ultra-intuitionism
- or extra-ultra-intuitionism
- or trans-ultra-intuitionism
- or pragma-ultra-intuitionism
- or lega-ultra-intuitionism
- or nega-ultra-intuitionism
- Or ? bi-nega-intuitionism

The reader who will have gone to the end of this work, who will have run through all the stages, which present themselves as a labyrinth rather than a gradual ascent, will have felt the emergence within him of what he never knew. was spoken in the modern world or in the ancient world, of the outlines of Science which will be expressed by 1'A lphysics.

The one who will work in Alphysics will be in one of the attitudes presented below:

The first: he constructs the form in a C. de G. and studies it, we will say that he supports the form.

The second: he constructs the form in a C. de C. but at the same time that he discovers it, that he creates it, he must become aware that he is creating it. Thus he will recreate it a second time. In this case it will still be operational. There will be both creation and nuptials, he will be a "Creator Word". He will begin to become a Man.

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René Huyghes, Forms and Forces, Edit. Flammarion, Paris (1971).
Petrus Talemarianus, Of Natural Architecture, Edit. Vega, Paris.

Annex IV

This work contains extracts or full articles from the journal ARK'ALL Communications; this was possible thanks to the authorizations of the ARK'ALL Foundation and those of the authors of these articles. Thank them for that. Let's give some additional indications for those who are interested in the review of the ARK'ALL Foundation. The ARK'ALL Communications review is made up of volumes which are themselves divided into fascicles. Their publication is not regular.

| | | | |
|-----------------------------------|--------------|-------------|---------------|
| have been published at this day : | Volume | I, booklets | 1-2-3 |
| | Volume II, | He | 1-2-3-4 |
| | Volume III, | !! | 1-2-3-4 |
| | Volume IV, | He | 1-2-3-4-5 |
| | Volume V, | " | 1-2-3-4-5 |
| | Volume VII, | H | 1-2-3-4-5 |
| | Volume VII, | 11 | 1-2-3-4-5 |
| | Volume VIII, | " | 1-2-3-4-5 (*) |

The ARK Foundation ¹ 'ALL has been created t m 1976. She has as a core the Asso-TOTARIS donation. This was given as justification: the Study of Intellectual and Material systems escaping the constraints of the Totelian Aris Logic and the excesses of the Analytical Method.

This Association was created at the request of J. Ravatin by three researchers: Théophile Kahn, Gérard Cordonnier and Jacques Ravatin. Little by little other researchers, isolated or belonging to already existing groups, came to join them, such as Marc Beigbeder who is also a member of the CIRLCA (International Center for Research on the Logic of Complementarity Antagonists) an association whose President is Stéphane Lupasco. This group, in the extension of the path opened by S. Lupasco, studies the implications and applications of contradictory dynamic complementarities in all fields of science, arts, human sciences... As soon as the TOTATIS Association has extended to the Foundation ARK'ALL, other groups have been created with very specific fields of study: let us quote for example PARAPHY whose interest is Parapsychology (telepathy, clairvoyance, duplication, psychokinesis ...), but apprehended in the spirit from ARK'ALL.

(*) You can obtain the booklets of the review (a few issues are out of print) by writing to S. Audebourg, 21 rue Louis Scocard, 91400 Orsay (France).

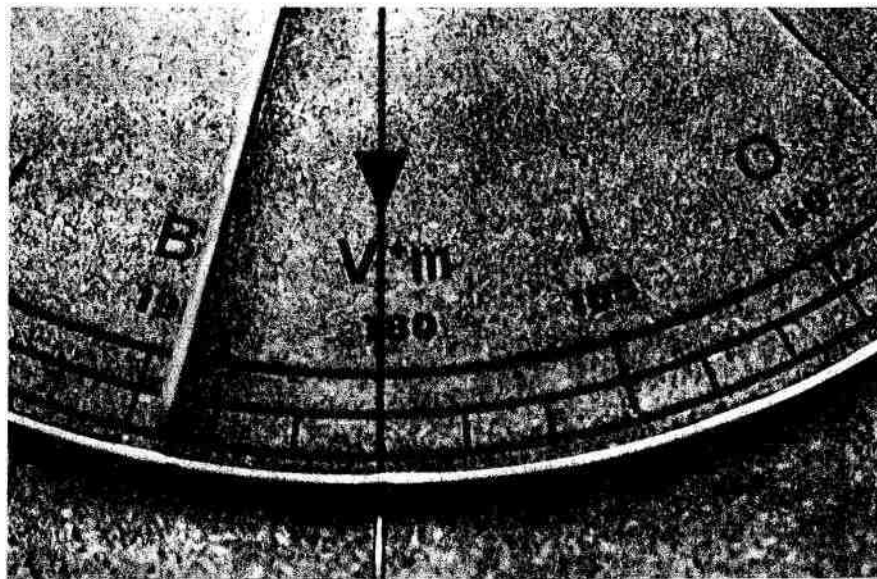
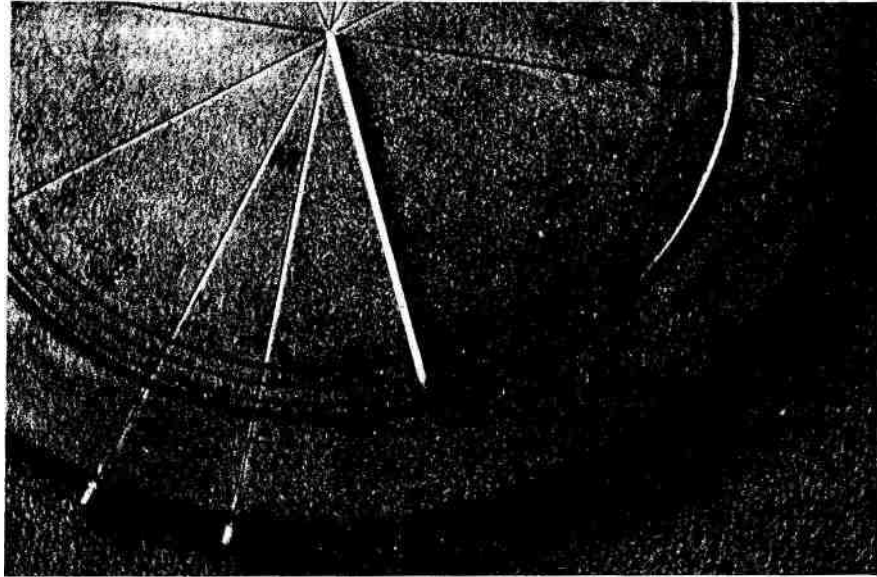
, (*)

In 1984, the ARKOLOGIE group was born, for which more details are given in the development of the work. There were still other groups in ARK'ALL; we will not give all the details. The ARK'ALL foundation is dedicated in particular to the study of the eifs, known to the ancient Egyptians, Incas, Mayas and many other civilizations, rediscovered in our century by Enel who was followed by Turenne and many other researchers mostly isolated; eifs are even more fundamental than electricity and electromagnetic waves; they are placed at all levels of existence; matter, energy, life, thought. We discover their importance in all areas of research and application, whether in physics, chemistry, architecture (housing), biology, medicine (homeopathy, acupuncture, anthrosophy).

One of ARK'ALL's missions is to open up all the ways of applying eifs in the fields of chemical and metallurgical industries, biology and medicine. Results, already obtained by researchers, have been presented in this work. The TOTARIS group, which studied the modulating processes of cultural vehicles (vehrels) managed by human communities, prompted ARK'ALL researchers to extend this work to mythology and folklore; for example, they establish the links between the eifs and the states associated with music; music therapy is thus seen from an entirely new angle. So in addition to their work in ARK'ALL, these researchers with so-called "Arkalian" thinking give a revival to disciplines bogged down in an overly Cartesian gangue, and often misunderstood.

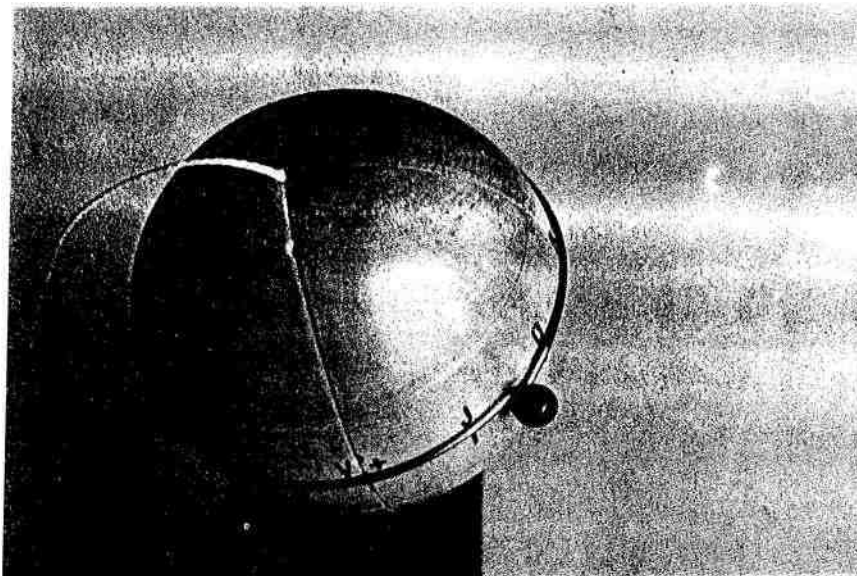
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(*) For all information on the ARKOLOGIE group, you can write to its President: Serge Hennemann, 77, rue de la République, 93200 Saint-Denis.



If it is an alternating current, connect with plugs

to N., and N., .
M E



Equatorial pendulum.

This pendulum is not identical to the one presented by La Foÿe, because the hole which goes to the center near M is chosen so that the ext of the pendulum is in dynamic duality with the ext of the person. .

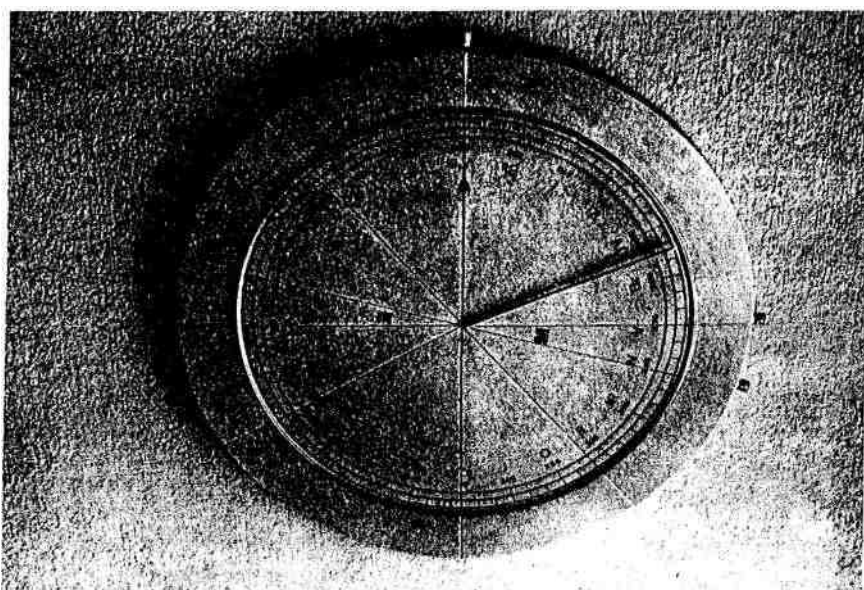
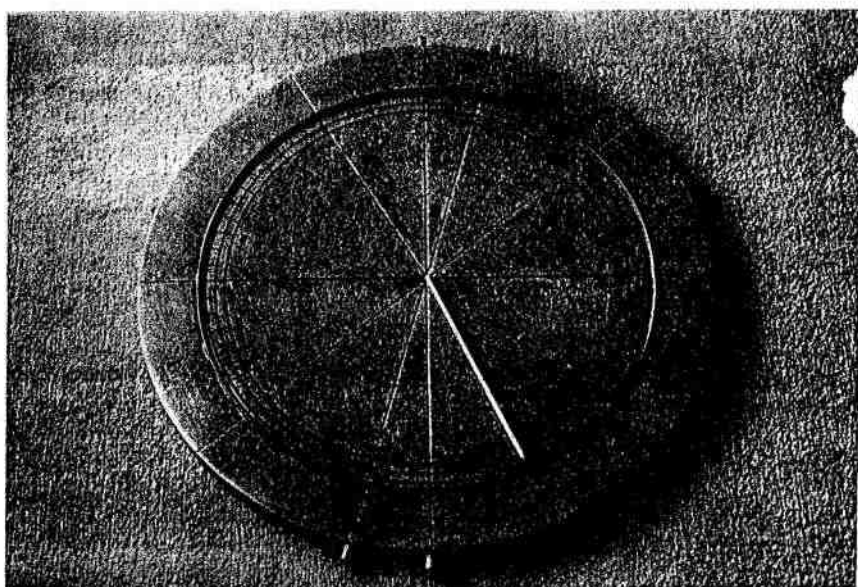
Disc with physical field.

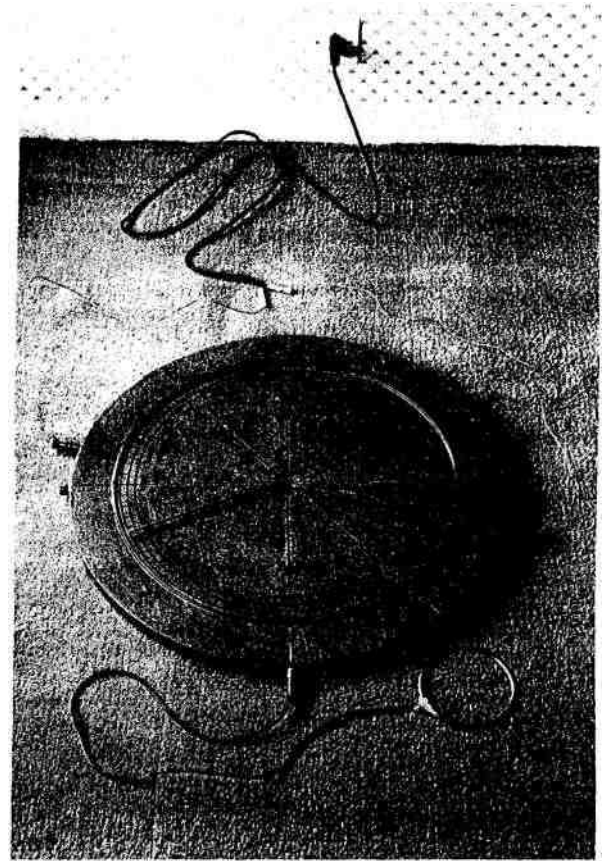
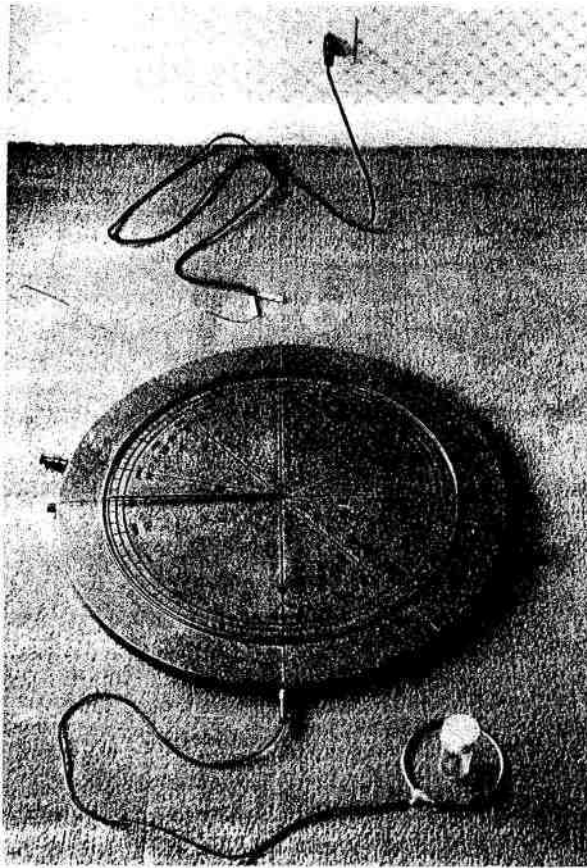
We connect the output film + (to recover the eif) to the .

It works like this but you can amplify the effects by connecting a direct current source to V., and V., or to M E

connecting the to a white body and the V., to a white body or E

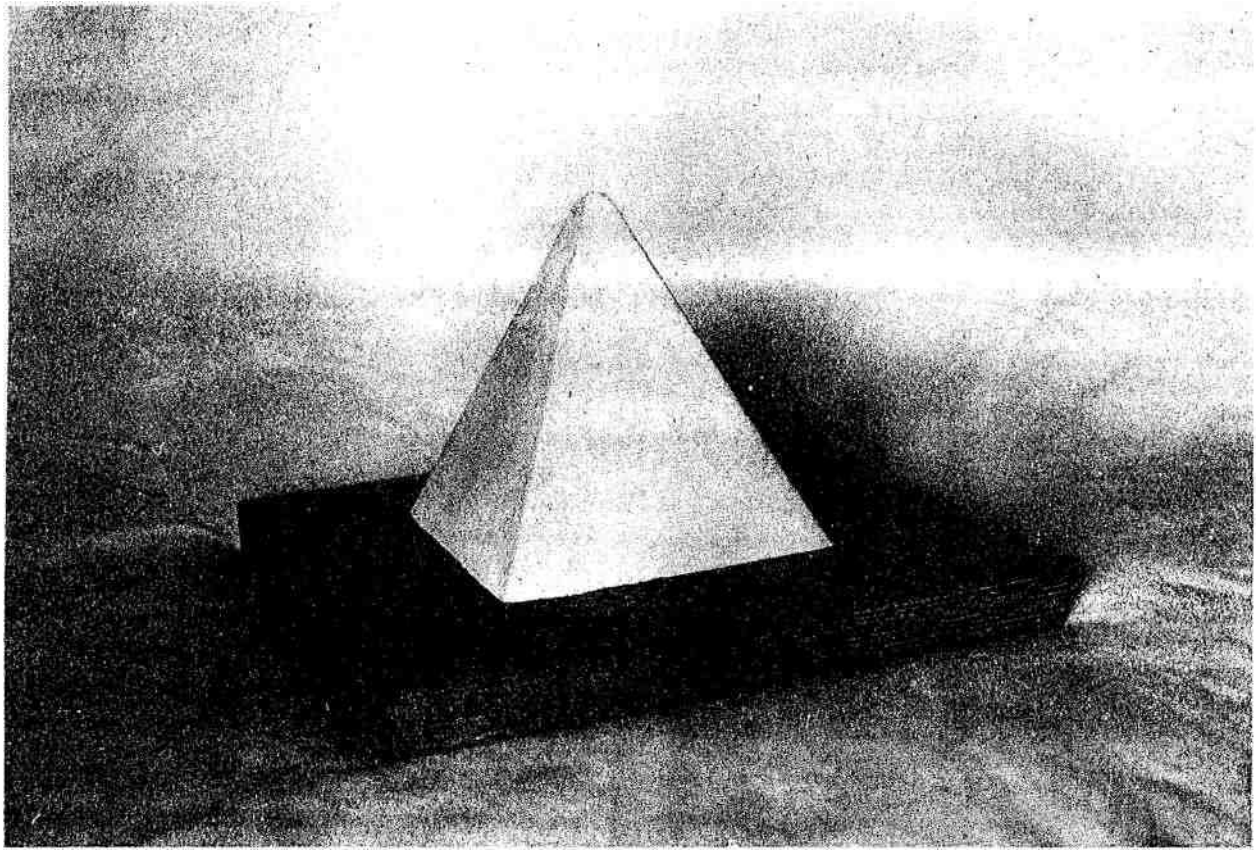
reciprocally (or more generally to 2 bodies of different colors or to two similar bodies but of different masses.



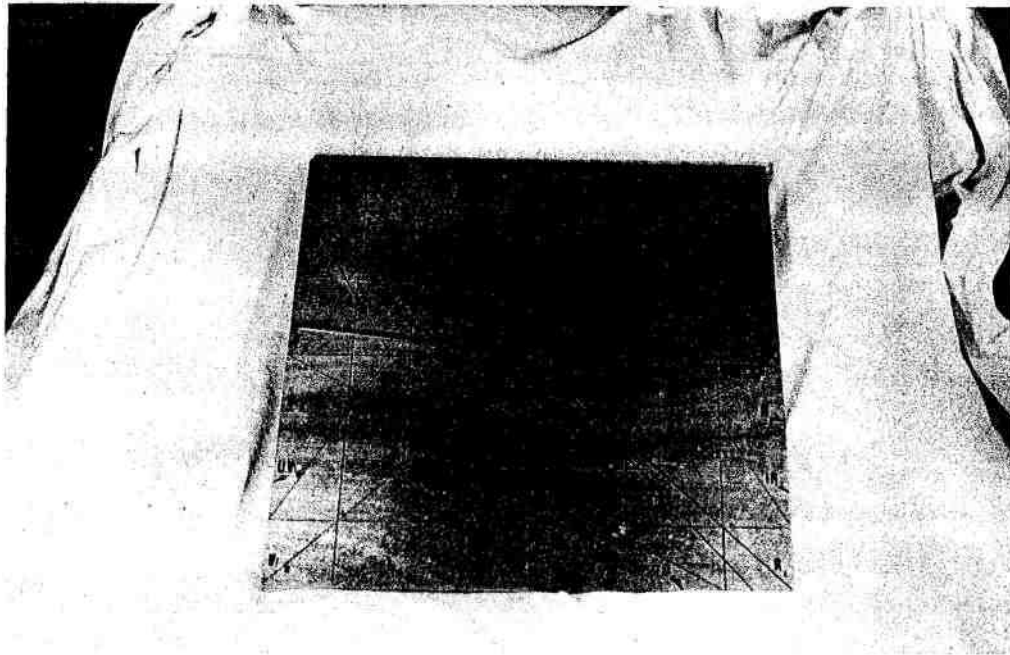


The ring connected to the output wire makes it possible to recover the eif and to detect it vertically, which makes it easy to load a product.

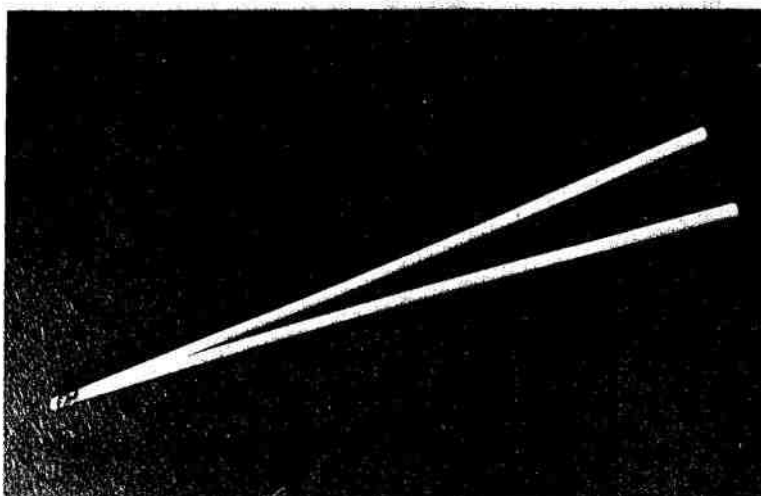
R. de Montercy noticed that if the disc is left to emit for 2 to 3 hours (that is to say with a needle), the indicated eif can change with the atmosphere. This can be avoided by adding 2 shapes to the disk. A disk must therefore be monitored. When not in use, it is a good idea to remove any needles.



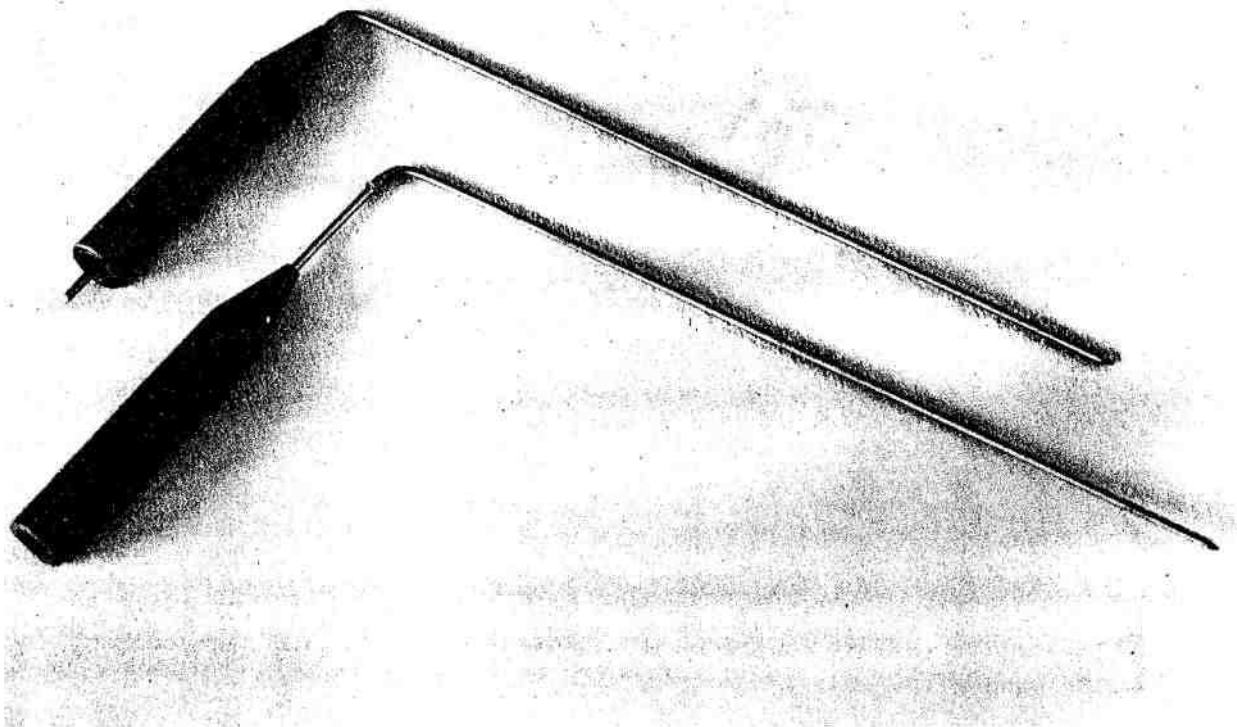
Square Emitter-Rebalancer surmounted by a pyramid connected to the Radiant number. Although the board is surmounted by a pyramid, the whole must be oriented to be effective. This kind of device does not saturate. Above all, this pyramid should not be replaced by another, whether homothetic or not.



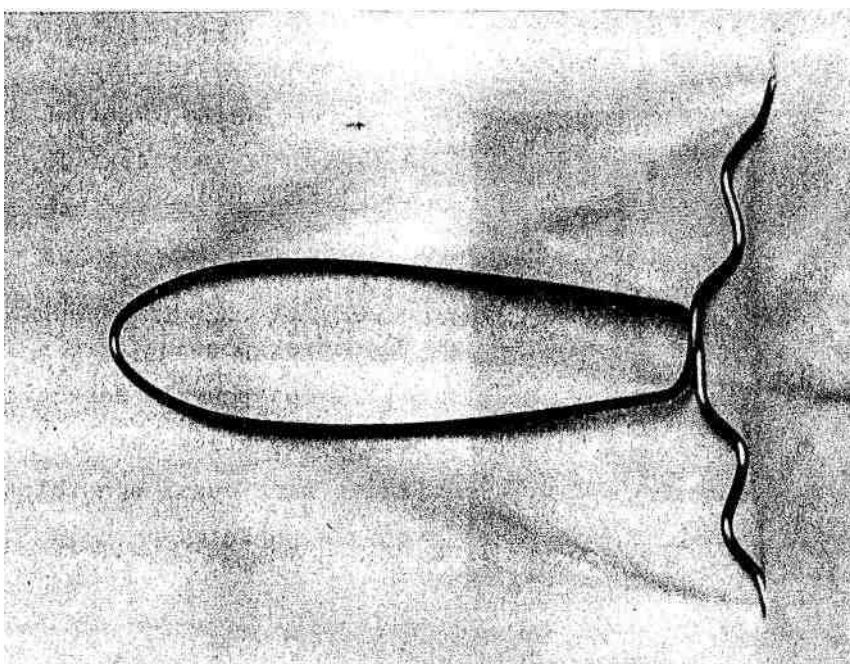
Square transmitter plate developed by J. Ravatin and produced by R. de Montery. It can generate the undifferentiated spectrum and the differentiated spectrum. It has all the possibilities of the disc but has many others and in particular, it is more stable. Without a needle and with a slight supplement, it can make an excellent transmitter-rebalancer. It can then give very high C. de T. and even α — “ as well as VC Inspired by this square plate, Montery has developed a new transmitter plate which advantageously replaces that of La Foÿe because it is, in particular, more stable. Ravatin drew from it, playing on the shape of the needles or the nature of the metals constituting them, a vectorial composition of the Eifs.



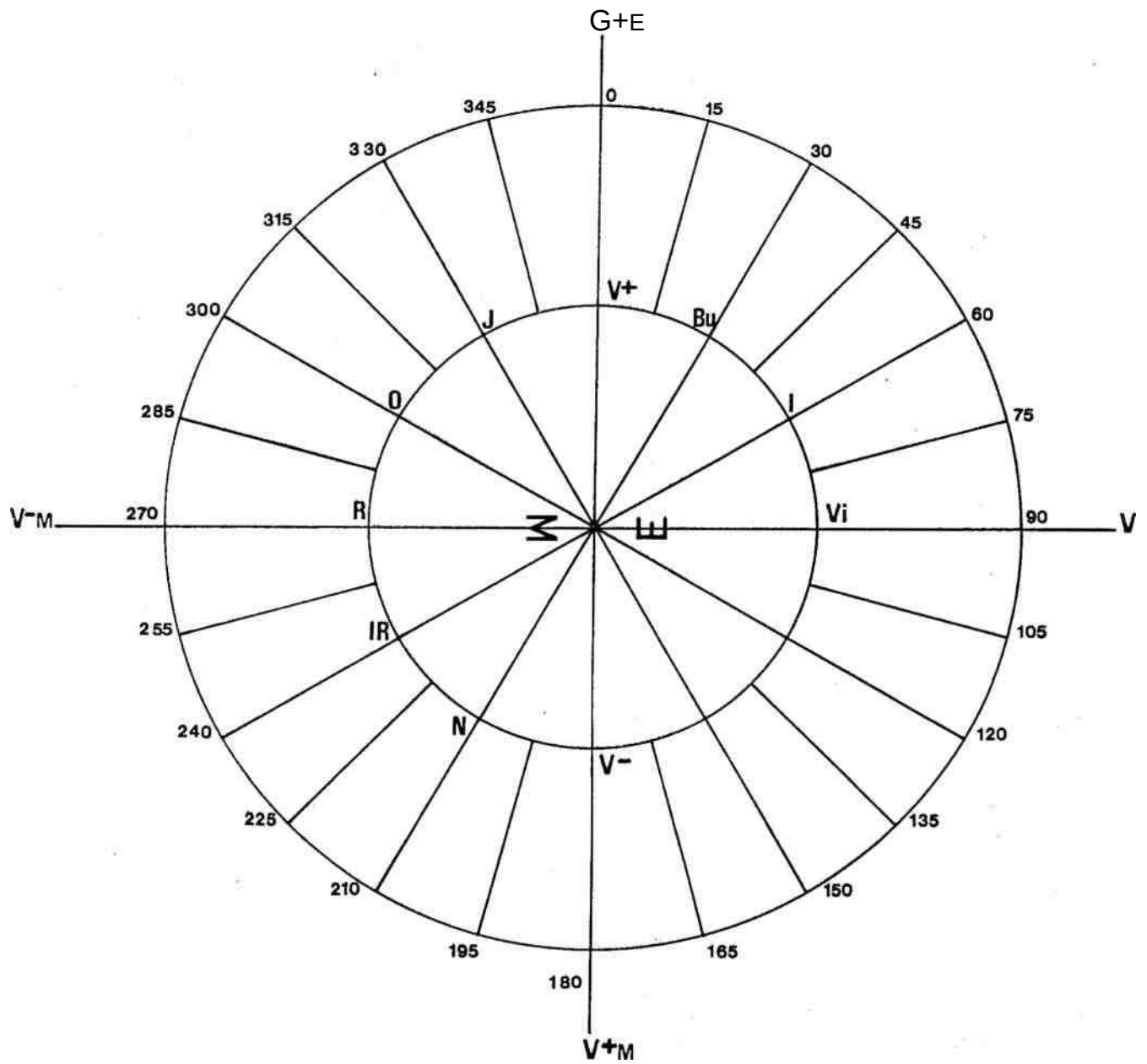
Rods that can be used for the detection of eifs, states, etc... They are as sensitive as the pendulum.



γ R. de Montercy.



Lobe-antenna used to detect the Hartmann array. This detector does not work on other networks.
All these devices are also made by P. Arrault and JP, Gazeau, architects of the ARKOLOGIE group.



Eifs at the BCM level shown on the transmitter disk.

This finds the two spectra: undifferentiated and differentiated.

Materials : It is good to start the work to use bobbins. It is a hollow ceramic or terracotta cylinder. It is useful to work with electrical resistance supports found in certain residential heating appliances. These are called barrels: they are hollow, with circular sections or in equilateral triangle with rounded tops. You can also take hollow cardboard cylinders but the eifs are less powerful and have a shorter range.

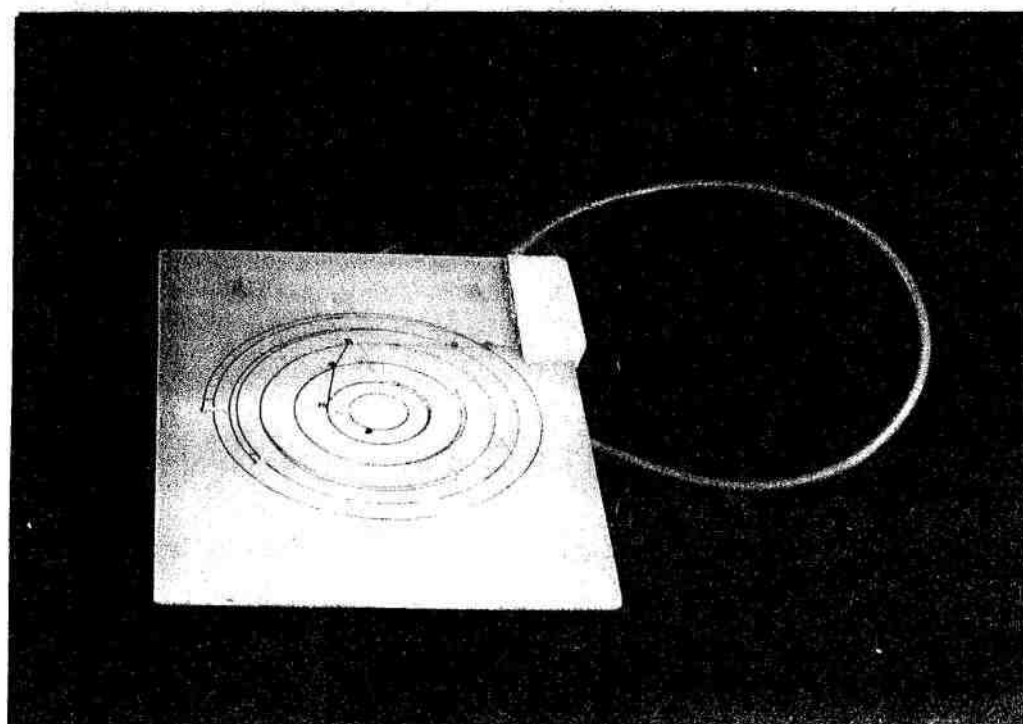
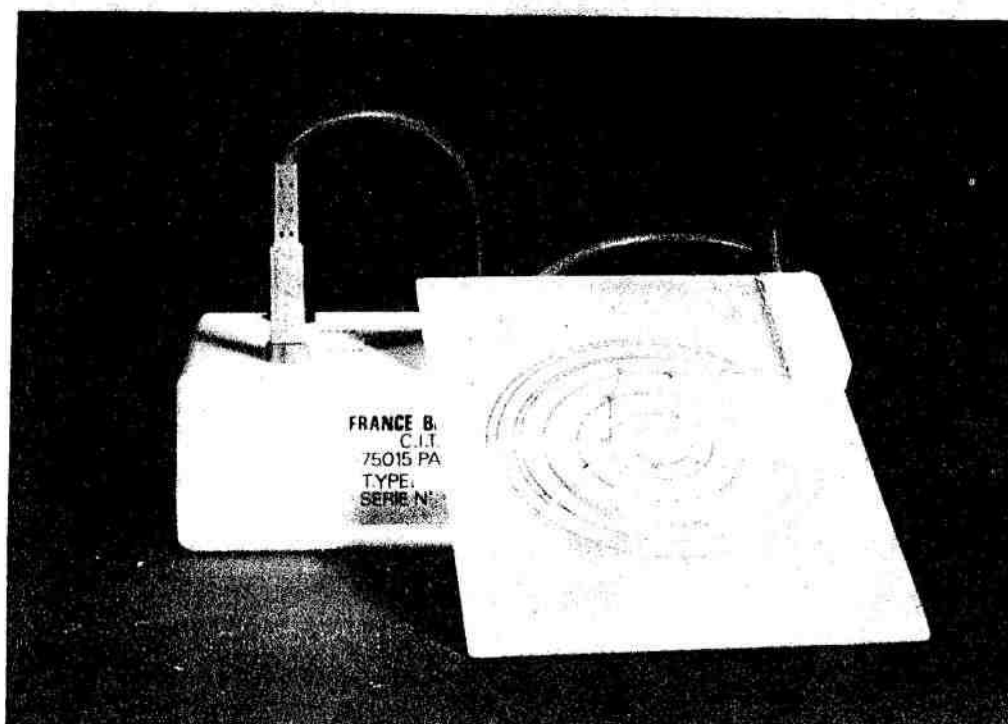
It is also good to have transmitter discs giving the undifferentiated spectrum and the differentiated spectrum. It is also necessary to have spherical and cylindrical pendulums necessary to detect the eifs, the states, the polarities. At the beginning, when the detection of the eifs is done without looking for their nature, a ball crossed by a cylindrical brass rod is enough. It is good, when the experimenter becomes sure of himself in the manipulation of the pendulum to have an Egyptian pendulum: this one is more sensitive. Nevertheless about the latter, do not buy just any » many copies have been made which are not in conformity. R. de Montercy showed that a good "Egyptian pendulum" must have 2 harmonics (the meaning of this expression is given in the book).

To get used to detecting eifs, we start at the BCM level; either a differentiated or undifferentiated spectrum, these eifs are detected at the end of the wire attached to the outlet of the circular or square plate.

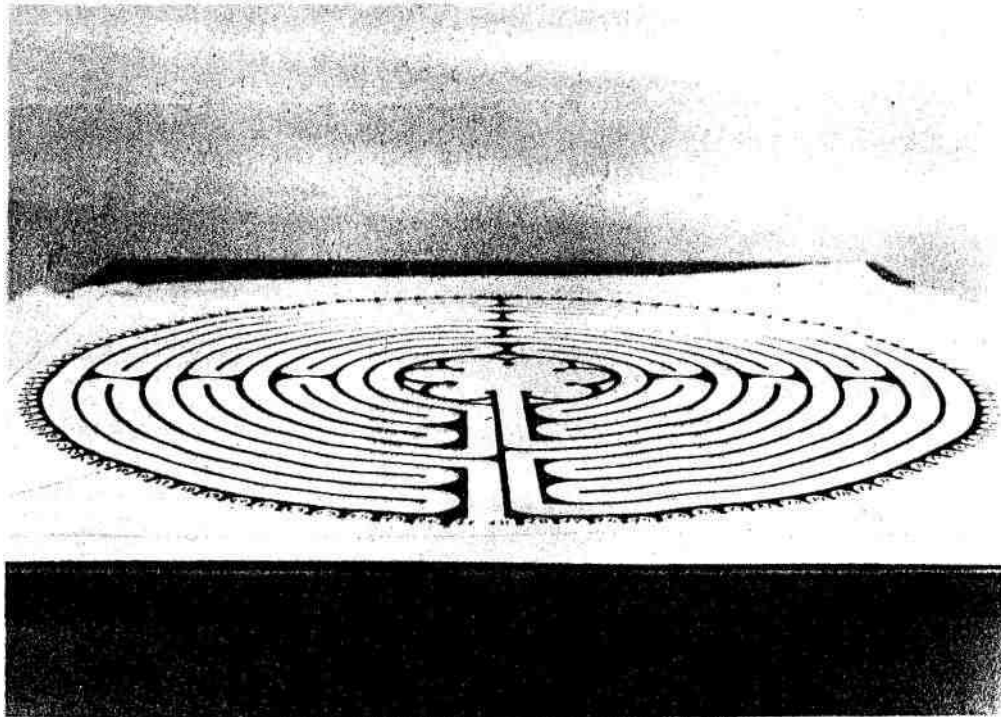
Everything, the material can be obtained from the ARKOLOGIE group.

Address: ARKOLOGY Group

Mr. Serge Hennemann, Expert-Architect 77, rue de la République 93200 Saint-Denis



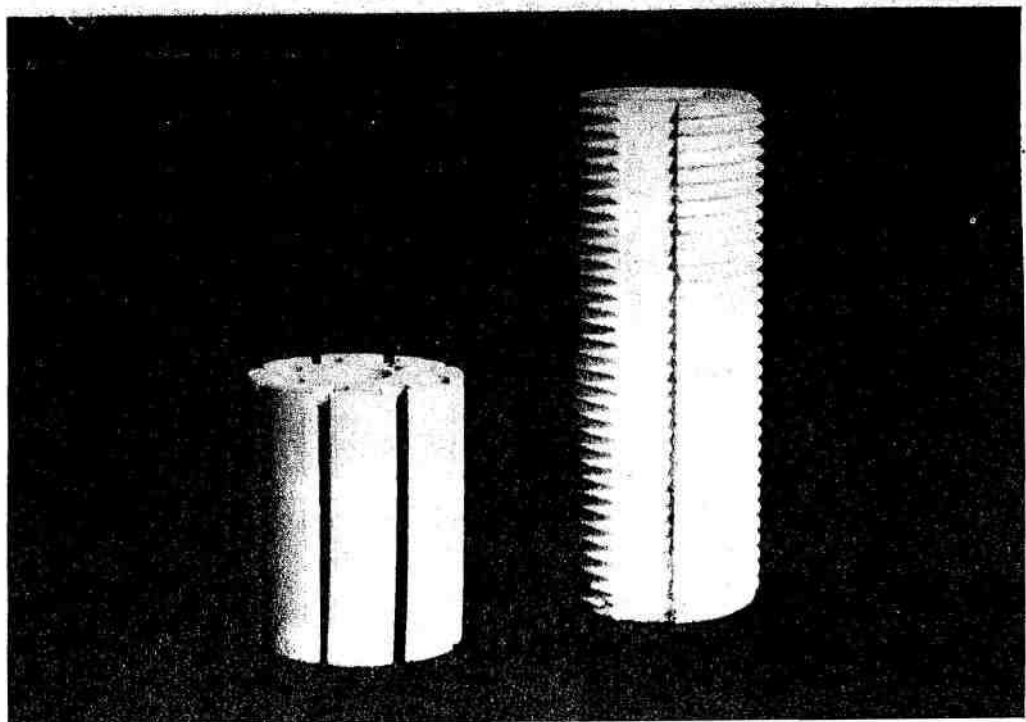
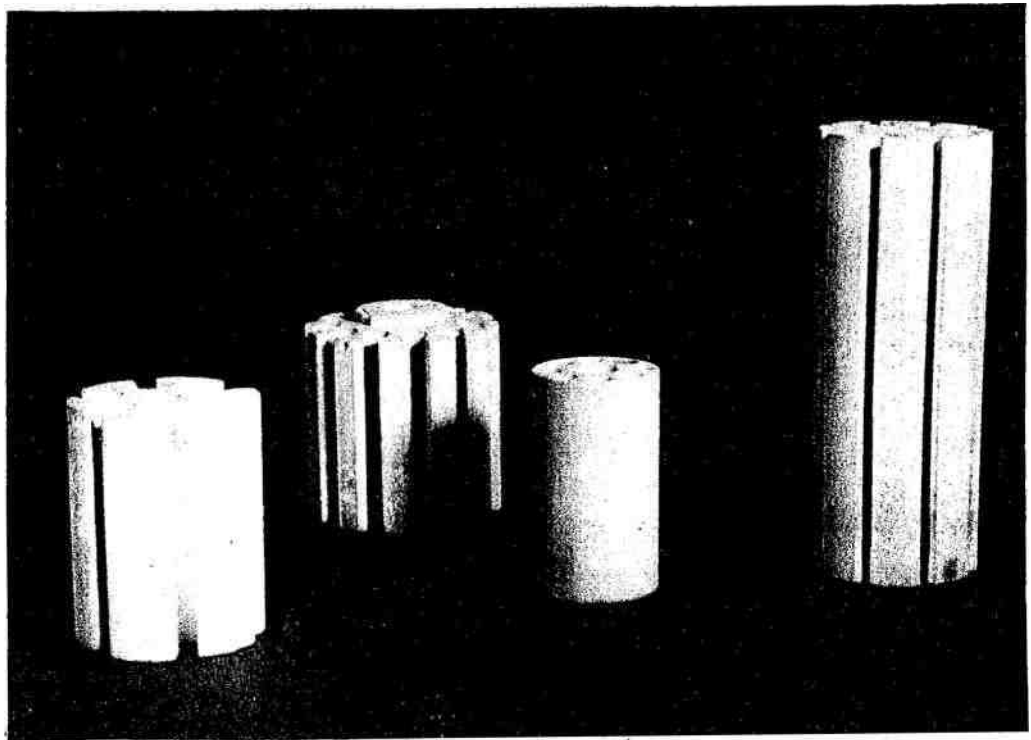
Transmitter antenna designed by David Franeric. An electronic assembly *has* been applied to it. This antenna then emits at high C. of T..



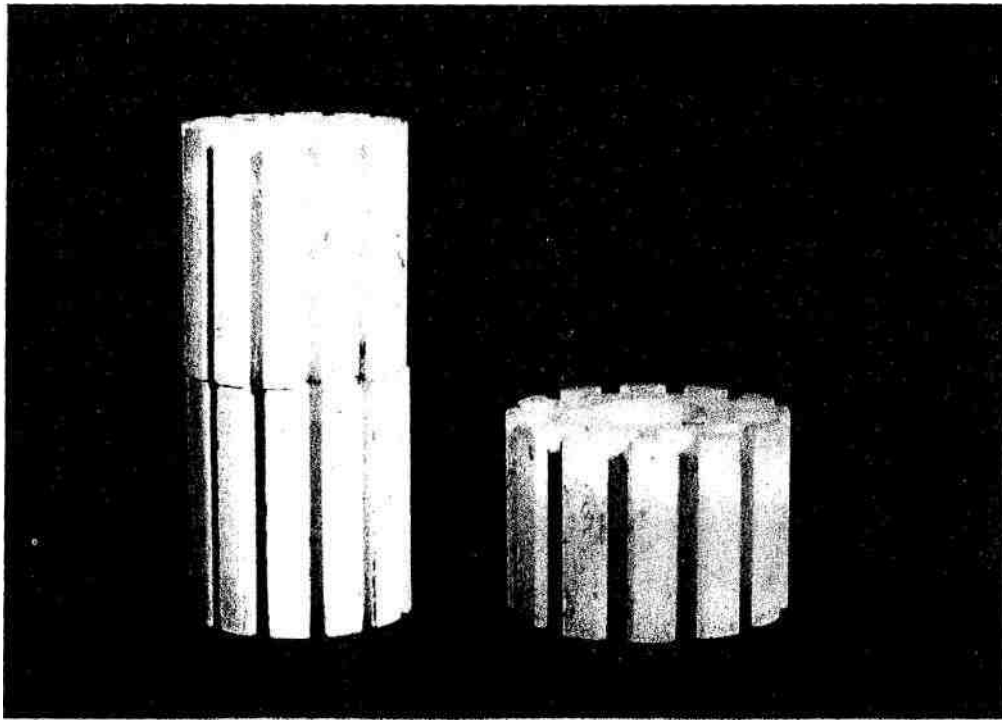
Labyrinth of the cathedral of Chartres. This form of τ about 80 cm in diameter reproduced on paper can be used for multiple manipulations. Its job is very delicate. A user's manual was imagined by JC Soënen; it is very effective. Its use can only be made by very knowledgeable people.



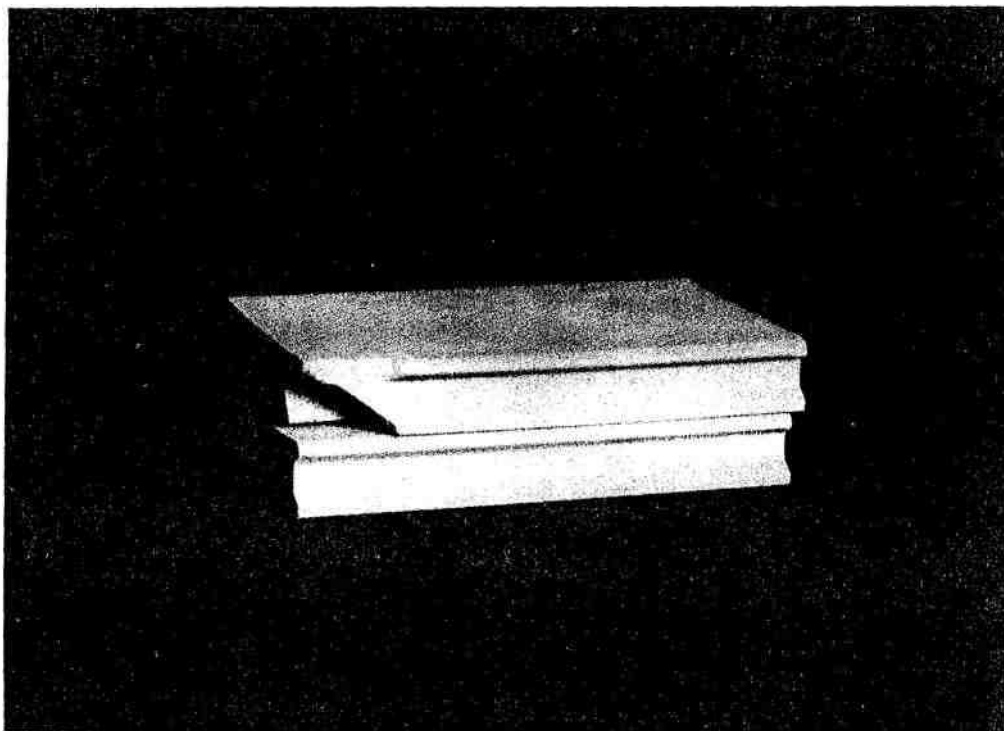
Labyrinth of the cathedral of Amiens



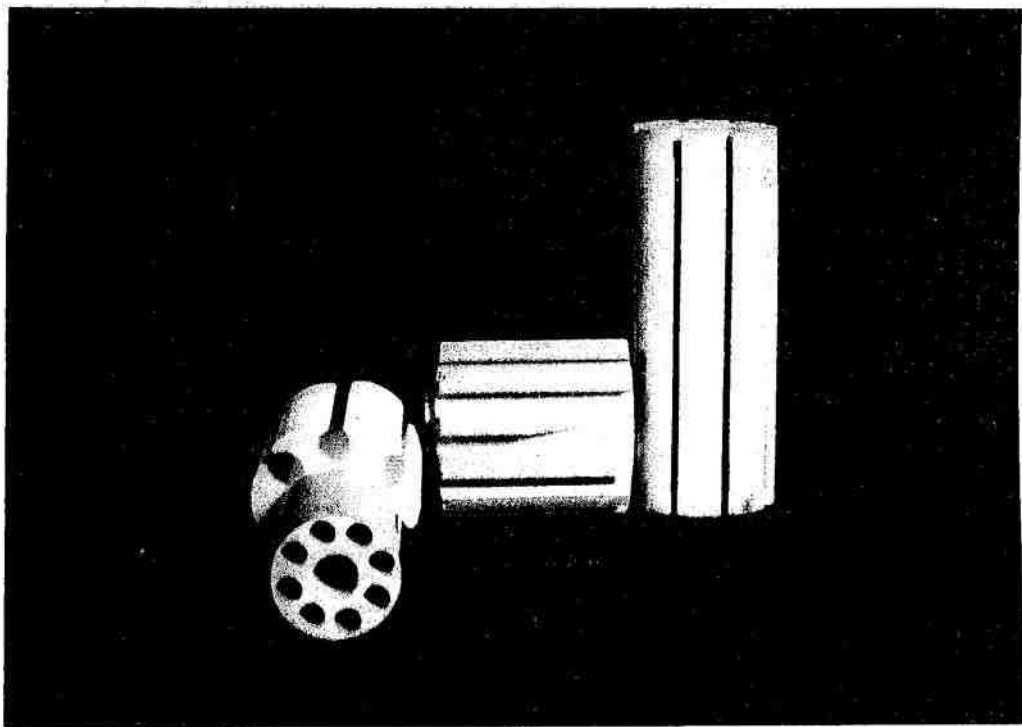
Different types of barrels



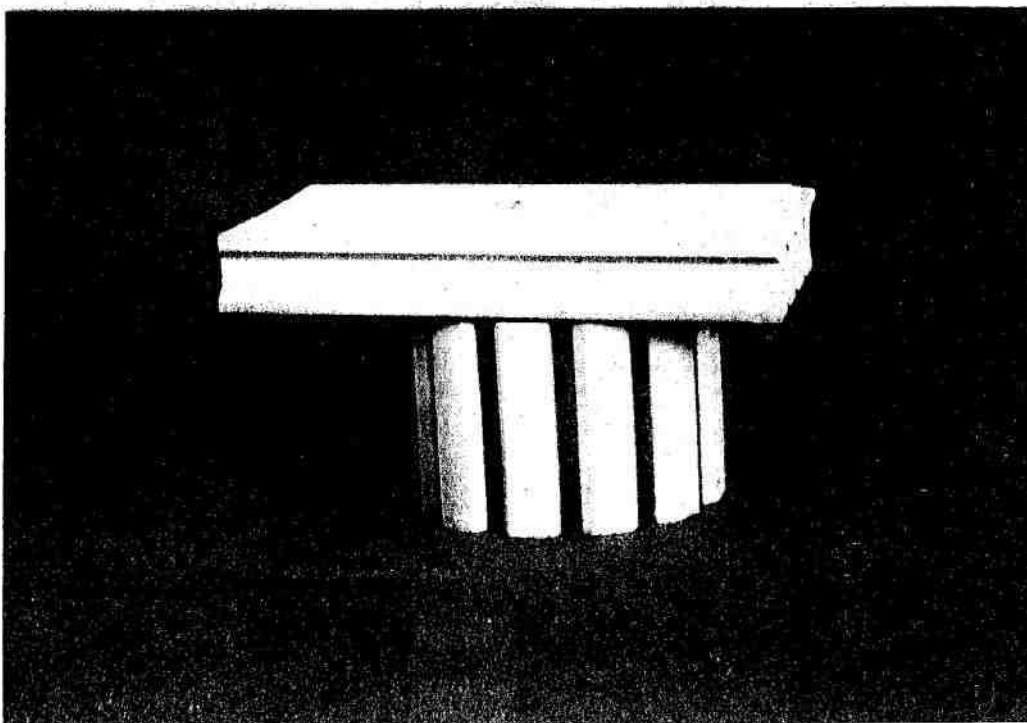
Barrels



Trays with holes



Different types of barrels



Barrel surmounted by a plate pierced with holes. The vertical eifs become horizontal and are transformed into other eifs. Depending on the direction of the plateau, their intensity and nature vary.

Exercise

that the b
others

more
with a

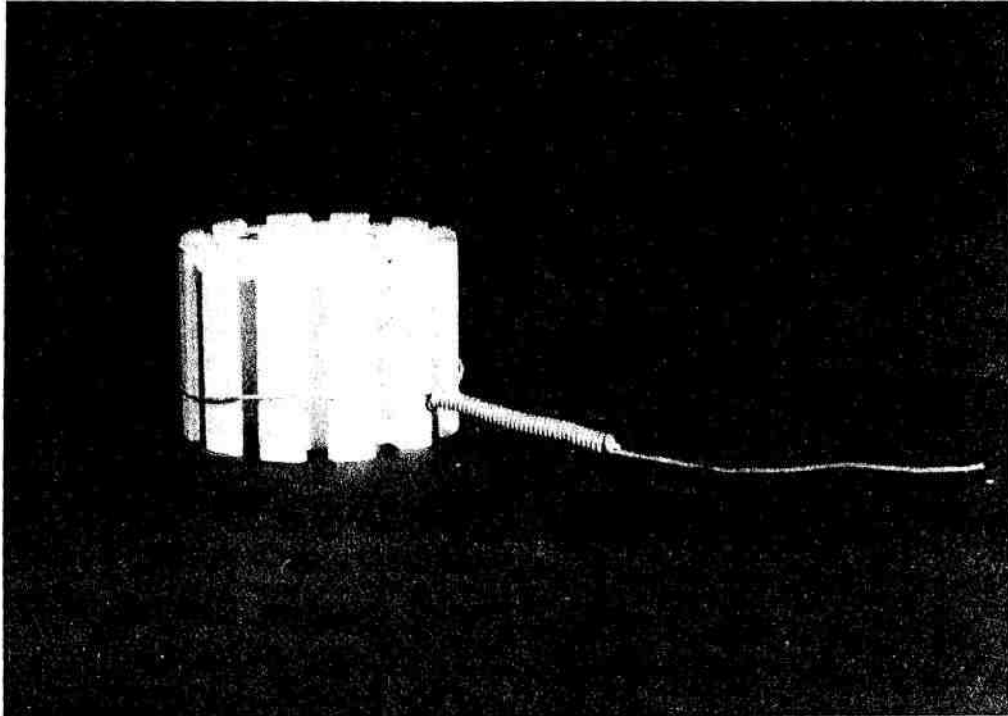
Bibliog

[1] Se

Platform-
style
exercise

1) Make
table, at
diff

2) On
un se< $\hat{1}$
les u



Barrel surrounded by a copper wire tightening it strongly. The detectable eifs along the axis then disappear and are detectable at the exit of the yarn. They can be transformed or not according to the direction of the turns, the nature of the wire etc...

~~Exercise {1}~~ On a large flock of birds turning on itself, check ■ that the flock has no ext., but the bird's ext is in transfer with those; others .

Verify that on human beings in levitation, often they do not have
i no more shin on the head or CV except in certain cases where they are detected with an artifice. On the other hand, they
still have their ext and their CP [1]

I

Bibliography

' [1] Serge Audebourg, ARK'ALL Commun., Vol.8, fasc.3, 43-44 (1964).

Exercise {2} Regarding amplification, put a barrel on a table i so that the axis is horizontal (table clear and without
drawer under the top).

1) Rotate the barrel at different angles in the plane of the table top. What do we observe? (we will not deal with the
nature of the eifs) by testing at different positions.

2) We put the barrel such that its axis is vertical and on this barrel we put a second whose axis coincides with the first.
We put 5 and 6 like this, on top of each other. What do we observe?

) them

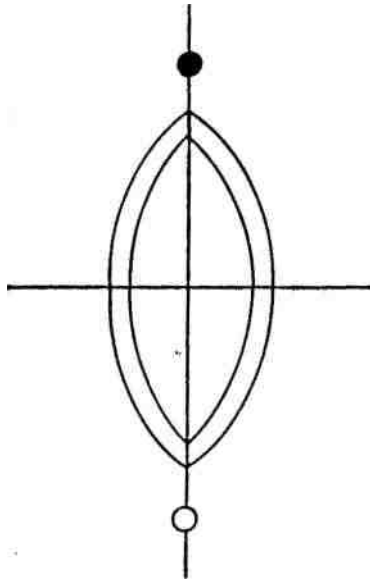
D

Problem {3} Among the ideograms of Easter Island, consider the almond; reproduce its outline internally and draw the 2 perpendicular axes of symmetry . Let's orient the largest with especially the north of form - south of form.

1) What is the eif'emitted by the tip at the north of form?

What can be detected vertically from the point of intersection of the 2 axes?

What is the level of equivalence?



Exercise {4} With a disc or square transmitters:

— Check that 1 needle placed at 320° causes the polarities to disappear,

— Check that the needle placed $\odot$ at 325° emits a CV, to a body placed in the extension $0-180^\circ$, that is to say at the exit of the V^+ ,

Exercise {5} If we send an eif in the axis of a solenoid whose helix is in the corkscrew direction, is it changed?

2) Same question if the solenoid has its helix in the opposite direction?

3) Is the solenoid an amplifier?

Exercise {6} We make a hemisphere out of lead. We put it on the pole.

- 1) What are the eifs on the axis?
- 2) If the half-sphere is made of wood or plaster, do we find the same thing?
- 3) The lead half-sphere is placed on the equatorial plane. What do we find?

Exercise {7} We consider a solid sphere, in baked plaster, unpainted, placed on a horizontal plane.

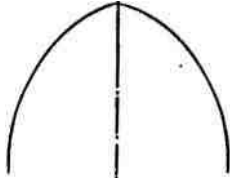
- 1) What are the eifs along the vertical axis?
- 2) What can be said if the sphere is made of lead?
- 3) If we color the sphere for example with a green or red color, what do we deduce?
- 4) Let us return to the sphere of the beginning. What is on the equatorial circle?

Problem (8) We consider an emitter of B (undifferentiated). It is sent by a wire on a wooden prism.

1) What do we observe on the other side of the prism?

2) We build a form in copper wire, for example as below (warhead)

and we place the N on the axis above, ^F
What do we observe in the inner axis?

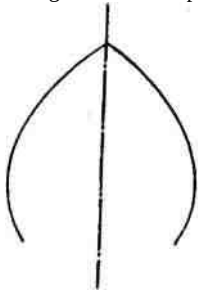


3) The eif detected is called according to Foÿe terminology: eif de Chartres (it is detected under a vertical warhead). What does it do on the eifs of the undifferentiated spectrum?

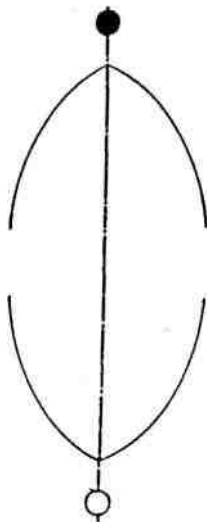
4) We throw it at a wooden prism. What do we observe on the other side of the prism?

5) By an appropriate assembly it is combined with the undifferentiated. What do we get?

6) Does an ogive of this shape give the eif of Chartres?



7) We realize the assembly below, that is to say 2 warheads having the same axis. We imposed the N. Do we still have the Chartres eif?



8) What are the M and E components? The component M is detected with the form $\frac{1}{2} \left(\frac{1}{2} + \frac{1}{2} \right)$ puts on a cylindrical pendulum.

The component E with the form



The Hebrew word (found by de la Foÿe) corresponds to the eif of Chartres (Haretz) the AR Ts (Earth)

| | | |
|-----------|----------|----------|
| D* | 1 | N |
| 27 | 20 | 1 |

Build the corresponding cannon. Check that oriented NS it emits the eif of rr Chartres.

Can the barrel be made direction independent?

Problem {9} Or. considers the emitter disc or square. We put 2 identical needles allowing each to emit in differentiated. One is at 320° and the other at 60° .

- 1) What do we get at V^* ?
- 2) We always put a needle at 60° and the other at 140° , What do we get back?
- 3) We now put 5 needles of the same length and length equal to that of the previous ones: at 320° , 60° , 105° , 140° , 285° . What do we get back. What does it remind?

Problem {10} 1) Consider a straight 1-Möbius strip and place a hollow cylinder inside the strip. What do we detect?

2) Same thing with a left 1-Möbius.

3) Same thing with a 3 or 5-Möbius right and then left.

Problem {11} We give ourselves the triangle of angles 40° , 60° , 80° .

1) Find the position to impose on the N so that it emits the right shin above, r
Does he issue the CV under these conditions?

2) We put it in a circle much larger than the triangle, the triangle
towards the center. The N is imposed on the circle. Check that we detect
shin above the triangle and the 4 components of CV

then the

3) We return to the conditions of 1). Is there a point in the triangle
who, pierced, makes the triangle emit the CV?

Problem {12} We consider again the triangle $40^\circ-60^\circ-80^\circ$ whose respective vertices are B,C,A , M the midpoint of BC , Cx the half-line of angle $(Cx,CB) = 15^\circ$. It intersects the perpendicular bisector at CB at T. From T we draw the straight lines Ty and Tz respectively, perpendicular to AC and AB • We pierce the plane at T . We see immediately that $(Tx, T\ddot{I}) = 105^\circ$, $(Ty, Tbf) = 60^\circ$ and $(TM, Tz) = 40^\circ$.

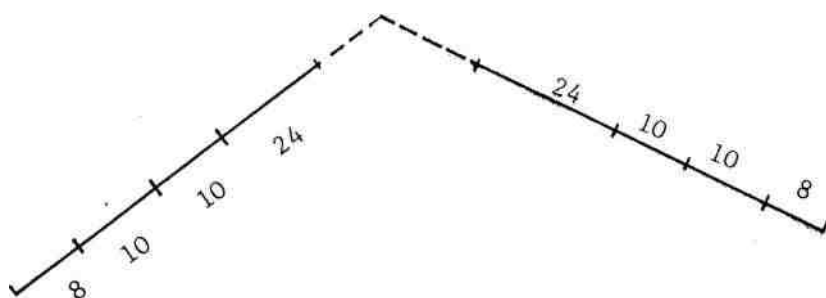
- 1) Observe that both inside the triangle and outside there are no polarities nor CV
- 2) Draw the line parallel to BC passing through T . What do we observe?
- 3) We draw the circle circumscribed to the triangle ABC and the circle with center T and radius CT . We can observe that the figure emits a CP By removing the triangle ABC, we still have the CP
- 4) By taking a pencil placed on the plane, the end not sharpened in T and directed along a radius (the length of the pencil greater than or equal to CT), it emits at S the eifs of the aura.
r
- 5) Can we have the undifferentiated spectrum?

Exercise {13}

1) Consider a circle surrounded by black with a black center. What does it emit vertically if placed in a horizontal plane?

2) Let's consider a black circle with a white center, what does it emit vertically if it is placed in a horizontal plane?
(Note: you have to be careful not to use to dye these circles black with a felt-tip pen; you wouldn't have the desired eifs.

3) Consider 2 guns 8-10-10-24 laid out in a horizontal plane as follows: 0



Two axes intersect at 0 . What do we detect at this point?

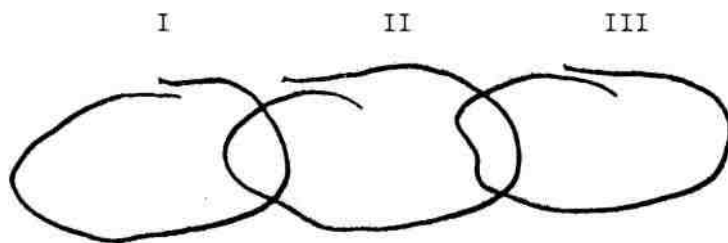
4) We take a wooden cylinder that we cover with the circle considered in 2).
What does it emit? Show that there is one position and only one in the plane, such that the cylinder capped by this circle will eliminate the V_{jl} in the center and will emit the node of life without V_{jl} by its axis.

5) What does the plane of the surface of the table on which we are working on emit? How to detect it?

Exercises {14}

- 1) Take a hollow cylinder, put it inside 1-Moebius D. Surround this 1-Moebius D with a 1-Moebius G. Do the search like the one previously presented.
- 2) Same but swapping bands D and G.
- 3) Repeat the 1st but replacing the band G by a cylinder.
- 4) Repeat the 2nd but replacing band D with a cylindrical band.
- 5) Reproduce with paper the shape below Moebius strip A to which an ear B has been added. Place a hollow cylinder or a ring inside B .
 - a) Is the ear twisted?
 - b) Search for eifs and states.

6) We consider the montages according to where I, II, III are either bands of



Moebius, or cylindrical bands
 , and among the Moebius
 we can have them D or G, or
 1, 3, 5, 7, 9, ...
 Inside we can measure shapes
 in one band, in two bands,
 whether or not.

As before, we will look for the eifs, the states

equivalence levels

if they exist, the outside etc... We can surround the whole .by a cylindrical band

dric or by a Moebius strip D or G and resume the search for the eifs,

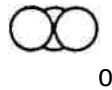
Exercises {15}

- 1) In a plane, we consider a winding road going to the horizon at a point fi . Study this figure,
- 2) Consider as Escher 2 vanishing points fi and $0'$ on the horizon line. Draw Labatchevsky parallels from these two points.
 - a) Knowing that we *have* two perspectives, determine in this case the states in fi and fi' . See possible transfers.
 - b) Can Q and Q' by this system be in dynamic duality?

The Spirals.

Let's say a few words about spirals; we have already seen some in this work with those of Cordonnier. They can be used as a metallic drawing or circuit through which an electric current is passed. Their direction of rotation is important. From the center outwards, they should turn clockwise. The center must not be marked, they must not start at this center, It is interesting that the center is a decalage. They are lockable. Additional shapes can be added to them. If we consider them alone, many spirals can be envisaged depending on the step, depending on whether they are constituted, whether traced on a plane, with a continuous line or not. The logarithmic spiral despite being invoked, because it is found in nature, is not the best. A spiral like certain forms, to be effective, must hook a cosmic phenomenon. Ravatin has developed a spiral which emits V, C, FV and something else as well as the high C . of T . and $\llcorner i$ 1° when an electric current passes through it.

Problem {16} We consider the life node ν . We impose the N as below



1) Find the eif it emits vertically.

2) The cH Y guns (living
and cH YH (live)

emit the life node. Check it.

Check with cH YYM (Lives).

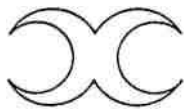
3) Check that an animal and a human have the life node (M) at the level from the navel. It is detected in a living human being at the level of the left temple. On the other hand, if a human being wears certain caps, kepis, or paints a red dot on his forehead, the knot of life passes in the middle of the forehead. If you have a helmet it is on the left temple but is detected with an artifice.

In some cases, the life knot can be detected all over the face. Often on a dead human being, it no longer comes out through the left temple but through the mouth. (*)

(*) It has nothing to do with the mathematical infinity sign
to the concept of leakage, that is to say cumulative or decalary.

The latter is connected

Problem {17} Consider the shape consisting of 2 crescents joined together ^k in **1** back. It corresponds to an eif ^(A) v ,



Make a cylindrical pendulum on which it is carried on the side surface .

- 1) Check that this eif is undifferentiated. Give the precise angle of emission when using the transmitter disc.
- 2) Check that we find Z^+ on the Star of David whose highest peak is pointed above, the fleur-de-lys of the kings of France, the Hermine of Brittany.
- 3) If the Star of David is vertical, not pointed, do we detect the Z^+ and where? If the fleur-de-lys, the ermine of Brittany are vertical in the usual gaze position, can the Z^+ be detected ?
- 4) If these shapes are in a horizontal plane how to have the Z ?
- 5) It corresponds to the Hebrew word DBR (Dabar)



20

Make a cannon 4 - 2 - 20; find by which extremity one detects Dabar.
(Should we orient to the N?) . If so what can be done to the canon to impose the r
 $N_F \bullet$

6) The $Z_+(E)$ corresponds to the form
Where is it found on the transmitter board? He

A (it

He

Same question for the $Z_+(M)$ which corresponds to the form (*)

(*) Discovered and named Z by J. de la Foÿe,

7) We make arrive by 2 wires, one in front of the other the following eifs

Bu and N

Vi and R

O and UVi

B c and J

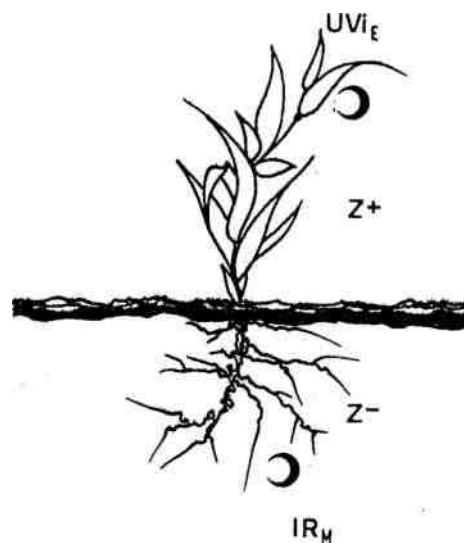
IR and I

V* and V'' M E

Show that we detect Z . To succeed in the experiment it is good to take
2 identical transmitter trays. Do not couple their output wires.

8) Check that this eif is emitted by the palm of the hand and the mouth when the eyes are open and by the top of the hand if the eyes are closed.

Exercise {18} 1) Detect on a plant the eifs marked on the drawing:



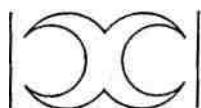
Note: we also have the CV and the right shin above.

2) What happens if we uproot the plant?

3) What happens if we add soil around the plant?

4) If you cut a plant at ground level, what happens?

Problem ^{*}{19} Consider the shape of the Z but inserted ^{4"}between 2 vertical lines



Make a cylindrical pendulum on which it is carried on the side surface. It corresponds to an eif,

1) Check that this eif is undifferentiated. Detect her on the board.

2) It corresponds to the Hebrew word Sh M chA (Shamch'ah) (listen-listen)

l at tb
Y6 13 21

Make a matching cannon.

4) To have Z_{-} and $Z_{\hbar}$ M we do as in the case of Z

5) Find the position of $Z_{,,}$ on the board and that of $Z_{,,} \bullet$ M E

(*) Discovered and named Z by J. de la Foÿe.

Exercise . {20} Do we give each the form opposite placed between the N and S . find _{rr} other our eif?

O

Exercises {21} 1) Look on photo albums for views of Delphi and Machupichu and check that the sites correspond to decalaires. Also check that you have C. de T.

2) Check what is announced below in the works cited in the bibliography[1]. These are bioions and cellos made by A. Stradivari. The dates given are the approximate dates of the position of the instruments.

1681: Owners Chanot-Chardon: violin known as violin without corners.

We find the node of life on the key no ext no shin no polarities no CV no C. de T.

1680 Paganini: violin

We find the QQ on the whole key
the shin near the key
an ext
polarities but weak
C. of T. 1 -----1 1 to 3 t ----- 3
resume

The instrument has the Fundamental Form of Life (FFV)

1716: Goldman: violin

has polarities
Right shin near the sidelines
QD on the whole fingerboard

RESU strong
ME \

All C. de T. and their FFV leak

1709: J. Thibaud - violin

no QQ
ext good
no polarities or shin
all C. de T. and their leak no FFV

1720: Piatti - cello

low ext
shin
no (X) on the key even by orienting 1¹ instrument no_polarities
1 i ----1 1
no FFV

1711: Duport - Cello

Ext very low shin
no (%) , no FFV
no polarities
1 —: 1 and 2 r—i 2 no shin

Yet this instrument has a very beautiful sound. The tests were carried out on photos. Photographic processes and (colour) reproduction may interfere with the test.

1677: "Sunrise": violin

no shin
Ext
no resume
no CX
no FFV
no C. of T.

but around 1800 he had FFV
all the C. de T.
QQ, Ext, CV but no shin

1698: Theonville: violin

Ext, no QQ
resume
FFV
all the C. of T.
and the shin

1710: Dacla

Ext
QQ on the whole key
resume
FFV
all the C. of T.
has a shin

Bibliography [1]: Violin Iconography of Antonio Stradivari, Edit. Herbert K.

Goodkind, 1972, New York.

Antonio Stradivari in Japan, Edit. Mikisaburo Oshio,
Tokyo 1984.

Exercise [22] A circuit in the form of a life node is made out of copper. 0 is a center of symmetry. These points A¹, A₂, A₂ are such that

is on the axis of

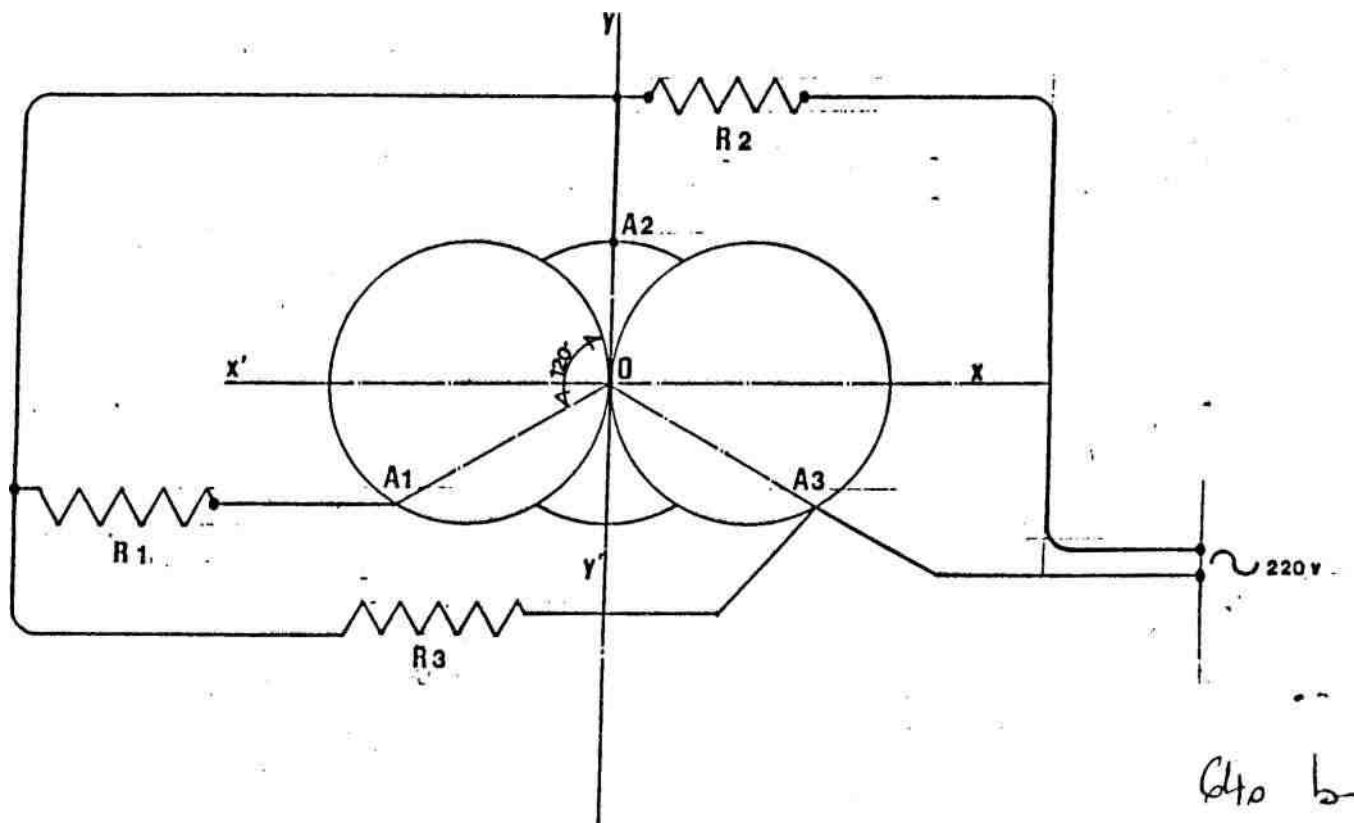
symmetry yy' and

$$(\hat{OA}^1, \hat{OA}^2) = 120^\circ. \text{ The resistors}$$

are

mounted as shown in the diagram and their values are $R_1 = R_2 = 5 \text{ k}\Omega$ and $R_3 = 100 \text{ k}\Omega$. The circuit is connected to 220 V alternating.

Check that we have eifs that come out vertically from the assembly. In addition, check that these eifs are of variable "intensity" and of variable nature, periodically.



Answers {2} 1) When the barrel axis is in the N~S_ direction. (North of form - South of form), the eifs are very low. Their intensity and their range follow as one turns and they are maximum when the axis is perpendicular to this direction. >

2) The barrel with a vertical axis has eifs following this axis above and below.

If we put a last barrel whose axis coincides with the first, the eifs are increased in power; their nature is preserved. If we put 3 or 4, there is still an increase.

Answers {3} 1) V_i w .
-----M

We find the vertical of the center of symmetry

The equivalence level is HhA RTs

Answers {5} The eif is changed to its symmetric with respect to the center of the transmitter platter.

2) The eif is unchanged.

3) Yes.

Answers {6J 1) Above the equatorial plane
below the pole, from the V .

on the axis, we find V_{M} in-

2) If it is made of wood or plaster found above the equatorial plane of the V_{E}
and below the pole of the .

The nature of the material comes into play.

Note that if the half-sphere (which is full) is not perfect, everything can change. Handling with the hemispheres is very delicate because a very slight deformation or a very slight variation in position changes absolutely everything.

3) We find V_{M} above the pole and V_{M} below when it is in M
lead.

Answers {1) 1) There is V at the upper pole and V at the lower pole.

2) The fact that the sphere is made of lead increases the range of the undifferentiated eifs above and below the sphere.

3) A sphere covered in green, red, blue, etc. paint. may emit something other than what was previously found. Each case is to be studied separately.

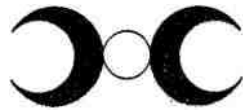
4) We will find eifs of the undifferentiated spectrum V^+ , J, O, R, IR, N, V, B,

UV., V_{I} , I, B. But the position of these eifs on the circle will change with
the position of the sun.

&

Answers {8} 1) We find the undifferentiated eifs (all the discrete spectrum)

2) An undifferentiated eif is detected. It is detected at 131° on the transmitter plate. To detect it on the axis of the ogive we can use a cylindrical pendulum with the shape below drawn on it.



3) It transforms them according to their symmetry with respect to the center of the board transmitter.

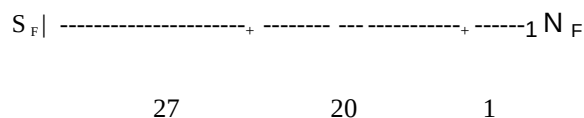
4) On the other side of the prism, the undifferentiated discrete spectrum is detected, but the eifs are in the reverse order of that found in the first question.

5) We put the vertical warhead about 3 cm from the transmitter plate, the needle of this indicating the BC. One detects behind the warhead the R and Vi.

6) Such a warhead, on which the N_F does not give the eif of Chartres. We detect when an eif is on its axis.

7) We no longer have the eif de Chartres but the N .

8) The M component is detected at $250^\circ 30'$ (you need a very fine needle) and the component E in the symmetrical position with respect to the center of the plate. One detects the eif of Chartres when it is placed as below compared ^{TO THE N}_{F SF -}



You can make the barrel independent of the direction by planting on the groove between the 20 and the 27 a small metal nail (height 3 to 5mm) if the barrel has a section of 1cm on the side or in diameter depending on whether it is at square or circular section.

Remark

De la Foÿe points out that when two sloping planks are supported freely on each other to form an angle, the Chartres eif is found at the point of contact.

Answers {9} 1) We recover V_M . This also makes it possible to load a witness better than with a single needle on the .

2) We have $V_{,,}$.
h

3) Initially, appears du and the whole spectrum on the right and left in M ,
then the UVi. After a while everything falls apart and appears the first £J
C. of T. 1 i ----i T

then 2 i——i 2

3 l -1 3

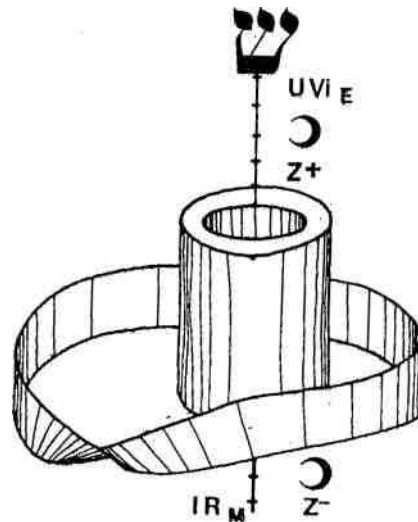
4 i -1 4

We don't go any further.

This is a reminder of what happens 3/4 hours after the Sprink device is powered up.

Answers {10} 1) The cylinder itself emits
cylinder begins to emit the components of the vital field
We have the shin right above the cylinder, in the axis. On
plane UVi,, , Chartres eif (M), Z +, and below
Chartres (M) and IR^

in its horizontal plane. the
normally placed,
detects above the
of the Z plane, the eif of



There are no polarities. We find what we detect on a healthy plant except that we do not have the life node at the base of the cylinder.

2) The shin is reversed above on the axis*. We also have UVi₁₂. Z is M replaced by another eif; ditto for the eif of Chartres (M). Similarly below Z is replaced by another eif. Ditto for the eif of Chartres (M), We have IRg.

The Chartres eif (M) above is replaced by an eif which is placed at 350° and below by an eif which is placed by symmetry in the spectrum E.

Z + and Z are replaced by an eif which is at 325°. In the horizontal plane on which the cylinder is placed, we find an answer but it is not at all a knot of life.

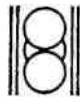
Above the plane, in the axis the 2 pendulums + and - held together do not react. We find the components of the CV but distributed as follows:

Form North

due t



East



South

There is a simple symmetry of the N and S.

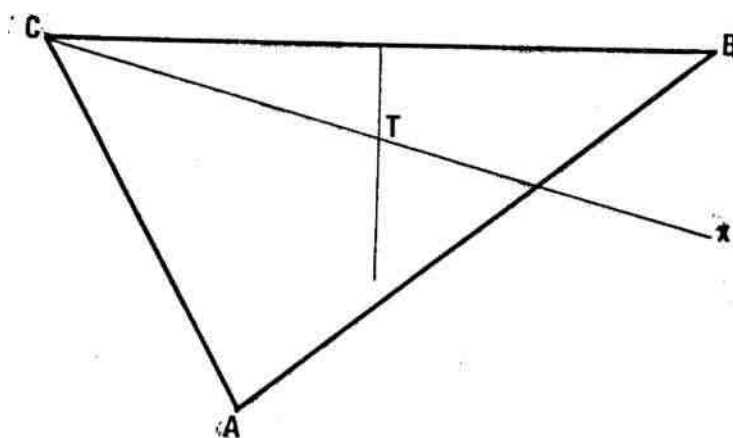
3) Switching to Mobius bands D or G (right or left) at 3,5,7 half-turns does not bring any changes.

Answers {11} 1) Black diagram for the N,, .

31

2) Nope

3) Yes. If A,B,C are the respective vertices of the angles 80° - 40° - 60° , and Cx the half-line forming an angle of 15° with CB , that is to say $(Cx, CB) = 15^\circ$. Let M be the midpoint of CB. The perpendicular bisector at M intersects Cx at T. T is the sought point.

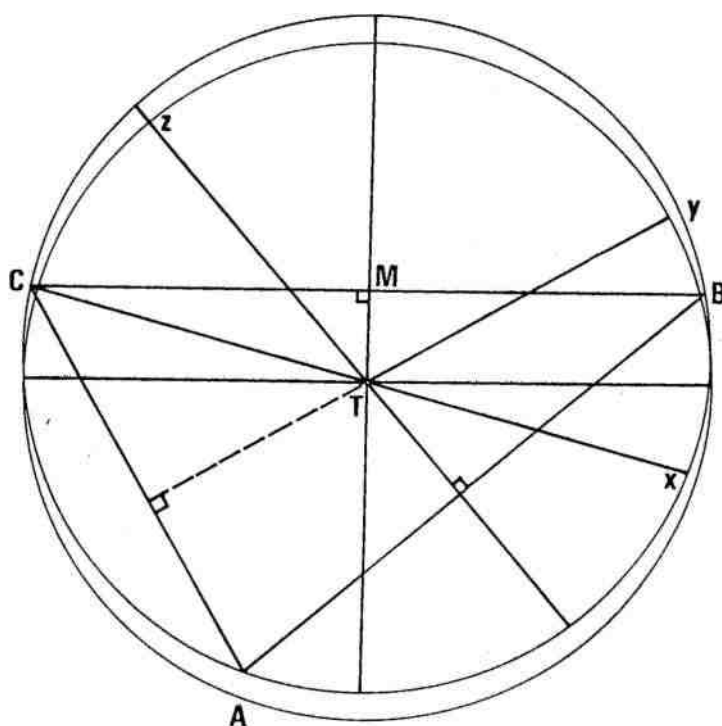


Answers {12} 2) There is a resume. The portion of the plane to the left of CA and delimited by extended BC and BA is negative. The portion of the plane to the right of BA and bounded by CA and extended CB is positive. The sense of T_y is positive in a portion that begins a little outside triangle ABC.

5) Yes.

If you want to have it with the same needle, you have to take it with a length less than or equal to TM .

Consider the homothetic triangle of ABC, in the homothety with center T and ratio k . We can have an undifferentiated eif with a needle of length less than or equal to the length of the segment carried by the ray issuing from T and of which one end is at T and the other obtained as the intersection of this ray with the side of the homothetic triangle of ABC in the dilation noted above.



Answers {13}

- 1) ∇M above and ∇E below.
- 2) ∇E above and ∇M below.
- 3) In 0 we detect with artifice a vertical ∇ , and without artifice the E knot of
- 4) The cylinder capped with the circle considered in 2) and emits above in its axis the ∇E below the ∇t .
- It is on an angle bisector constituted by the 2 axes of the guns.
- 5) The surface of the table meets M ch A Sh Ph (26-21-16-13) when we walk above a ring.

Answers {16} 1) It is placed at 200° in the differentiated spectrum in M and for E, at 20° .

2) The cH Y cannon with a small point in the groove separating the 8 from the 10, emits the node of life (M) at the end 10 and (E) at the end 8. (The answer is weaker).

the cannon cH YH emits life node (M) at end 10 and (E) at 1 end to avoid 5 (the 5 at N). It is necessary to plant a point between the 8 and the 10 orientation to the N.

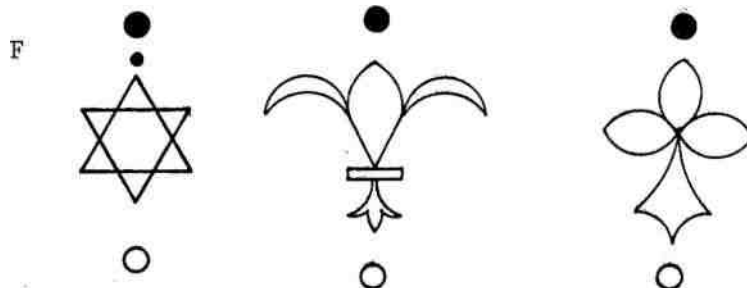
the F

cannon cH YYM emits the life node (M) at the 2 ends of the cannon, whatever the orientation.

Answers {17} 1) 130

3) Yes and vertically above these shapes.

4) For the star of David, the fleur-de-lys and the ermine of Brittany, we impose



On the Star of David it is detected vertically from the point. On the other 2 shapes, it is found in the plane on the axis of symmetry of the shapes, above them.

5) $s_F \quad i \quad \overset{20}{\text{-----}} \rightarrow \overset{24}{\text{---}} \quad ' \quad \text{---} \quad ' \quad N \quad F$

We detect Dabar on the side of the end 4. If we plant a small point in the groove between 2 and 20, we impose the N,, .. It is good to carry on the barrel £ the numbers 4-2-20 not to work on 2 - 1 - 10 . We detect the Z^+ at 250° and the Z^* by symmetry with respect to the center of the transmitter plate. M E

Answer {18} 2) The life node disappears immediately. Z^+ and Z^- still detect each other a little and slowly disappear, as does the eif of Chartres (M).

3) The node of life, if the level of the ground is not too high compared to the initial level, is always detected on the surface of the ground. If the level is too high, it is no longer detected.

4) The life node, Z and Z disappear.

Replies ' {19}

1) 220°

2) $S_F > \overset{16}{\text{-----}+} \overset{13}{\text{-----}T} \overset{21}{\text{-----}H_{NF}}$

We detect when it is properly oriented on the side of 21. By planting a small point in the groove marking 13 and 21, we make it independent of the orientation.

4) Z is detected at 290° on the transmitter plate. Z_ to the symmetric with respect to the center.

Answer {20} It has a differentiated eif at 230° . It has at E , when placed between 4 vertical lines, an eif at 50° .

Index of new terms and expressions and their abbreviations for certain

Aether (Æ)

primer

Ancron

Cumulative shift arithmetic (ACD)

Atmosphere

aureolar

Breakout of the Global

Coherence field (C. of C.)

usual (rational) field of coherence

other field of coherence

physical field

Vital field (VC)

Psychic Field (CP)

Field of Taofel (C. of T.)

Location path

F1 -

Overlap (F overlapped by F' : F)

Creation - from the past into the future

- from the past in the past

Cumulative

Cumulative shift

Shifted

Relocation - relocated

Procedure

Dynamic duality (Dd) (A in Dd with B: A> -----1 B)

Dynamism

Collapse

Eif (EF)

Emergence

"Mass broadcasts"

"Mass emissions"

Enel (£')

Winding

envelope of eifs

Shape envelope

Equimsey (F in equimsey with F': F/F')

wandering

Etallance

State

Space-Time (ET)

Ext

Form

Form and background

Plated form

Form that depends on the moment

Intestable testable form

Unlocatable form

Open form

Form-formless

Form that contains its time

Barred form

Formeriade "decalco" type shape

Fractant

parent

Slippage

Overall

Globallance

Great Gnosis

Gate

Form harmonic

immersion

Local

Location

Macron A <-----
 A *i* --B (A begets B)

Memoid

Nebular

BCM level

Equivalence level

Number crossed out

Fractal Object (OF)

Technical object (OT) aesthetic (OE) phenist (OP)

Obs.

Observer

Sheaf of Thought

Magfeu Pansy (M-type)

Structure transportation principle

Polaror (Pol)

Radius of Coherence (R. of C.)

RBB

Reflection

Scuba and form

Continuous spectrum

differentiated undifferentiated

On in

Support

Non-Cartesian System (SNC)

TAG system

AG type system

Route

Transfer

Véhrel (Cultural vehicle)

Voiloc (Location Neighborhood)

Square Hebrew: Numerical Values

| | kabbalistic | by G. Bardet | | kabbalistic of G. Bardet | |
|-------------|-------------|--------------|----|--------------------------|-----|
| K | 1 | 1 | | 30 | 12 |
| 3 | 2 | 2 | ⸏ | 40 | 13 |
| | 3 | 3 | ⸒ | 50 | 14 |
| “ | 4 | 4 | ⸐ | 60 | 15 |
| H | 5 | 5 | ⸑ | 70 | 16 |
| | 6 | 6 | ⸒ | 80 | 17 |
| T | 7 | 7 | | 90 | 18 |
| n | 8 | 8 | | 100 | 19 |
| ⸐ | 9 | 9 | n | 200 | 20 |
| | 10 | 10 | | 300 | 21 |
| ⸏ | 20 | 11 | n | 400 | 22 |
| ⸐ | | | | | |
| ⸐ | 500 | 600 | ⸐ | 700 | 800 |
| kabbalistic | | | | | 900 |
| G.Bardet | 23 | 24 | 25 | 26 | 27 |

In Kabbalah there is a classification of letters according to 3 plans. This has not been respected here. The values assigned to the 5 finals are optional. The plan concept should be reviewed according to the appearance of this work. But we can't treat it; this would require too much development. Letters are

pointed. This is necessary as explained earlier in the book. In the domain of eifs, it is Bardet's numerical values that give satisfactory results because the notion of number usually understood is used. If we took kabbalistic values, we would have to completely reject the notion of number; for example 2 is not $1+1$, the same for the tens and the hundreds.

A few Hebrew words (modified by Bardet and La Foÿe) carried on clocks. The reason for their modification is cited in the book.

LIVING

Y cH

EIFS OF CHARTRES

pN
Ts R hA

TSOROR LIFE

rrn
HY cH

liix
D „ D

SHADAIM

Of i»
MYD Sh

EHAD = ONE

"N
OT

DEMONIC POWERS

THE NIGHT

לילה
H L Y L

Ph Sh Ph chA M cH hA

SHATAN

HELOHIM

NT Sh

MYHL hA

MAGIC

t]O
E

A, K Sh Ph must be added

K.Sh.Ph.

KoShcPho

. K Sh Ph „

and there are many more

IT WILL SPOUT WATER

B'uijcr

MYMW hA Ts Y

(THE) SPIRIT

nin

cH WR

RUAH (SPIRITUAL)

THE BREATH OF LIFE

rrr^

HY cH Sh Ph NL

(VITAL - equivalence level)

EARTH

Ts R hA H

HAARETZ (PHYSICS -
equivalence level)

TALK - TALK

TO LISTEN

RB D

DABAR (THE VERB)

chA M Sh

YHWH

HiIT 1

HW HY

(THE) LIGHT

“B

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